Calculus

Author: Star
Email: wusida@westlake.edu.cn

Instructors: Prof. Yabin Zhang
Dr. Sean Brendon Lynch
Emails: zhangyabin@westlake.edu.cn
sblynch@westlake.edu.cn

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Chapter 1 Preparations: Ten Axioms

There are many ways to introduce the real-number system. One popular method is to begin with the positive integers $1, 2, 3, \ldots$ and use them as building blocks to construct a more comprehensive system having the properties desired. Briefly, the idea of this method is to take the positive integers as undefined concepts, state some axioms concerning them, and then use the positive integers to build a larger system consisting of the positive rational numbers (quotients of positive integers). The positive rational numbers, in turn, may then be used as a basis for constructing the positive irrational numbers (real numbers like $\sqrt{2}$ and π that are not rational). The final step is the introduction of the negative real numbers and zero. The most difficult part of the whole process is the transition from the rational numbers to the irrational numbers.

The point of view we shall adopt here is nonconstructive. We shall start rather far out in the process, taking the real numbers themselves as undefined objects satisfying a number of properties that we use as axioms. That is to say, we shall assume there exists a set $\mathbb{R}$ of objects, called real numbers, which satisfy the 10 axioms listed in the next few sections. All the properties of real numbers can be deduced from the axioms in the list. When the real numbers are defined by a constructive process, the properties we list as axioms must be proved as theorems.

1 Fields and the Field Axioms

Definition 1.1.1 (Field Axioms) A **field** $\mathcal{K}$ is a set equipped with addition and multiplication, denoted by $(\mathcal{K}, +, \cdot)$, satisfying the following axioms:

1. Associative Laws: let $a, b, c \in \mathcal{K}$,

$$a + b + c = a + (b + c), \qquad a \cdot b \cdot c = a \cdot (b \cdot c).$$

2. Existence of Identity Elements: let $a \in \mathcal{K}$,

$$a + 0_{\mathcal{K}} = a, \qquad a \cdot 1_{\mathcal{K}} = a,$$

   where $0_{\mathcal{K}}$ and $1_{\mathcal{K}}$ is called the additive identity and multiplicative identity.

3. Existence of Negative: for any $a \in \mathcal{K}$, there exists one element in $\mathcal{K}$ denoted by $(-a)$ such that

$$a + (-a) = 0_{\mathcal{K}}.$$

4. Existence of Reciprocal: for any $a \in \mathcal{K}$ and $a \neq 0_{\mathcal{K}}$, there exists one element in $\mathcal{K}$ denoted by a^{-1} such that

$$a \cdot a^{-1} = 1_{\mathcal{K}}.$$

5. Commutative Laws: let $a, b \in \mathcal{K}$,

$$a + b = b + a, \qquad a \cdot b = b \cdot a.$$

6. Distributive Law: let $a, b, c \in \mathcal{K}$,

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

Remark 1.1.2 Closure is contained in the defined addition and multiplication.

Notation 1.1.3 For $a, b \in \mathcal{K}$, we usually write $a - b$ to replace $a + (-b)$ and write $\frac{a}{b}$ or a/b to replace $a \cdot b^{-1}$ when $b \neq 0_{\mathcal{K}}$.

Theorem 1.1.4 (Cancellation Law) Let $a, b, c \in \mathcal{K}$, if $a + c = b + c$, then $a = b$.

Corollary 1.1.5 $0_{\mathcal{K}}$ and $1_{\mathcal{K}}$ in **Field Axiom 2** is unique.

Example 1.1.6 $\mathbb{C}, \mathbb{R}, \mathbb{Q}$ are fields, but $\mathbb{N}$ is not a field because $2 \in \mathbb{N}$, but its reciprocal $\frac{1}{2} \notin \mathbb{N}$. This is the main difference between $\mathbb{Q}$ and $\mathbb{N}$.

Definition 1.1.7 Let $\mathcal{F}$ be a field and a set $\mathcal{K} \subseteq \mathcal{F}$. We shall say $\mathcal{K}$ a **subfield** of $\mathcal{F}$ when $\mathcal{K}$ satisfies the followings:

1. If x, y are elements of $\mathcal{K}$, then $x + y \in \mathcal{K}$ and $x \cdot y \in \mathcal{K}$.
2. If $x \in \mathcal{K}$, then $(-x) \in \mathcal{K}$. If furthermore $x \neq 0$, $x^{-1} \in \mathcal{K}$.
3. $0_{\mathcal{K}}$ and $1_{\mathcal{K}}$ are in $\mathcal{K}$.

Remark 1.1.8 Definition 1.1.7 provides a simpler way to prove a set to be a field.

Proposition 1.1.9 Let $\mathcal{K}$ be a field and A, B are subfields of $\mathcal{K}$, then $A \cap B$ is also a subfield of $\mathcal{K}$.

Proposition 1.1.10 Let $\mathcal{K}$ be a field and A, B are subfields of $\mathcal{K}$, then $A \cup B$ is also a subfield of $\mathcal{K}$ if and only if $A \subseteq B$ or $B \subseteq A$.

Example 1.1.11 $\mathbb{R}$ is a subfield of $\mathbb{C}$.

2 The Order Axioms

Definition 1.2.1 (Order Axioms) We shall assume that there exists a set $\mathbb{R}^+ \subset \mathbb{R}$, called the set of positive real numbers, which satisfies the followings:

1. For any $x, y \in \mathbb{R}^+$, we have $x + y \in \mathbb{R}^+$ and $x \cdot y \in \mathbb{R}^+$.

2. For any real $x \neq 0$, either $x \in \mathbb{R}^+$ or $(-x) \in \mathbb{R}^+$, but not both.
3. $0 \notin \mathbb{R}^+$.

Corollary 1.2.2 $\mathbb{R}^+$ is non-empty since $1 \in \mathbb{R}^+$.

Notation 1.2.3 For any $a, b \in \mathbb{R}$, we define the symbols $>$, $<$, $\geq$, $\leq$ as below:

$$x > y \text{ means that } x - y \text{ is positive.}$$

$$x < y \text{ means that } y > x.$$

$$x \geq y \text{ means that } x > y \text{ or } x = y.$$

$$x \leq y \text{ means that } y \geq x.$$

Theorem 1.2.4 (Trichotomy Law) For any $a, b \in \mathbb{R}$, exactly one of the three relations holds: $a > b$, $a < b$ and $a = b$.

Remark 1.2.5 Theorem 1.2.4 is always used to prove real number equality, while another commonly-used way is to prove both $a \geq b$ and $a \leq b$ hold. The latter one is similar to the method of proving set equality.

Theorem 1.2.6 (Transitive Law) For any $a, b, c \in \mathbb{R}$, if $a < b$ and $b < c$, then $a < c$.

3 The Completeness Axiom

Definition 1.3.1 Let S be an nonempty set of real numbers. For any $x \in S$, if we have $x \leq B$, then we say S is **bounded above** by B and B is an **upper bound** of S. Furthermore, if $B \in S$, then we say B the **maximum** of S, denoted by $\max S$.
Similarly, we can define a **lower bound** and a **minimum** of a nonempty set. A minimum of S is denoted by $\min S$.

Definition 1.3.2 A number B is called a **least upper bound** (a **supremum**) of a nonempty set S, denoted by $\sup S$, if B satisfies the followings:

1. B is an upper bound of S.
2. There's no number smaller than B being an upper bound of S.

Similarly, we can define a **greatest lower bound** (a **infimum**) of a nonempty set. An infimum of S is denoted by $\inf S$.

Axiom 1.3.3 (Completeness Axiom) Every nonempty set S of real numbers which is bounded above has a supremum, i.e., there exists such a number B that $B = \sup S$.

Remark 1.3.4 The completeness axiom indicates a crucial difference between $\mathbb{Q}$ and $\mathbb{R}$ because $\mathbb{Q}$ is not complete. For instance, it's impossible to find a least upper bound in $\mathbb{Q}$ for the set $\{x \in \mathbb{Q} : x^2 < 2\}$.

Definition 1.3.5 A set S of real numbers is said to be **bounded** if there exists a real number $M \geq 0$ such that

$$|x| \leq M \quad \text{for all } x \in S.$$

Theorem 1.3.6 Every nonempty set of real numbers which is bounded below has a infimum.

Theorem 1.3.7 The set $\mathbb{N}^+$ of positive integers $1, 2, 3...$ is not bounded above.

Theorem 1.3.8 For any $x \in \mathbb{R}$, there exists an positive integer n such that $n > x$.

Theorem 1.3.9 If $x > 0$ and y is an arbitrary real number, there exists an positive integer n such that $nx > y$.

Theorem 1.3.10 If three real numbers x, y, a satisfy:

$$a \leq x \leq a + \frac{y}{n}$$

for every integer $n \geq 1$, then $x = a$.

Proposition 1.3.11 Suppose x and y are arbitrary real numbers. If $x < y$, then there exists a rational number r such that $x < r < y$. Moreover, there exists infinite many different rational numbers $r_1, r_2, ...$ such that $x < r_i < y$ for all $i \in \mathbb{Z}^+$.

Remark 1.3.12 Proposition 1.3.11 is often described by saying that rational numbers are **dense** in the real numbers.

Proposition 1.3.13 Suppose x and y are arbitrary real numbers. If $x < y$, then there exists a irrational number z such that $x < z < y$. Moreover, there exists infinite many different irrational numbers $z_1, z_2, ...$ such that $x < z_i < y$ for all $i \in \mathbb{Z}^+$.

Remark 1.3.14 Proposition 1.3.13 is often described by saying that irrational numbers are **dense** in the real numbers.

Proposition 1.3.15 If x is a rational number, $x \neq 0$, and y is an irrational number, then $x + y, x - y, xy, \frac{x}{y}, \frac{y}{x}$ are all irrational.

Theorem 1.3.16 Let S be an nonempty set of real numbers. If S has a supremum, then there exists an element $x \in S$ such that for any $h \in \mathbb{R}^+$:

$$x > \sup S - h$$

If S has an infimum, then there exists an element $y \in$ S such that for any $h \in \mathbb{R}^+$:

$$y < \inf S + h$$

Remark 1.3.17 Theorem 1.3.16 shows two properties of supremum (infimum) vividly and precisely. On one hand, it's greater (smaller) than or equal to any element of S. On the other hand, it's smaller (greater) than or equal to any upper bounds (lower bounds). This theorem also shows the uniqueness of supremum (infimum).

Definition 1.3.18 Let A, B be two sets. We define the **sum** of A and B as below:

$$A + B := \{x \,|\, x = a + b, a \in A, \, b \in B\}$$

Theorem 1.3.19 Let A, B be two non-empty sets of real numbers. If both A and B are bounded above, then:

$$\sup(A + B) = \sup A + \sup B$$

If both A and B are bounded below, then:

$$\inf(A + B) = \inf A + \inf B$$

Theorem 1.3.20 Let A, B be two non-empty sets of real numbers. If for any element $x \in A$ and $y \in B$, we have $x \leq y$. Then:

$$\sup A \leq \inf B$$

4 Inductive Sets and Induction

Definition 1.4.1 We call a set S an **inductive set** if it satisfies the followings:

1. $1 \in S$.
2. if $x \in$ S, then $x + 1 \in S$.

Definition 1.4.2 Let I be an inductive set, we define the set of all **positive integers** $\mathbb{Z}^+$ as the intersection of all inductive subsets of I:

$$\mathbb{Z}^+ := \bigcap \{A \subseteq I : A \text{ is inductive}\}.$$

Theorem 1.4.3 (Principle of Mathematical Induction) Let S be a set of positive integers which has the following two properties:

1. $1 \in S$.
2. If an integer $k \in S$, then so is $k + 1$.

Then every positive integer is in the set S.

Remark 1.4.4 When we're proving by induction, we're applying Theorem 1.4.3 to the set $S := \{n \in \mathbb{Z}^+ : A(n + n_1 - 1) \text{ is true}\}$.

Method 1.4.5 (First Method of Proof by Induction) Let $A(n)$ be an assertion involving an integer n. We conclude that $A(n)$ is true for every $n \geq n_1$ if we can perform the following two steps:

1. Prove that $A(n_1)$ is true.
2. Let k be an arbitrary but fixed integer $\geq n_1$. Assume that $A(k)$ is true and prove that $A(k+1)$ is true

Remark 1.4.6 If we assume that $A(n)$ is true for every integer $n \leq k$ and prove that $A(k+1)$ is true, this method is called the second method of proof by induction.

Theorem 1.4.7 (Well-Ordered Principle) Every nonempty set of positive integers has a smallest member.

Proof. We shall prove by contradiction.
Let T a nonempty set of positive integers which doesn't has a smallest member, we define a set S:

$$S := \{x : x < t \text{ for all } t \in T\}$$

Then $1 \notin T$ because 1 is the smallest positive integer. Thus, $1 \in S$.
For a positive integer k, if $k \in S$, that is, $k < t$ for all $t \in T$. We shall consider $k+1$. For these two statements $k+1 \in S$ and $k+1 \notin S$, only one of them holds. If we could prove both of them are false, then we reach a contradiction.

- If $k+1 \notin S$, then there exists $t_1 \in T$ such that $k+1 \geq t_1$. Since T has no smallest member, there exists $t_2 < t_1$ and we duduce $k+1 > t_2$, that is, $k \geq t_2$, which contradicts with $k > t$ for all $t \in T$.
- If $k+1 \in S$, then we shall perform the principle of induction to S, that is, every positive integer is in S. Since T is nonempty, there exists one positive integer $t \in T$. However, this t is also in S, then we deduce $t < t$ by the definition of S, which is impossible.

Hence, the nonempty set T must have a smallest member. □

Example 1.4.8 We introduce one typical example of proving by well-ordered principle.
Let n be a positive integer.

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

It's trivial that when $n = 1$, the equality holds. Then there're two possible statements:

- For all positive integer, the equality holds.
- There exists at least one positive integer such that the equality doesn't hold.

We shall prove statement 2 is wrong so that statement 1 naturally holds. We apply the well-ordered principle to the collection of all positive integer that satisfy statement 2, thus we get one positive integer $k > 1$ and for $k-1$ the equality holds. Now we add k to both sides of equality and find the equality also holds for k, which leads to a contradiction. Thus statement 1 holds.

Remark 1.4.9 Theorem 1.4.7 is one important property of positive integers since it doesn't hold for arbitrary sets of integers. For example, $\mathbb{Z}$ doesn't have smallest member.

Notation 1.4.10 We define the summation notation as below:

1. $\displaystyle\sum_{i=1}^{1} a_i = a_1$.

2. $\sum_{i=1}^{n+1} a_i = \left(\sum_{i=1}^{n} a_i\right) + a_{n+1}, n \geq 1.$

Example 1.4.11 We shall see another example of inductive definition. Suppose x a positive real number and consider the following inductive definition:

- Let a_0 denote the largest integer $\leq x$.
- Having chosen a_0, we let a_1 denote the largest integer such that $a_0 + \frac{a_1}{10} \leq x$.
- More generally, having chosen $a_0, a_1, \cdots, a_{n-1}$, we let a_n denote the largest integer such that

$$\sum_{i=0}^{n} \frac{a_i}{10^i} \leq x.$$

Define set S to be the set of all numbers obtained in the above way, i.e.,

$$S := \left\{\sum_{i=0}^{n} \frac{a_i}{10^i} \;:\; n \in \mathbb{Z}^+\right\} \cup \{0\}$$

We have the following propositions:

1. S is bounded above and $x = \sup S$.
2. Let $a_0, a_1, a_2, \cdots$ be as the definition of S. We define the notation $a_0.a_1a_2a_3\cdots$ to be the **decimal expansion** of x. Then if we replace all "$\leq$" with "$<$", then one real number might have two different decimal expansions. This is merely a reflection of the fact that two different sets of of real numbers can have the same supremum.
3. Note that x satisfies the following inequality:

$$\sum_{i=0}^{n-1} \frac{a_i}{10^i} + \frac{a_n}{10^n} < x < \sum_{i=0}^{n-1} \frac{a_i}{10^i} + \frac{a_n + 1}{10^n}$$

   This gives us a way to approximate x from above and below, which could also serve as a way to construct irrational numbers from rational numbers.

5 Square Root

Theorem 1.5.1 Every nonnegative real number a has one unique square root, denoted by $a^{1/2}$ or $\sqrt{a}$.

Proof. If $a = 0$, then 0 is clearly the unique nonnegative number whose square is 0. Hence we assume $a > 0$. Define

$$S := \{x \in \mathbb{R} : x^2 \leq a\}.$$

We shall define $\sqrt{a}$ as the least upper bound of S, and then prove that its square is exactly a.
First, S is nonempty since $0 \in S$. Also, $a + 1$ is an upper bound for S. Indeed, if $x > a + 1$, then

$$x^2 > (a+1)^2 > a,$$

so $x \notin S$. Hence no element of S can be greater than $a+1$.
Therefore, by the completeness axiom, S has a supremum. Let

$$b := \sup S.$$

We prove that $b^2 = a$.
First suppose, for contradiction, that $b^2 > a$. The idea is to move slightly to the left of b, but still remain above all elements of S. Choose

$$t := \frac{b^2 - a}{2b} > 0, \qquad c := b - t.$$

Then $c < b$, and

$$c^2 = (b-t)^2 = b^2 - 2bt + t^2 > b^2 - 2bt = b^2 - (b^2 - a) = a.$$

Thus every $x \in S$ must satisfy $x < c$. Therefore c is an upper bound for S, but $c < b$, contradicting $b = \sup S$. Hence $b^2 > a$ is impossible.
Next suppose, for contradiction, that $b^2 < a$. The idea is to move slightly to the right of b, while still staying inside S. Choose

$$t := \min\left(1, \frac{a - b^2}{2b+1}\right) > 0, \qquad c := b + t.$$

Then $c > b$. Since $t \le 1$, we have $t^2 \le t$. Hence

$$c^2 = (b+t)^2 = b^2 + 2bt + t^2 \le b^2 + (2b+1)t \le b^2 + (a - b^2) = a.$$

Therefore $c \in S$. But this contradicts the fact that b is an upper bound for S, since $c > b$. Hence $b^2 < a$ is impossible.
By trichotomy, the only remaining possibility is

$$b^2 = a.$$

The uniqueness is trivial by the uniqueness of supremum. □

Remark 1.5.2 This proof shows the beauty of the completeness axiom.

6 Absolute Value and Inequalities

Definition 1.6.1 We define the **absolute value** of a real number x as below:

$$|x| := \begin{cases} x, \ x \ge 0 \\ -x, \ x < 0 \end{cases}$$

Theorem 1.6.2 If $a \ge 0$, $|x| \le a$ if and only if $-a \le x \le a$.

Theorem 1.6.3 (Triangle Inequality) If $x, y \in \mathbb{R}$, then

$$|x + y| \leq |x| + |y|.$$

The equality sign holds if and only if $xy \geq 0$.

Remark 1.6.4 Theorem 1.6.3 could be generalized to vectors or complex numbers. Taking $x = a-b,\ y = b-c$, then we have $|a - c| \leq |a - b| + |b - c|$, which is more commonly used.

Theorem 1.6.5 For arbitrary real numbers $a_1, a_2, \cdots, a_n$, we have:

$$\left|\sum_{i=1}^{n} a_i\right| \leq \sum_{i=1}^{n} |a_i|$$

Theorem 1.6.6 (Cauchy-Schwarz Inequality) For arbitrary real numbers $a_1, a_2, \cdots, a_n$ and $b_1, b_2, \cdots, b_n$, we have:

$$\left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right) \geq \left(\sum_{i=1}^{n} a_i b_i\right)^2$$

The equality sign holds if and only if there exists a real number x such that $a_i x + b_i = 0$ for any $i \in \{1, 2, \cdots n\}$.

Remark 1.6.7 The following two inequalities are more commonly used:

$$\left(\sum_{i=1}^{n} \frac{a_i^2}{b_i}\right)\left(\sum_{i=1}^{n} b_i\right) \geq \left(\sum_{i=1}^{n} a_i\right)^2$$
$$\left(\sum_{i=1}^{n} \frac{a_i}{b_i}\right)\left(\sum_{i=1}^{n} a_i b_i\right) \geq \left(\sum_{i=1}^{n} a_i\right)^2$$

Chapter 2 Integral Calculus

The integral is often introduced as a tool for calculating areas, volumes, and other quantities. But in a rigorous treatment, we cannot simply assume that every function can be integrated. Before we write down an integral and start computing, we should first ask:

Is the function we intend to integrate actually integrable?

This chapter answers this question in a gradual way. We start from the geometric idea of area and introduce the ordinate set of a nonnegative function. Then we define the integral of step functions, whose values are directly determined by finite sums of rectangle areas. For more general bounded functions, we approximate the function from below and above by step functions. The gap between these two approximations gives rise to the lower integral and the upper integral. A function is integrable precisely when these two quantities agree.

After establishing the definition, we study basic criteria for integrability, such as monotonicity, and prove the fundamental algebraic properties of the integral. We then apply the theory to compute elementary integrals and to describe the area between two graphs.

1 Basic Concepts

Definition 2.1.1 Let f be a correspondence. If each element of $\mathscr{D}_f$ has at most one image under f, we say that f is a **function**.

Remark 2.1.2 Refer to the Fundamental Algebra and Analysis if you're confused about the definition of correspondence, domain and image.

Definition 2.1.3 Suppose $f : \mathbb{R} \to \mathbb{R}$ is a function and $f(x) \geq 0$ for any $x \in [a, b]$. We define the set

$$\mathcal{S} := \left\{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b \text{ and } 0 \leq y \leq f(x)\right\}$$

to be the **ordinate set** of f.

Remark 2.1.4 Definitely $\mathcal{S}$ defined above is a plane region. Thus, we can naturally raise these questions: can we compute the area of $\mathcal{S}$, or more fundamentally, what is the area of $\mathcal{S}$, does it even make sense to talk about the area of $\mathcal{S}$? These questions motivate the axiomatic definition of area and measurable sets in $\mathbb{R}^2$. (Here a measurable set is a set that we can assign an area to.)

Definition 2.1.5 We assume that there exists $\mathscr{M}$ a collection of subsets of $\mathbb{R}^2$ and a function A with $\operatorname{Dom}(A) = \mathscr{M}$ and $\operatorname{Im}(A) \in \mathbb{R}$ such that the following properties all hold:

1. Nonnegative property: For any $S \in \mathscr{M}$, $A(S) \geq 0$.
2. Additive property: If $S, T \in \mathscr{M}$, then $S \cap T, S \cup T \in \mathscr{M}$. Additionally, $A(S \cup T) = A(S) + A(T) - A(S \cap T)$.
3. Difference property: If $S, T \in \mathscr{M}$ and $S \subseteq T$, then $T - S$ (also $T \backslash S$) $\in \mathscr{M}$. Additionally, $A(T - S) = A(T) - A(S)$.
4. Invariance under congruence: If $S \in \mathscr{M}$ and if $T \in \mathbb{R}^2$ is congruent to S, then $T \in \mathscr{M}$ and $A(T) = A(S)$.
5. Choice of scale: Every rectangle R is in $\mathscr{M}$. If the adjecent edges of R are of length w and h, then $A(R) = wh$.
6. Exhaustion Property: Suppose $Q \subseteq \mathbb{R}^2$. If there exists a unique $c \in \mathbb{R}$ such that $A(S) \leq c \leq A(T)$ for all step regions S and T with $S \subseteq Q \subseteq T$, then $Q \in \mathscr{M}$ and $A(Q) = c$.

We call such A an area function and each $S \in \mathscr{M}$ a **measurable set**.

Definition 2.1.6 (Additional) Some definitions are lost above and here we add it.

1. Two sets S and T in $\mathbb{R}^2$ are called **congruent** if there exists a **one-to-one correspondence** (or a **bijection**) between S and T such that the distance between points are preserved, i.e., if the two points $s_1 = (a, b),\ s_2 = (c, d)$ in S are mapped to the two points $t_1 = (x, y),\ t_2 = (p, q)$ in T under such a bijection, then we must have
$$\sqrt{(a-c)^2 + (b-d)^2} = \sqrt{(x-p)^2 + (y-q)^2}.$$
2. A set $R \subseteq \mathbb{R}^2$ is called a **rectangle** if it is congruent to a set of the form
$$\{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq w \text{ and } 0 \leq y \leq h\},$$
where $w \geq 0$ and $h \geq 0$.
3. A set $S \subseteq \mathbb{R}^2$ is called a **step region** if it is the union of a finite collection of adjacent rectangles whose bases all rest on the x-axis.

Theorem 2.1.7 Here are some theorems about some basic cases.

1. $S_1 := \{(a, b)\}$ where $a, b \in \mathbb{R}$ is measurable and $A(S_1) = 0$.
2. $S_2 := \{(a_1, b_1), (a_2, b_2), \cdots, (a_n, b_n)\}$ where $n \in \mathbb{Z}^+$ and $a_i, b_i \in \mathbb{R}$ for $i = 1, 2, \cdots, n$ is measurable with $A(S_2) = 0$.
3. If S_3 is the union of a finite collection of line segments in $\mathbb{R}^2$, then S_3 is measurable with $A(S_3) = 0$.
4. Suppose S_4 is the triangle with vertices $(0, 0), (w, 0), (\frac{w}{2}, h)$ where $w, h > 0$, S_4 is measurable and $A(S_4) = \frac{1}{2}wh$.

Definition 2.1.8 A **partition** P of a given closed interval $[a, b]$ is a collection of points $x_0, x_1, x_2, \cdots, x_n$ satisfying:
$$a = x_0 < x_1 < x_2 < \cdots < x_n = b$$
and we use the symbol
$$P = \{x_0, x_1, x_2, \cdots, x_n\}$$

to designate this partition. We call $[x_{k-1}, x_k]$ the kth closed subinterval and (x_{k-1}, x_k) the k-th open interval.

Remark 2.1.9 Though we use set notation to designate a partition, the order **does** matter.

Definition 2.1.10 A function s, whose domain is a closed interval $[a, b]$, is called a **step function** if there is a partition $P = \{x_0, x_1, \cdots, x_n\}$ of $[a, b]$ such that for each $k = 1, 2, \cdots, n$, there is a real number s_k satisfying

$$s(x) = s_k \quad \text{if} \quad x_{k-1} < x < x_k.$$

Example 2.1.11 We define a function $D(x)$ on $\mathbb{R}$:

$$D(x) := \begin{cases} 1, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

which is called the **Dirichlet Function**. It's easy to verify that $D(x)$ is not a step function. Even if we confine Dirichlet Function on $[0, 1]$, it is not a step function.

Remark 2.1.12 At each of the endpoints x_{k-1} and x_k the function must have some well-defined value, but this need not be the same as s_k.

Definition 2.1.13 For a given partition P of $[a, b]$, a **refinement** is a new partition P' satisfying $P \subseteq P'$ (order should be ensured). Then P' is said to be **finer** than P.

Definition 2.1.14 For two partitions P_1, P_2 of $[a, b]$, the union of P_1 and P_2 is called the **common refinement** of P_1 and P_2.

Theorem 2.1.15 The sum and product of two given step functions are also step functions.

2 Integral for Step Functions

Definition 2.2.1 Let s be a step function defined on $[a, b]$, and let $P = x_0, x_1, \cdots, x_n$ be a partition of $[a, b]$ such that:

$$s(x) = s_k \quad \text{if} \quad x_{k-1} < x < x_k, \quad k = 1, 2, \cdots, n$$

The **integral** of s from a to b, is defined as:

$$\int_a^b s(x)\,\mathrm{d}x := \sum_{k=1}^{n} s_k \cdot (x_k - x_{k-1})$$

Remark 2.2.2 The definition is constructed so that the integral of a nonnegative step function is equal to the area of its ordinate set.

Theorem 2.2.3 Here are some properties of the integral of a step function. If not stated, $s(x)$ and $t(x)$ are step functions well defined on $[a, b]$.

1. Linearity Property:

$$\int_a^b [c_1 s(x) + c_2 t(x)]\,\mathrm{d}x = c_1 \int_a^b s(x)\,\mathrm{d}x + c_2 \int_a^b t(x)\,\mathrm{d}x.$$

   for every real c_1 and c_2.

2. Comparison Theorem: if $s(x) < t(x)$ for every x in $[a, b]$, then

$$\int_a^b s(x)\,\mathrm{d}x < \int_a^b t(x)\,\mathrm{d}x.$$

3. Additivity with respect to the interval of integration:

$$\int_a^c s(x)\,\mathrm{d}x + \int_c^b s(x)\,\mathrm{d}x = \int_a^b s(x)\,\mathrm{d}x \quad \text{if} \quad a < c < b.$$

4. Invariance under translation:

$$\int_a^b s(x)\,\mathrm{d}x = \int_{a+c}^{b+c} s(x-c)\,\mathrm{d}x \quad \text{for every real } c.$$

5. Expansion or Contraction of the interval of integration:

$$\int_{ka}^{kb} s\left(\frac{x}{k}\right)\,\mathrm{d}x = k \int_a^b s(x)\,\mathrm{d}x \quad \text{for every } k > 0.$$

Remark 2.2.4 Since the proofs of properties above are quite similar, we shall only prove Property 7 here as examples.

Proof of Property 7. Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$ such that

$$s(x) = s_k \quad \text{if} \quad x_{k-1} < x < x_k \ (k = 1, 2, \ldots, n).$$

Let

$$t(x) = s\left(\frac{x}{k}\right) \quad \text{if} \quad ka \le x \le kb.$$

Then

$$t(x) = s_i \quad \text{if} \quad kx_{i-1} < x < kx_i.$$

Hence $P' = \{kx_0, kx_1, \ldots, kx_n\}$ is a partition of $[ka, kb]$, and t is a step function whose integral is

$$\int_{ka}^{kb} t(x)\,\mathrm{d}x = \sum_{i=1}^{n} s_i \cdot (kx_i - kx_{i-1}) = k \sum_{i=1}^{n} s_i \cdot (x_i - x_{i-1}) = k \int_a^b s(x)\,\mathrm{d}x.$$

□

Definition 2.2.5 If $a > b$, then

$$\int_a^b s(x)\,\mathrm{d}x := -\int_b^a s(x)\,\mathrm{d}x.$$

3 Integral for More General Functions

Definition 2.3.1 If f is a real function, we say f is **bounded** on $[a, b]$ if there exists a number $M > 0$ such that

$$|f(x)| \leq M \quad \text{for every } x \in [a, b].$$

Particularly, if $f \leq M$ for some real M, we say f is **bounded above** on $[a, b]$. If $f \geq m$ for some real m, we say f is **bounded below** on $[a, b]$.

Definition 2.3.2 If f is a function defined and bounded on $[a, b]$, and s, t are arbitrary step functions satisfying

$$s(x) \leq f(x) \leq t(x) \quad \text{for every } x \in [a, b].$$

And for every pair of s and t, there exists one and only one number I such that

$$\int_a^b s(x)\,\mathrm{d}x \leq I \leq \int_a^b t(x)\,\mathrm{d}x \quad \text{for every } x \in [a, b],$$

then I is defined as the **integral** of f and denoted by

$$I := \int_a^b f(x)\,\mathrm{d}x.$$

When such I exists, we say f is integrable on $[a, b]$.

Remark 2.3.3 This definition is quite similar to exhaustion property. We will see they work together in the following text.

Definition 2.3.4 Let f be a real function bounded on $[a, b]$ and s, t be two step functions satisfying $s \leq f \leq t$, we define

$$S := \left\{ \int_a^b s(x)\,\mathrm{d}x \,\middle|\, s \leq f \right\}, \quad T := \left\{ \int_a^b t(x)\,\mathrm{d}x \,\middle|\, f \leq t \right\}.$$

Since f is bounded, S and T are nonempty sets. By $s \leq t$, we have

$$\int_a^b s(x)\,\mathrm{d}x \leq \sup S \leq \inf T \leq \int_a^b t(x)\,\mathrm{d}x.$$

We say $\sup S$ the **lower integral** of f and $\inf T$ the **upper integral** of f denoted by

$$\underline{I}(f) = \sup \left\{ \int_a^b s(x)\,\mathrm{d}x \,\middle|\, s \leq f \right\}, \qquad \overline{I}(f) = \inf \left\{ \int_a^b t(x)\,\mathrm{d}x \,\middle|\, f \leq t \right\}.$$

Theorem 2.3.5 Every function f which is bounded on $[a,b]$ has a lower integral and a upper integral satisfying

$$\int_a^b s(x)\,\mathrm{d}x \le \underline{I}(f) \le \overline{I}(f) \le \int_a^b t(x)\,\mathrm{d}x.$$

for all step functions s and t with $s \le f \le t$. The function f is integrable if and only if when

$$\int_a^b f(x)\,\mathrm{d}x = \underline{I}(f) = \overline{I}(f).$$

Remark 2.3.6 Seemingly, all bounded functions are integrable. However, this is not true. Otherwise we won't introduce such complex notations above. Here's a counter example showing bounded functions can also be non-integrable.

Example 2.3.7 Recall the definition of Dirichlet Function. We shall confine it on $[0,1]$. By the dense property of rational numbers and irrational numbers, it's easy to prove

$$\underline{I}(f) = 0 < 1 = \overline{I}(f) \quad \text{so} \quad \underline{I}(f) \ne \overline{I}(f).$$

Hence, the Dirichlet Function is non-integrable.

Remark 2.3.8 So far, we have only introduced the definition of the integral for bounded functions. The example of the Dirichlet function shows that not every bounded function is integrable. However, this still leaves several deeper questions open: under what conditions can an unbounded function be integrated? And more generally, what are the structural features of non-integrable functions?
These questions require a more advanced theory of integration and will be studied in later courses. In this course, we shall mainly focus on functions that are integrable and on how to use their integrals.

Theorem 2.3.9 Let f be a nonnegative function, integrable on an interval $[a,b]$, and let Q denote the ordinate set of f over $[a,b]$. Then Q is measurable and

$$A(Q) = \int_a^b f(x)\,\mathrm{d}x.$$

Remark 2.3.10 Note that we only say ordinate sets of some particular functions are measurable, but at this stage we don't know what kind of ordinate sets are not measurable.

Proposition 2.3.11 Let

$$\widetilde{Q} := \left\{(x,y) \in \mathbb{R}^2 \mid a \le x \le b,\ 0 \le y < f(x)\right\},$$

then $\widetilde{Q}$ is also measurable and $A\left(\widetilde{Q}\right) = A(Q)$.

Proposition 2.3.12 Let

$$S := \left\{(x,y) \in \mathbb{R}^2 \mid a \le x \le b,\ y = f(x)\right\},$$

then S is also measurable and $A(S) = 0$.

Remark 2.3.13 After we define the integral of more general functions, there comes two questions naturally: which bounded functions are integrable, and given that a function f is integrable, how do we compute the integral of f?

For the first question, we will only give partial answers requiring only elementary ideas because a complete answer needs more prerequisites. For the second question, we'll introduce some techniques.

Definition 2.3.14 Let f be a function defined on a set S. We define the following notions:

1. f is said to be **increasing** on S if $f(x) \leq f(y)$ for every $x, y \in S$ with $x < y$.
2. f is said to be **strictly increasing** on S if $f(x) < f(y)$ for every $x, y \in S$ with $x < y$.
3. f is said to be **decreasing** on S if $f(x) \geq f(y)$ for every $x, y \in S$ with $x < y$.
4. f is said to be **strictly decreasing** on S if $f(x) > f(y)$ for every $x, y \in S$ with $x < y$.
5. f is called **monotonic** on S if it is either increasing or decreasing on S.
6. f is called **strictly monotonic** on S if it is either strictly increasing or strictly decreasing on S.
7. f is called **piecewise monotonic** on $[a, b]$ if there exists a partition $P = \{a = x_0 < x_1 < \cdots < x_n = b\}$ such that f is monotonic on each open subinterval (x_{i-1}, x_i), for $i = 1, \ldots, n$.

Remark 2.3.15 Note that a monotonic function needn't to be "continuous" intuitively. We will introduce the concept of continuity in Chapter 4.

Theorem 2.3.16 If a function f is monotonic on $[a, b]$, then f is integrable.

Proof. We shall prove the theorem for increasing functions. Assume f is increasing on $[a, b]$ and let $P = \{x_0, x_1, \cdots, x_n\}$ be a partition satisfying

$$x_k - x_{k-1} = \frac{b-a}{n}, \quad k = 1, 2, \cdots, n.$$

and s, t be step functions satisfying:

$$s(x) = f(x_{k-1}) \quad \text{and} \quad t(x) = f(x_k), \quad \text{if} \quad x_{k-1} < x < x_k, \quad k = 1, 2, \cdots, n.$$

At the subdivision points, we define s_n and t_n so as to preserve the relation $s_n(x) \leq f(x) \leq t_n(x)$ throughout $[a, b]$. It suffices to prove

$$\underline{I}(f) = \overline{I}(f).$$

First we compute

$$\begin{aligned}
\int_a^b t(x)\,\mathrm{d}x - \int_a^b s(x)\,\mathrm{d}x &= \sum_{k=1}^n f(x_k)(x_k - x_{k-1}) - \sum_{k=1}^n f(x_{k-1})(x_k - x_{k-1}) \\
&= \frac{b-a}{n}\left(f(b) - f(a)\right) \\
&= \frac{C}{n}, \quad \text{where} \quad C := (b-a)(f(b) - f(a)) > 0
\end{aligned}$$

Since we have

$$\int_a^b s(x)\,\mathrm{d}x \le \underline{I}(f) \le \overline{I}(f) \le \int_a^b t(x)\,\mathrm{d}x,$$

thus

$$0 \le \overline{I}(f) - \underline{I}(f) \le \int_a^b t(x)\,\mathrm{d}x - \int_a^b s(x)\,\mathrm{d}x = \frac{C}{n}$$

holds for a positive C and every positive integer n. Hence we deduce

$$\underline{I}(f) = \overline{I}(f).$$

□

Remark 2.3.17 Note that the proof for decreasing functions is analogous because it suffices to exchange $s(x)$ and $t(x)$. Here we see the magic power of Archimedian properties of real numbers again.

Theorem 2.3.18 Assume f is increasing on $[a, b]$. Let

$$x_k = a + \frac{b-a}{n} \cdot k, \quad k = 0, 1, \cdots, n.$$

If I is any number satisfying the inequalities

$$\frac{b-a}{n} \sum_{k=0}^{n-1} f(x_k) \le I \le \frac{b-a}{n} \sum_{k=1}^{n} f(x_k),$$

for every integer $n \ge 1$, then

$$I = \int_a^b f(x)\,\mathrm{d}x.$$

Remark 2.3.19 This inspires us to compute the integral of a general function by approaching from above and below.

Theorem 2.3.20 Here are some properties of integrals of general functions. If not stated, f and g are functions integrable on $[a, b]$.

1. Linearity: Suppose c_1, c_2 constants, then

$$\int_a^b [c_1 f(x) + c_2 g(x)]\,\mathrm{d}x = c_1 \int_a^b f(x)\,\mathrm{d}x + c_2 \int_a^b g(x)\,\mathrm{d}x.$$

2. Additivity:

$$\int_a^b f(x)\,\mathrm{d}x + \int_b^c f(x)\,\mathrm{d}x = \int_a^c f(x)\,\mathrm{d}x.$$

3. Invariance under translation: For every real c,

$$\int_a^b f(x)\,\mathrm{d}x = \int_{a+c}^{b+c} f(x-c)\,\mathrm{d}x.$$

4. Expansion or Contraction: If $k \neq 0$,

$$\int_a^b f(x)\,\mathrm{d}x = \frac{1}{k}\int_{ka}^{kb} f\left(\frac{x}{k}\right)\,\mathrm{d}x.$$

5. Comparison: If $g(x) \leq f(x)$ for every x in $[a, b]$,

$$\int_a^b g(x)\,\mathrm{d}x \leq \int_a^b f(x)\,\mathrm{d}x.$$

Proof. We shall prove linearity only, since proofs of other properties are analogous. We decompose the linearity into two parts:

(A)

$$\int_a^b [f(x) + g(x)]\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x + \int_a^b g(x)\,\mathrm{d}x$$

(B)

$$\int_a^b cf(x)\,\mathrm{d}x = c\int_a^b f(x)\,\mathrm{d}x$$

To prove (A), let

$$I(f) = \int_a^b f(x)\,\mathrm{d}x, \quad I(g) = \int_a^b g(x)\,\mathrm{d}x$$

. We shall prove

$$\underline{I}(f+g) = \overline{I}(f+g) = I(f) + I(g).$$

Let s_1, s_2 be arbitrary step functions below f, g respectively, then

$$I(f) = \sup\left\{\int_a^b s_1(x)\,\mathrm{d}x \,\middle|\, s_1 \leq f\right\}, \quad I(g) = \sup\left\{\int_a^b s_2(x)\,\mathrm{d}x \,\middle|\, s_2 \leq g\right\}.$$

So

$$\begin{aligned} I(f) + I(g) &= \sup\left\{\int_a^b s_1(x)\,\mathrm{d}x \,\middle|\, s_1 \leq f\right\} + \sup\left\{\int_a^b s_2(x)\,\mathrm{d}x \,\middle|\, s_2 \leq g\right\} \\ &= \sup\left\{\int_a^b [s_1(x) + s_2(x)]\,\mathrm{d}x \,\middle|\, s_1 \leq f, s_2 \leq g\right\} \end{aligned}$$

Since $s_1 \leq f, s_2 \leq g$, we have $s_1(x) + s_2(x) \leq f(x) + g(x)$. Thus

$$\underline{I}(f+g) \text{ is an upper bound of } \left\{\int_a^b [s_1(x) + s_2(x)]\,\mathrm{d}x \,\middle|\, s_1 \leq f, s_2 \leq g\right\}.$$

And an upper bound cannot be less than the least upper bound, i.e. the supremum, so

$$I(f) + I(g) \leq \underline{I}(f+g).$$

Similarly we can deduce that

$$I(f) + I(g) \geq \overline{I}(f+g).$$

Together with

$$\underline{I}(f+g) \leq \overline{I}(f+g),$$

we reach the equality

$$\underline{I}(f+g) = \overline{I}(f+g) = I(f+g).$$

To prove (B), it's trivial when $c = 0$. Now we focus on the case when $c > 0$. First we shall prove one property of supremum and infemum: if c is a positive real number, then we have

$$\sup\{cx \,|\, x \in A\} = c \cdot \sup\{x \,|\, x \in A\}, \quad \inf\{cx \,|\, x \in A\} = c \cdot \inf\{x \,|\, x \in A\}.$$

By definition,

$$\forall x \in A,\ x \leq \sup A \Rightarrow cx \leq c \cdot \sup A.$$

Assume $c \cdot \sup A$ is not the supremum of the set $\{cx \,|\, x \in A\}$, since it's a nonempty set and bounded above, there exists a positive h such that $c \cdot \sup A - h$ is the supremum, then

$$\forall x \in A, cx \leq c \cdot \sup A - h \Rightarrow x \leq \sup A - \frac{h}{c}.$$

Then the right-hand side of the inequality should be the new supremum of A, which leads to a contradiction. Thus we prove

$$\sup\{cx \,|\, x \in A\} = c \cdot \sup\{x \,|\, x \in A\}.$$

It's similar to prove infimum's property. Back to our proof, let s, s_1, t, t_1 be step functions with $s_1 = cs, t_1 = ct$. Then

$$\underline{I}(cf) = \sup\left\{\int_a^b s_1(x)\,\mathrm{d}x \,\middle|\, s_1 \leq cf\right\} = \sup\left\{c\int_a^b s(x)\,\mathrm{d}x \,\middle|\, s \leq f\right\} = cI(f)$$

$$\overline{I}(cf) = \inf\left\{\int_a^b t_1(x)\,\mathrm{d}x \,\middle|\, t_1 \leq cf\right\} = \inf\left\{c\int_a^b t(x)\,\mathrm{d}x \,\middle|\, t \leq f\right\} = cI(f).$$

Therefore, $\underline{I}(cf) = \overline{I}(cf) = cI(f)$. For the case when $c < 0$, by using

$$\sup\{cx \,|\, x \in A\} = c \cdot \inf\{x \,|\, x \in A\}, \quad \inf\{cx \,|\, x \in A\} = c \cdot \sup\{x \,|\, x \in A\},$$

it's analogous to finish the proof. □

Theorem 2.3.21 If p is a positive integer and $b > 0$, we have

$$\int_0^b x^p\,\mathrm{d}x = \frac{b^{p+1}}{p+1}.$$

Moreover, we can deduce the equality also holds for $b \leq 0$. Therefore, it can be generalized that

$$\int_a^b x^p\,\mathrm{d}x = \frac{b^{p+1} - a^{p+1}}{p+1}, \qquad a, b \in \mathbb{R},\ p \in \mathbb{Z}^+.$$

We can also use the special symbol

$$P(x)\Big|_a^b := P(b) - P(a).$$

Then the integral could be rewritten as

$$\int_a^b x^p \, \mathrm{d}x = \frac{x^{p+1}}{p+1}\Big|_a^b = \frac{b^{p+1} - a^{p+1}}{p+1}.$$

Definition 2.3.22 Let f, g be two functions and $f \leq g$ on $[a, b]$. We call such set S the **region between two graphs** if S is defined as

$$S := \left\{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, f(x) \leq y \leq g(x)\right\}.$$

Theorem 2.3.23 Assume two functions f, g are integrable with $f \leq g$ on $[a, b]$. Then the region S between their graphs is measurable and its area $A(S)$ is given by the integral

$$A(S) = \int_a^b [g(x) - f(x)] \, \mathrm{d}x.$$

Remark 2.3.24 This theorem didn't require f, g to be nonnegative.

Theorem 2.3.25 If p is a positive integer and $a, b > 0$, then we have

$$\int_a^b x^{\frac{1}{p}} \, \mathrm{d}x = \left(\frac{p}{p+1}\right)\left(b^{\frac{p+1}{p}} - a^{\frac{p+1}{p}}\right).$$

Chapter 3 Applications of Integration

Integration is a powerful tool for calculating areas, volumes, and many other quantities. However, before we use an integral to represent or compute such a quantity, there is a more fundamental issue to settle: is the function we intend to integrate actually integrable?

1 More about Area

Definition 3.1.1 A **circular disk** of radius r is the set of all points inside or on the boundary of a circle of radius r.

Theorem 3.1.2 A circular disk is measurable and its area $A(r)$ is given by the integral

$$A(r) = 2\int_{-r}^{r} \sqrt{r^2 - x^2}\,\mathrm{d}x.$$

Definition 3.1.3 We define the number π to be the area of a unit disk, i.e., the radius of which is exactly 1. Symbolically:

$$\pi := 2\int_{-1}^{1} \sqrt{1 - x^2}\,\mathrm{d}x.$$

Corollary 3.1.4 The area of a circular disk with radius r is given by

$$A(r) = \pi r^2.$$

Theorem 3.1.5 Let f be nonnegative and integrable on $[a, b]$ and let $\mathcal{S}$ be its ordinate set. We define a new function g denoted by

$$g(x) := kf\left(\frac{x}{k}\right) \qquad \text{with } x \in [ka, kb],\ k > 0$$

and its ordinate set $k\mathcal{S}$. Then the area of $k\mathcal{S}$ is given by

$$A(k\mathcal{S}) = k^2 A(\mathcal{S}).$$

Remark 3.1.6 In fact, this new g's shape is "similar" to f. Thus, we call this definition a similarity transformation.

2 Integration formulas for the sine and cosine

We omit deriving properties of sine and cosine in this lecture note since I'm lazy. However, with the following theorems and lemmas, we can derive the integral formulas for sine and cosine.

Theorem 3.2.1 For $0 < x < \frac{1}{2}\pi$, we have

$$0 < \cos x < \frac{\sin x}{x} < \frac{1}{\cos x}.$$

Lemma 3.2.2 Here are several inequalities important in deriving integral formulas for sine and cosine.

1. For any $x \in \mathbb{R}$,

$$2\sin\frac{x}{2}\sum_{k=1}^{n}\cos(kx) = \sin\left(\left(n+\frac{1}{2}\right)x\right) - \sin\frac{x}{2}.$$

2. For any $\theta \in \mathbb{R}$,

$$2\sin\theta\sum_{k=1}^{n}\cos(2k\theta) = \sin(2n+1)\theta - \sin\theta$$
$$2\sin\theta\sum_{k=0}^{n-1}\cos(2k\theta) = \sin(2n-1)\theta + \sin\theta.$$

3. Suppose θ satisfies $0 < 2n\theta < \frac{1}{2}\pi$, we have

$$\sin(2n+1)\theta - \sin\theta < \frac{\sin\theta}{\theta}\sin(2n\theta).$$

4. Suppose θ satisfies $0 < 2n\theta < \frac{1}{2}\pi$. First we have

$$\frac{\cos(n-1)\theta}{\cos(n\theta)} > \frac{\sin\theta}{\theta}$$

and then we can deduce

$$\frac{\sin\theta}{\theta}\sin(2n\theta) < \sin(2n-1)\theta + \sin\theta.$$

Theorem 3.2.3 For any $0 < a \leq \frac{1}{2}\pi$, we have

$$\frac{a}{n}\sum_{k=1}^{n}\cos\frac{ka}{n} < \sin a < \frac{a}{n}\sum_{k=0}^{n-1}\cos\frac{ka}{n}.$$

Theorem 3.2.4 For every real a we have

$$\int_0^a \cos x \,\mathrm{d}x = \sin a, \qquad \int_0^a \sin x \,\mathrm{d}x = 1 - \cos a.$$

More generally, for every real a, b, we have

$$\int_a^b \cos x \,\mathrm{d}x = \sin x \Big|_a^b, \qquad \int_a^b \sin x \,\mathrm{d}x = (-\cos x) \Big|_a^b.$$

3 Polar Coordinates

Definition 3.3.1 The polar coordinates of a point P in the plane are an ordered pair (r, θ) where r is the distance from P to the origin O and θ is the angle formed by the line segment OP and the positive x-axis. Here $r \geq 0$ and $\theta \in \mathbb{R}$. They are related to the rectangular coordinates (x, y) of P by the equations

$$x = r\cos\theta, \quad y = r\sin\theta.$$

Here we call r the **radial distance** of P and θ a **polar angle**.

Remark 3.3.2 Note that we call θ "a" polar angle instead of "the" polar angle because $\theta + 2n\pi$ also represents the same angle for any integer n. But the radial distance r is uniquely determined.
If $r = 0$, then by convention θ can be any real number.

Definition 3.3.3 A **curve** is said to be given in polar coordinates if there is a nonnegative function f defined on an interval I such that every point P on the curve has polar coordinates (r, θ) satisfying

$$r = f(\theta), \quad \theta \in I.$$

And this equation is called the **polar equation** of the curve. Similarly, the **graph** of f in polar coordinates is the set of all points P with polar coordinates (r, θ) satisfying the above equation.

Remark 3.3.4 In some textbooks, the variable r in polar coordinates is allowed to be negative: when $r < 0$, the point (r, θ) is identified with the point $(-r, \theta + \pi)$. Note that the points (r, θ) and $(-r, \theta)$ are symmetric with respect to the origin. The additional definition allows the existence of inner loops in the graph of a polar equation. But in this lecture note, we only consider nonnegative functions in polar coordinates.

Example 3.3.5 Sketch the curves for

$$f(\theta) = 1 + c\sin\theta \quad \text{where} \quad \theta \in [0, 2\pi],\ c \in \{-2, 2\}$$

under two different defintion and compare them.

Definition 3.3.6 Let f be a nonnegative function defined on an interval $[a, b]$, where $0 \leq b - a \leq 2\pi$. The set of all points P in the plane with polar coordinates (r, θ) satisfying

$$0 \le r \le f(\theta), \quad \theta \in [a, b]$$

is called the **radial set** determined by f.

Definition 3.3.7 Similarly as in rectangular coordinates, a nonnegative function s defined on $[a, b]$ is called a **step function in polar coordinates** if there exists a partition $P = \{\theta_0, \theta_1, \cdots, \theta_n\}$ of $[a, b]$ such that

$$s(\theta) = s_k \quad \text{where} \quad \theta \in (\theta_{k-1}, \theta_k), \quad k = 1, 2, \cdots, n.$$

And the area of this sector is given by

$$A = \frac{1}{2} s_k^2 (\theta_k - \theta_{k-1}).$$

Since $b - a < 2\pi$, there is no overlapping between different sectors. The area of the radial set determined by s is given by

$$A(S) = \frac{1}{2} \sum_{k=1}^{n} s_k^2 (\theta_k - \theta_{k-1}) = \frac{1}{2} \int_a^b s^2(\theta) \, \mathrm{d}\theta.$$

Theorem 3.3.8 Let f be a nonnegative function on $[a, b]$, where $0 \le b - a \le 2\pi$. Assume the radial set S determined by f is measurable. If f^2 is integrable, then the area $A(S)$ is given by the integral

$$A(S) = \frac{1}{2} \int_a^b f^2(\theta) \, \mathrm{d}\theta.$$

4 Volume

Similarly, we can use axiomatic definitions to define volume like we did for area.

Definition 3.4.1 We assume there exists a class $\mathcal{A}$ of solids and a set function V, whose domain is $\mathcal{A}$, with the following properties:

1. Nonnegative property: For each set S in $\mathcal{A}$ we have $V(S) \ge 0$.
2. Additive property: If S and T are in $\mathcal{A}$, then $S \cup T$ and $S \cap T$ are in $\mathcal{A}$, and we have
$$V(S \cup T) = V(S) + V(T) - V(S \cap T).$$
3. Difference property: If S and T are in $\mathcal{A}$ with $S \subseteq T$, then $T - S$ is in $\mathcal{A}$, and we have
$$V(T - S) = V(T) - V(S).$$
4. Cavalieri's principle.: If S and T are two Cavalieri solids in $\mathcal{A}$ with $A(S \cap F) \le A(T \cap F)$ for every plane F perpendicular to a given line, then $V(S) \le V(T)$.
5. Choice of scale.: Every box B is in $\mathcal{A}$. If the edges of B have lengths a, b, and c, then
$$V(B) = abc.$$

6. Every convex set is in $\mathcal{A}$.

Definition 3.4.2 (Additional) Some definitions are lost above and here we add it.

1. A solid S in $\mathcal{A}$ is called a **Cavalieri solid** if for every plane F perpendicular to a given line L, the area $A(S \cap F)$ of the cross section of S by F is well-defined, i.e., the cross section is measurable.
2. A **box** or **rectangular parallelepiped** is any set congruent to a set of the form

$$\{(x, y, z) \in \mathbb{R}^3 \mid 0 \leq x \leq a,\ 0 \leq y \leq b,\ 0 \leq z \leq c\},$$

where nonnegative real numbers a, b, c are called the lengths of the edges of the box.
3. A solid S in $\mathcal{A}$ is called a **convex set** if for every two points $P, Q \in S$, the line segment PQ is also contained in S.
4. A plane set is call **bounded** if it is a subset of some square in the plane.

Corollary 3.4.3 Here's some propositions for some basic cases.

1. The empty set $\varnothing$ is in $\mathcal{A}$ and $V(\varnothing) = 0$.
2. Monotonicity property: If S and T are in $\mathcal{A}$ with $S \subseteq T$, then $V(S) \leq V(T)$.
3. Every bounded plane set S in $\mathcal{A}$ has volume zero, i.e., $V(S) = 0$.

Remark 3.4.4 Note that the Cavalieri's principle can be applied twice: if $A(S \cap F) = A(T \cap F)$ for every plane F perpendicular to a given line, then $V(S) = V(T)$. In Chinese, we call this property Zu Geng's principle.

Definition 3.4.5 A **right cylindrical solid** is a set congruent to a set S of the form

$$S = \left\{(x, y, z) \in \mathbb{R}^3 \mid (x, y) \in B,\ a \leq z \leq b\right\},$$

where B is a bounded plane set and a, b are real numbers with $a \leq b$. The base of the right cylindrical solid is the set B, and its height is $b - a$.

Theorem 3.4.6 Let S be a right cylindrical solid with base B and height h. Then S is in $\mathcal{A}$ and its volume is given by

$$V(S) = A(B) \cdot h = \int_a^b A_S(z)\,\mathrm{d}z$$

More generally, let R be a Cavalieri solid in $\mathcal{A}$ with a cross-sectional area function A_R which is integrable on an interval $[a, b]$ and zero outside $[a, b]$. Then the volume of R is given by

$$V(R) = \int_a^b A_R(z)\,\mathrm{d}z.$$

Theorem 3.4.7 Let f and g be nonnegative integrable functions on $[a, b]$ satisfying $g(x) \leq f(x)$ for all $x \in [a, b]$. If the region bounded by $y = f(x)$ and $y = g(x)$ is revolved about the x-axis, then the resulting solid of revolution (if it belongs to $\mathcal{A}$) has volume

$$V = \pi \int_a^b \left[f^2(x) - g^2(x)\right] \mathrm{d}x.$$

In particular, if $g(x) = 0$, the volume is

$$V = \pi \int_a^b f^2(x) \, \mathrm{d}x.$$

Example 3.4.8 Let $f(x) = \sqrt{r^2 - x^2}$ on $[-r, r]$, then the solid is a sphere of radius r, and its volume is given by

$$V = \pi \int_{-r}^{r} (r^2 - x^2) \, \mathrm{d}x = \frac{4}{3}\pi r^3.$$

5 Indefinite Integral

Definition 3.5.1 Let f be a real-valued function integrable on $[a, b]$. The **average value** of f, denoted by $\operatorname{avg}(f)$, on $[a, b]$ is defined to be

$$\operatorname{avg}(f) = \frac{1}{b-a} \int_a^b f(x) \, \mathrm{d}x.$$

Remark 3.5.2 Note that the definition of average value of a function is consistent with the definition of average value of a finite set of numbers, i.e., the discrete average. You might not see how this definition works in this section, but it will be useful in later section.

Theorem 3.5.3 (Weighted Mean Value Theorem for Integrals) Assume f and g are continuous on $[a, b]$. If g never changes sign on $[a, b]$, then for some $c \in [a, b]$, we have

$$\int_a^b f(x)g(x) \, \mathrm{d}x = f(c) \int_a^b g(x) \, \mathrm{d}x.$$

Corollary 3.5.4 (Mean Value Theorem for Integrals) Let $g(x) \equiv 1$, we have the **Mean Value Theorem for Integrals**:

$$\int_a^b f(x) \, \mathrm{d}x = f(c)(b-a).$$

Remark 3.5.5 The proof is in Chapter 4.

Definition 3.5.6 Let f be a real-valued function integrable on every closed interval contained in an open interval I. An **indefinite integral** of f on I is the function F defined by

$$F(x) = \int_a^x f(t) \, \mathrm{d}t, \quad x \in I,$$

where a is an arbitrary point in I.

Remark 3.5.7 Indeed, a different choice of a will only lead to a different indefinite integral by a constant, since

$$\int_b^x f(t)\,\mathrm{d}t = \int_a^x f(t)\,\mathrm{d}t - \int_a^b f(t)\,\mathrm{d}t.$$

Theorem 3.5.8 Suppose f integrable and nonnegative on $[a, b]$, then the indefinite integral $A(f)$ is increasing.

Definition 3.5.9 A function g is said to be **convex** on an interval $[a, b]$ if for every $x_1, x_2 \in [a, b]$ and $0 \leq \lambda \leq 1$, we have

$$g(\lambda x_1 + (1-\lambda)x_2) \leq \lambda g(x_1) + (1-\lambda)g(x_2).$$

Similarly, g is said to be **concave** on $[a, b]$ if the above inequality is reversed, i.e.,

$$g(\lambda x_1 + (1-\lambda)x_2) \geq \lambda g(x_1) + (1-\lambda)g(x_2).$$

Remark 3.5.10 Geometrically, a function g is convex on $[a, b]$ if the line segment joining any two points on the graph of g lies above or on the graph between these two points. Similarly, g is concave on $[a, b]$ if the line segment joining any two points on the graph of g lies below or on the graph between these two points.

Theorem 3.5.11 Let f be integrable on $[a, b]$ and let F be an indefinite integral of f on $[a, b]$. Then:

1. if f is increasing on $[a, b]$, then F is convex on $[a, b]$;
2. if f is decreasing on $[a, b]$, then F is concave on $[a, b]$.

Chapter 4 Limits and Continuity of functions

1 Basic Concepts

Definition 4.1.1 Any open interval containing $p \in \mathbb{R}$ as the midpoint is called a **neighborhood** of p. Suppose $r > 0$, then

$$N(p) = (p - r, p + r) = \{x \in \mathbb{R} : |x - p| < r\}$$

is a neighborhood of p with **radius** r. Moreover, a **punctured neighborhood** is the deleted neighborhood given by removing its midpoint.

Notation 4.1.2 We can also denote $N(p)$ as $N(p\,; r)$.

Definition 4.1.3 Let f be a function defined in some neighborhood of p except possibly not defined at $x = p$. Suppose $A \in \mathbb{R}$. The symbol

$$\lim_{x \to p} f(x) = A \quad [\text{or} \quad f(x) \to A \quad \text{as} \quad x \to p]$$

reads "the limit of $f(x)$ as x approaches p is equal to A". The symbol means that for every neighborhood $N_1(A)$, there exists some neighborhood $N_2(p)$ such that

$$f(x) \in N_1(A) \quad \text{whenever} \quad x \in N_2(p) \quad \text{and} \quad x \neq p.$$

Remark 4.1.4 In this definition, we only consider A to be a real number and f to be a real-valued function. Besides, the limit has nothing to do with what happens at $x = p$.

Definition 4.1.5 Equivalently, the limit can be defined as

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ |f(x) - A| < \varepsilon \text{ whenever } 0 < |x - p| < \delta,$$

which is called the "$\varepsilon - \delta$ definition of limit".

Remark 4.1.6 You've seen and you will see infinitely many "whenever" which all mean the same thing: "for all ...".

Corollary 4.1.7 The following symbols are all equivalent:

$$\lim_{x\to p} f(x) = A,$$
$$\lim_{x\to p} [f(x) - A] = 0,$$
$$\lim_{x\to p} |f(x) - A| = 0,$$
$$\lim_{h\to 0} f(p+h) = A,$$
$$\lim_{h\to 0} f(p-h) = A.$$

Definition 4.1.8 Let f be a function defined on (p, b) for some $b > p$ and $A \in \mathbb{R}$. The symbol

$$\lim_{x\to p^+} f(x) = A$$

means that for every neighborhood $N_1(A)$, there exists some neighborhood $N_2(p)$ such that

$$f(x) \in N_1(A) \quad \text{whenever} \quad x \in N_2(p) \quad \text{and} \quad x > p,$$

which is called the **right-side limit**.
Similarly, we can define what is called the **left-side limit**. Let f be a function defined on (a, p) for some $a < p$ and $A \in \mathbb{R}$. The symbol

$$\lim_{x\to p^-} f(x) = A$$

means that for every neighborhood $N_1(A)$, there exists some neighborhood $N_2(p)$ such that

$$f(x) \in N_1(A) \quad \text{whenever} \quad x \in N_2(p) \quad \text{and} \quad x < p.$$

Remark 4.1.9 If we have

$$\lim_{x\to p} f(x) = A,$$

then it's easy to verify that both

$$\lim_{x\to p^+} f(x) = A \quad \text{and} \quad \lim_{x\to p^-} f(x) = A$$

hold. Furthermore, the converse still holds.

Definition 4.1.10 A function f is said to be **continuous** at $p \in \mathbb{R}$ if it satisfies two conditions:

1. f is defined at p.
2. $\lim_{x\to p} f(x) = f(p)$.

Definition 4.1.11 Let f be a function and $a \in \mathbb{R}$. We say that f has a **removable discontinuity** at a if the two-sided limit

$$\lim_{x\to a} f(x) = L \in \mathbb{R}$$

exists and is finite, but either $f(a)$ is not defined or $f(a) \neq L$. Equivalently, redefining $f(a) := L$ makes f

continuous at a.

Example 4.1.12 Let

$$f(x) := \begin{cases} 1, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then f has a removable discontinuity at $x = 0$.

Definition 4.1.13 Let f be a function and $a \in \mathbb{R}$. We say that f has a **jump discontinuity** at a if both one-sided limits exist and are finite,

$$\lim_{x \to a^-} f(x) = L_- \in \mathbb{R}, \qquad \lim_{x \to a^+} f(x) = L_+ \in \mathbb{R},$$

but $L_- \neq L_+$. The jump size is $|L_+ - L_-|$.

Example 4.1.14 Let

$$f(x) = x - [x],$$

then f has a jump discontinuity at every $x = k$ where k is an integer.

Definition 4.1.15 Let f be a function and $a \in \mathbb{R}$. We say that f has an **infinite discontinuity** at a if either

$$\lim_{x \to a^-} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \to a^+} f(x) = \pm\infty$$

holds. Equivalently, f is unbounded on every punctured neighborhood of a.

Example 4.1.16 Let

$$f(x) := \begin{cases} \dfrac{1}{x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then $\lim_{x \to p} f(x)$ **does not exist** and so for the left-side and right-side limit. Though we haven't define what "∞" is, we should learn it intuitively yet.

Definition 4.1.17 Let f be a function and $a \in \mathbb{R}$. We say that f has an **oscillatory discontinuity** at a if at least one of the one-sided limits

$$\lim_{x \to a^-} f(x), \qquad \lim_{x \to a^+} f(x)$$

fails to exist, and the failure is *not* because the limit diverges to $+\infty$ or $-\infty$. Equivalently, f oscillates without approaching any finite limit on every punctured neighborhood of a.

Example 4.1.18 Let

$$f(x) = \begin{cases} \sin\left(\dfrac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then f has an oscillatory discontinuity at $x = 0$. Indeed, neither one-sided limit

$$\lim_{x\to 0^-} \sin\left(\frac{1}{x}\right), \qquad \lim_{x\to 0^+} \sin\left(\frac{1}{x}\right)$$

exists, because the function oscillates infinitely often between -1 and 1 as $x \to 0$ without approaching any limiting value.

Theorem 4.1.19 Let f and g be functions such that

$$\lim_{x\to p} f(x) = A, \qquad \lim_{x\to p} g(x) = B.$$

Then we have the following properties:

1. $\lim_{x\to p}[f(x) + g(x)] = A + B$,
2. $\lim_{x\to p}[f(x) - g(x)] = A - B$,
3. $\lim_{x\to p}[f(x) \cdot g(x)] = A \cdot B$,
4. If $B \neq 0$, then $\lim_{x\to p} \dfrac{f(x)}{g(x)} = \dfrac{A}{B}$.

Theorem 4.1.20 Let f and g be continuous at p. Then

$$f + g,\ f - g,\ fg$$

are also continuous at p. If we know $g(p) \neq 0$, then $\dfrac{f}{g}$ is also continuous at p.

Theorem 4.1.21 (Squeeze Theorem) Suppose that $f(x) \leq g(x) \leq h(x)$ for all $x \neq p$ in some neighborhood $N(p)$ and additionally

$$\lim_{x\to p} f(x) = \lim_{x\to p} h(x) = A,$$

then

$$\lim_{x\to p} g(x) = A.$$

Proof. For every $\varepsilon > 0$, there exists some neighborhood $N_1(p)$ such that

$$f(x) \in N(A\,;\varepsilon), \quad h(x) \in N(A\,;\varepsilon)$$

whenever $x \in N_1(p)$ and $x \neq p$. Let $N_2(p)$ be the neighborhood such that

$$f(x) \leq g(x) \leq h(x)$$

holds whenever $x \in N_2(p)$ and $x \neq p$. Now let $N_3(p) = N_1(p) \cap N_2(p)$, then for every $x \in N_3(p)$ and $x \neq p$, we have

$$A - \varepsilon < f(x) \leq g(x) \leq h(x) < A + \varepsilon,$$

i.e., $g(x) \in N(A\,;\varepsilon)$. Thus by definition we have

$$\lim_{x\to p} g(x) = A.$$

$\square$

Remark 4.1.22 When we choose f and h to apply the squeeze theorem, they're usually not unique as long as they work well.

Theorem 4.1.23 Assume f is integrable on $[a,x]$ for every $x \in [a,b]$ and define

$$F(x) = \int_a^x f(t)\,\mathrm{d}t, \quad x \in [a,b].$$

Then F is continuous at every $x \in (a,b)$.

Remark 4.1.24 This theorem provides us a way to prove the continuity of some complicated functions constructed by integrals.
We can replace "continuous at every $x \in (a,b)$" in the conclusion by "continuous at every $x \in [a,b]$" with the following definition of one-sided continuity.

Definition 4.1.25 A function f is said to be **right-side continuous** or **continuous from the right** at p if it satisfies two conditions:

1. f is defined at p.
2. $\lim_{x\to p^+} f(x) = f(p)$.

Similarly, a function f is said to be **left-side continuous** or **continuous from the left** at p if it satisfies two conditions:

1. f is defined at p.
2. $\lim_{x\to p^-} f(x) = f(p)$.

Remark 4.1.26 Moreover, the above theorems all hold for one-sided limits and one-sided continuity.

Example 4.1.27 Here are some basic examples.

1. Note that
$$\sin x = \int_0^x \cos t\,\mathrm{d}t, \quad \cos x = 1 - \int_0^x \sin t\,\mathrm{d}t,$$
we can conclude that both $\sin x$ and $\cos x$ are continuous on $\mathbb{R}$.

2. Define
$$f(x) := \begin{cases} \dfrac{\sin x}{x}, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

Then f is continuous at $x = 0$ since

$$0 < \cos x < \frac{\sin x}{x} < \frac{1}{\cos x} \quad \text{for} \quad 0 < |x - 0| < \frac{\pi}{2}.$$

and by squeeze theorem we have

$$\lim_{x\to 0} \frac{\sin x}{x} = 1 = f(0).$$

Moreover, f is continuous on $\mathbb{R}$.

2 Major Theorems for Continuous functions

Theorem 4.2.1 Assume v is continuous at p and u is continuous at $q = v(p)$. Then the composite function $f = u \circ v$ is continuous at p.

Lemma 4.2.2 Let f be continuous at c and suppose that $f(c) \neq 0$. Then there exists $\delta > 0$ such that for the neighborhood $N(c\,;\delta)$ we have

$$\forall x \in N(c\,;\delta),\ f(x)\,f(c) > 0.$$

Proof. Since f is continuous at c, for $\varepsilon = \dfrac{|f(c)|}{2} > 0$, there exists some $\delta > 0$ such that

$$|f(x) - f(c)| < \varepsilon = \frac{|f(c)|}{2}$$

holds whenever $x \in N(c\,;\delta)$. Thus we have

$$-\frac{|f(c)|}{2} < f(x) - f(c) < \frac{|f(c)|}{2},$$

i.e.,

$$f(c) - \frac{|f(c)|}{2} < f(x) < f(c) + \frac{|f(c)|}{2}.$$

If $f(c) > 0$, then we have

$$\frac{f(c)}{2} < f(x) < \frac{3f(c)}{2},$$

which implies $f(x) > 0$. If $f(c) < 0$, then we have

$$\frac{3f(c)}{2} < f(x) < \frac{f(c)}{2},$$

which implies $f(x) < 0$. In both cases, we have $f(x)f(c) > 0$. □

Theorem 4.2.3 (Bolzano's Theorem) Let f be a function continuous on the closed interval $[a, b]$. If $f(a)$ and $f(b)$ have opposite signs, then there exists some $c \in (a, b)$ such that

$$f(c) = 0.$$

Proof. Without loss of generality, we assume $f(a) < 0$ and $f(b) > 0$. Let

$$S = \{x \in [a, b] : f(x) \leq 0\}.$$

Since $a \in S$, S is nonempty. And since $S \subseteq [a, b]$, S is bounded above by b. By the completeness axiom of real numbers, let

$$c = \sup S.$$

We shall prove that $f(c) = 0$. Suppose on the contrary that $f(c) \neq 0$. Then by the above lemma, there exists some $\delta > 0$ such that for every $x \in N(c\,;\delta)$, we have

$$f(x)f(c) > 0.$$

If $f(c) > 0$, then for every $x \in N(c\,;\delta)$, we have $f(x) > 0$. Note that there exists some $x_1 \in S$ such that

$$c - \delta < x_1 < c \quad \text{and} \quad f(x_1) > 0,$$

since $c = \sup S$. But we know $x_1 \in S$ and thus $f(x_1) \leq 0$, which leads to a contradiction.
If $f(c) < 0$, then for every $x \in N(c\,;\delta)$, we have $f(x) < 0$. Note that there exists some $x_3 \in [a, b] \setminus S$ such that

$$c < x_3 < c + \delta, \quad \text{and} \quad f(x_3) < 0,$$

since $c = \sup S$. But we know $x_3 \notin S$ and thus $f(x_3) > 0$, which leads to a contradiction.
Hence we must have $f(c) = 0$. □

Remark 4.2.4 One-sided continuity should be considered at the endpoints a and b when applying Bolzano's Theorem. Otherwise, the conclusion might not hold.

Theorem 4.2.5 (Intermediate Value Theorem (IVT)) Let f be continuous at each point of a closed interval $[a, b]$. For two arbitrary points $x_1, x_2 \in [a, b]$ with $f(x_1) \neq f(x_2)$ and any number L between $f(x_1)$ and $f(x_2)$, there exists some c between x_1 and x_2 such that

$$f(c) = L.$$

Remark 4.2.6 The Intermediate Value Theorem is a immediate consequence of Bolzano's Theorem. Note such c above might not be unique.

Proposition 4.2.7 If n is a positive integer and if $a > 0$, then there is exactly one positive real number x such that

$$x^n = a.$$

Remark 4.2.8 Of course we can derive the existence and uniqueness of nth root of any real number by using the completeness axiom of real numbers. But here using the Intermediate Value Theorem is more efficient.

Theorem 4.2.9 Assume f is strictly increasing and continuous on an interval $[a, b]$. Let $c = f(a)$ and $d = f(b)$ and let g be the inverse function of f defined on $[c, d]$. Then

1. g is strictly increasing on $[c, d]$,
2. g is continuous on $[c, d]$.

Proof. 1. For any $y_1, y_2 \in [c, d]$ with $y_1 < y_2$, let

$$x_1 = g(y_1), \quad x_2 = g(y_2).$$

Since f is strictly increasing, we have

$$y_1 = f(x_1) < f(x_2) = y_2,$$

which implies that $x_1 < x_2$. Thus g is strictly increasing on $[c, d]$.

2. Choose arbitrary $y_0 \in (c, d)$. To prove g is continuous at y_0, we need to show that for every $\varepsilon > 0$, there exists some $\delta > 0$ such that

$$|g(y) - g(y_0)| < \varepsilon \quad \text{whenever} \quad |y - y_0| < \delta.$$

Let

$$x_0 = g(y_0), \quad f(x_0) = y_0.$$

Suppose $\varepsilon > 0$ is given. Since f is increasing, we let

$$\delta = \min\{f(x_0 + \varepsilon) - f(x_0), f(x_0) - f(x_0 - \varepsilon)\} > 0.$$

Then we have

$$f(x_0 - \varepsilon) < y < f(x_0 + \varepsilon) \quad \text{whenever} \quad |y - y_0| < \delta.$$

Since g is strictly increasing, we have

$$x_0 - \varepsilon < g(y) < x_0 + \varepsilon,$$

which implies that

$$|g(y) - g(y_0)| < \varepsilon.$$

□

Remark 4.2.10 Suppose $f(x) = x^2$ on $[-a, a]$ where $a > 0$ is not invertible, but $f(x)$ is invertible on $[-a, 0]$ or on $[0, a]$. In some textbooks, we express "the inverse" of $f(x)$ on $[0, a^2]$ as "double function".

Definition 4.2.11 Suppose f is a real-valued function on a set $S \subseteq \mathbb{R}$. Then f is said to have an **absolute maximum** on S if

$$\exists c \in S \, \forall x \in S, \; f(x) \leq f(c).$$

And $f(c)$ is calle the **absolute maximum value** of f on S. Similarly, f is said to have an **absolute minimum** on S if

$$\exists d \in S \, \forall x \in S, \; f(x) \geq f(d).$$

And $f(d)$ is calle the **absolute minimum value** of f on S.

Lemma 4.2.12 Suppose f is continuous on $[a, b]$, then f is bounded on $[a, b]$.

Proof. We shall prove by contradiction and using the method of **successive bisection**. Suppose f is unbounded on $[a, b]$. Let

$$c = \frac{a+b}{2},$$

then we know among the two subintervals $[a, c]$ and $[c, b]$, at least on one of the two we know f is unbounded. Pick the subinterval on which f is unbounded (if f is unbounded on both, then we pick the left one, i.e., $[a, c]$) and denote it by $[a_1, b_1]$. We continue this process and keep bisecting the chosen subintervals to get a sequence of subintervals:

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots [a_n, b_n] \supseteq \cdots$$

where n is any positive integer and on which f is unbounded. Note that the length of $[a_n, b_n]$ is given by

$$b_n - a_n = \frac{b-a}{2^n}$$

Let

$$A := \{a, a_1, a_2, \cdots, a_n, \cdots\} \neq \varnothing$$

and A is bounded above by b. Then we let $\alpha = \sup A$. Since $\alpha \in [a, b]$ and f is continuous at α, there exists some $\delta > 0$ such that given $\varepsilon = 1$, we have

$$|f(x) - f(\alpha)| < 1 \quad \text{whenever} \begin{cases} x \in (\alpha - \delta, \alpha + \delta), & \text{if } \alpha \in [a, b] \\ x \in [\alpha, \alpha + \delta), & \text{if } \alpha = a \\ x \in (\alpha - \delta, \alpha], & \text{if } \alpha = b \end{cases}$$

By triangle inequality we have

$$|f(x)| - |f(\alpha)| \leq |f(x) - f(\alpha)| < 1.$$

So we know f is bounded on $\begin{cases} (\alpha - \delta, \alpha + \delta), & \text{if } \alpha \in (a, b) \\ [\alpha, \alpha + \delta), & \text{if } \alpha = a \\ (\alpha - \delta, \alpha], & \text{if } \alpha = b \end{cases}$

Since

$$a \leq a_1 \leq a_2 \leq \cdots \leq a_n \leq \cdots,$$

we know there exists some k larger enough such that

$$\begin{cases} [a_k, b_k] \subseteq (\alpha - \delta, \alpha + \delta), & \text{if } \alpha \in (a, b) \\ [a_k, b_k] \subseteq [\alpha, \alpha + \delta), & \text{if } \alpha = a \\ [a_k, b_k] \subseteq (\alpha - \delta, \alpha], & \text{if } \alpha = b \end{cases}$$

This means f unbounded on $[a_k, b_k]$ but

$$|f(x)| \leq 1 + |f(\alpha)|, \quad \forall x \in [a_k, b_k],$$

which is a contradiction. Thus f is bounded on $[a, b]$. $\square$

Theorem 4.2.13 (Extreme Value Theorem (EVT)) Suppose f is continuous on $[a, b]$, then there exists $c \in [a, b]$ and $d \in [a, b]$ such that
$$\forall x \in [a, b] \quad f(d) \leq f(x) \leq f(c).$$

Proof. By the above lemma, since f is continuous on $[a, b]$, it is bounded on $[a, b]$. Thus the set
$$S := \{f(x) : x \in [a, b]\}$$
is a nonempty bounded subset of $\mathbb{R}$, and therefore has a supremum
$$M := \sup S.$$
We first prove that f attains the value M. Suppose, towards a contradiction, that $f(x) < M$ for every $x \in [a, b]$. Then for all $x \in [a, b]$ the quantity $M - f(x)$ is positive. Define a new function
$$g(x) := \frac{1}{M - f(x)}, \qquad x \in [a, b].$$
Because f is continuous and $M - f(x) > 0$ everywhere, the function g is also continuous and bounded on $[a, b]$; that is, there exists $K > 0$ such that
$$g(x) \leq K, \qquad \forall x \in [a, b].$$
Substituting the definition of g, we obtain
$$\frac{1}{M - f(x)} \leq K \qquad \Longrightarrow \qquad f(x) \leq M - \frac{1}{K}, \qquad \forall x \in [a, b].$$
Thus $M - \dfrac{1}{K}$ is an upper bound for S, but
$$M - \frac{1}{K} < M,$$
which contradicts the definition of $M = \sup S$. Therefore our assumption was false, and there must exist $c \in [a, b]$ such that $f(c) = M$.
To obtain the minimum, apply the same argument to the continuous function $-f$ on $[a, b]$. There exists $d \in [a, b]$ such that
$$-f(d) = \sup\{-f(x) : x \in [a, b]\} \quad \Longrightarrow \quad f(d) = \min\{f(x) : x \in [a, b]\}.$$
Thus f attains both its maximum and minimum on $[a, b]$, completing the proof. □

Remark 4.2.14 Actually if we define
$$\sup f := \sup\{f(x) : x \in [a, b]\} \quad \text{and} \quad \inf f := \inf\{f(x) : x \in [a, b]\}$$
then in the above theorem we have
$$c = \sup f, \quad d = \inf f.$$

Corollary 4.2.15 Combining **EVT** and **IVT**, we have the following : If f is continuous on $[a, b]$, then the range of f on $[a, b]$ is $[\inf f, \sup f]$.

Definition 4.2.16 Suppose f is continuous on $[a, b]$, we call $(\sup f - \inf f)$ the **leap** or **span** of f on $[a, b]$.

Proposition 4.2.17 If f is continuous on $[a, b]$ and $[c, d] \subseteq [a, b]$, then the leap on $[c, d]$ cannot exceed the leap of f on $[a, b]$.

Theorem 4.2.18 Suppose f is continuous on $[a, b]$. Then for any $\varepsilon > 0$, there exists a partition $P = \{x_0, x_1, \cdots, x_n\}$ of $[a, b]$ such that the leap of f in every closed interval $[x_{k-1}, x_k]$ is less than ε.

Proof. We shall prove by contradiction. Recall how we use the method of successive bisection to prove that a continuous function on a closed interval is bounded. Now we do similarly to choose a sequence of subintervals:

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots [a_n, b_n] \supseteq \cdots$$

where n is any positive integer and on which the leap of f is at least ε_0. Let

$$\alpha = \sup\{a, a_1, a_2, \cdots, a_n, \cdots\} \in [a, b]$$

with

$$a \leq a_1 \leq a_2 \leq \cdots \leq a_n \leq \cdots.$$

By continuity, there exists some $\delta > 0$ such that in an interval

$$\begin{cases} (\alpha - \delta, \alpha + \delta), & \text{if } \alpha \in (a, b) \\ [\alpha, \alpha + \delta), & \text{if } \alpha = a \\ (\alpha - \delta, \alpha], & \text{if } \alpha = b \end{cases}$$

the leap of f is less than ε_0. Note that the length of $[a_n, b_n]$ is given by

$$b_n - a_n = \frac{b - a}{2^n}.$$

So there exists some k larger enough such that

$$\begin{cases} [a_k, b_k] \subseteq (\alpha - \delta, \alpha + \delta), & \text{if } \alpha \in (a, b) \\ [a_k, b_k] \subseteq [\alpha, \alpha + \delta), & \text{if } \alpha = a \\ [a_k, b_k] \subseteq (\alpha - \delta, \alpha], & \text{if } \alpha = b \end{cases}$$

This means the leap of f on $[a_k, b_k]$ is less than ε_0, which is a contradiction. Thus we can keep bisecting until the leap of f on every subinterval is less than ε_0. □

Theorem 4.2.19 Suppose f is continuous on $[a, b]$, then f is integrable on $[a, b]$.

Proof. Obviously f is bounded on $[a, b]$, so f has both an upper integral $\overline{I}(f)$ and a lower integral $\underline{I}(f)$. We shall prove that

$$\overline{I}(f) = \underline{I}(f),$$

which implies that f is integrable on $[a, b]$.
Choose an integer $N \geq 1$ and let $\varepsilon = 1/N$. By the small–span theorem for continuous functions, for this ε there exists a partition

$$P = \{x_0, x_1, \ldots, x_n\}$$

of $[a, b]$ into n subintervals such that the span of f in every subinterval is less than ε. Denote by $M_k(f)$ and $m_k(f)$ the absolute maximum and minimum values of f on the kth subinterval $[x_{k-1}, x_k]$. Then

$$M_k(f) - m_k(f) < \varepsilon \qquad \text{for each } k = 1, \ldots, n.$$

Now define two step functions s_n and t_n on $[a, b]$ by

$$s_n(x) = \begin{cases} m_k(f), & \text{if } x_{k-1} < x \leq x_k, \\ m_1(f), & x = a, \end{cases} \qquad t_n(x) = \begin{cases} M_k(f), & \text{if } x_{k-1} \leq x < x_k, \\ M_n(f), & x = b. \end{cases}$$

Then clearly

$$s_n(x) \leq f(x) \leq t_n(x), \qquad \forall\, x \in [a, b].$$

Moreover,

$$\int_a^b s_n = \sum_{k=1}^n m_k(f)(x_k - x_{k-1}), \qquad \int_a^b t_n = \sum_{k=1}^n M_k(f)(x_k - x_{k-1}).$$

The difference of these two integrals is

$$\int_a^b t_n - \int_a^b s_n < \varepsilon \sum_{k=1}^n (x_k - x_{k-1}) = \varepsilon(b - a) = \frac{b - a}{N}.$$

On the other hand, the upper and lower integrals of f satisfy

$$\int_a^b s_n \leq \underline{I}(f) \leq \overline{I}(f) \leq \int_a^b t_n.$$

Hence we obtain

$$0 \leq \overline{I}(f) - \underline{I}(f) \leq \int_a^b t_n - \int_a^b s_n < \frac{b - a}{N}.$$

Since this holds for every integer $N \geq 1$, we must have

$$\overline{I}(f) = \underline{I}(f).$$

Thus f is integrable on $[a, b]$. □

Theorem 4.2.20 If f is continuous on $[a, b]$, then

$$\int_a^b f(x)\,\mathrm{d}x = f(c)(b - a) \quad \text{for some} \quad c \in [a, b],$$

i.e., there exists $c \in [a, b]$ such that

$$f(c) = \operatorname{avg}(f) = \frac{1}{b - a} \int_a^b f(x)\,\mathrm{d}x.$$

Proof. Let $m = \inf f$ and $M = \sup f$. Since $m \leq f(x) \leq M$ for all $x \in [a, b]$, we have

$$m = \frac{1}{b-a} \int_a^b m \, \mathrm{d}x \leq \frac{1}{b-a} \int_a^b f(x) \, \mathrm{d}x \leq \frac{1}{b-a} \int_a^b M \, \mathrm{d}x = M.$$

By **EVT** and **IVT**, we know there exists $c \in [a, b]$ such that $f(c) = \operatorname{avg}(f)$. $\square$

Remark 4.2.21 You can also expand the above theorem to what is so-called **Weighted Mean Value Theorem for Integrals** mentioned in Chapter 3. The proof is left as an exercise (since it's quite analogous to the above proof).

Chapter 5 Differential Calculus

1 Basic Concepts

Definition 5.1.1 Let f be a function. If

$$\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=L\in\mathbb{R},$$

then we call it the **derivative** of f and denote it by $f'(x)$. Furthermore, the n-th derivative of f is defined by

$$f^{(n)}(x)=\left[f^{(n-1)}(x)\right]' \quad \text{for} \quad n=1,2,3,\cdots$$

and $f^{(0)}(x)=f(x)$.

Example 5.1.2 We're going to prove some basic derivatives using the definition.

1. Suppose n is a positive integer and $x\geq 0$. Consider $f(x)=x^n$, then

$$\begin{aligned} f'(x) &= \lim_{h\to 0}\frac{(x+h)^n-x^n}{h} \\ &= \lim_{h\to 0}\sum_{k=0}^{n-1}(x+h)^k x^{n-1-k} \\ &= \sum_{k=0}^{n-1}x^k x^{n-1-k} \\ &= nx^{n-1}. \end{aligned}$$

2. Consider $f(x)=\sin x$, then

$$\begin{aligned} f'(x) &= \lim_{h\to 0}\frac{\sin(x+h)-\sin x}{h} \\ &= \lim_{h\to 0}\frac{2\sin\left(\frac{h}{2}\right)\cos\left(x+\frac{h}{2}\right)}{h} \\ &= \left[\lim_{h\to 0}\frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}}\right]\left[\lim_{h\to 0}\cos\left(x+\frac{h}{2}\right)\right] \\ &= \cos x. \end{aligned}$$

3. Consider $f(x) = \cos x$, then

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{\cos(x+h) - \cos x}{h} \\ &= \lim_{h \to 0} \frac{-2 \sin\left(\frac{h}{2}\right) \sin\left(x + \frac{h}{2}\right)}{h} \\ &= -\left[\lim_{h \to 0} \frac{\sin\left(\frac{h}{2}\right)}{\frac{h}{2}}\right] \left[\lim_{h \to 0} \sin\left(x + \frac{h}{2}\right)\right] \\ &= -\sin x. \end{aligned}$$

4. Suppose n is a positive integer and $x > 0$. Consider $f(x) = x^{\frac{1}{n}}$ and for convenience let

$$u = (x+h)^{\frac{1}{n}} \quad \text{and} \quad v = x^{\frac{1}{n}}.$$

Hence we have

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{(x+h)^{\frac{1}{n}} - x^{\frac{1}{n}}}{h} \\ &= \lim_{h \to 0} \frac{u - v}{u^n - v^n} \\ &= \lim_{u \to v} \frac{1}{u^{n-1} + u^{n-2}v + \cdots + uv^{n-2} + v^{n-1}} \\ &= \frac{1}{nx^{\frac{n-1}{n}}} \\ &= \frac{1}{n} x^{\frac{1}{n} - 1}. \end{aligned}$$

For the case $x = 0$, we can see $f(x)$ is defined at $x = 0$ but $f'(x)$ is not defined.

Theorem 5.1.3 If function f has a derivative at x, then f is continuous at x.

Proof. Let

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

So

$$\lim_{h \to 0} f(x+h) = \lim_{h \to 0} \left[\frac{f(x+h) - f(x)}{h} \cdot h + f(x)\right] = f'(x) \cdot 0 + f(x) = f(x).$$

□

Remark 5.1.4 The converse of the above theorem is not true, i.e., if f is continuous at x, then f does not necessarily have a derivative at x. The absolute value function $f(x) = |x|$ is a typical counterexample since f is continuous at 0 but $f'(0)$ does not exist.

Moreover, even if f is differentiable at x, f' may not be continuous at x. For example, we define

$$f(x) = \begin{cases} x^2 \sin \dfrac{1}{x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}, \qquad f'(x) = \begin{cases} 2x \sin \dfrac{1}{x^2} - \dfrac{2}{x} \cos \dfrac{1}{x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}.$$

Then f is differentiable on $\mathbb{R}$ but f' is not continuous at 0 since $\lim\limits_{x \to 0} f'(x)$ does not exist while $f'(0) = 0$.

Theorem 5.1.5 Suppose f and g are two functions defined on a common interval. If both f adn g have derivatives at $x = a$, then so do the following functions:

$$f + g, f - g, fg, \frac{f}{g} \text{ (provided that } g(a) \neq 0).$$

Additionally, their derivatives satisfy the following:

(1) $(f + g)' = f' + g'$,

(2) $(f - g)' = f' - g'$,

(3) $(fg)' = f'g + fg'$,

(4) $\left(\dfrac{f}{g}\right)' = \dfrac{f'g - fg'}{g^2}$.

Proof. We only prove (3) and (4) here.

(3) We have

$$\begin{aligned} (fg)'(a) &= \lim_{h \to 0} \frac{f(a+h)g(a+h) - f(a)g(a)}{h} \\ &= \lim_{h \to 0} \left[g(a+h) \cdot \frac{f(a+h) - f(a)}{h} + f(a) \cdot \frac{g(a+h) - g(a)}{h} \right] \\ &= g(a) \cdot f'(a) + f(a) \cdot g'(a). \end{aligned}$$

(4) We have

$$\begin{aligned} \left(\frac{f}{g}\right)'(a) &= \lim_{h \to 0} \frac{\dfrac{f(a+h)}{g(a+h)} - \dfrac{f(a)}{g(a)}}{h} \\ &= \lim_{h \to 0} \frac{f(a+h)g(a) - f(a)g(a+h)}{hg(a+h)g(a)} \\ &= \lim_{h \to 0} \left[\frac{g(a) \cdot [f(a+h) - f(a)]}{hg(a+h)g(a)} - \frac{f(a) \cdot [g(a+h) - g(a)]}{hg(a+h)g(a)} \right] \\ &= \frac{g(a) \cdot f'(a)}{g^2(a)} - \frac{f(a) \cdot g'(a)}{g^2(a)} \\ &= \left. \frac{f'g - fg'}{g^2} \right|_{x=a}. \end{aligned}$$

□

Remark 5.1.6 Using these properties and induction, we can expand

$$(x^n)' = nx^{n-1}$$

for all rational numbers n.

Notation 5.1.7 In fact, the following notations are also commonly used for derivatives:

1. Leibniz's notation:
$$\frac{\mathrm{d}y}{\mathrm{d}x}, \frac{\mathrm{d}^2y}{\mathrm{d}x^2}, \frac{\mathrm{d}^3y}{\mathrm{d}x^3}, \cdots, \frac{\mathrm{d}^ny}{\mathrm{d}x^n},$$
where $\mathrm{d}x$ and $\mathrm{d}y$ are called **differentials**.
2. Lagrange's notation:
$$f', f'', f''', \cdots, f^{(n)};$$
3. Newton's notation (mainly used in physics):
$$\dot{y}, \ddot{y}, \dddot{y}, \cdots;$$
4. Arbogast's notation:
$$Df, D^2f, D^3f, \cdots, D^nf,$$
where D is called the **differential operator**. It's easy to verify that D is linear.

2 Chain Rule for Differentiation

Theorem 5.2.1 Suppose u and v are functions and the function f is given by $f = u \circ v$. Suppose $v'(x)$ and $u'(y)$ where $y = v(x)$ both exist. Then $f'(x)$ exists and is given by

$$f'(x) = u'(v(x)) \cdot v'(x).$$

Proof. For convenience, let

$$y = v(x), \quad \Delta y = v(x+h) - v(x).$$

Let

$$g(t) = \begin{cases} 0, & \text{if } t = 0, \\ \dfrac{u(y+t) - u(y)}{t} - u'(y), & \text{if } t \neq 0, \end{cases}$$

since we know for sure

$$\lim_{t\to 0} g(t) = \lim_{t\to 0}\left[\frac{u(y+t)-u(y)}{t} - u'(y)\right] = 0.$$

Then we have

$$u(y+t) - u(y) = \begin{cases} 0\left[u'(y) + g(t)\right], & \text{if } t = 0 \\ t\left[u'(y) + g(t)\right], & \text{if } t \neq 0 \end{cases} = t\left[u'(y) + g(t)\right]$$

no matter $t = 0$ or not. Thus plugging in $t = \Delta y$, we know

$$f'(x) = \lim_{h \to 0} \frac{u(y + \Delta y) - u(y)}{h} = \lim_{h \to 0} \frac{\Delta y}{h} \left[u'(y) + g(\Delta y) \right].$$

Note that

$$\lim_{h \to 0} \frac{\Delta y}{h} = v'(x)$$

as discussed before. Also we know g is continuous at 0, so

$$\lim_{h \to 0} \left[g(\Delta y) + u'(y) \right] = g(0) + u'(y) = u'(y).$$

Hence we have

$$f'(x) = u'(v(x)) \cdot v'(x).$$

$\square$

Remark 5.2.2 In Leibniz's notation, suppose $y = v(x)$ and $z = u(y)$, then we write

$$\frac{\mathrm{d}y}{\mathrm{d}x} := v'(x) \quad \text{and} \quad \frac{\mathrm{d}z}{\mathrm{d}y} := u'(y).$$

We also write

$$\frac{\mathrm{d}z}{\mathrm{d}x} := f'(x) \quad \text{where} \quad f = u \circ v.$$

Then the chain rule can be written as

$$\frac{\mathrm{d}z}{\mathrm{d}x} = \frac{\mathrm{d}z}{\mathrm{d}y} \cdot \frac{\mathrm{d}y}{\mathrm{d}x} \not\equiv \frac{\mathrm{d}z}{\cancel{\mathrm{d}y}} \cdot \frac{\cancel{\mathrm{d}y}}{\mathrm{d}x}.$$

Here, $\not\equiv$ means "not identically equal to", which indicates that the differentials $\mathrm{d}y$ are *not* actually quantities that can be cancelled out.

3 Theorems of Derivatives

Definition 5.3.1 A function f defined on a set $S \subseteq \mathbb{R}$ is said to have a **relative maximum** at $c \in S$ if there exists some open interval I containing c such that

$$\forall x \in I \cap S, \ f(x) \leq f(c).$$

Similarly, we can define **relative minimum** of f at $d \in S$.

Remark 5.3.2 We also say f has an **extremum** at c if it has either a relative maximum or a relative minimum at c.

Theorem 5.3.3 Suppose function f is defined on an open interval (a, b) containing c. If f has a relative extremum at c and if $f'(c)$ exists, then

$$f'(c) = 0.$$

Proof. Let

$$g(x)=\begin{cases}\dfrac{f(x)-f(c)}{x-c}, & \text{if } x\in(a,b), x\neq c,\\ f'(c), & \text{if } x=c.\end{cases}$$

It suffices to show that $g(c)=0$. Note

$$\lim_{x\to c}\frac{f(x)-f(c)}{x-c}=f'(c)$$

since $f'(c)$ exists, thus g is continuous at c. Now suppose $g(c)>0$, then there exists some $\delta>0$ such that

$$g(x)>0,\quad \forall x\in(c-\delta,c+\delta)\subseteq(a,b).$$

If $x\in(c,c+\delta)$, then from $g(x)>0$ we derive $f(x)>f(c)$. If $x\in(c-\delta,c)$, then from $g(x)>0$ we derive $f(x)<f(c)$. Note this contradicts to f having an extremum at c. Similarly, we can prove that $g(c)<0$ also leads to a contradiction. Thus we must have $g(c)=0$, i.e., $f'(c)=0$. □

Theorem 5.3.4 (Rolle's Theorem) Let f be continuous on $[a,b]$ and differentiable on (a,b). If $f(a)=f(b)$, then there exists some $c\in(a,b)$ such that

$$f'(c)=0.$$

Proof. Suppose $f'(x)\neq 0$ for all $x\in(a,b)$. By **EVT**, we know

$$\sup f=f(\alpha)\quad\text{and}\quad\inf f=f(\beta)\quad\text{for some}\quad\alpha,\beta\in[a,b].$$

Note we know f have extremum at α and β. If $\alpha,\beta\in(a,b)$, then by the above theorem we have

$$f'(\alpha)=f'(\beta)=0,$$

which contradicts our assumption. Thus $\alpha,\beta\in\{a,b\}$ and $f(\alpha)=f(\beta)$. Thus we know f is constant on $[a,b]$, which implies

$$f'(x)=0,\quad\forall x\in(a,b),$$

contradicting our assumption again. Then we complete the proof. □

Theorem 5.3.5 (Lagrange's Mean Value Theorem) Let f be continuous on $[a,b]$ and differentiable on (a,b). Then there exists some $c\in(a,b)$ such that

$$f'(c)=\frac{f(b)-f(a)}{b-a}.$$

Proof. Define a new function

$$g(x)=f(x)-\left[\frac{f(b)-f(a)}{b-a}(x-a)+f(a)\right].$$

Clearly g is continuous on $[a,b]$ and differentiable on (a,b). Note that

$$\begin{cases} g(a) = f(a) - f(a) = 0, \\ g(b) = f(b) - \left[\dfrac{f(b)-f(a)}{b-a}(b-a) + f(a)\right] = 0. \end{cases}$$

By **Rolle's Theorem**, there exists some $c \in (a,b)$ such that

$$g'(c) = f'(c) - \frac{f(b)-f(a)}{b-a} = 0,$$

which implies

$$f'(c) = \frac{f(b)-f(a)}{b-a}.$$

□

Theorem 5.3.6 (Cauchy's Mean Value Theorem) Suppose f and g are both continuous on $[a,b]$ and differentiable on (a,b), then there exists some $c \in (a,b)$ such that

$$f'(c)\left[g(b) - g(a)\right] = g'(c)\left[f(b) - f(a)\right].$$

Proof. Define a new function

$$h(x) = f(x)\left[g(b) - g(a)\right] - g(x)\left[f(b) - f(a)\right].$$

Clearly h is continuous on $[a,b]$ and differentiable on (a,b). Note that

$$\begin{cases} h(a) = f(a)\left[g(b) - g(a)\right] - g(a)\left[f(b) - f(a)\right], \\ h(b) = f(b)\left[g(b) - g(a)\right] - g(b)\left[f(b) - f(a)\right]. \end{cases} \Rightarrow h(b) - h(a) = 0.$$

By **Rolle's Theorem**, there exists some $c \in (a,b)$ such that

$$h'(c) = f'(c)\left[g(b) - g(a)\right] - g'(c)\left[f(b) - f(a)\right] = 0,$$

which implies

$$f'(c)\left[g(b) - g(a)\right] = g'(c)\left[f(b) - f(a)\right].$$

□

Theorem 5.3.7 Suppose f is continuous on $[a,b]$ and differentiable on (a,b). Then:

(1) If $f'(x) = 0$ for all $x \in (a,b)$, then f is constant on $[a,b]$;

(2) If $f'(x) > 0$ for all $x \in (a,b)$, then f is strictly increasing on $[a,b]$;

(3) If $f'(x) < 0$ for all $x \in (a,b)$, then f is strictly decreasing on $[a,b]$.

Theorem 5.3.8 Suppose f is continuous on $[a, b]$ and differentiable on (a, b) and $c \in (a, b)$. Then:

(1) If $f'(x) > 0$ for all $x \in (a, c)$ and $f'(x) < 0$ for all $x \in (c, b)$, then f has a relative maximum at c;

(2) If $f'(x) < 0$ for all $x \in (a, c)$ and $f'(x) > 0$ for all $x \in (c, b)$, then f has a relative minimum at c.

Remark 5.3.9 The above two cases still hold even if $f'(c)$ is not well-defined. (We still require f to be continuous at c.)

Definition 5.3.10 Suppose f is continuous on $[a, b]$ and differentiable on (a, b) and $c \in (a, b)$. If $f'(c) = 0$, then we call c a **critical point** of f.

Theorem 5.3.11 Suppose f has a second-derivative f'' defined on (a, b) and $c \in (a, b)$ is a critical point of f. Then:

(1) If $f''(c) > 0$, then f has a relative minimum at c;

(2) If $f''(c) < 0$, then f has a relative maximum at c;

(3) If $f''(c) = 0$, then the test is inconclusive, i.e., f may have a relative maximum, a relative minimum or neither at c.

Theorem 5.3.12 Suppose f is continuous on $[a, b]$ and differentiable on (a, b). Then:

1. If f' is increasing on (a, b), or equivalently $f''(x) \geq 0$ for all $x \in (a, b)$ provided that f'' is well-defined, then f is convex on $[a, b]$.
2. If f' is decreasing on (a, b), or equivalently $f''(x) \leq 0$ for all $x \in (a, b)$ provided that f'' is well-defined, then f is concave on $[a, b]$.

Remark 5.3.13 Note that a flat line is both convex and concave (in our text book's definition).

Chapter 6 Integration and Differentiation

1 Two Fundamental Theorems for Calculus

Theorem 6.1.1 (First Fundamental Theorem of Calculus (FTC1)) Suppose f is a function that is integrable on $[a, x]$ for every $x \in [a, b]$. Let $c \in [a, b]$ and define function F on $[a, b]$ by

$$F(x) = \int_c^x f(t)\,\mathrm{d}t.$$

Then for any $x \in (a, b)$, if f is continuous at x, then $F'(x)$ exists and

$$F'(x) = f(x) \quad \text{or} \quad \frac{\mathrm{d}}{\mathrm{d}x}\left(\int_c^x f(t)\,\mathrm{d}t\right) = f(x).$$

Proof. Let $x \in (a, b)$ be fixed. Consider the quotient in the definition of $F'(x)$:

$$\begin{aligned}
\frac{F(x+h) - F(x)}{h} &= \frac{1}{h}\left(\int_c^{x+h} f(t)\,\mathrm{d}t - \int_c^x f(t)\,\mathrm{d}t\right) \\
&= \frac{1}{h}\int_x^{x+h} f(t)\,\mathrm{d}t \\
&= \frac{1}{h}\int_x^{x+h} [f(x) + (f(t) - f(x))]\,\mathrm{d}t \\
&= f(x) + \frac{1}{h}\int_x^{x+h} (f(t) - f(x))\,\mathrm{d}t
\end{aligned}$$

Note it suffices to show

$$G(h) := \frac{1}{h}\int_x^{x+h} (f(t) - f(x))\,\mathrm{d}t \to 0 \quad \text{as} \quad h \to 0.$$

Given any $\varepsilon > 0$, since f is continuous at x, there exists some $\delta > 0$ such that

$$|f(t) - f(x)| < \frac{\varepsilon}{2}, \quad \forall t \in (x - \delta, x + \delta) \subseteq (a, b).$$

Claim: Suppose f is a function integrable on $[a, b]$, then we have

$$\int_a^b |f(t)|\mathrm{d}t \geq \left|\int_a^b f(t)\mathrm{d}t\right|.$$

Proof. Note that

$$|f(x)| \geq \max\{f(x), -f(x)\}.$$

So we have

$$\int_a^b |f(t)|\mathrm{d}t \geq \max\left\{\int_a^b f(x)\mathrm{d}t, -\int_a^b f(x)\mathrm{d}t\right\} = \left|\int_a^b f(t)\mathrm{d}t\right|.$$

Thus the claim holds. $\square$

Case 1: Suppose $0 < h < \delta$, then $[x-h, x+h] \subseteq (x-\delta, x+\delta)$. Thus we have

$$|G(h)| = \left|\frac{1}{h}\int_x^{x+h} (f(t) - f(x))\,\mathrm{d}t\right| \leq \frac{1}{h}\int_x^{x+h} |f(t) - f(x)|\mathrm{d}t < \frac{\varepsilon}{2} < \varepsilon.$$

So $G(h) \to 0$ as $h \to 0^+$.
Case 2: Suppose $-\delta < h < 0$, then analogously we have

$$\lim_{h\to 0^-} G(h) = 0.$$

Together we have $G(h) \to 0$ as $h \to 0$. Thus we complete the proof. $\square$

Definition 6.1.2 A function P is called a **primitive** (or an **antiderivative**) on a function f on an open interval I if $P'(x) = f(x)$ for any $x \in I$.

Remark 6.1.3 Note that if P is a primitive of f on I, then P is not unique since for any constant C, we have $(P + C)' = P' = f$.

Theorem 6.1.4 (Second Fundamental Theorem of Calculus (FTC2)) Assume f is continuous on an open interval I and let P be a primitive of f on I. Then for each $x, c \in I$, we have

$$\int_c^x f(t)\,\mathrm{d}t = P(x) - P(c).$$

Remark 6.1.5 We have

$$\int_a^b f(x)\,\mathrm{d}x = P(b) - P(a)$$

provided that P is a primitive of f on some open interval containing $[a, b]$.

Remark 6.1.6 If F is a primitive of f on I but f is not necessarily continuous, then f is not necessarily integrable. For example, we define

$$F(x) = \begin{cases} x^2 \sin\frac{1}{x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}, \qquad f(x) = F'(x) = \begin{cases} 2x\sin\frac{1}{x^2} - \frac{2}{x}\cos\frac{1}{x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Then F is a primitive of f on $\mathbb{R}$. However, f is not integrable in the neighborhood of 0 since f is not bounded

near 0.
The converse is also not true, i.e., if f is integrable on I, then f does not necessarily have a primitive on I. For example, we define

$$f(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Then f is integrable on $[0, 1]$ but f does not have a primitive.

Notation 6.1.7 Leibniz introduced the notation **indefinite integral** to denote a general primitive of f or the common format shared by all primitive of f:

$$\int f(x) \, \mathrm{d}x := \int_c^x f(t) \, \mathrm{d}t + C$$

where C is an arbitrary constant. In practice, we often just write

$$\int f(x) \, \mathrm{d}x = P(x) + C$$

where $P(x)$ it self *does not* have any constant term.

Remark 6.1.8 In some textbooks, since we often write

$$C = \int_d^c f(t) \, \mathrm{d}t$$

and then

$$\int f(x) \, \mathrm{d}x = \int_c^x f(t) \, \mathrm{d}t + C = \int_d^x f(t) \, \mathrm{d}t,$$

the defintion of indefinite integral is generalized so that Leibniz's notation is also called an indefinite integral.

2 Integration Techniques

Method 6.2.1 (Integration by Substitution) Suppose $u = g(x)$ is a differentiable function whose range is an interval I and f is continuous on I. Then

$$\int f(g(x)) g'(x) \, \mathrm{d}x = \int f(u) \, \mathrm{d}u.$$

Theorem 6.2.2 Suppose g has a continuous derivative g' on an open interval I. Let J be the set of values taken by g on I and assume that f is continuous on J. Then for any $x, c \in I$, we have

$$\int_c^x f(g(t)) g'(t) \, \mathrm{d}t = \int_{g(c)}^{g(x)} f(u) \, \mathrm{d}u.$$

Proof. Since g is continuous, by **IVT** we know $J = g(I)$ is a well-defined interval. Thus we can define

$$F(u) = \int_{g(c)}^{u} f(w)\,\mathrm{d}w, \quad u \in J.$$

Note that f is continuous on J, we know F is differentiable on J. By **FTC1**, we have $F'(u) = f(u)$. Define

$$H(t) = F(g(t)), \quad t \in I$$

with

$$H'(t) = F'(g(t))\,g'(t) = f(g(t))\,g'(t), \quad t \in I.$$

By **FTC2**, we know

$$\begin{aligned}\int_c^x f(g(t))g'(t)\,\mathrm{d}t &= \int_c^x H'(t)\,\mathrm{d}t = H(x) - H(c) = F(g(x)) - F(g(c)) \\ &= \int_{g(c)}^{g(x)} f(w)\,\mathrm{d}w - \int_{g(c)}^{g(c)} f(w)\,\mathrm{d}w = \int_{g(c)}^{g(x)} f(w)\,\mathrm{d}w.\end{aligned}$$

Note that the integral value is independent of integral variable. Hence, we complete the proof. □

Method 6.2.3 (Integration by Parts) Suppose $u = u(x)$ and $v = v(x)$ are two differentiable functions on an open interval I. Then

$$\int u(x)v'(x)\,\mathrm{d}x = u(x)v(x) - \int v(x)u'(x)\,\mathrm{d}x.$$

For definite integrals, we have

$$\int_a^b u(x)v'(x)\,\mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b v(x)u'(x)\,\mathrm{d}x.$$

Remark 6.2.4 With Leibniz's notation, since equivalently (in a sense of notation but it's rigorous by defining what is so called differential later) we have

$$\mathrm{d}u = u'(x)\,\mathrm{d}x \quad \text{and} \quad \mathrm{d}v = v'(x)\,\mathrm{d}x,$$

we can write integration by parts as

$$\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u.$$

Leibniz's notation is often more convenient and aesthetically pleasing.

Example 6.2.5 Recall

$$\int x^n\,\mathrm{d}x = \frac{1}{n+1}x^{n+1} + C, \quad \text{does not hold when } n = -1.$$

Now we use integration by parts to try solve the case $n = -1$:

$$\begin{aligned}\int \frac{1}{x}\,\mathrm{d}x &= \int 1\cdot\frac{1}{x}\,\mathrm{d}x \\ &= x\cdot\frac{1}{x} - \int x\cdot\left(-\frac{1}{x^2}\right)\mathrm{d}x + C \\ &= 1 + \int \frac{1}{x}\,\mathrm{d}x + C.\end{aligned}$$

Now we see that the constant term C is actually important, otherwise we would derive something nonsense. And we know it's not a wise choice to use integration by parts here.

Theorem 6.2.6 Suppose g is continuous on $[a, b]$ and f has a continuous derivative f' on $[a, b]$. If f' does not change sign on $[a, b]$, then there exists some $c \in [a, b]$ such that

$$\int_a^b f(x)g(x)\,\mathrm{d}x = f(a)\int_a^c g(x)\,\mathrm{d}x + f(b)\int_c^b g(x)\,\mathrm{d}x.$$

Proof. Let

$$G(x) = \int_a^x g(t)\,\mathrm{d}t.$$

Since g is continuous on $[a, b]$, by the **FTC1**, we know $G'(x) = g(x)$ for any $x \in (a, b)$. By **integration by parts**, we have

$$\begin{aligned}\int_a^b f(x)g(x)\,\mathrm{d}x &= \int_a^b f(x)G'(x)\,\mathrm{d}x \\ &= [f(x)G(x)]_a^b - \int_a^b G(x)f'(x)\,\mathrm{d}x \\ &= f(b)G(b) - f(a)\cdot 0 - \int_a^b G(x)f'(x)\,\mathrm{d}x \\ &= f(b)\int_a^b g(t)\,\mathrm{d}t - \int_a^b G(x)f'(x)\,\mathrm{d}x\end{aligned}$$

Using the **MVT**, since f' does not change sign on $[a, b]$, there exists some $c \in [a, b]$ such that

$$\int_a^b G(x)f'(x)\,\mathrm{d}x = G(c)\int_a^b f'(x)\,\mathrm{d}x = \left(\int_a^c g(t)\,\mathrm{d}t\right)[f(b) - f(a)].$$

Thus we have

$$\begin{aligned}\int_a^b f(x)g(x)\,\mathrm{d}x &= f(b)\int_a^b g(t)\,\mathrm{d}t - \left(\int_a^c g(t)\,\mathrm{d}t\right)[f(b) - f(a)] \\ &= f(b)\int_c^b g(t)\,\mathrm{d}t + f(a)\int_a^c g(t)\,\mathrm{d}t.\end{aligned}$$

□

Chapter 7 Transcendental Functions

1 The Logarithm

Definition 7.1.1 If x is a positive real number, then the **natural logarithm** of x is defined by

$$\ln x = \int_1^x \frac{1}{t} \, \mathrm{d}t.$$

Theorem 7.1.2 The logarithm function has the following properties:

(1) $\ln 1 = 0$;

(2) $(\ln x)' = \frac{1}{x}$ for all $x > 0$;

(3) $\ln(xy) = \ln x + \ln y$ for all $x, y > 0$;

(4) $\ln x^r = r \ln x$ for all $x > 0$ and all positive integers r;

(5) $\ln \frac{1}{x} = -\ln x$ for all $x > 0$;

(6) $f(x) = \ln x$ is strictly increasing and unbounded on $(0, \infty)$;

(7) $f(x) = \ln x$ is concave on $(0, \infty)$.

Theorem 7.1.3 For every real number b, there exists a unique positive real number a such that $\ln a = b$.

Proof. Suppose $b > 0$ and pick positive integer n so that $\ln 2^n = n \ln 2 > b$. Since $f(x) = \ln x$ is continuous and strictly increasing on $[1, 2^n]$, by the **IVT**, there exists some unique $a \in (1, 2^n)$ such that $\ln a = b$.
The case $b = 0$ is trivial since $\ln 1 = 0$ and the case $b < 0$ can be proved similarly. $\square$

Definition 7.1.4 Particularly, we denote the unique positive real number e such that

$$\ln e = 1,$$

which is called Euler's number or the **natural number**.

Remark 7.1.5 Suppose $f(x) = c \ln x$ for some fixed constant c. We know it preserves all properties of $\ln x$. Now we know there exists a unique positive real number α such that

$$c \ln \alpha = 1, \quad (\text{or } f(\alpha) = 1, \alpha \neq 1) \implies c = \frac{1}{\ln \alpha}.$$

With this α notation, we can rewrite

$$f(x) = c\ln x = \frac{\ln x}{\ln \alpha},$$

which motivates us to define logarithm with base α.

Definition 7.1.6 Suppose $b > 0$ and $b \neq 1$. For any $x > 0$, the **logarithm with base** b is defined by

$$\log_b x = \frac{\ln x}{\ln b}.$$

We usually call b the **base** of the logarithm.

Remark 7.1.7 Note that $\log_e x = \ln x$.

Remark 7.1.8 By injectivity of logarithm function, we know for any $b > 0, u \in \mathbb{R}$ and $b \neq 1$, there exists a unique $x > 0$ such that $\log_b x = u$. We denote this unique x by

$$b^u.$$

If we adopt the convention that $1^u := 1$, then the power b^u is well-defined for any $b > 0$ and $u \in \mathbb{R}$.

Example 7.1.9 Here we see how logarithm helps us to integrate some functions.

1. $$I = \int \tan x \,\mathrm{d}x = \int \frac{\sin x}{\cos x}\,\mathrm{d}x = -\int \frac{(\cos x)'}{\cos x}\,\mathrm{d}x = -\ln|\cos x| + C.$$

2. $$I = \int \ln x \,\mathrm{d}x = \int 1 \cdot \ln x \,\mathrm{d}x = x\ln x - \int x \cdot \frac{1}{x}\,\mathrm{d}x = x\ln x - x + C.$$

3. $$I_1 = \int \sin(\ln x)\,\mathrm{d}x = \int \sin(\ln x)\cdot 1\,\mathrm{d}x = x\sin(\ln x) - \int x \cdot \frac{1}{x}\cos(\ln x)\,\mathrm{d}x.$$

   Now we need to calculate $\int \cos(\ln x)\,\mathrm{d}x$:

   $$I_2 = \int \cos(\ln x)\,\mathrm{d}x = \int \cos(\ln x)\cdot 1\,\mathrm{d}x = x\cos(\ln x) + \int x \cdot \frac{1}{x}\sin(\ln x)\,\mathrm{d}x.$$

   Thus we have

   $$\int \sin(\ln x)\,\mathrm{d}x = \frac{1}{2}\left[x\sin(\ln x) - x\cos(\ln x)\right] + C.$$
   $$\int \cos(\ln x)\,\mathrm{d}x = \frac{1}{2}\left[x\cos(\ln x) + x\sin(\ln x)\right] + C.$$

Remark 7.1.10 Since

$$\ln x = \int_1^x \frac{1}{t}\,\mathrm{d}t \quad \text{for all } x > 0,$$

using substitution $u = 1 - t$ we have

$$\ln(1-x) = \int_1^{1-x} \frac{1}{t}\,\mathrm{d}t = -\int_0^x \frac{1}{1-u}\,\mathrm{d}u \quad \text{for all } x < 1.$$

We use a polynomial series to approximate $\dfrac{1}{1-u}$:

$$-\ln(1-x) = \int_0^x \left(1 + u + \frac{u^2}{1-u}\right)\mathrm{d}u = x + \frac{x^2}{2} + \int_0^x \frac{u^2}{1-u}\,\mathrm{d}u.$$

Note when x is near 0, the polynomial $P(x) = x + \dfrac{x^2}{2}$ is a good approximation of $-\ln(1-x)$. This motivates us to introduce the following theorem.

Theorem 7.1.11 Let

$$P_n(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots + \frac{x^n}{n} = \sum_{k=1}^{n} \frac{x^k}{k}$$

denote the polynomial of degree n. Then for any $x < 1$ and any positive integer n, we have

$$-\ln(1-x) = P_n(x) + \int_0^x \frac{t^n}{1-t}\,\mathrm{d}t.$$

Remark 7.1.12 We denote the last term by

$$E_n(x) := \int_0^x \frac{t^n}{1-t}\,\mathrm{d}t$$

and call it the **error term**. Now we study the error term and see how good the approximation is.

Theorem 7.1.13 With the same setup as above, we have the following:

(1) If $0 < x < 1$, then

$$\frac{x^{n+1}}{n+1} \leq E_n(x) \leq \frac{1}{(1-x)} \cdot \frac{x^{n+1}}{(n+1)}.$$

(2) If $x < 0$, then

$$0 < (-1)^{n+1} E_n(x) \leq \frac{|x|^{n+1}}{n+1}.$$

Remark 7.1.14 (1) If $|x|$ is "very small", then $E_n(x)$ is also "very small" (no matter what n is).

(2) If $|x|$ is fixed and n is "very large", then $E_n(x)$ is also "very small" and thus $P_n(x)$ is a good approximation of $-\ln(1-x)$.

2 Exponential Function

Definition 7.2.1 For any real number x, the **exponential function** or the **antilogorithm** is defined by

$$y = e^x = \text{the unique positive real number such that } \ln y = x.$$

Notation 7.2.2 We use the notation $\exp(x)$ to denote e^x sometimes.

Remark 7.2.3 Note that

$$\mathrm{Dom}(\exp) = \mathbb{R}, \quad \mathrm{Im}(\exp) = (0, +\infty).$$

Since $\ln x$ is continuous and strictly increasing on $(0, +\infty)$ and $\exp x$ is the inverse of $\ln x$, we know $\exp(x)$ is also continuous and strictly increasing on $\mathbb{R}$.

Theorem 7.2.4 The exponential function has the following properties:

(1) $e^0 = 1$;

(2) $(e^x)' = e^x$ for all $x \in \mathbb{R}$;

(3) $e^{x+y} = e^x \cdot e^y$ for all $x, y \in \mathbb{R}$;

Proof. We only prove (2) here. By the definition of derivative, we have

$$(e^x)' = \lim_{h \to 0} \frac{e^{x+h} - e^x}{h} = e^x \lim_{h \to 0} \frac{e^h - 1}{h}.$$

It suffices to show the limit is 1. Note that

$$k := e^h - 1 \implies h = \ln(k+1).$$

Thus we have

$$\lim_{h \to 0} \frac{e^h - 1}{h} = \lim_{k \to 0} \frac{k}{\ln(k+1)} = \lim_{k \to 0} \frac{k}{\ln(1+k) - \ln 1} = \frac{1}{(\ln(1+k))'}\bigg|_{k=0} = 1.$$

□

Definition 7.2.5 For any $a > 0$ and $x \in \mathbb{R}$, we also define $a^x := e^{x \ln a}$.

Method 7.2.6 Using exponential function, we can rewrite integration by substitution as

$$\int e^{f(x)} f'(x)\,\mathrm{d}x = e^{f(x)} + C.$$

Definition 7.2.7 The **hyperbolic sine** and **hyperbolic cosine** functions are defined by

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}.$$

Similarly, we define the **hyperbolic tangent** function by

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

Theorem 7.2.8 Suppose f is strictly increasing and continuous on $[a, b]$ and let g be the inverse of f. If the derivative $f'(x)$ exists and is non-zero on (a, b), then the derivative $g'(y)$ also exists and is non-zero at the corresponding $y = f(x)$. Moreover, we have

$$g'(y) = \frac{1}{f'(g(y))} \quad \text{or} \quad \frac{\mathrm{d}}{\mathrm{d}y} f^{-1}(y) = \frac{1}{f'(f^{-1}(y))}.$$

3 Inverse of Trigometric Functions

Note that the usual trigonometric functions are not invertible on their entire domains (due to periodicity). Therefore, in order to define their "inverses", we consider them restricted to some interval on which they are monotonic.

Definition 7.3.1 Here are the definitions of the three most common inverse trigonometric functions:

1. To define the **inverse sine function**, we consider $f(x) = \sin x$ for $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, we denote the inverse function by $g(y) = \arcsin y$ for $y \in [-1, 1]$.
2. To define the **inverse cosine function**, we consider $f(x) = \cos x$ for $x \in [0, \pi]$, we denote the inverse function by $g(y) = \arccos y$ for $y \in [-1, 1]$.
3. To define the **inverse tangent function**, we consider $f(x) = \tan x$ for $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, we denote the inverse function by $g(y) = \arctan y$ for $y \in \mathbb{R}$.

Theorem 7.3.2 The derivatives of the inverse trigonometric functions are given by:

(1)

$$g'(y) = \frac{1}{f'(g(y))} = \frac{1}{\cos(\arcsin y)} = \frac{1}{\sqrt{1 - \sin^2(\arcsin y)}} = \frac{1}{\sqrt{1 - y^2}}.$$

(2)

$$g'(y) = \frac{1}{f'(g(y))} = \frac{-1}{\sin(\arccos y)} = \frac{-1}{\sqrt{1 - \cos^2(\arccos y)}} = \frac{-1}{\sqrt{1 - y^2}}.$$

(3)

$$g'(y) = \frac{1}{f'(g(y))} = \frac{1}{\sec^2(\arctan y)} = \frac{1}{1 + \tan^2(\arctan y)} = \frac{1}{1 + y^2}.$$

Theorem 7.3.3 The integrals involving inverse trigonometric functions are given by:

(1)

$$\int \arcsin x \, \mathrm{d}x = x \arcsin x + \int \frac{x^2}{\sqrt{1 - x^2}} \, \mathrm{d}x = x \arcsin x + \sqrt{1 - x^2} + C.$$

(2)

$$\int \arccos x \, \mathrm{d}x = x \arccos x - \int \frac{x^2}{\sqrt{1 - x^2}} \, \mathrm{d}x = x \arccos x - \sqrt{1 - x^2} + C.$$

(3)

$$\int \arctan x \, dx = x \arctan x - \int \frac{x^2}{1+x^2} \, dx = x \arctan x - x + \ln\left(1+x^2\right) + C.$$

Remark 7.3.4 As for $\operatorname{arcsec} x$, $\operatorname{arccsc} x$ and $\operatorname{arccot} x$, we define them to be:

1. $$\operatorname{arcsec} x := \arccos \frac{1}{x}, \quad |x| \geq 1;$$

2. $$\operatorname{arccsc} x := \arcsin \frac{1}{x}, \quad |x| \geq 1;$$

3. $$\operatorname{arccot} x := \frac{\pi}{2} - \arctan x, \quad x \in \mathbb{R}$$

4 Integration by Partial Fractions

Definition 7.4.1 We call a rational function $r(x) = \dfrac{f(x)}{g(x)}$ **proper** if the degree of $f(x)$ is strictly less than that of $g(x)$. Otherwise, we call it **improper**.

Remark 7.4.2 For an improper rational function, we can always use polynomial division to find polynomial $Q(x)$ and $R(x)$:

$$\frac{f(x)}{g(x)} = Q(x) + \frac{R(x)}{g(x)}$$

where $\dfrac{R(x)}{g(x)}$ is a proper rational function. Here $Q(x)$ is called the **quotient** and $R(x)$ is called the **remainder**.

Definition 7.4.3 A quadratic function/factor of the form $x^2 + bx + c$ is called **irreducible** over $\mathbb{R}$ if it cannot be further factored as $(x + x_1)(x + x_2)$ where $x_1, x_2 \in \mathbb{R}$, i.e. its discriminant $b^2 - 4c < 0$.

Theorem 7.4.4 Every proper rational function can be expressed as a finite sum of a collection of terms each of which takes the form

$$\frac{A}{(x+a)^k}, \quad \frac{Bx+C}{(x^2+bx+c)^m}$$

where A, B, C, a, b, c are real numbers, k, m are positive integers, and $x^2 + bx + c$ is irreducible over $\mathbb{R}$.

Method 7.4.5 Suppose $r(x) = \dfrac{f(x)}{g(x)}$ is proper.

Case 1: $g(x)$ can be factorized into a collection of distinct linear factors. For example, suppose

$$g(x) = (x - x_1)(x - x_2) \cdots (x - x_n) \quad \text{where } x_i \neq x_j \text{ for } i \neq j \in \{1, 2, \ldots, n\}$$

Then

$$\frac{f(x)}{g(x)} = \frac{A_1}{x - x_1} + \frac{A_2}{x - x_2} + \cdots + \frac{A_n}{x - x_n} \quad \text{for some } A_1, A_2, \ldots, A_n \in \mathbb{R}.$$

and
$$\int \frac{f(x)}{g(x)} \mathrm{d}x = A_1 \ln|x - x_1| + A_2 \ln|x - x_2| + \cdots + A_n \ln|x - x_n| + C.$$

Case 2: $g(x)$ can still be factorized into a collection of linear terms, but some of these linear terms are repeated. For example, suppose $(x-a)$ is repeated p times where $p \geq 2$ is an integer, then the corresponding term in the partial fraction decomposition becomes the following sum

$$\frac{B_1}{(x-a)} + \frac{B_2}{(x-a)^2} + \cdots + \frac{B_p}{(x-a)^p} \quad \text{for some } B_1, B_2, \ldots, B_p \in \mathbb{R}.$$

Case 3: $g(x)$ contains non-repeated irreducible quadratic factors (in addition to linear factors) $x^2 + bx + c$ where $b^2 - 4c < 0$. Then in the partial fraction decomposition, we will have the term

$$\frac{Ax + B}{x^2 + bx + c} \quad \text{for some } A, B \in \mathbb{R}.$$

Now we integrate this term:

$$\int \frac{Ax + B}{x^2 + bx + c} \, \mathrm{d}x = \frac{A}{2} \ln|x^2 + bx + c| + \frac{2B - Ab}{\sqrt{4c - b^2}} \arctan\left(\frac{2x + b}{\sqrt{4c - b^2}}\right) + C.$$

Case 4: $g(x)$ contains repeated irreducible quadratic factors of the form $(x^2 + bx + c)^p$ where $p \geq 2$ is an integer. Then in the partial fraction decomposition, we will have the term

$$\frac{A_1 x + B_1}{x^2 + bx + c} + \frac{A_2 x + B_2}{(x^2 + bx + c)^2} + \cdots + \frac{A_p x + B_p}{(x^2 + bx + c)^p} \quad \text{for some } A_i, B_i \in \mathbb{R}.$$

And we do similarly as Case 3 to integrate each term.

Chapter 8 Polynomial Approximations to Functions

With the help of a special identity $1 - x^n = (1-x)(1+x+x^2+\cdots+x^{n-1})$, we can find a polynomial approximation to $-\ln(1-x)$ in the previous section. However, this may not help to find a polynomial approximation to other functions, which motivates us to find a more general method to find polynomial approximations to functions.

1 The Taylor Polynomial for a Function

Theorem 8.1.1 Let f be a function with derivatives of order n at the point $x = a$. Then there exists a unique polynomial P of degree at most n such that $P(a) = f(a), P'(a) = f'(a), \ldots, P^{(n)}(a) = f^{(n)}(a)$. Additionally, this polynomial is given by

$$P(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

and it is referred to as the **Taylor polynomial of degree** n **for** f **at** $x = a$.

Notation 8.1.2 We often denote P in the above theorem by $T_n(f)$ or $T_n(f; a)$ to emphasize the degree and the center of the Taylor polynomial.

Example 8.1.3 Here are some examples of Taylor polynomials:

(1) The Taylor polynomial of degree n for $f(x) = e^x$ at $x = 0$ is given by

$$T_n(f) = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} = \sum_{k=0}^{n} \frac{x^k}{k!}.$$

(2) The Taylor polynomial of degree $2n+1$ for $f(x) = \sin x$ at $x = 0$ is given by

$$T_{2n+1}(f) = x - \frac{x^3}{3!} + \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} = \sum_{k=0}^{n} (-1)^k \frac{x^{2k+1}}{(2k+1)!}.$$

(3) The Taylor polynomial of degree $2n$ for $f(x) = \cos x$ at $x = 0$ is given by

$$T_{2n}(f) = 1 - \frac{x^2}{2!} + \cdots + (-1)^n \frac{x^{2n}}{(2n)!} = \sum_{k=0}^{n} (-1)^k \frac{x^{2k}}{(2k)!}.$$

Theorem 8.1.4 Here are some properties of the Taylor polynomial:

(1) Linearity: if $c_1, c_2 \in \mathbb{R}$, then
$$T_n(c_1 f + c_2 g) = c_1 T_n(f) + c_2 T_n(g).$$

(2) Differentiation:
$$(T_n(f))' = T_{n-1}(f').$$

(3) Integration: if $F(x) = \displaystyle\int_a^x f(t)\,\mathrm{d}t$, then
$$T_{n+1}(F) = \int_a^x T_n(f(t))\,\mathrm{d}t.$$

(4) Substitution: if $g(x) = f(cx)$ where $c \in \mathbb{R}$ is a constant, then
$$T_n(g)(x; a) = T_n(f)(cx; ca).$$

Particularly, if $a = 0$, then $T_n(g)(x) = T_n(f)(cx)$.

Example 8.1.5 With the help of the properties above, we continue to find the Taylor polynomial of degree n for more functions at $x = 0$:

(4) The Taylor polynomial of degree $2n$ for $f(x) = \cosh x$ at $x = 0$ is given by
$$T_{2n}(f) = 1 + \frac{x^2}{2!} + \cdots + \frac{x^{2n}}{(2n)!} = \sum_{k=0}^{n} \frac{x^{2k}}{(2k)!}.$$

(5) The Taylor polynomial of degree $2n-1$ for $f(x) = \sinh x$ at $x = 0$ is given by
$$T_{2n-1}(f) = x + \frac{x^3}{3!} + \cdots + \frac{x^{2n-1}}{(2n-1)!} = \sum_{k=0}^{n-1} \frac{x^{2k+1}}{(2k+1)!}.$$

(6) Recall
$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \frac{x^{n+1}}{1-x} := P_n(x) + \frac{x^{n+1}}{1-x} \quad \text{for all } x \neq 1.$$

Let $h(x) = x^n g(x)$ where $g(x) = \dfrac{x}{1-x}$, then we have
$$f(x) = h(x) + P_n(x) \quad \text{for all } x \neq 1.$$

Note that by checking the derivatives of f and P_n at $x = 0$, we have
$$f^{(k)}(0) = P_n^{(k)}(0) \quad \text{for all } k \in \{0, 1, \ldots, n\}.$$

Thus we have $T_n(x) = P_n(x)$. By $\displaystyle\int_0^x \frac{1}{1-t}\mathrm{d}t = -\ln(1-x)$, we have

$$T_{n+1}\left[-\ln(1-x)\right] = \int_0^x T_n\left(\frac{1}{1-t}\right)\mathrm{d}t = \int_0^x P_n(t)\mathrm{d}t = \sum_{k=1}^{n+1}\frac{x^k}{k}.$$

Theorem 8.1.6 Let P_n be a polynomial of degree $n \geq 1$. Let f and g be two functions with derivatives of order n at $x = 0$ and assume that

$$f(x) = P_n(x) + x^n g(x),$$

where $g(x) \to 0$ as $x \to 0$. Then P_n is the Taylor polynomial $T_n(f)$ at 0.

Remark 8.1.7 the Taylor polynomial is the unique polynomial approximation whose error is $x^n g(x)$ as $x \to 0$. Here, $x^n g(x)$ is called the **Peano form of the error term** of the Taylor polynomial P_n.

Example 8.1.8 (7) We start again by

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \frac{x^{n+1}}{1-x}$$

but replace x by $(-x^2)$:

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - \cdots + (-1)^n x^{2n} + \frac{(-1)^{n+1}x^{2n+1}}{1+x^2} \quad \forall x \in \mathbb{R}.$$

Analogously, we let $P_{2n}(x) = 1 - x^2 + x^4 - \cdots + (-1)^n x^{2n}$ and $g(x) = \dfrac{(-1)^{n+1}x}{1+x^2}$, then we have

$$\frac{1}{1+x^2} = P_{2n}(x) + x^{2n}g(x) \quad \text{for all } x \in \mathbb{R}.$$

By the above theorem, we have $T_{2n}\left(\dfrac{1}{1+x^2}\right) = P_{2n}(x)$. By $\displaystyle\int_0^x \frac{1}{1+t^2}\mathrm{d}t = \arctan x$, we have

$$T_{2n+1}(\arctan x) = \int_0^x T_{2n}\left(\frac{1}{1+t^2}\right)\mathrm{d}t = \int_0^x P_{2n}(t)\mathrm{d}t = \sum_{k=0}^{n}\frac{(-1)^k x^{2k+1}}{2k+1}.$$

Definition 8.1.9 We call the function $E_n(x) := f(x) - T_n(f(x))$ the **error term** or the **remainder** of the Taylor polynomial $T_n(f)$.

Example 8.1.10 (8) Suppose f has continuous second derivative f'' in some neighborhood of a. Then in this neighborhood of a, we know:

$$\begin{aligned}
E_1(x) &= f(x) - T_1(f) \\
&= f(x) - [f(a) + f'(a)(x-a)] \\
&= \int_a^x (f'(t) - f'(a))\,\mathrm{d}t \quad \text{(IBP with } u = f'(t) - f'(a), v = t - x) \\
&= [f'(t) - f'(a)]\,(t-x)\Big|_{t=a}^{x} - \int_a^x f''(t)(t-x)\,\mathrm{d}t \\
&= \int_a^x f''(t)(x-t)\,\mathrm{d}t.
\end{aligned}$$

If, additionally, we know $m \le f''(t) \le M$ for any t in the considered neighborhood of a, then we have

$$\frac{m(x-a)^2}{2} \le E_1(x) = \int_a^x f''(t)(x-t)\,\mathrm{d}t \le \frac{M(x-a)^2}{2}.$$

By the squeeze theorem, we have $\lim\limits_{x\to a} E_1(x) = 0$. This example motivates us to find a more general formula for the error term E_n as a integral.

Theorem 8.1.11 Assume f has a continuous derivative of order $n+1$ in some interval containing a. Then for every x in this interval, we have the Taylor formula:

$$f(x) = T_n(f) + E_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{1}{n!}\int_a^x f^{(n+1)}(t)(x-t)^n\,\mathrm{d}t.$$

If, additionally, there exists m and M such that $m \le f^{(n+1)}(t) \le M$ for any t in the considered interval, then we have the following bounds/estimates for the error term:

$$\frac{m(x-a)^{n+1}}{(n+1)!} \le E_n(x) = \frac{1}{n!}\int_a^x f^{(n+1)}(t)(x-t)^n\,\mathrm{d}t \le \frac{M(x-a)^{n+1}}{(n+1)!} \ (\forall x > a).$$

and

$$\frac{m(a-x)^{n+1}}{(n+1)!} \le (-1)^{n+1}E_n(x) \le \frac{M(a-x)^{n+1}}{(n+1)!} \quad \forall x < a.$$

Remark 8.1.12 Here, $E_n(x) = \dfrac{1}{n!}\displaystyle\int_a^x f^{(n+1)}(t)(x-t)^n\,\mathrm{d}t$ is called the **integral form of the error term** of the Taylor polynomial $T_n(f)$.

Example 8.1.13 We go back to the example of $f(x) = e^x = \displaystyle\sum_{k=0}^{n} \frac{x^k}{k!} + E_n(x)$ and $a = 0$. Note for any $x \in [b, c]$, we know $e^b \le e^x \le e^c$. Particularly, if $b = 0$ and $c = x = 1$, then we have

$$\frac{1}{(n+1)!} \le E_n(1) \le e \cdot \frac{1}{(n+1)!} < \frac{3}{(n+1)!}.$$

Thus we have $e = \displaystyle\sum_{k=0}^{n} \frac{1}{k!} + E_n(1)$.

Remark 8.1.14 Recall the error term

$$E_n(x) = \frac{1}{n!}\int_a^x f^{(n+1)}(t)(x-t)^n \, \mathrm{d}t.$$

Note that for any $t \in [a, x]$, the term $(x-t)^n$ never changes sign. Back in Theorem 8.1.11, we require $f^{(n+1)}$ to be continuous on the considered interval. By the weighted Mean Value Theorem for integrals, we can find some $c \in [a, x]$ such that

$$\begin{aligned} E_n(x) &= \frac{1}{n!}\int_a^x f^{(n+1)}(t)(x-t)^n \, \mathrm{d}t \\ &= \frac{1}{n!} f^{(n+1)}(c) \int_a^x (x-t)^n \, \mathrm{d}t \\ &= \frac{1}{(n+1)!} f^{(n+1)}(c)(x-a)^{n+1} \quad \text{for some } c \in [a, x]. \end{aligned}$$

This is called the **Lagrange form of the error term**.
Suppose $f^{(n+1)}$ is continous on some interval $[a-r, a+r]$ where $r > 0$, then $f^{(n+1)}$ is bounded on $[a-r, a+r]$ and there exists $M > 0$ such that $|f^{(n+1)}(t)| \leq M$ for any $t \in [a-r, a+r]$. Then

$$|E_n(x)| \leq M \cdot \frac{|x-a|^{n+1}}{(n+1)!} \quad \text{for any } x \in [a-r, a+r].$$

For $x \neq a$, we know

$$0 \leq \left|\frac{E_n(x)}{(x-a)^n}\right| \leq \frac{M}{(n+1)!}|x-a|.$$

By the squeeze theorem, we know $\dfrac{E_n(x)}{(x-a)^n} \to 0$ as $x \to a$. We say that the error term E_n is of small order that $(x-a)^n$ as $x \to a$, and we often write

$$E_n(x) = o\left((x-a)^n\right) \quad \text{as } x \to a.$$

Here, $E_n(x)$ is called the **Peano form of the error term** of the Taylor polynomial $T_n(f)$.

Definition 8.1.15 Assume $g(x) \neq 0$ for $x \neq a$ in some interval containing a. The notation $f(x) = o(g(x))$ as $x \to a$ means that $\lim\limits_{x\to a} \dfrac{f(x)}{g(x)} = 0$.

Example 8.1.16 We can rewrite Taylor's formula in o-notation as

$$f(x) = T_n(f) + o\left((x-a)^n\right) \quad \text{as } x \to a.$$

Theorem 8.1.17 We deduce some useful properties of the o-notation: as $x \to a$,

(1) $o(g(x)) \pm o(g(x)) = o(g(x))$.

(2) $o(cg(x)) = o(g(x))$ for any real constant $c \neq 0$.

(3) $f(x) \cdot o(g(x)) = o(f(x)g(x))$ for $f(x) \neq 0$.

(4) $o(o(g(x))) = o(g(x))$.

(5) $\dfrac{1}{1+g(x)} = 1 - g(x) + o(g(x))$ if $g(x) \to 0$ as $x \to a$.

Proof. We only prove (1) and (5) here since the others are analogous.

(1) Let $f_1(x) = o(g(x))$ and $f_2(x) = o(g(x))$, then $\lim\limits_{x\to a} \dfrac{f_1(x)}{g(x)} = 0$ and $\lim\limits_{x\to a} \dfrac{f_2(x)}{g(x)} = 0$. Thus we have

$$\lim_{x\to a} \frac{f_1(x) \pm f_2(x)}{g(x)} = \lim_{x\to a} \frac{f_1(x)}{g(x)} \pm \lim_{x\to a} \frac{f_2(x)}{g(x)} = 0.$$

Thus $o(g(x)) + o(g(x)) = f_1(x) + f_2(x) = o(g(x))$.

(5) Recall

$$\frac{1}{1+g(x)} = 1 - g(x) + \frac{g(x)^2}{1+g(x)} \quad \text{for any } g(x) \neq -1.$$

Thus we have

$$\lim_{x\to a} \frac{\frac{1}{1+g(x)} - (1-g(x))}{g(x)} = \lim_{x\to a} \frac{g(x)^2}{(1+g(x))g(x)} = \lim_{x\to a} \frac{g(x)}{1+g(x)} = 0.$$

Thus we have $\dfrac{1}{1+g(x)} = 1 - g(x) + o(g(x))$.

□

Example 8.1.18 We know

$$\cos x = 1 - \frac{x^2}{2!} + \cdots + \frac{(-1)^n x^{2n}}{(2n)!} + o(x^{2n+1}) \quad \text{as } x \to 0.$$

Let $g(x) = \cos x - 1$, then $g(x) \to 0$ as $x \to 0$. By the properties of o-notation, we have

$$\frac{1}{\cos x} = \frac{1}{1+g(x)} = 1 - g(x) + o(g(x)) = 1 + \frac{x^2}{2} + o(x^2) + o(x^3) \quad \text{as } x \to 0.$$

Note $o(x^2) + o(x^3) = o(x^2)$, so we have

$$\tan x = \frac{\sin x}{\cos x} = \left(x - \frac{x^3}{3!} + o(x^4)\right)\left(1 + \frac{x^2}{2} + o(x^3)\right) = x + \frac{x^3}{3} + o(x^3) \quad \text{as } x \to 0.$$

Example 8.1.19 With Taylor's formula, we can calculate the following limit:

1. $\lim\limits_{x\to 0} \dfrac{a^x - b^x}{x}$ where $a, b > 0$.
   Note that
   $$e^x = 1 + x + o(x) \quad \text{as } x \to 0.$$
   Thus we have
   $$a^x - b^x = e^{x \ln a} - e^{x \ln b} = x(\ln a - \ln b) + o(x) \quad \text{as } x \to 0.$$
   By definition of o-notation, we have
   $$\lim_{x\to 0} \frac{a^x - b^x}{x} = \ln a - \ln b.$$

2. $\lim\limits_{x\to 0} \dfrac{\ln(1+ax)}{x}$ where $a \in \mathbb{R}$.

If $a \neq 0$, note that

$$\ln(1+ax) = ax + o(x) \quad \text{as } x \to 0.$$

Thus we have

$$\frac{\ln(1+ax)}{x} = a + o(1) \quad \text{as } x \to 0 \implies \lim_{x\to 0} \frac{\ln(1+ax)}{x} = a.$$

But, what if $a = 0$?

2 Indeterminate Forms and L'Hôpital's Rule

Theorem 8.2.1 Assume f and g have derivatives $f'(x)$ and $g'(x)$ for every x in an open interval (a,b) and suppose that $\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = 0$. If $g'(x) \neq 0$ for every $x \in (a,b)$ and the limit $\lim_{x\to a^+} \frac{f'(x)}{g'(x)}$ exists and have value L, then we have

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = \frac{f'(x)}{g'(x)} = L.$$

Proof. Note that L'ôpital's Rule does require f, g, f', g' to be well-defined for $x \in (a,b)$. So we shall "extend" the definition of f and g "a bit" by letting

$$F(x) = \begin{cases} f(x) & \text{if } x \in (a,b) \\ 0 & \text{if } x = a \end{cases} \quad \text{and} \quad G(x) = \begin{cases} g(x) & \text{if } x \in (a,b) \\ 0 & \text{if } x = a \end{cases}$$

We can apply Cauchy's Mean Value Theorem to the interval $[a,x]$ for F and G to get

$$\frac{F(x) - F(a)}{G(x) - G(a)} = \frac{F'(c)}{G'(c)} \quad \text{for some } c \in (a,x).$$

Simply note that $F(a) = G(a) = 0$, we have

$$\frac{f(x)}{g(x)} = \frac{F(x)}{G(x)} = \frac{F'(c)}{G'(c)} = \frac{f'(c)}{g'(c)} \quad \text{for some } c \in (a,x).$$

Note $g(x) \neq 0$ for $x \in (a,b)$ otherwise $G(x) = 0 = G(a)$ and by Role's Theorem, there exists some $d \in (a,x)$ such that $G'(d) = 0 = g'(d)$, which is a contradiction. Thus as $x \to a^+$, we have $c \to a^+$ and

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = \frac{f'(x)}{g'(x)} = L.$$

□

Remark 8.2.2 The "left-sided limit" and the "two-sided limit" versions of L'Hôpital's Rule also hold provided that the derivatives are defined in the needed open interval.
More versions of L'Hôpital's Rule can be derived for indeterminate forms of the type ∞/∞, $0 \cdot \infty$, $\infty - \infty$, 0^0, 1^∞ and ∞^0 or for "limit notation" like $\lim_{x\to\infty} \frac{f(x)}{g(x)}$, $\lim_{x\to a} \frac{f(x)}{g(x)} = \infty$ or $\lim_{x\to\infty} \frac{f(x)}{g(x)} = -\infty$.

Example 8.2.3 We use L'Hôpital's Rule to calculate some previous limits:

(1) Calculate $\lim\limits_{x\to 0}\dfrac{a^x-b^x}{x}$ where $a,b>0$. Note that

$$\lim_{x\to 0} a^x = \lim_{x\to 0} b^x = 1.$$

Thus we have

$$\lim_{x\to 0}\frac{a^x-b^x}{x} = \lim_{x\to 0}\frac{a^x\ln a - b^x\ln b}{1} = \ln a - \ln b.$$

(2) Calculate $\lim\limits_{x\to 0}\dfrac{\ln(1+ax)}{x}$ where $a\in\mathbb{R}$. We talk about the case $a=0$ here. Note that

$$\lim_{x\to 0}\ln(1+ax) = \ln 1 = 0.$$

Thus we have

$$\lim_{x\to 0}\frac{\ln(1+ax)}{x} = \lim_{x\to 0}\frac{a}{1+ax} = a.$$

(3) $\lim\limits_{x\to 0}\dfrac{x-\tan x}{x-\sin x} = \lim\limits_{x\to 0}\dfrac{1-\sec^2 x}{1-\cos x} = -\lim\limits_{x\to 0}\dfrac{\cos x+1}{\cos^2 x} = -2.$

Definition 8.2.4 We introduce extended definitions of limits and "limits with infinity" here:

- The symbol
$$\lim_{x\to\infty} f(x) = L \quad \text{or} \quad f(x)\to L \quad \text{as } x\to\infty$$
means that for every $\varepsilon>0$, there exists some $M>0$ such that
$$|f(x)-L|<\varepsilon \quad \text{for any } x>M.$$
- The symbol $\lim\limits_{x\to a} f(x)=\infty$ means that for every $M>0$, there exists some $\delta>0$ such that $f(x)>M$ whenever $0<|x-a|<\delta$. Similarly, the symbol $\lim\limits_{x\to a} f(x)=-\infty$, $\lim\limits_{x\to a^+} f(x)=\pm\infty$ and $\lim\limits_{x\to a^-} f(x)=\pm\infty$ can be defined analogously.
- The symbol $\lim\limits_{x\to\infty} f(x)=\infty$ means that for every $M>0$, there exists some $N>0$ such that $f(x)>M$ whenever $x>N$. Similarly, the symbol $\lim\limits_{x\to\infty} f(x)=-\infty$ can be defined analogously.

Remark 8.2.5 Note, if $g(t)=f(1/t)$ and $\lim\limits_{t\to 0^+} g(t)=A$, then $\lim\limits_{x\to\infty} f(x)=A$. Similarly, if $\lim\limits_{t\to 0^-} g(t)=B$, then $\lim\limits_{x\to-\infty} f(x)=B$.

Remark 8.2.6 With the extended definitions of limits, we can also apply L'Hôpital's Rule to calculate limits like $\lim\limits_{x\to\infty}\dfrac{f(x)}{g(x)}$ provided that the derivatives are defined in the needed open interval and the limit $\lim\limits_{x\to\infty}\dfrac{f'(x)}{g'(x)}$ exists.

Theorem 8.2.7 Suppose $a,b>0$ then $\lim\limits_{x\to+\infty}\dfrac{(\ln x)^b}{x^a}=0$ and $\lim\limits_{x\to+\infty}\dfrac{x^b}{e^{ax}}=0$.

Proof. Suppose $c > 0$ and $t \geq 1$. Then $\dfrac{1}{t} \leq \dfrac{1}{t^{1-c}}$. For any $x > 1$, we have

$$0 < \ln x = \int_1^x \frac{1}{t} \mathrm{d}t \leq \int_1^x \frac{1}{t^{1-c}} \mathrm{d}t = \frac{x^c - 1}{c} < \frac{x^c}{c}.$$

Thus we have

$$0 < \frac{(\ln x)^b}{x^a} < \frac{x^{cb-a}}{c^b} \quad \text{for any } x > 1.$$

Since c is arbitrary, we choose $c = \dfrac{a}{2b}$ to get $x^{cb-a} = x^{-\frac{a}{2}} \to 0$ as $x \to +\infty$. By the squeeze theorem, we have $\displaystyle\lim_{x\to+\infty} \frac{(\ln x)^b}{x^a} = 0.$

For the second limit, we let $t = e^x$ and we have

$$\lim_{x\to+\infty} \frac{x^b}{e^{ax}} = \lim_{t\to+\infty} \frac{(\ln t)^b}{t^a} = 0.$$

□

Example 8.2.8

$$\lim_{x\to 0} x^x = \lim_{x\to 0} e^{x \ln x} = e^{\lim_{x\to 0} x \ln x} = e^0 = 1.$$

Example 8.2.9

$$\lim_{x\to\infty} x^{1/x} = \lim_{x\to 0} \frac{1}{x^x} = \frac{1}{\lim\limits_{x\to 0} x^x} = \frac{1}{1} = 1.$$

Chapter 9 Sequences and Series

1 Sequences and Partial Sums

Definition 9.1.1 A function $f : \mathbb{N} \to \mathbb{R}$ is called a **sequence**. The function value $f(n)$ is called the n-th term of the sequence. We often denote such a sequence by $\{a_n\}$ or $f(n)$ where $a_n := f(n)$ for any $n \in \mathbb{N}$.

Definition 9.1.2 A sequence $\{f(n)\}$ is said to have a limit $L \in \mathbb{R}$, or **converges** to L, if for every $\varepsilon > 0$, there exists some $N \in \mathbb{N}$ such that $|f(n) - L| < \varepsilon$ for any $n > N$ and in this case we say the sequence is **convergent**. If a sequence is not convergent, then we say it is **divergent**.

Definition 9.1.3 Let $\{f(n)\}$ be a sequence. We define the following notions:

1. The sequence $\{f(n)\}$ is **increasing** if $f(n) \leq f(n+1)$ for all $n \in \mathbb{N}$.
2. The sequence $\{f(n)\}$ is **decreasing** if $f(n) \geq f(n+1)$ for all $n \in \mathbb{N}$.
3. The sequence $\{f(n)\}$ is **monotonic** if it is either increasing or decreasing.
4. The sequence $\{f(n)\}$ is **bounded** if there exists some $M > 0$ such that $|f(n)| \leq M$ for all $n \in \mathbb{N}$.
5. The sequence $\{f(n)\}$ is **unbounded** if it is not bounded.

Theorem 9.1.4 A monotonic sequence converges if and only if it is bounded.

Proof. If $\{f(n)\}$ is increasing, then it converges. Additionally, if $S = \{f(n) : n \in \mathbb{N}\}$, then $f(n) \to \sup S = L$. For any $\varepsilon > 0$, $L - \varepsilon$ is not an upper bound of S, so there exists some $N \in \mathbb{N}$ such that $f(N) > L - \varepsilon$. For any $n \geq N$, we have $f(n) \geq f(N) > L - \varepsilon$ and $f(n) \leq L$, so $|f(n) - L| < \varepsilon$. Thus $\{f(n)\}$ converges to L. The case for decreasing sequence is analogous. □

Corollary 9.1.5 Let $f : [0, 1] \to \mathbb{R}$ be monotone increasing and bounded. Then

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} f\left(\frac{k}{n}\right) = \int_0^1 f(x)\,\mathrm{d}x.$$

Proof. For each $n \in \mathbb{N}$, partition $[0, 1]$ into n equal subintervals:

$$x_k = \frac{k}{n}, \qquad k = 0, 1, \ldots, n.$$

Define

$$s_n = \frac{1}{n}\sum_{k=0}^{n-1} f(x_k), \qquad t_n = \frac{1}{n}\sum_{k=1}^{n} f(x_k).$$

Since f is monotone increasing, for every $x \in [x_{k-1}, x_k]$, we have $f(x_{k-1}) \le f(x) \le f(x_k)$. Hence

$$\frac{1}{n} f(x_{k-1}) \le \int_{x_{k-1}}^{x_k} f(x)\,\mathrm{d}x \le \frac{1}{n} f(x_k).$$

Summing from $k = 1$ to n, we get

$$s_n \le \int_0^1 f(x)\,\mathrm{d}x \le t_n.$$

Moreover,

$$t_n - s_n = \frac{1}{n}\left(\sum_{k=1}^{n} f(x_k) - \sum_{k=0}^{n-1} f(x_k)\right) = \frac{f(1) - f(0)}{n}.$$

Therefore,

$$0 \le t_n - \int_0^1 f(x)\,\mathrm{d}x \le t_n - s_n = \frac{f(1) - f(0)}{n} \to 0.$$

Hence

$$\lim_{n\to\infty} t_n = \int_0^1 f(x)\,\mathrm{d}x.$$

Since

$$t_n = \frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right),$$

we conclude that

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} f\left(\frac{k}{n}\right) = \int_0^1 f(x)\,\mathrm{d}x.$$

□

Example 9.1.6

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} \frac{1}{1 + k/n} = \int_0^1 \frac{1}{1+x}\,\mathrm{d}x = \log(1+x)\Big|_0^1 = \log 2.$$

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} \frac{n^2}{n^2+k^2} = \lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} \frac{1}{1+(k/n)^2} = \int_0^1 \frac{1}{1+x^2}\,\mathrm{d}x = \arctan x\Big|_0^1 = \frac{\pi}{4}.$$

Definition 9.1.7 For any sequence $\{a_n\}$, we can generate another sequence $\{S_n\}$ by defining

$$S_n = a_1 + a_2 + \cdots + a_n = \sum_{k=1}^{n} a_k \quad \text{for any } n \in \mathbb{N}.$$

We call S_n the partial sums of the first n terms of $\{a_n\}$ and $\{S_n\}$ is called an **(infinite) series**.

Notation 9.1.8 The following notations are often used for series:

$$\sum a_n \equiv a_1 + a_2 + \cdots \equiv \sum_{n=1}^{\infty} a_n := \lim_{n\to\infty} S_n = \lim_{n\to\infty} \sum_{k=1}^{n} a_k.$$

If $S_n \to S$ as $n \to \infty$, then we say the series $\{S_n\}$ converges and has the sum S. In this case, we often write $\sum_{n=1}^{\infty} a_n = S$ instead of $S_n \to S$. A series that does not converge is divergent and it has no sum.

Remark 9.1.9 We define infinite series as the limit of partial sums. Ordinary rules for finite sums do not necessarily hold for infinite series due to the limit operation.

Example 9.1.10 We list some typical examples of series here:

1. If $a_n = \dfrac{1}{n}$, then
$$S_n = \sum_{k=1}^{n} \frac{1}{k} > \int_1^{n+1} \frac{1}{t} \mathrm{d}t = \ln(n+1) \to +\infty \quad \text{as } n \to +\infty.$$
This divergent series is called the **harmonic series**.
2. If $a_n = \dfrac{1}{2^{n-1}}$, then
$$S_n = \sum_{k=1}^{n} \frac{1}{2^{k-1}} = 2\left(1 - \frac{1}{2^n}\right) \to 2 \quad \text{as } n \to +\infty.$$
This kind of series that the ratio of two adjacent terms is a constant is called a **geometric series**.

Theorem 9.1.11 Let $\sum a_n$ and $\sum b_n$ be two convergent series and $\alpha, \beta \in \mathbb{R}$. Then the series $\sum(\alpha a_n + \beta b_n)$ converges and its sum is given by

$$\sum_{n=1}^{\infty}(\alpha a_n + \beta b_n) = \alpha \sum_{n=1}^{\infty} a_n + \beta \sum_{n=1}^{\infty} b_n.$$

Corollary 9.1.12 If $\sum a_n$ converges and $\sum b_n$ diverges, then $\sum(a_n + b_n)$ diverges.

Theorem 9.1.13 If $\{b_n\}$ is a sequence and another sequence $\{a_n\}$ satisfies $a_n = b_n - b_{n+1}$ for any $n \in \mathbb{N}$, then we call the series $\sum a_n$ a **telescoping series** and $\sum a_n$ converges if and only if $\{b_n\}$ converges. Furthermore, if $\{b_n\}$ converges to L as $n \to \infty$, then we have

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty}(b_n - b_{n+1}) = b_1 - L.$$

Theorem 9.1.14 Let $a_n = x^n$ for $n \in \mathbb{N}$, then the partial sum $S_n = \sum_{k=1}^{n} x^k$ satisfies

$$S_n = \begin{cases} n+1 & \text{if } x = 1 \\ \dfrac{1-x^n}{1-x} & \text{if } x \neq 1 \end{cases}$$

We call $\{S_n\}$ the **geometric series** and $\{S_n\}$ converges if and only if $|x| < 1$. When $\{S_n\}$ converges, we have

$$\sum_{n=1}^{\infty} x^n = \frac{1}{1-x}.$$

Example 9.1.15 We can do some replacement to get some typical examples of geometric series:

(1) Replace x by x^2 in the geometric series, we have

$$1 + x^2 + x^4 + \cdots = \frac{1}{1-x^2} \quad \text{for } |x| < 1.$$

(2) For $|x| < 1$, we multiply both sides by x:

$$x + x^3 + x^5 + \cdots = \frac{x}{1-x^2}.$$

(3) Replace x by $-x$:

$$1 - x + x^2 - x^3 + \cdots = \frac{1}{1+x} \quad \text{for } |x| < 1.$$

Definition 9.1.16 We call a series of the form

$$\sum_{n=0}^{\infty} a_n x^n$$

a **power series** in x and the number a_n is called the coefficient of x^n in the power series.

Example 9.1.17 Suppose f is a function with derivatives of arbitrary order in a neighborhood of $x = 0$. Let $a_n = \dfrac{f^{(n)}(0)}{n!}$ for $n \in \mathbb{N}$, then the power series with coefficients a_n is called the **Taylor series** of f at $x = 0$. Since

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n + E_n(x),$$

Thus if $E_n(x) \to 0$ as $n \to \infty$ for any x in some neighborhood of 0, then the Taylor series converges to $f(x)$.

2 Convergence Tests for Series

In this section, we introduce several tests and some necessary conditions for convergence of series.

Theorem 9.2.1 If $\sum a_n$ converges, then $\lim\limits_{n\to\infty} a_n = 0$.

Proof. Suppose $S_n = \sum\limits_{k=1}^{n} a_k$. Then $a_n = S_n - S_{n-1}$. Since $\{S_n\}$ converges, we have $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} S_n - \lim\limits_{n\to\infty} S_{n-1} = 0$. □

Remark 9.2.2 This is a necessary condition for convergence of series but not a sufficient condition. For example, the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges but $\lim_{n\to\infty} \frac{1}{n} = 0$.

Theorem 9.2.3 Suppose $a_n \geq 0$ for any $n \in \mathbb{N}$. Then the series $\sum a_n$ converges if and only if the sequence of partial sums $\{S_n\}$ is bounded.

Proof. We know $\{S_n\}$ is increasing and by the monotone convergence theorem, $\{S_n\}$ converges if and only if $\{S_n\}$ is bounded. □

Remark 9.2.4 This is a necessary and sufficient condition for convergence test.

Theorem 9.2.5 (Comparison Test) Suppose $a_n \geq 0$ and $b_n \geq 0$ for any $n \in \mathbb{N}$. If there exists some $c > 0$ such that $a_n \leq cb_n$ for all n, then the convergence of $\sum b_n$ implies the convergence of $\sum a_n$. (or equivalently, the divergence of $\sum a_n$ implies the divergence of $\sum b_n$).

Proof. Suppose $S_n = \sum_{k=1}^{n} a_k$ and $T_n = \sum_{k=1}^{n} b_k$. Then we have $S_n \leq cT_n$ for any $n \in \mathbb{N}$. If $\{T_n\}$ converges, then $\{S_n\}$ is bounded and thus $\{S_n\}$ converges. □

Example 9.2.6 Recall that we already have $\sum \frac{1}{n^2+n}$ convergent. Note that we have

$$\lim_{n\to\infty} \frac{1/n^2}{1/(n^2+n)} = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right) = 1 > 0$$

Thus $\sum \frac{1}{n^2}$ converges. We define **Riemann zeta function** by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

We have $\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. We define the domain $\mathrm{Dom}\,(\zeta)$ to be the set of all $s \in \mathbb{R}$ such that $\zeta(s)$ converges.

Theorem 9.2.7 (Integral Test) Suppose f is a positive decreasing function defined for all $x \geq 1$. For each positive integer n, let $S_n = \sum_{k=1}^{n} f(k)$ and $t_n = \int_1^n f(x)\mathrm{d}x$. Then the two sequences $\{S_n\}$ and $\{t_n\}$ both converge or both diverge.

Proof. Geometrically, we have

$$S_n - f(1) = \sum_{k=2}^{n} f(k) \leq \int_1^n f(x)\mathrm{d}x = t_n \leq \sum_{k=1}^{n-1} f(k) = S_{n-1}.$$

Equivalently, we have

$$t_n + f(n) \leq S_n \leq t_n + f(1).$$

□

Example 9.2.8 Let $f(x) = x^{-s}$, then

$$t_n = \int_1^n x^{-s} \mathrm{d}x = \begin{cases} \dfrac{n^{1-s} - 1}{1-s} & \text{if } s \neq 1 \\ \ln n & \text{if } s = 1 \end{cases}$$

If $s > 1$, then $n^{1-s} \to 0$ as $n \to +\infty$, so $t_n \to \dfrac{1}{s-1}$.
If $s \leq 1$, then t_n diverges.
Recall the definition of Riemann zeta function $\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$, we have $\zeta(s)$ converges if and only if $s > 1$ and $\zeta(s)$ diverges if $s \leq 1$.

Example 9.2.9

$$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^s}$$

converges if and only if $s > 1$.

Remark 9.2.10 The integral test only tells us about the convergence or divergence of the series but does not give us the value of the sum when the series converges (they are not necessarily equal)

Theorem 9.2.11 (Root Test) Suppose $a_n \geq 0$ for any $n \in \mathbb{N}$ and additionally, $a_n^{1/n} \to r$ as $n \to \infty$ for some $r \in \mathbb{R}$. Then

(1) If $r < 1$, then $\sum a_n$ converges.

(2) If $r > 1$, then $\sum a_n$ diverges.

(3) If $r = 1$, then the test is inconclusive/fails.

Proof. If $r < 1$, then there exists some x such that $r < x < 1$. Since $a_n^{1/n} \to r$, there exists some $N \in \mathbb{N}$ such that $a_n^{1/n} < x$ for any $n > N$. Thus we have $a_n < x^n$ for any $n > N$. By the comparison test, we have $\sum a_n$ converges.
If $r > 1$, then there exists $\widetilde{N}$ such that $a_n^{1/n} > 1$ for all $n \geq \sqrt{\widetilde{N}}$. Then $a_n > 1$ for all $n \geq \widetilde{N}$ and thus $\lim\limits_{n\to\infty} a_n$ cannot be 0 and $\sum a_n$ diverges.
If $r = 1$, it suffices to give two examples.
Example 1: Suppose $a_n = \left(\dfrac{1}{n^2}\right)^{1/n}$, then

$$\lim_{n\to\infty} \left(\frac{1}{n^2}\right)^{1/n} = \lim_{x\to 0} (x^x)^2 = 1.$$

However, we know $\sum a_n = \zeta(2)$ is convergent.

Example 2: Simply letting $b_n = 1$ gives a valid example.

$\square$

Theorem 9.2.12 (Ratio Test) Suppose $a_n > 0$ for all $n \geq 1$ and additionally $\dfrac{a_{n+1}}{a_n} \to r$ as $n \to \infty$. Then

(1) If $r < 1$, then $\sum a_n$ converges.

(2) If $r > 1$, then $\sum a_n$ diverges.

(3) If $r = 1$, then the test is inconclusive/fails.

Definition 9.2.13 A series of the form $\displaystyle\sum_{n=1}^{\infty}(-1)^{n-1}a_n$ or $\displaystyle\sum_{n=1}^{\infty}(-1)^n a_n$ where $a_n > 0$ is called an **alternating** series.

Theorem 9.2.14 (Leibniz Test (Alternating Series Test)) If $\{a_n\}$ is a decreasing positive sequence with $\lim\limits_{n\to\infty} a_n = 0$, then the alternating series $\displaystyle\sum_{n=1}^{\infty}(-1)^{n-1}a_n$ converges. In the case where it does converge, let S to denote the sum and S_n denote the partial sum, then

$$0 \leq (-1)^n(S - S_n) = a_{n+1} - a_{n+2} + a_{n+3} + \cdots \leq a_{n+1} \quad \text{for all } n \geq 1.$$

Proof. Note

$$S_{2n+2} - S_{2n} = a_{2n+1} - a_{2n+2} \geq 0$$

and

$$S_{2n+1} - S_{2n-1} = a_{2n+1} - a_{2n} \leq 0.$$

For $\{S_{2n}\}$,

$$S_2 \leq S_{2n} = a_1 - a_2 + a_3 - \cdots - a_{2n-2} + a_{2n-1} - a_{2n} \leq a_1.$$

For $\{S_{2n-1}\}$,

$$S_2 \leq S_2 + (a_3 - a_4) + \cdots + (a_{2n-3} - a_{2n-2}) + a_{2n-1} = S_{2n-1} \leq S_1.$$

By the monotone convergence theorem, we know $S_{2n} \to S_e$ and $S_{2n-1} \to S_o$ where $S_e, S_o \in \mathbb{R}$ as $n \to \infty$. Since

$$S_e - S_o = \lim_{n\to\infty}(S_{2n} - S_{2n-1}) = \lim(-1)^{2n-1}a_{2n} = 0,$$

we find $S_e = S_o := S$. And $S_{2n} \to S, S_{2n-1} \to S$ as $n \to \infty$ tells us $S_n \to S$ as $n \to \infty$. Then $\displaystyle\sum_{n=1}^{\infty}(-1)^{n-1}a_n$ converges.

Since $\{S_{2n}\}$ increases and $\{S_{2n-1}\}$ decreases, we have

$$S_{2n} \leq S_{2n+2} \leq S \leq S_{2n+1} \leq S_{2n-1}.$$

Thus,

$$0 \leq S - S_{2n} \leq S_{2n+1} - S_{2n} = a_{2n+1}$$

$$0 \leq S_{2n-1} - S \leq S_{2n-1} - S_{2n} = a_{2n}.$$

Conbining the two sets of inequalities we get

$$0 \le (-1)^k (S - S_k) \le a_{k+1}.$$

□

Example 9.2.15 The harmonic series

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots \to \infty.$$

However the alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \cdots$$

converges. This motivates to think about the relationship between $\sum |a_n|$ and $\left|\sum a_n\right|$.

Definition 9.2.16 $\sum a_n$ is called **absolutely convergent** if $\left|\sum a_n\right|$ converges. $\sum a_n$ is called **conditionally convergent** if $\sum a_n$ converges but $\left|\sum a_n\right|$ diverges.

Theorem 9.2.17 If $\sum |a_n|$ converges, then $\sum a_n$ also converges with

$$\left|\sum_{n=1}^{\infty} a_n\right| \le \sum_{n=1}^{\infty} |a_n|$$

Proof. Let $b_n = a_n + |a_n| \ge 0$. We will show $\sum b_n$ converges. Note that

$$b_n = \begin{cases} 0 & \text{if } a_n < 0 \\ 2a_n & \text{if } a_n \ge 0 \end{cases}$$

and thus $0 \le b_n \le 2|a_n|$. Thus using the comparison test, we have $\sum b_n$ converges. Since $\sum a_n = \sum b_n - \sum |a_n|$, we have $\sum a_n$ congerges.

Since $\left|\sum_{k=1}^{n} a_k\right| \le \sum_{k=1}^{n} |a_k|$ for all n, taking the limit as $n \to \infty$ (provided that they both exist), we get

$$\left|\sum_{n=1}^{\infty} a_n\right| \le \sum_{n=1}^{\infty} |a_n|.$$

□

Example 9.2.18 The alternating harmonic series converges conditionally. The series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$ converges absolutely.

Theorem 9.2.19 If $\sum a_n$ and $\sum b_n$ both are absolutely convergent, then $\sum(\alpha a_n + \beta b_n)$ is also absolutely convergent for any $\alpha, \beta \in \mathbb{R}$.

$$\sum_{k=1}^{n} |\alpha a_n + \beta b_n| \le \alpha \left(\sum_{k=1}^{n} |a_n|\right) + \beta \left(\sum_{k=1}^{n} |b_n|\right) \le \alpha \sum_{k=1}^{\infty} |a_n| + \beta \sum_{k=1}^{\infty} |b_n| =: S.$$

Since all partial sum are bounded above by S, we have $\sum |\alpha a_n + \beta b_n|$ converges and thus $\sum(\alpha a_n + \beta b_n)$ is absolutely convergent.

Example 9.2.20 Let $\{a_n\}$ be a sequence defined by

$$a_{2n-1} = \frac{1}{n} \qquad \text{and} \qquad a_{2n} = \int_n^{n+1} \frac{\mathrm{d}x}{x} \qquad \text{for } n = 1, 2, 3, \ldots.$$

By Leibniz's rule, the series $\sum_{n=1}^{\infty} (-1)^{n-1} a_n$ converges. We denote the sum of this series by S and denote the partial sums by s_n. Note

$$\begin{aligned} s_{2n-1} &= 1 - \int_1^2 \frac{\mathrm{d}x}{x} + \frac{1}{2} - \int_2^3 \frac{\mathrm{d}x}{x} + \cdots + \frac{1}{n-1} - \int_{n-1}^{n} \frac{\mathrm{d}x}{x} + \frac{1}{n} \\ &= 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \int_1^n \frac{\mathrm{d}x}{x} \\ &= 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \log n \to S \quad \text{as } n \to \infty. \end{aligned}$$

Then we have

$$\lim_{n\to\infty} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n} - \log n\right) = S.$$

We call this S the **Euler-Mascheroni constant**. Now consider the partial sum the partial sum of the alternating harmonic series $t_m = \sum_{k=1}^{m} \frac{(-1)^{k-1}}{k}$. In Example 9.2.15, we show t_m converges and now we use the above result to find the sum of t_m. Note that

$$\begin{aligned} t_{2n} &= \sum_{k=1}^{n} \frac{1}{2k-1} - \sum_{k=1}^{n} \frac{1}{2k} = \left(\sum_{k=1}^{2n} \frac{1}{k} - \sum_{k=1}^{n} \frac{1}{2k}\right) - \sum_{k=1}^{n} \frac{1}{2k} = \sum_{k=1}^{2n} \frac{1}{k} - \sum_{k=1}^{n} \frac{1}{k} \\ &= \big(\log(2n) + S + o(1)\big) - \big(\log n + S + o(1)\big) = \log 2 + o(1) \quad \text{as } n \to \infty. \end{aligned}$$

Hence, we have $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \log 2$. Now, consider the consider the following series obtained by rearranging the terms in the above series (the rule is to take two positive terms and then one negative term):

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} + \cdots.$$

Let p_n denote the partial sum of this rearranged series. Then we have

$$\begin{aligned} p_{3m} &= \sum_{k=1}^{2m} \frac{1}{2k-1} - \sum_{k=1}^{m} \frac{1}{2k} = \left(\sum_{k=1}^{4m} \frac{1}{k} - \sum_{k=1}^{2m} \frac{1}{2k}\right) - \frac{1}{2}\sum_{k=1}^{m} \frac{1}{k} = \sum_{k=1}^{4m} \frac{1}{k} - \frac{1}{2}\sum_{k=1}^{2m} \frac{1}{k} - \frac{1}{2}\sum_{k=1}^{m} \frac{1}{k} \\ &= \big(\log(4m) + C + o(1)\big) - \frac{1}{2}\big(\log(2m) + C + o(1)\big) - \frac{1}{2}\big(\log m + C + o(1)\big) = \frac{3}{2}\log 2 + o(1) \text{ as } m \to \infty. \end{aligned}$$

Thus, we show $p_m \to \dfrac{3}{2}\log 2$ as $m \to \infty$. This phenomenon that rearrangement of the terms of a convergent series may alter its sum motivates us to investigate the relationship between rearrangement and convergence of series.

Theorem 9.2.21 Let $\sum a_n$ be an absolutely convergent series having the sum S. Then any rearrangement of $\sum a_n$ also converges to S.

Proof. Let $\sum b_n$ be a rearrangement, say $b_n = a_{\sigma(n)}$. First we note that $\sum b_n$ converges absolutely because $\sum |b_n|$ is a series of nonnegative terms whose partial sums are bounded above by the number $\sum |a_n|$ (well-defined as $\sum a_n$ converges absolutely).
To prove that $\sum b_n$ also has sum S, we introduce

$$B_n = \sum_{k=1}^{n} b_k, \qquad A_n = \sum_{k=1}^{n} a_k, \qquad A_n^* = \sum_{k=1}^{n} |a_k|, \qquad \text{and} \qquad S^* = \sum_{k=1}^{\infty} |a_k|.$$

Now $A_n \to S$ and $A_n^* \to S^*$ as $n \to \infty$. Therefore, given any $\epsilon > 0$, there is an N such that

$$|A_N - S| < \frac{\epsilon}{2} \qquad \text{and} \qquad |A_N^* - S^*| < \frac{\epsilon}{2}.$$

For this N we can choose M so that

$$\{1, 2, \ldots, N\} \subseteq \{\sigma(1), \sigma(2), \ldots, \sigma(M)\}.$$

This is possible because the range of σ includes all the positive integers. If $n \geq M$, we have

$$|B_n - S| = |B_n - A_N + A_N - S| \leq |B_n - A_N| + |A_N - S| \leq |B_n - A_N| + \frac{\epsilon}{2}.$$

But we also have

$$|B_n - A_N| = \left| \sum_{k=1}^{n} b_k - \sum_{k=1}^{N} a_k \right| = \left| \sum_{k=1}^{n} a_{\sigma(k)} - \sum_{k=1}^{N} a_k \right|.$$

The terms $a_1, \ldots, a_N$ cancel in the subtraction, so we have

$$|B_n - A_N| \leq |a_{N+1}| + |a_{N+2}| + \cdots = |A_N^* - S^*| < \frac{\epsilon}{2}.$$

Combining the above two inequalities, we see that $|B_n - S| < \epsilon$ for all $n \geq M$, which means that $B_n \to S$ as $n \to \infty$. This proves that the rearranged series $\sum b_n$ has sum S. □

Remark 9.2.22 This is a very important result. It says that if a series is absolutely convergent, then rearrangement of the terms does not alter its sum.

Lemma 9.2.23 Given a series $\sum a_n$, we seperate its terms by sign and define

$$a_n^+ = \frac{a_n + |a_n|}{2}, \qquad a_n^- = \frac{a_n - |a_n|}{2}.$$

If $\sum a_n$ conditionally converges, then $\sum a_n^+$ and $\sum a_n^-$ both diverge.

Theorem 9.2.24 (Riemann's Rearrangement Theorem) Let $\sum a_n$ be a conditionally convergent series and let $S \in \mathbb{R}$ be any real number. Then there exists a rearrangement $\sum b_n$ of $\sum a_n$ which converges to S.

Proof. We shall follow the notation in Lemma 9.2.23. Both series $\sum a_n^+$ and $\sum a_n^-$ diverge since $\sum a_n$ is conditionally convergent. We rearrange $\sum a_n$ as follows:
Take, in order, just enough positive terms a_n^+ so that their sum exceeds S. If p_1 positive terms are required, we have

$$\sum_{n=1}^{p_1} a_n^+ > S \qquad \text{but} \qquad \sum_{n=1}^{q} a_n^+ \leq S \qquad \text{if } q < p_1.$$

This is always possible since the partial sums of $\sum a_n^+$ tend to $+\infty$. To this sum we add just enough negative terms a_n^-, say n_1 negative terms, so that the resulting sum is less than S. This is possible since the partial sums of a_n^- tend to $-\infty$. Thus, we have

$$\sum_{n=1}^{p_1} a_n^+ + \sum_{n=1}^{n_1} a_n^- < S \qquad \text{but} \qquad \sum_{n=1}^{p_1} a_n^+ + \sum_{n=1}^{m} a_n^- \geq S \qquad \text{if } m < n_1.$$

Now we repeat the process, adding just enough new positive terms to make the sum exceed S, and then just enough new negative terms to make the sum less than S. Continuing in this way, we obtain a rearrangement $\sum b_n$. Each partial sum of $\sum b_n$ differs from S by at most one term a_n^+ or a_n^-. But $a_n \to 0$ as $n \to \infty$ since $\sum a_n$ converges, so the partial sums of $\sum b_n$ tend to S. This proves that the rearranged series $\sum b_n$ converges and has sum S, as asserted. □

Remark 9.2.25 Riemann's rearrangement theorem not only tells us that the sum of a conditionally convergent series can be changed by rearrangement, but also tells us that we can rearrange the terms of a conditionally convergent series to make it converge to any real number we want.

Lemma 9.2.26 (Abel's Partial Sum Formula) Suppose two sequences $\{a_n\}$ and $\{b_n\}$. Let $A_n = \sum_{k=1}^{n} a_k$. Then

$$\sum_{k=1}^{n} a_k b_k = A_n b_{n+1} + \sum_{k=1}^{n} A_k (b_k - b_{k+1}).$$

Proof. Let $A_0 = 0$, then $a_k = A_k - A_{k-1}$ and

$$\begin{aligned}\sum_{k=1}^{n} a_k b_k = \sum_{k=1}^{n} A_k b_k - \sum_{k=1}^{n} A_{k-1} b_k &= \sum_{k=1}^{n} A_k b_k - \left[\sum_{k=1}^{n} A_k b_{k+1} - A_n b_{n+1}\right] \\ &= A_n b_{n+1} + \sum_{k=1}^{n} A_k (b_k - b_{k+1}).\end{aligned}$$

□

Remark 9.2.27 If both $\{A_n b_{n+1}\}$ and $\sum_{k=1}^{n} A_k (b_k - b_{k+1})$ converge, then $\sum_{k=1}^{n} a_k b_k$ converges.

Theorem 9.2.28 (Dirichlet's Test) Suppose $\sum a_n$ is a series whose partial sum $\{A_n\}$ is bounded. Suppose $\{b_n\}$ is a decreasing sequence with $b_n \to 0$ as $n \to \infty$. Then the series $\sum a_n b_n$ converges.

Proof. Let $M > 0$ be such that $|A_n| \leq M$ for any $n \in \mathbb{N}$. We tackle the convergence pf $\{A_n b_{n+1}\}$ and $\sum_{k=1}^{n} A_k(b_k - b_{k+1})$ seperately.

- Note that $\{A_n b_{n+1}\}$ converges becuase $0 \leq |A_n b_{n+1}| \leq M|b_{n+1}| \to 0$ as $n \to \infty$.
- For the convergence $\sum A_k(b_k - b_{k+1})$, we have

$$0 \leq |A_k(b_k - b_{k+1})| \leq M|b_k - b_{k+1}| = M(b_k - b_{k+1}).$$

  $\sum(b_k - b_{k+1})$ converges since it is a telescoping series with $\lim_{n\to\infty} b_n$ exists. Thus $\sum A_k(b_k - b_{k+1})$ converges by the comparison test.

By Abel's partial sum formula, we have $\sum a_n b_n$ converges. □

Theorem 9.2.29 (Abel's Test) If $\sum a_n$ converges and $\{b_n\}$ is monotonic and convergent, then $\sum a_n b_n$ converges.

Remark 9.2.30 If $\{b_n\}$ is decreasing and $b_n \to 0$ as $n \to \infty$. Taking $a_n = x^n$, we know the partial sum $A_n = \sum_{k=1}^{n} x^k$ is bounded for any $x \in [-1, 1)$. Particularly, when $x = -1$, Dirichlet's test says that $\sum_{n=1}^{\infty} a_n b_n = \sum_{n=1}^{\infty} (-1)^n b_n$ converges, which is simply the alternating series test or the Leibniz test.

Lemma 9.2.31 (Raabe's test) For a positive sequence $\{a_n\}$, if there is an $r > 0$ and an $N \geq 1$ such that

$$\frac{a_{n+1}}{a_n} \leq 1 - \frac{1}{n} - \frac{r}{n} \qquad \text{for all } n \geq N,$$

then $\sum a_n$ converges.

Theorem 9.2.32 (Gauss's test) Let $\sum a_n$ be a series of positive terms. If there is an $N \geq 1$, an $s > 1$, and an $M > 0$ such that

$$\frac{a_{n+1}}{a_n} = 1 - \frac{A}{n} + \frac{f(n)}{n^s} \qquad \text{for } n \geq N,$$

where

$$|f(n)| \leq M \qquad \text{for all } n,$$

then $\sum a_n$ converges if $A > 1$ and diverges if $A \leq 1$.

Example 9.2.33

$$\sum_{n=1}^{\infty} \left(\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n} \right)^k$$

converges if $k > 2$ and diverges if $k \leq 2$. Note that for this example the ratio test fails but Gauss's test works.

3 Improper Integrals

Definition 9.3.1 We define

$$\int_a^{\infty} f(x)\mathrm{d}x := \lim_{b\to\infty}\int_a^b f(x)\mathrm{d}x, \qquad \int_{-\infty}^{b} f(x)\mathrm{d}x := \lim_{a\to-\infty}\int_a^b f(x)\mathrm{d}x.$$

They are referred to as **improper integrals of the first kind**. We say the improper integral $\int_a^{\infty} f(x)\mathrm{d}x$ converges if the limit exists and diverges otherwise. If the limit does exist and equal to A, then we call A the value of the improper integral and write $\int_a^{\infty} f(x)\mathrm{d}x = A$.

If $\int_{-\infty}^{c} f(x)\mathrm{d}x$ and $\int_c^{\infty} f(x)\mathrm{d}x$ both converge for some $c\in\mathbb{R}$, then we say the improper integral $\int_{-\infty}^{\infty} f(x)\mathrm{d}x$ also converges. And we say $\int_{-\infty}^{\infty} f(x)\mathrm{d}x$ diverges if at least one of $\int_{-\infty}^{c} f(x)\mathrm{d}x$ and $\int_c^{\infty} f(x)\mathrm{d}x$ diverges.

Remark 9.3.2 Note that

$$\int_{-\infty}^{\infty} f(x)\mathrm{d}x = \int_{-\infty}^{c} f(x)\mathrm{d}x + \int_c^d f(x)\mathrm{d}x + \int_d^{\infty} f(x)\mathrm{d}x \quad (\forall c < d \in \mathbb{R})\,.$$

Combining the first two terms and last two terms gives us

$$\int_{-\infty}^{c} f(x)\mathrm{d}x + \int_c^{\infty} f(x)\mathrm{d}x = \int_{-\infty}^{d} f(x)\mathrm{d}x + \int_d^{\infty} f(x)\mathrm{d}x.$$

Thus we just need to find one point $c\in\mathbb{R}$ such that $\int_{-\infty}^{c} f(x)\mathrm{d}x$ and $\int_c^{\infty} f(x)\mathrm{d}x$ both converge to determine the convergence of $\int_{-\infty}^{\infty} f(x)\mathrm{d}x$.

Example 9.3.3 We introduces some typical examples of improper integrals:

(1) For a real s,

$$\int_1^{\infty} x^{-s}\mathrm{d}x = \begin{cases} \left.\dfrac{x^{1-s}}{1-s}\right|_1^{\infty} = \dfrac{1}{s-1} & \text{if } s > 1 \\ \ln x\Big|_1^{\infty} = +\infty & \text{if } s = 1 \\ \left.\dfrac{x^{1-s}}{1-s}\right|_1^{\infty} = +\infty & \text{if } s < 1 \end{cases}.$$

(2) For a real positive a,

$$\int_{-\infty}^{\infty} e^{-a|x|}\mathrm{d}x = \frac{2}{a}$$

(3) $\int_{-\infty}^{\infty} x\mathrm{d}x$ diverges since $\int_c^{\infty} x\mathrm{d}x$ and $\int_{-\infty}^{c} x\mathrm{d}x$ diverges for any $c\in\mathbb{R}$.
In this case, we cannot cancel the two divergent integrals like

$$\lim_{b\to\infty}\left(\int_{-b}^{0} x\mathrm{d}x + \int_0^b x\mathrm{d}x\right) = \lim_{b\to\infty}\int_{-b}^{b} x\mathrm{d}x = 0.$$

Theorem 9.3.4 Suppose $f(x) \geq 0$ for all $x \geq a$ and the proper integral $\int_a^b f(x)\mathrm{d}x$ exists for any $b \geq a$. Then the improper integral $\int_a^{\infty} f(x)\mathrm{d}x$ converges if and only if there is an $M > 0$ such that $\int_a^b f(x)\mathrm{d}x \leq M$ for any $b \geq a$.

Theorem 9.3.5 Suppose $f(x) \geq 0$ and $g(x) \geq 0$ for all $x \geq a$ and the proper integrals $\int_a^b f(x)\mathrm{d}x$ and $\int_a^b g(x)\mathrm{d}x$ exist for any $b \geq a$. If there exists some $c > 0$ such that $f(x) \leq cg(x)$ for all $x \geq a$, then the convergence of $\int_a^{\infty} g(x)\mathrm{d}x$ implies the convergence of $\int_a^{\infty} f(x)\mathrm{d}x$. (or equivalently, the divergence of $\int_a^{\infty} f(x)\mathrm{d}x$ implies the divergence of $\int_a^{\infty} g(x)\mathrm{d}x$). Particularly, if $\int_a^{\infty} g(x)\mathrm{d}x$ converges, then $\int_a^{\infty} f(x)\mathrm{d}x \leq c\int_a^{\infty} g(x)\mathrm{d}x$.

Theorem 9.3.6 Suppose that $f(x) > 0$ and $g(x) > 0$ for all $x \geq a$ and the proper integrals $\int_a^b f(x)\mathrm{d}x$ and $\int_a^b g(x)\mathrm{d}x$ exist for any $b \geq a$. If $\lim\limits_{x\to\infty} \dfrac{f(x)}{g(x)} = c > 0$, then $\int_a^{\infty} f(x)\mathrm{d}x$ converges if and only if $\int_a^{\infty} g(x)\mathrm{d}x$ converges.

Remark 9.3.7 Consider the same set-up in Theorem 9.3.6 but with $f(x) \geq 0$ and $\lim\limits_{x\to+\infty} \dfrac{f(x)}{g(x)} = c = 0$. Intuitively the conclusion should be weaker: the convergence of $\int_a^{\infty} g(x)\mathrm{d}x$ implies the convergence of $\int_a^{\infty} f(x)\mathrm{d}x$ but the converse may not hold. A trivial example is $f(x) = 0$ and $g(x) = x$.

Definition 9.3.8 Suppose f is defined on $(a, b]$ where $b > a$ and the proper integral $\int_x^b f(t)\mathrm{d}t$ exists for any $x \in (a, b]$. Then the symbol $\int_{a^+}^b f(t)\mathrm{d}t$ means $\lim\limits_{x\to a^+} \int_x^b f(t)\mathrm{d}t$ and is referred to as an **improper integral of the second kind**. We say the improper integral $\int_{a^+}^b f(t)\mathrm{d}t$ converges if the limit exists and diverges otherwise. If the limit does exist and equal to A, then we call A the value of the improper integral and write $\int_{a^+}^b f(t)\mathrm{d}t = A$.

Definition 9.3.9 The symbol $\int_a^{b^-} f(t)\mathrm{d}t$ is defined similarly. If both $\int_{a^+}^c f(t)\mathrm{d}t$ and $\int_c^{b^-} f(t)\mathrm{d}t$ converge, then we say $\int_{a^+}^{b^-} f(t)\mathrm{d}t$ also converges and write

$$\int_{a^+}^{b^-} f(t)\mathrm{d}t = \int_{a^+}^{c} f(t)\mathrm{d}t + \int_c^{b^-} f(t)\mathrm{d}t.$$

4 Sequences and Series of Functions

Definition 9.4.1 Suppose for each positive integer n, f_n is a function. We then call $\{f_n\}$ a **sequence of functions**. If there exists a function f and a set $S \subseteq \mathbb{R}$ such that

$$f(x) = \lim_{n\to\infty} f_n(x) \quad \text{for all } x \in S,$$

then we say f **the limit function of** $\{f_n\}$ **on** S and we say that $\{f_n\}$ **converges pointwise to** f **on** S. We often simply write

$$f_n \to f \text{ pointwise on } S.$$

Remark 9.4.2 Here we introduce the "$\varepsilon - N$" definition for the pointwise convergence of sequence of functions:

$$\forall x \in S, \forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } |f_n(x) - f(x)| < \varepsilon \text{ for all } n > N.$$

Note that the choice of N depends on x and ε because when x takes different values, we actually have different sequences $\{f_n(x)\}$.(it's a sequence of real numbers now!!)

Example 9.4.3 Let $f_n(x) = x^n$ for $x \in [0, 1]$ and all $n \in \mathbb{N}$. Then $\{f_n\}$ converges pointwise to the function

$$f(x) = \begin{cases} 0 & \text{if } x \in [0, 1) \\ 1 & \text{if } x = 1 \end{cases}.$$

Note f_n is continuous on $[0, 1]$ but the limit function f is not continuous on $[0, 1]$. Thus f doesn't preserve the continuity of f_n, which implies that the pointwise definition is not strong enough. This is because there exists a particular $x = 1$ such that $f_n(1) = 1$ for all $n \in \mathbb{N}$ and thus the convergence is not uniform. However, our definition treats all $x \in [0, 1]$ equally. This motivates us to introduce the concept of uniform convergence.

Definition 9.4.4 A sequence of functions $\{f_n\}$ is said to **converge uniformly** to a function f on a set S if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall x \in S, |f_n(x) - f(x)| < \varepsilon \text{ for all } n > N.$$

We often simply write

$$f_n \to f \text{ uniformly on } S.$$

Remark 9.4.5 Note that the choice of N only depends on ε but not on x. Thus we can choose a single N for all $x \in S$ to make sure $|f_n(x) - f(x)| < \varepsilon$ for all $n > N$. In other words, the convergence is uniform in the sense that it is uniform across all $x \in S$.

Theorem 9.4.6 If $f_n \to f$ uniformly on $S \subseteq \mathbb{R}$ and for each $n \geq 1$, f_n is continuous at $x = a \in S$, then then limit function f is also continuous at $x = a$.

Proof. We need to show that $\lim_{x\to a} f(x) = f(a)$.
Given $\varepsilon > 0$, since $f_n \to f$ uniformly on S, there exists an $N \in \mathbb{N}$ such that $|f_n(x) - f(x)| < \varepsilon/3$ for all $x \in S$ and all $n \geq N$.

Since f_N is continuous at $x = a$, there exists some $\delta > 0$ such that $|f_N(x) - f_N(a)| < \varepsilon/3$ for all $x \in S$ with $|x - a| < \delta$. Note we've chosen fixed N first and then δ is also fixed.
For such x and $n \geq N$, we have

$$|f(x) - f(a)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(a)| + |f_N(a) - f(a)| < \varepsilon/3 \times 3 = \varepsilon.$$

□

Proposition 9.4.7 Particularly, if $f_n = \sum_{k=1}^{n} u_k$ is the sequence of partial sums of another function sequence $\{u_n\}$, then the above theorem can be written as:
If a series of function $\sum u_n$ converges uniformly to a sum function f on $S \subseteq \mathbb{R}$ and for each $n \geq 1$, u_n is continuous at $x = a \in S$, then the sum function f is also continuous at $x = a$.
We often just write the conclusion as

$$\lim_{x\to a} \sum_{n=1}^{\infty} u_n(x) = \lim_{x\to a} f(x) = f(a) = \sum_{n=1}^{\infty} u_n(a) = \sum_{n=1}^{\infty} \lim_{x\to a} u_n(x).$$

Theorem 9.4.8 (Weierstrass M-Test) Suppose $f_n \to f$ uniformly on $[a, b]$ and for each $n \geq 1$, f_n is continuous on $[a, b]$. Define another function $\{g\}$ as $g_n(x) = \int_a^x f_n(t)\mathrm{d}t$ for $x \in [a, b]$ and function g as $g(x) = \int_a^x f(t)\mathrm{d}t$ for $x \in [a, b]$. Then $g_n \to g$ uniformly on $[a, b]$ we have

$$\lim_{n\to\infty} \int_a^x f_n(t)\mathrm{d}t = \int_a^x \lim_{n\to\infty} f_n(t)\mathrm{d}t.$$

Proof. Given $\varepsilon > 0$, there is an N such that $|f_n(x) - f(x)| < \varepsilon/(b - a)$ for all $x \in [a, b]$ whenever $n \geq N$. Then for any $x \in [a, b]$ we have

$$\begin{aligned} |g_n(x) - g(x)| &= \left| \int_a^x f_n(t)\mathrm{d}t - \int_a^x f(t)\mathrm{d}t \right| \\ &\leq \int_a^x |f_n(t) - f(t)|\mathrm{d}t \\ &< \int_a^x \frac{\varepsilon}{b - a}\mathrm{d}t = \frac{\varepsilon}{b - a}(x - a) \leq \varepsilon. \end{aligned}$$

□

Example 9.4.9 Suppose $f_n(x) = nxe^{-nx^2}$ for $n = 1, 2, \ldots$ and x real, then

$$\frac{1}{2} = \lim_{n\to\infty} \int_0^1 f_n(x)\,\mathrm{d}x \neq \int_0^1 \lim_{n\to\infty} f_n(x)\,\mathrm{d}x = 0.$$

The equality fails because the convergence of $\{f_n\}$ is not uniform on $[0, 1]$.

Proposition 9.4.10 Particularly, if $f_n = \sum_{k=1}^{n} u_k$ where $\{u_n\}$ is another function sequence, then the above theorem can be written as:

Suppose a series of functions $\sum u_n$ converges uniformly to a sum function f on $[a,b]$ and for each $k \geq 1$, u_k is continuous on $[a,b]$. Then we have

$$g_n(x) = \int_a^x \sum_{k=1}^{n} u_k(t)\mathrm{d}t, \qquad g(x) = \int_a^x \sum_{k=1}^{\infty} u_k(t)\mathrm{d}t$$

Then $g_n \to g$ uniformly on $[a,b]$ and

$$\lim_{n\to\infty} \int_a^x f_n(t)\mathrm{d}t = \int_a^x \lim_{n\to\infty} f_n(t)\mathrm{d}t.$$

Note we have the left hand side as

$$\lim_{n\to\infty} \int_a^x f_n(t)\mathrm{d}t = \lim_{n\to\infty} \int_a^x \sum_{k=1}^{n} u_k(t)\mathrm{d}t = \lim_{n\to\infty} \sum_{k=1}^{n} \int_a^x u_k(t)\mathrm{d}t = \sum_{k=1}^{\infty} \int_a^x u_k(t)\mathrm{d}t,$$

while the right hand side is

$$\int_a^x \lim_{n\to\infty} f_n(t)\mathrm{d}t = \int_a^x \sum_{k=1}^{\infty} u_k(t)\mathrm{d}t.$$

This also suggests that we can integrate term-by-term for a uniformly convergent series of functions, i.e.

$$\sum_{k=1}^{\infty} \int_a^x u_k(t)\mathrm{d}t = \int_a^x \sum_{k=1}^{\infty} u_k(t)\mathrm{d}t.$$

Example 9.4.11 Let $f(x) = \sum_{n=1}^{\infty} \frac{\sin nx}{n^2}$ for $x \in \mathbb{R}$. Note f is continuous on $[0,\pi]$ and the series $\sum_{n=1}^{\infty} (\sin nx)/n^2$ converges uniformly on $[0,\pi]$. Then we can integrate term-by-term to get

$$\begin{aligned}\int_0^\pi f(x)\,\mathrm{d}x &= \int_0^\pi \sum_{n=1}^{\infty} \frac{\sin(nx)}{n^2}\,\mathrm{d}x = \sum_{n=1}^{\infty} \int_0^\pi \frac{\sin(nx)}{n^2}\,\mathrm{d}x = \sum_{n=1}^{\infty} \frac{1}{n^2} \int_0^\pi \sin(nx)\,\mathrm{d}x \\ &= \sum_{n=1}^{\infty} \frac{1}{n^2} \left[-\frac{\cos(nx)}{n}\right]_0^\pi = \sum_{n=1}^{\infty} \frac{1}{n^2} \cdot \frac{1}{n} \cdot (1-(-1)^n) = 2\sum_{k=1}^{\infty} \frac{1}{(2k-1)^3}.\end{aligned}$$

Remark 9.4.12 We do not have such nice property (i.e. differentiating term-by-term) for general uniformly convergent series. For example, let $f_n(x) = \frac{\sin nx}{n^2}$. Note that $\sum f_n$ is convergent since $0 \leq |f_n(x)| \leq \frac{1}{n^2}$ and $\sum \frac{1}{n^2}$ converges. Then by the Weierstrass M-test (which will be introduced later), we have $\sum f_n$ converges uniformly.

Term-by-term differentiation gives us $\sum f_n'(x) = \sum \frac{\cos nx}{n}$, which diverges at $x = 0$ since $\sum \frac{1}{n}$ diverges.

Theorem 9.4.13 Given a series of functions $\sum u_n$ that converges pointwise to a function f on $S \subseteq \mathbb{R}$. If there is a convergent series of positive numbers $\sum M_n$ such that $|u_n(x)| \leq M_n$ for all $x \in S$ and all $n \in \mathbb{N}$, then $\sum u_n$ converges uniformly to f on S.

Proof. Given $\varepsilon > 0$, we need to show that there is an $N \in \mathbb{N}$ such that $\left|\sum_{k=1}^{n} u_k(x) - f(x)\right| < \varepsilon$ for all $x \in S$ whenever $n \geq N$.

Suppose $\sum_{k=1}^{\infty} M_k = A$, then for the given $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that $\left|\sum_{k=1}^{n} M_k - A\right| < \varepsilon$ for all $n \geq N$.

$$\left|f(x) - \sum_{k=1}^{n} u_k(x)\right| = \left|\sum_{k=n+1}^{\infty} u_k(x)\right| \leq \sum_{k=n+1}^{\infty} |u_k(x)| \leq \left|\sum_{k=n+1}^{\infty} M_k\right| = \left|A - \sum_{k=1}^{n} M_k\right| < \varepsilon.$$

holds for any $x \in S$ and all $n \geq N$. Thus $\sum u_n$ converges uniformly to f on S. □

5 Power Series

Recall in Definition 9.1.16, we have defined the power series $\sum_{n=0}^{\infty} a_n x^n$. Now we will discuss the convergence of power series.

Definition 9.5.1 In general, a series of the form $\sum_{n=0}^{\infty} a_n (x-a)^n$ is also called a **power series** in $(x-a)$ or a **power series centered at** $x = a$. We call the a the center and a_n the coefficient of the power series.

Example 9.5.2

1. For the power series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$, ratio test gives us

$$\left|\frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n}\right| = \frac{|x|}{n+1} \to 0 < 1 \quad \text{for } x \neq 0$$

as $n \to \infty$. The case where $x = 0$ is trivial. Thus $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ converges for all $x \in \mathbb{R}$.

2. For the power series $\sum_{n=0}^{\infty} n^2 3^n x^n$, root test gives us

$$\sqrt[n]{|n^2 3^n x^n|} = 3|x| \quad \text{for } x \in \mathbb{R} \text{ as } n \to \infty$$

Thus $\sum_{n=0}^{\infty} n^2 3^n x^n$ converges for $|x| < \frac{1}{3}$ and diverges for $|x| > \frac{1}{3}$. The case where $|x| = \frac{1}{3}$ is inconclusive and we need to check it manually. When $x = \frac{1}{3}$, we have $\sum_{n=0}^{\infty} n^2$ which diverges. When $x = -\frac{1}{3}$, we have $\sum_{n=0}^{\infty} (-1)^n n^2$ also diverges.

Theorem 9.5.3 If a power series $\sum a_n x^n$ converges for a particular $x_1 \neq 0$, then

(a) the power series converges absolutely for every x with $|x| < |x_1|$;

(b) the power series converges uniformly on every interval $[-R, R]$ with $0 < R < |x_1|$.

Proof. We first prove (b). Note the necessary condition for the convergence of $\sum a_n x^n$ is that $a_n x^n \to 0$ as $n \to \infty$. Particularly, there is an $N \in \mathbb{N}$ such that $|a_n x_1^n| < 1$ whenever $n \geq N$. Then for $x \in [-R, R]$ with $0 < R < |x_1|$, we have

$$0 \leq |a_n x^n| = |a_n||x|^n = |a_n x_1^n| \cdot \left|\frac{x}{x_1}\right|^n < \left|\frac{x}{x_1}\right|^n \leq \left|\frac{R}{x_1}\right|^n < 1 \quad \text{for } n \geq N.$$

Then by the comparison test, we have $\sum a_n x^n$ converges absolutely on $[-R, R]$. Also by the Weierstrass M-test(in this case, $M_n = \left|\frac{R}{x_1}\right|^n$), we have $\sum a_n x^n$ converges uniformly on $[-R, R]$.
Now we prove (a). For any $|x| < |x_1|$, we can find some R such that $|x| < R < |x_1|$. Then by (b), we have $\sum a_n x^n$ converges absolutely for every x with $|x| < R$. In particular, $\sum a_n x^n$ converges absolutely for every x with $|x| < |x_1|$. □

Theorem 9.5.4 If a power series $\sum a_n x^n$ converges for a particular $x_1 \neq 0$ and diverges for a particular $x_2 \neq 0$, then there is a unique r such that

(c) the power series converges absolutely for every x with $|x| < r$;

(d) the power series diverges for every x with $|x| > r$.

We call r the **radius of convergence** of the power series since this r is unique.

Proof. We prove (d) first. Let $A = \{|x| : \sum a_n x^n \text{ converges at } x\}$. Note $|x_1| \in A$ and $|x_2| \notin A$. By (a) we know $|x_2|$ is an upper bound of A. Let $r = \sup A$, then r is the unique number. Note $0 < |x_1| \leq r \leq |x_2|$. Then for any $|x| > r$, $|x| \notin A$ and thus $\sum a_n x^n$ diverges at such x.
Now we prove (c). For $0 < |x| < r$, $|x|$ is not an upper bound of A and there exists $|z| \in A$ such that $0 < |x| < |z| < r$. By (a), we have $\sum a_n x^n$ converges absolutely for every x with $|x| < |z|$. In particular, $\sum a_n x^n$ converges absolutely for every x with $|x| < r$. □

Remark 9.5.5

- If the power series $\sum a_n x^n$ converges for all $x \in \mathbb{R}$, then we say the radius of convergence is $r = +\infty$.
- If the power series $\sum a_n x^n$ diverges for all $x \neq 0$, then we say the radius of convergence is $r = 0$.

Remark 9.5.6 Theorem 9.5.3 and Theorem 9.5.4 also hold for power series of the form $\sum_{n=0}^{\infty} a_n (x-a)^n$ by simply replacing x with $x - a$ in the proofs so the power series centers at $x = a$ instead of $x = 0$. Let r be the **radius of convergence**, we often call $(a - r, a + r)$ the **interval of convergence** of the power series $\sum_{n=0}^{\infty} a_n (x-a)^n$. Note the convergence at the endpoints $x = a \pm r$ is inconclusive and we need to check it manually.

Definition 9.5.7 On the interval of convergence of the power series $\sum_{n=0}^{\infty} a_n (x-a)^n$, we write $f(x) = \sum_{n=0}^{\infty} a_n (x-a)^n$ and call f the **sum function**. We also say that the power series **represent the function** f in its interval of convergence and we call it the **power series expansion** of f at $x = a$.

Remark 9.5.8 By Theorem 9.5.3 and Theorem 9.5.4, given a power series $\sum_{n=0}^{\infty} a_n(x-a)^n$, let r be the radius of convergence, we know the power series

- converges absolutely on $(a-r, a+r)$;
- converges uniformly on $[a-R, a+R]$ for any $0 < R < r$;

Thus the sum function f is continuous on $(a-r, a+r)$. We also know we can integrate the power series term-by-term on $[a-r, a+r]$ to obtain the integral of the sum function f. For any $x \in (a-r, a+r)$ we have

$$\int_a^x f(t)\mathrm{d}t = \int_a^x \sum_{n=0}^{\infty} a_n(t-a)^n \mathrm{d}t = \sum_{n=0}^{\infty} a_n \int_a^x (t-a)^n \mathrm{d}t = \sum_{n=0}^{\infty} a_n \frac{(x-a)^{n+1}}{n+1}.$$

Theorem 9.5.9 Suppose $f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n$ on $(a-r, a+r)$ where r is the radius of convergence. Then

(a) $\sum_{n=1}^{\infty} na_n(x-a)^{n-1}$ also has the same interval convergence $(a-r, a+r)$

(b) The derivative $f'(x)$ exists for each $x \in (a-r, a+r)$ and $f'(x) = \sum_{n=1}^{\infty} na_n(x-a)^{n-1}$ for each $x \in (a-r, a+r)$.

Proof. Suppose $a = 0$. We start with (a). For any $x \in (0, r)$, we know for any $h > 0$ with $0 < x < x+h < r$,

$$f(x+h) = \sum_{n=0}^{\infty} a_n(x+h)^n, \qquad \frac{f(x+h)-f(x)}{h} = \sum_{n=0}^{\infty} a_n \frac{(x+h)^n - x^n}{h}.$$

Let $g_n(x) = x^n$, by the **MVT** to g_n on $[x, x+h]$ (note g_n is continuous on $[x, x+h]$ and differentiable on $(x, x+h)$), there is some $c_n \in (x, x+h)$ such that

$$g_n(x+h) - g_n(x) = hg'(c_n) \quad \text{where } x < c_n < x+h.$$

Thus we have

$$\frac{f(x+h)-f(x)}{h} = \sum_{n=0}^{\infty} a_n \frac{(x+h)^n - x^n}{h} = \sum_{n=0}^{\infty} a_n g'(c_n) = \sum_{n=1}^{\infty} na_n c_n^{n-1} \quad \text{converges absolutely.}$$

Note that $\sum_{n=1}^{\infty} na_n x^{n-1}$ satisfies

$$0 \le |na_n x^{n-1}| < |na_n c_n^{n-1}| \quad \text{for all } n \ge 1.$$

Thus $\sum_{n=1}^{\infty} na_n x^{n-1}$ converges absolutely for any $x \in (-r, r)$. Similarly, we know it actually converges absolutely for any $x \in (-r, 0)$ and thus for any $x \in (-r, r)$.

Now we prove $\sum_{n=1}^{\infty} na_n(x-a)^{n-1}$ diverges for any x with $|x| > r$. Suppose $\sum_{n=1}^{\infty} na_n x^{n-1}$ converges absolutely on $(-r_2, r_2)$ where $r_2 > r$ and $0 \le |a_n x^n| < |n a_n x^{n-1}|$ when $n > r_2$. For any $x \in (r, r_2)$, we have $\sum_{n=0}^{\infty} a_n x^n$ diverges and thus $\sum_{n=1}^{\infty} n a_n x^{n-1}$ also diverges. Thus the differentiated series $\sum_{n=1}^{\infty} n a_n x^{n-1}$ share the same interval of convergence as the orginal series $\sum_{n=0}^{\infty} a_n x^n$.

Now we prove (b). Let $D(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$, then for any $x \in (-r, r)$,

$$\int_0^x D(t)\mathrm{d}t = \sum_{n=1}^{\infty} a_n \int_0^x n t^{n-1}\mathrm{d}t = \sum_{n=1}^{\infty} a_n x^n = f(x) - a_0.$$

Note $D(x)$ is continuous on $(-r, r)$. By the **FTC1**, we have $D(x) = f'(x)$ for any $x \in (-r, r)$. □

Example 9.5.10 Recall

$$1 + x + x^2 + \cdots = \frac{1}{1-x} \quad r = 1. \tag{1}$$

Differentiating (1) term-by-term gives us

$$1 + 2x + 3x^2 + \cdots = \frac{1}{(1-x)^2} \quad r = 1. \tag{2}$$

Replace x by $-x$ in (1) gives us

$$1 - x + x^2 - \cdots = \frac{1}{1+x} \quad r = 1. \tag{3}$$

Integrating (3) term-by-term gives us

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots = \ln(1+x) \quad r = 1. \tag{4}$$

Replace x by x^2 in (3) gives us

$$1 - x^2 + x^4 - \cdots = \frac{1}{1+x^2} \quad r = 1. \tag{5}$$

Integrating (5) term-by-term gives us

$$x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots = \arctan x \quad r = 1. \tag{6}$$

For (4), when $x = 1$, we have

$$1 - \frac{1}{2} + \frac{1}{3} - \cdots = \ln 2.$$

For (6), when $x = 1$, we have

$$1 - \frac{1}{3} + \frac{1}{5} - \cdots = \frac{\pi}{4}.$$

Remark 9.5.11 Suppose $f(x) = \sum_{n=0}^{\infty} a_n (x-a)^n$ on its interval of convergence $(a-r, a+r)$. For any $x \in (a-r, a+r)$, $f'(x), f''(x), \cdots, f^{(k)}(x)$ are all well-defined (and can be obtained by keeping differentiating the power series) for any positive integer k. We say that f is infinitely differentiable on $(a-r, a+r)$. Note

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k+1)a_n(x-a)^{n-k} \quad \text{for any } k \geq 1.$$

In particular,

$$f^{(k)}(a) = k!a_k \Rightarrow a_k = \frac{f^{(k)}(a)}{k!} \quad \text{for any } k \geq 0.$$

Then $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$ for any $x \in (a-r, a+r)$. In other words, the power series expansion of f at $x = a$ is actually the Taylor series expansion of f at $x = a$ and thus unique.

Theorem 9.5.12 If two power series $\sum_{n=0}^{\infty} a_n(x-a)^n$ and $\sum_{n=0}^{\infty} b_n(x-a)^n$ have the same sum function f in some neighbourhood of $x = a$, then two series are equal (term by term); in fact, we have $a_n = b_n = \frac{f^{(n)}(a)}{n!}$ for all $n \geq 0$.

Proposition 9.5.13 If we know a function f can be represented by a power series $\sum_{n=0}^{\infty} a_n(x-a)^n$ in $(a-r, a+r)$, then its sequence of Taylor polynomial $T_n f(x; a)$ converges absolutely on $(a-r, a+r)$ and converges uniformly on $[a-R, a+R]$ for any $0 < R < r$.

Remark 9.5.14 Suppose f is infinitely differentiable in some open neighbourhood of $x = a$ and consider the power series (Taylor series) $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$. Note

- the taylor series does not necessarily converge: $f(x) = \frac{1}{1-x}$ is infinitely differentiable and it's Taylor series at $x = 0$ is $\sum_{n=0}^{\infty} x^n$, which diverges for $|x| \geq 1$.
- if the taylor series converges, it does not necessarily converge to f:

$$f(x) = \begin{cases} \exp\left\{-\frac{1}{x^2}\right\} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

  Note f is infinitely differentiable at $x = 0$ and $T_n f(x) \equiv 0$ for all $n \in \mathbb{N}$. Thus the Taylor series of f at $x = 0$ converges to the zero function, which is not equal to f itself except for $x = 0$.

Combining with the Taylor formula with error terms, we know

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n \to f(x) \Leftrightarrow E_n(x) \to 0 \text{ as } n \to \infty.$$

Theorem 9.5.15 Suppose f is infinitely differentiable in an open interval $I = (a-r, a+r)$ and there is a constant $A > 0$ such that $|f^{(n)}(x)| \leq A^n$ for $n \geq 1$ and for any $x \in I$. Then the Taylor series generated by f at

$x = a$ converges to f for any $x \in I$.

Proof. Recall that if f has continous derivative of order $n+1$ in some interval containing a, then the Taylor formula is the following

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + E_n(x) \quad \text{where } E_n(x) = \frac{1}{n!}\int_a^x (x-t)^n f^{(n+1)}(t)\mathrm{d}t.$$

We do a change of variable $t = x + (a-x)u$. Then $\begin{cases} t = a \Leftrightarrow u = 1 \\ t = x \Leftrightarrow u = 0 \\ \mathrm{d}t = (a-x)\mathrm{d}u \end{cases}$ Thus the error term can be written as

$$E_n(x) = \frac{1}{n!}(x-a)^{n+1}\int_0^1 u^n f^{(n+1)}(x+(a-x)u)\mathrm{d}u.$$

Note

$$\begin{aligned} 0 \le |E_n(x)| &\le \frac{1}{n!}|x-a|^{n+1}\left|\int_0^1 u^n f^{(n+1)}(x+(a-x)u)\mathrm{d}u\right| \\ &\le \frac{1}{n!}|x-a|^{n+1}\int_0^1 u^n |f^{(n+1)}(x+(a-x)u)|\mathrm{d}u \\ &\le \frac{1}{n!}|x-a|^{n+1}A^{n+1}\int_0^1 u^n \mathrm{d}u = \frac{(A|x-a|)^{n+1}}{(n+1)!} \to 0 \text{ as } n \to \infty. \end{aligned}$$

By the squeeze theorem, we know $E_n(x) \to 0$ as $n \to \infty$ and thus the Taylor series generated by f at $x = a$ converges to f for any $x \in I$. □

Example 9.5.16

1. For $f(x) = e^x$, $|f^{(n)}(x)| = |e^x| \le A^n$ when $x \in (-r, r)$ and let $A = e^r$. Since r can be any positive real number, actually
$$1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \cdots = e^x \qquad \text{for } x \in \mathbb{R}.$$
2. For $f(x) = \sin x$, $|f^{(n)}(x)| \le 1$ for any $n \ge 1$ and any $x \in \mathbb{R}$. Thus
$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \cdots \qquad \text{for } x \in \mathbb{R}.$$
3. For $f(x) = \cos x$, $|f^{(n)}(x)| \le 1$ for any $n \ge 1$ and any $x \in \mathbb{R}$. Thus
$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^n \frac{x^{2n}}{(2n)!} + \cdots \qquad \text{for } x \in \mathbb{R}.$$

Example 9.5.17 Power series may be used to solve differential equations. For example, consider the following second order ordinary differential equation $y = y(x)$:

$$(1-x)y'' = -2y.$$

We can try to find a power series solution of the form $y = \sum_{n=0}^{\infty} a_n x^n$. Then we have

$$(1-x)\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} = \sum_{n=0}^{\infty}\left[(n+1)(n+2)a_{n+2} - n(n-1)a_n\right]x^n = \sum_{n=0}^{\infty} -2a_n x^n.$$

Thus we have the following recursive relation for the coefficients a_n:

$$a_{n+2} = \frac{n-2}{n+2}a_n \quad \text{for } n \geq 0.$$

Comparing the coefficients of x^n on both sides, we have

$$\begin{cases} a_2 = -a_0 & \\ a_{2n} = 0 \cdot a_2 = 0 & \text{for } n \geq 2 \\ a_{2n+1} = \dfrac{-1}{(2n+1)(2n-1)}a_1 & \text{for } n \geq 1 \end{cases}$$

Thus we have

$$y = a_0(1-x^2) + a_1\sum_{n=0}^{\infty}\frac{-1}{(2n+1)(2n-1)}x^{2n+1} \quad \text{for } x \in (-1,1).$$

Thus the series gives a family of solutions to the differential equation. However, this is not necessarily all the solutions to the differential equation since we only consider power series solutions. In other words, can we have some solutions not infinitely differentiable but still satisfying the differential equation? We'll only introduce several basic theorems about differential equations here.

Definition 9.5.18 Assume P and Q are given functions continuous on some open interval I. A differential equation of the form

$$y' + P(x)y = Q(x),$$

is called a **first-order linear differential equation**.

Theorem 9.5.19 Assume P is continuous on an open interval I. Choose any point a in I and let b be any real number. Then there is one and only one function $y = f(x)$ which satisfies the initial-value problem

$$y' + P(x)y = 0, \qquad \text{with} \qquad f(a) = b,$$

on the interval I. This function is given by the formula

$$f(x) = be^{-A(x)}, \qquad \text{where} \qquad A(x) = \int_a^x P(t)\mathrm{d}t$$

Theorem 9.5.20 Assume P and Q are continuous on an open interval I. Choose any point a in I and let b be any real number. Then there is one and only one function $y = f(x)$ which satisfies the initial-value problem

$$y' + P(x)y = Q(x), \qquad \text{with} \qquad f(a) = b,$$

on the interval I. This function is given by the formula

$$f(x) = be^{-A(x)} + e^{-A(x)} \int_a^x Q(t)e^{A(t)}\mathrm{d}t,$$

where $A(x) = \displaystyle\int_a^x P(t)\mathrm{d}t$.

Chapter 10 Vector Algebra and Analytic Geometry

1 Lines and Planes

Remark 10.1.1 We shall use the words "point" and "vector"(a vector with initial point at the origin $O = (0, 0, \dots, 0)$ and ending point at the point P) interchangeably to denote an element $P = (p_1, p_2, \cdots, p_n) \in \mathbb{R}^n$.

Definition 10.1.2 Suppose P is a given point and A is a given nonzero vector in $\mathbb{R}^n (n \geq 2)$. The set of all points of the form $P + tA$ where $t \in \mathbb{R}$ is called the **line** passing through P and parallel to A. We denote this line by $L(P, A)$ and write $L(P, A) = \{P + tA : t \in \mathbb{R}\}$. A point Q is said to be **on the line** $L(P, A)$ if $Q \in L(P, A)$. We often call A a **direction vector** of line $L(P, A)$.

Theorem 10.1.3 Two lines $L(P, A)$ and $L(P, B)$ through the same point P are equal sets if and only if the dirction vectors A and B are parallel, i.e. there is some $\lambda \in \mathbb{R}$ such that $A = \lambda B$ for some $\lambda \in \mathbb{R}_{\neq 0}$.

Theorem 10.1.4 Two lines $L(P, A)$ and $L(Q, A)$ with the same direction vector A are equal if and only if $Q \in L(P, A)$.

Theorem 10.1.5 Given a line L_1 and a point Q not on L_1, there is one and only one line L_2 containing Q and parallel to L_1.

Theorem 10.1.6 If $P \neq Q$ are two points, then there is one and only one line containing both P and Q. And this line is simply the set $\{P + t(Q - P) : t \in \mathbb{R}\}$.

Theorem 10.1.7 We call a function $X : \mathbb{R} \to \mathbb{R}^n$ (with $n \geq 2$) a **vector-valued function** (of a real variable). If P is a point and A is a nonzero vector in $\mathbb{R}^n$, then $X(t) = P + tA$ is a vector-valued function whose image is the line $L(P, A)$. Here t is often called a parameter and X is called a **vector/parametric equation** of the line $L(P, A)$.

Definition 10.1.8 Note that distinct values of the parameter t correspond to distinct points on the line. If $t_1 < t_2 < t_3$, then we say $X(t_2)$ is **between** $X(t_1)$ and $X(t_3)$ on the line $L(P, A)$. A pair of points P, Q are called **congruent** to another pair of points P', Q' if $|P - Q| = |P' - Q'|$ where $|\cdot|$ is the Euclidean distance in $\mathbb{R}^n$. We also call $|P - Q|$ the **distance between** P and Q.

Remark 10.1.9 If $P = (p_1, p_2, \cdots, p_n)$ and $A = (a_1, a_2, \cdots, a_n) \neq 0$, then the vector equation $X = (x_1, x_2, \cdots, x_n)$ of the line $L(P, A)$ is equivalent to (a set of) n **scalar parametric equations** of the line $L(P, A)$:

$$\begin{cases} x_1 = p_1 + ta_1 \\ x_2 = p_2 + ta_2 \\ \cdots \\ x_n = p_n + ta_n \end{cases}$$

Particularly in $\mathbb{R}^2$, solving the scalar parametric equations gives us

$$a_2(x_1(t) - p_1) = a_1(x_2(t) - p_2) \Leftrightarrow N \cdot (X(t) - P) = 0 \quad \text{where } N := (a_2, -a_1).$$

Here N is called a **normal vector** of the line $L(P, A)$ since it is perpendicular to the direction vector A. Furthermore, if $a_1 \neq 0$, we can rewrite the scalar parametric equations as

$$(x_2(t) - p_2) = \frac{a_2}{a_1}(x_1(t) - p_1)$$

The value $\dfrac{a_2}{a_1}$ is called the **slope** of the line.

Theorem 10.1.10 Suppose L is a line in $\mathbb{R}^2$ consisting of all points x satisfying $x \cdot N = P \cdot N$ where P is a given point on L and N is a given nonzero vector for L. Let

$$d := \frac{|P \cdot N|}{\|N\|}.$$

Then $|x| \geq d$ and $|x| = d$ if and only if x is the projection of P onto N, i.e. $x = \dfrac{P \cdot N}{\|N\|^2} N$.

Definition 10.1.11 Suppose P is a given point and A and B are two **linearly independent** vectors in $\mathbb{R}^n (n \geq 3)$. The set $M := \{P + sA + tB : s, t \in \mathbb{R}\}$ is called the **plane** passing through P and parallel to A and B. A point Q is said to be **on the plane** M if $Q \in M$.

Theorem 10.1.12 Two planes $M_1 = \{P + sA + tB\}$ and $M_2 = \{P + sC + tD\}$ through the same point P are equal sets if and only if $\operatorname{span}\{A, B\} = \operatorname{span}\{C, D\}$.

Theorem 10.1.13 Two planes $M_1 = \{P + sA + tB\}$ and $M_2 = \{Q + sA + tB\}$ with the same direction vectors A and B are equal if and only if $Q \in M_1$.

Definition 10.1.14 Two planes $M_1 = \{P + sA + tB\}$ and $M_2 = \{Q + sC + tD\}$ are called **parallel** if $\operatorname{span}\{A, B\} = \operatorname{span}\{C, D\}$ but $Q \notin M_1$. We also say that a vector X is **parallel** to a plane M if $X \in \operatorname{span}\{A, B\}$.

Remark 10.1.15 Similarly, a line is parallel to a plane if a direction vector of the line is parallel to the plane. Note a line parallel to a plane may be on the plane.

Theorem 10.1.16 Given a plane M and a point Q not on M, there is one and only one plane M' containing Q and parallel to M.

Theorem 10.1.17 If P, Q and R are three non-collinear points, i.e., not on the same line, then there is one and only one plane containing all three points. And this plane is simply the set $\{P + s(Q - P) + t(R - P) : s, t \in \mathbb{R}\}$.

Definition 10.1.18 A function $X : \mathbb{R}^2 \to \mathbb{R}^n$ (with $n \geq 3$) is called a **vector-valued function** of two real variables. If P is a point and A and B are two linearly independent vectors in $\mathbb{R}^n$, then $X(s, t) = P + sA + tB$ is a vector-valued function whose image is the plane $M = \{P + sA + tB : s, t \in \mathbb{R}\}$. Here (s, t) is often called parameters and X is called a **vector/parametric equation** of the plane M.

Remark 10.1.19 Similarly a plane can also be described by a **scalar equation** of the form $X \cdot N = P \cdot N$ where P is a given point on the plane and N is a given nonzero vector perpendicular to the plane, i.e. N is a normal vector of the plane. Suppose $N = (a, b, c)$, then $N \cdot (X(s, t) - P) = 0$ is equivalent to

$$d = ap_1 + bp_2 + cp_3, \quad ax_1(s, t) + bx_2(s, t) + cx_3(s, t) = d.$$

2 Cross Product

Definition 10.2.1 Let $A = (a_1, a_2, a_3)$ and $B = (b_1, b_2, b_3) \in \mathbb{R}^3$. Their **cross product** $A \times B$ is defined to be

$$A \times B = (a_2b_3 - a_3b_2, a_3b_1 - a_1b_3, a_1b_2 - a_2b_1).$$

Theorem 10.2.2 For $A, B, C \in \mathbb{R}^3$ and $k \in \mathbb{R}$, we have

(1) $A \times B = -B \times A$;

(2) $A \times (B + C) = A \times B + A \times C$ and $(B + C) \times A = B \times A + C \times A$;

(3) $(kA) \times B = k(A \times B)$ and $A \times (kB) = k(A \times B)$;

(4) $A \cdot (A \times B) = 0, B \cdot (A \times B) = 0$;

(5) $\|A \times B\|^2 = \|A\|^2\|B\|^2 - (A \cdot B)^2$;

(6) $A \times B = 0$ if and only if A and B are linearly dependent;

(7) $\|A \times B\| = \|A\|\|B\| \sin\theta$ where θ is the angle between A and B.

Theorem 10.2.3 Suppose A and B are two linearly independent vectors in $\mathbb{R}^3$. Then

1. A, B and $A \times B$ are linearly independent;
2. Any $N \in \mathbb{R}^3$ such that $N \cdot A = 0$ and $N \cdot B = 0$ must be parallel to $A \times B$, i.e. $N \in \operatorname{span}\{A \times B\}$.

Remark 10.2.4 The direction of $A \times B$ is determined by the right-hand rule, which corresponds to the common 3D-coordinate system.

Definition 10.2.5 Let $A, B, C \in \mathbb{R}^3$. $A \cdot B \times C := A \cdot (B \times C)$ is called the **scalar triple product** of A, B and C.

Theorem 10.2.6 Suppose $A, B, C \in \mathbb{R}^3$. Then

(1) A, B and C are linearly dependent if and only if $A \cdot B \times C = 0$;

(2) $A \cdot B \times C = B \cdot C \times A = C \cdot A \times B$.

(3) $A \times B \cdot C = A \cdot B \times C$.

Definition 10.2.7 Let $M = \{P + sA + tB : s, t \in \mathbb{R}\} \in \mathbb{R}^3$ be the plane through P and spanned by linearly independent vectors A and B. A vector $N \in \mathbb{R}^3$ is said to be perpendicular to the plane M if N is perpendicular to both A and B, i.e. $N \cdot A = 0$ and $N \cdot B = 0$. If, additionally, $N \neq 0$, then N is called a **normal vector** of the plane M.

Theorem 10.2.8 Given a plane $M = \{P + sA + tB : s, t \in \mathbb{R}\}$ in $\mathbb{R}^3$ where P is a point and A and B are linearly independent vectors. Let $N = A \times B$, then $N \neq 0$ and N is a normal vector of the plane M. Let $M' := \{X \in \mathbb{R}^3 : N \cdot (X - P) = 0\}$, then $M' = M$.

Remark 10.2.9 If $N = (a, b, c) \neq 0$, $P = (x_1, x_2, x_3)$ and $X = (x, y, z)$, then the scalar equation $N \cdot (X - P) = 0$ of the plane M can be rewritten as

$$ax + by + cz = ax_1 + bx_2 + cx_3 \quad \text{or} \quad a(x - x_1) + b(y - y_2) + c(z - z_3) = 0.$$

Theorem 10.2.10 Given a plane M through a point P and with a normal vector N, let $d := \dfrac{|P \cdot N|}{\|N\|}$. Then for any $X \in M$, we have $|X| \geq d$ and $|X| = d$ if and only if X is the projection of P onto N, i.e. $X = \dfrac{P \cdot N}{\|N\|^2} N$.

Remark 10.2.11 Suppose point Q is not on M, then for any $X \in M$, the minimum value of $\|X - Q\|$ is $\dfrac{|(Q - P) \cdot N|}{\|N\|}$ and is achieved when $(X - Q)$ is the projection of $(Q - P)$ onto N. We call $\dfrac{|(Q - P) \cdot N|}{\|N\|}$ the **distance** from point Q to the plane M.

Chapter 11 Multivariable Calculus

1 Vector-valued functions

Definition 11.1.1 Suppose $f_1, \cdots, f_n$ are functions fron $\mathbb{R}$ to $\mathbb{R}$. We call

$$\boldsymbol{F}(t) = (f_1(t), f_2(t), \cdots, f_n(t))$$

a **vector-valued function** and we define the **limits,derivatives** and **integrals** by the following:

$$\begin{aligned}\lim_{t\to p} \boldsymbol{F}(t) &= \left(\lim_{t\to p} f_1(t), \cdots, \lim_{t\to p} f_n(t)\right),\\ \boldsymbol{F}'(t) &= (f_1'(t), \cdots, f_n'(t)),\\ \int_a^b \boldsymbol{F}(t)\mathrm{d}t &= \left(\int_a^b f_1(t)\mathrm{d}t, \cdots, \int_a^b f_n(t)\mathrm{d}t\right).\end{aligned}$$

Theorem 11.1.2 Suppose $\boldsymbol{F}$ and $\boldsymbol{G}$ are vector-valued functions from $\mathbb{R}$ to $\mathbb{R}^n$ and u is a real-valued function from $\mathbb{R}$ to $\mathbb{R}$. If F, G and u are differentiable on in interval, then so are $F + G$, uF and $F \cdot G$ and we have

(1) $(\boldsymbol{F} + \boldsymbol{G})' = \boldsymbol{F}' + \boldsymbol{G}'$;

(2) $(u\boldsymbol{F})' = u'\boldsymbol{F} + u\boldsymbol{F}'$;

(3) $(\boldsymbol{F} \cdot \boldsymbol{G})' = \boldsymbol{F}' \cdot \boldsymbol{G} + \boldsymbol{F} \cdot \boldsymbol{G}'$.

For the case $n = 3$, then $\boldsymbol{F} \times \boldsymbol{G}$ is also differentiable with $(\boldsymbol{F} \times \boldsymbol{G})' = \boldsymbol{F}' \times \boldsymbol{G} + \boldsymbol{F} \times \boldsymbol{G}'$.

Proposition 11.1.3 Suppose $\boldsymbol{F}$ is a vector-valued function from $\mathbb{R}$ to $\mathbb{R}^n$. If $\boldsymbol{F}$ is differentiable and be constant length/norm on an open interval I, i.e., $\|\boldsymbol{F}(t)\| = c$ for any $t \in I$, then $\boldsymbol{F} \cdot \boldsymbol{F}' = 0$ on I. Namely, $\boldsymbol{F}'(t)$ is always perpendicular to $\boldsymbol{F}(t)$ for any $t \in I$.

Proof. Let $g(t) := \boldsymbol{F}(t) \cdot \boldsymbol{F}(t) = \|\boldsymbol{F}(t)\|^2 = c^2$ for any $t \in I$. Then $g'(t) = 2\boldsymbol{F}(t) \cdot \boldsymbol{F}'(t) = 0$ for any $t \in I$. Thus $\boldsymbol{F}(t) \cdot \boldsymbol{F}'(t) = 0$ for any $t \in I$. □

Theorem 11.1.4 Suppose $\boldsymbol{F} : \mathbb{R} \to \mathbb{R}^n$ and $u : \mathbb{R} \to \mathbb{R}$. Let $\boldsymbol{G} := \boldsymbol{F} \circ u$. Then:

(1) If u is continuous at t and $\boldsymbol{F}$ is continuous at $u(t)$, then $\boldsymbol{G}$ is continuous at t.

(2) If the derivative $u'(t)$ and $\boldsymbol{F}'(u(t))$ exist, then $\boldsymbol{G}'(t)$ exists and $\boldsymbol{G}'(t) = u'(t)\boldsymbol{F}'(u(t))$.

Theorem 11.1.5 Suppose $\boldsymbol{F} : \mathbb{R} \to \mathbb{R}^n$ is continuous on $[a, b]$, then:

(1) $\boldsymbol{F}$ is integrable on $[a, b]$;

(2) For any $c \in [a, b]$, we can define the primitive or indefinite integral function $\boldsymbol{A}(x) := \int_c^x \boldsymbol{F}(t)\mathrm{d}t$ for any $x \in [a, b]$. Then $\boldsymbol{A}$ is differentiable on (a, b) and $\boldsymbol{A}'(x) = \boldsymbol{F}(x)$ for any $x \in (a, b)$.

(3) If the derivative $\boldsymbol{F}'(t)$ is continuous on some open interval I, then for every c and x in I, we have

$$\boldsymbol{F}(x) = \boldsymbol{F}(c) + \int_c^x \boldsymbol{F}'(t)\mathrm{d}t.$$

Theorem 11.1.6 Suppose $\boldsymbol{F}$ and $\boldsymbol{G}$ are from $\mathbb{R}$ to $\mathbb{R}^n$ and are integrable on $[a, b]$. Then :

(1) for any $\alpha, \beta \in \mathbb{R}$, $\alpha\boldsymbol{F} + \beta\boldsymbol{G}$ is also integrable on $[a, b]$ with

$$\int_a^b (\alpha\boldsymbol{F} + \beta\boldsymbol{G})(t)\mathrm{d}t = \alpha \int_a^b \boldsymbol{F}(t)\mathrm{d}t + \beta \int_a^b \boldsymbol{G}(t)\mathrm{d}t;$$

(2) Let $C \in \mathbb{R}^n$, then the dot product $C \cdot \boldsymbol{F}$ is also integrable on $[a, b]$ with

$$\int_a^b (C \cdot \boldsymbol{F})(t)\mathrm{d}t = C \cdot \int_a^b \boldsymbol{F}(t)\mathrm{d}t.$$

(3) If $\|\boldsymbol{F}\|$ is also integrable on $[a, b]$, then

$$\left\| \int_a^b \boldsymbol{F}(t)\mathrm{d}t \right\| \leq \int_a^b \|\boldsymbol{F}(t)\|\mathrm{d}t.$$

Proof. We only prove (3) here. Let $\boldsymbol{A} := \int_a^b \boldsymbol{F}(t)\mathrm{d}t$. The case where $\|\boldsymbol{A}\| = 0$ is trivial. Suppose $\|\boldsymbol{A}\| > 0$. Then

$$\|\boldsymbol{A}\|^2 = \boldsymbol{A} \cdot \boldsymbol{A} = \int_a^b \boldsymbol{F}(t) \cdot \boldsymbol{A}\mathrm{d}t \leq \int_a^b \|\boldsymbol{F}(t)\|\|\boldsymbol{A}\|\mathrm{d}t = \|\boldsymbol{A}\| \int_a^b \|\boldsymbol{F}(t)\|\mathrm{d}t.$$

Thus $\left\| \int_a^b \boldsymbol{F}(t)\mathrm{d}t \right\| = \|\boldsymbol{A}\| \leq \int_a^b \|\boldsymbol{F}(t)\|\mathrm{d}t.$ □

2 Curves

Definition 11.2.1 Suppose $\boldsymbol{X}$ is a vector-valued function from $\mathbb{R}$ to $\mathbb{R}^n$ with $n = 2$ or $n = 3$. Suppose the domain of $\boldsymbol{X}$ is an interval $D \subseteq \mathbb{R}$. If $\boldsymbol{X}$ is continuous on D, then we call the range/image of $\boldsymbol{X}$ a **curve** in $\mathbb{R}^n$. We also say that the curve is **described** (parametrically) by the function $\boldsymbol{X}$ with parameter $t \in D$. We also call D a **parameter interval**.

Remark 11.2.2 The curve described by $\boldsymbol{X}$ is invarient under the change of parameter. For example, if $u : \mathbb{R} \to \mathbb{R}$ and $I = \{u(t) : t \in J\}$ for some interval J, then $\boldsymbol{Y} := \boldsymbol{X} \circ u$ also describes the same curve as $\boldsymbol{X}$. We say $\boldsymbol{X}$ and $\boldsymbol{Y}$ are **equivalent** in describing the curve C.

Definition 11.2.3 Let C be a curve described by $\boldsymbol{X}$ (note $\boldsymbol{X}$ is continuous). If the derivative $\boldsymbol{X}'(t)$ exists and $\boldsymbol{X}'(t) \neq 0$, then the line through point $\boldsymbol{X}(t)$ with direction vector $\boldsymbol{X}'(t)$ is called the **tangent line** of the curve C at point $\boldsymbol{X}(t)$.

Remark 11.2.4 Though the derivative of $\boldsymbol{X}$ and $\boldsymbol{Y} = \boldsymbol{X} \circ u$ are different, the tangent lines of the curve C at point $\boldsymbol{X}(t) = \boldsymbol{Y}(s)$ are the same since $\boldsymbol{Y}'(s) = u'(s)\boldsymbol{X}'(u(s))$ is parallel to $\boldsymbol{X}'(u(s))$. Thus the tangent line of a curve at a point is well-defined and does not depend on the choice of parameter.

Definition 11.2.5 Consider a motion (of a particle) described by a vector-valued function $\boldsymbol{X} : \mathbb{R} \to \mathbb{R}^n$ with $n = 2$ or $n = 3$. The derivative $\boldsymbol{X}'(t)$ is called the **velocity** at time t. $\|\boldsymbol{X}'(t)\|$ is called the **speed** at time t. The second derivative $\boldsymbol{X}''(t)$ is called the **acceleration** at time t. In some textbooks, we use $\boldsymbol{r}(t)$ for $\boldsymbol{X}(t)$, $\boldsymbol{v}(t)$ for $\boldsymbol{X}'(t)$, $v(t)$ for $\|\boldsymbol{X}'(t)\|$ and $\boldsymbol{a}(t)$ for $\boldsymbol{X}''(t)$.

Example 11.2.6 Suppose $\boldsymbol{r}(t) = (r\cos(f(t)), r\sin(f(t)), bf(t))$ with $r > 0$ and $b \neq 0$ fixed. This is called a **helix/helical curve** in $\mathbb{R}^3$. Note a helix is not contained in any plane in $\mathbb{R}^3$ but it is contained fully in a cylinder/cylindrical surface in $\mathbb{R}^3$.

Definition 11.2.7 Consider a curve described by $\boldsymbol{X}(t)$. We define the **unit tangent vector** $\boldsymbol{T}$ as

$$\boldsymbol{T}(t) := \frac{\boldsymbol{X}'(t)}{\|\boldsymbol{X}'(t)\|} \quad \text{whenever } \boldsymbol{X}'(t) \neq 0.$$

Note that $\|\boldsymbol{T}(t)\| = 1$ and thus $\boldsymbol{T}(t) \cdot \boldsymbol{T}'(t) = 0$. We define the **principal normal vetor** $\boldsymbol{N}$ as

$$\boldsymbol{N}(t) := \frac{\boldsymbol{T}'(t)}{\|\boldsymbol{T}'(t)\|} \quad \text{whenever } \boldsymbol{T}'(t) \neq 0.$$

Note $\|\boldsymbol{N}(t)\| = 1$ and $\boldsymbol{N}(t) \cdot \boldsymbol{T}(t) = 0$. Furthermore, we call the plane through the point $\boldsymbol{X}(t)$ and spanned by $\boldsymbol{T}(t)$ and $\boldsymbol{N}(t)$ the **osculating plane** of the curve at point $\boldsymbol{X}(t)$.

Remark 11.2.8 Consider three points $\boldsymbol{X}(t_1), \boldsymbol{X}(t_2)$ and $\boldsymbol{X}(t_3)$ on the curve. As t_2 and t_3 approach t_1, the plane determined by $\boldsymbol{X}(t_1), \boldsymbol{X}(t_2)$ and $\boldsymbol{X}(t_3)$, supposing the three points are not collinear, approaches the osculating plane at point $\boldsymbol{X}(t_1)$. Note if the curve is a planar curve, then the osculating plane is the fixed plane containing the curve.

Theorem 11.2.9 For a motion described by $\boldsymbol{r}(t)$, the acceleration $\boldsymbol{a}(t) := \boldsymbol{r}''(t)$ is a linear combination of $\boldsymbol{T}(t)$ and $\boldsymbol{T}'(t)$ given by

$$\boldsymbol{a}(t) = v'(t)\boldsymbol{T}(t) + v(t)\boldsymbol{T}'(t).$$

If, additionally $\boldsymbol{T}'(t) \neq 0$, then

$$\boldsymbol{a}(t) = v'(t)\boldsymbol{T}(t) + v(t)\|\boldsymbol{T}'(t)\|\boldsymbol{N}(t).$$

Remark 11.2.10 We call $v'(t)$ and $v(t)\|\boldsymbol{T}'(t)\|$ the tangential and normal components of the acceleration $\boldsymbol{a}(t)$, respectively. In some textbook, the **binormal vector** $\boldsymbol{B}$ is also introduced as $B(t) := \boldsymbol{T}(t) \times \boldsymbol{N}(t)$. Note $\|\boldsymbol{B}(t)\| = 1$ and $\boldsymbol{T}(t), \boldsymbol{N}(t)$ and $\boldsymbol{B}(t)$ together provide a moving reference frame for $\mathbb{R}^3$ along the curve.

Definition 11.2.11 For a curve in $\mathbb{R}^2$ and $\mathbb{R}^3$ described by $\boldsymbol{r}(t)$ for $t \in [a, b]$ (Note $\boldsymbol{r}$ is required to be continuous on $[a, b]$). Let $P = \{t_0, t_1, \ldots, t_n\}$ be a partition of $[a, b]$. We denote the polygon whose vertices are $\boldsymbol{r}(t_0), \boldsymbol{r}(t_1), \ldots, \boldsymbol{r}(t_n)$ by $\Pi(P)$ and denote the **length of polygon** by $\|\Pi(P)\|$. Then

$$\|\Pi(P)\| = \sum_{i=1}^{n} \|\boldsymbol{r}(t_i) - \boldsymbol{r}(t_{i-1})\|.$$

If there exists an $M > 0$ such that $\|\Pi(P)\| \leq M$ for any partition P of $[a, b]$, then we say the curve is said to be **rectifiable** and its arc length, denoted by $\Lambda(a, b)$, is defined as

$$\Lambda(a, b) := \sup\{\|\Pi(P)\| : P \text{ is a partition of } [a, b]\}.$$

If there's no such M, then the curve is said to be **non-rectifiable**.

Theorem 11.2.12 For a curve in $\mathbb{R}^2$ or $\mathbb{R}^3$ described by $\boldsymbol{r}(t)$ for $t \in [a, b]$. If $\boldsymbol{v}(t) := \boldsymbol{r}'(t)$ is continuous on $[a, b]$, then the curve is rectifiable and its arc length satisfies

$$\Lambda(a, b) \leq \int_a^b \|\boldsymbol{v}(t)\| \mathrm{d}t.$$

Proof. Suppose $P = \{t_0, t_1, \ldots, t_n\}$ is a partition of $[a, b]$. Then

$$\|\Pi(P)\| = \sum_{i=1}^{n} \|\boldsymbol{r}(t_i) - \boldsymbol{r}(t_{i-1})\| = \sum_{i=1}^{n} \left\| \int_{t_{i-1}}^{t_i} \boldsymbol{v}(t) \mathrm{d}t \right\| \leq \sum_{i=1}^{n} \int_{t_{i-1}}^{t_i} \|\boldsymbol{v}(t)\| \mathrm{d}t = \int_a^b \|\boldsymbol{v}(t)\| \mathrm{d}t.$$

Since $\Lambda(a, b)$ is the least upper bound, we have $\Lambda(a, b) \leq \displaystyle\int_a^b \|\boldsymbol{v}(t)\| \mathrm{d}t.$ □

Theorem 11.2.13 For a rectifiable curve in $\mathbb{R}^2$ or $\mathbb{R}^3$ described by $\boldsymbol{r}(t)$ for $t \in [a, b]$. If $a < c < b$ and C_1 and C_2 are the curves corresponding to $\boldsymbol{r}(t)$ on $[a, c]$ and $[c, b]$, respectively, then C_1 and C_2 are both rectifiable and $\Lambda(a, b) = \Lambda(a, c) + \Lambda(c, b)$.

Proof. Suppose P_1 and P_2 are partitions of $[a, c]$ and $[c, b]$, respectively. Let $P = P_1 \cup P_2$ be the joint partition. Then $\|\Pi(P_1)\| + \|\Pi(P_2)\| = \|\Pi(P)\| \leq \Lambda(a, b)$. Thus $\|\Pi(P_1)\|$ and $\|\Pi(P_2)\|$ are both bounded above by $\Lambda(a, b)$ for any partitions P_1 and P_2. Thus C_1 and C_2 are both rectifiable.

- For a fixed P_2, $\|\Pi(P_1)\| \leq \Lambda(a, b) - \|\Pi(P_2)\|$ holds for any partition P_1 of $[a, c]$ and there by

$$\Lambda(a, c) \leq \Lambda(a, b) - \|\Pi(P_2)\| \implies \|\Pi(P_2)\| \leq \Lambda(a, b) - \Lambda(a, c).$$

- since the above inequality holds for any partition P_2 of $[c, b]$, we have $\Lambda(a, b) \geq \Lambda(a, c) + \Lambda(c, b)$.

We still need to show $\Lambda(a, b) \leq \Lambda(a, c) + \Lambda(c, b)$. Suppose P is a partition of $[a, b]$. Adding in c into P, we obtain a partition P_1 of $[a, c]$ and a partition P_2 of $[c, b]$ such that

$$\|\Pi(P)\| \leq \|\Pi(P_1)\| + \|\Pi(P_2)\| \leq \Lambda(a, c) + \Lambda(c, b).$$

Thus $\Lambda(a,b) \leq \Lambda(a,c) + \Lambda(c,b)$. Combining the two inequalities, we have $\Lambda(a,b) = \Lambda(a,c) + \Lambda(c,b)$. □

Definition 11.2.14 Suppose a curve in $\mathbb{R}^2$ or $\mathbb{R}^3$ is described by $\boldsymbol{r}(t)$ for $t \in [a,b]$. We define the **arc-length function** s as

$$s(t) := \begin{cases} 0, & t = a; \\ \Lambda(a,t), & t \in (a,b]. \end{cases}$$

Theorem 11.2.15 For any rectifiable curve, the arc-length function s is monotonically increasing on $[a,b]$.

Theorem 11.2.16 Suppose a rectifiable curve described by $\boldsymbol{r}(t)$ for $t \in [a,b]$. Let s denote the arc-length function. If the speed $v(t) = \|\boldsymbol{r}'(t)\|$ is continuous on $[a,b]$, then the derivative $s'(t)$ exists and is given by $s'(t) = v(t)$ for any $t \in [a,b]$.

Proof. Suppose $h > 0$, then

$$\frac{s(t+h)-s(t)}{h} \leq \frac{1}{h}\int_t^{t+h} v(u)\mathrm{d}u \to v(t) \quad \text{as } h \to 0^+.$$

We also have

$$\frac{s(t+h)-s(t)}{h} \geq \left\|\frac{\boldsymbol{r}(t+h)-\boldsymbol{r}(t)}{h}\right\| \to v(t) \quad \text{as } h \to 0^+.$$

By the squeeze theorem, we know $\lim\limits_{h\to 0^+} \dfrac{s(t+h)-s(t)}{h} = v(t)$ and thus $s'(t) = v(t)$. The case where $h < 0$ can be proved similarly. □

Remark 11.2.17 Note

$$s(t_2) - s(t_1) = \int_{t_1}^{t_2} v(t)\mathrm{d}t,$$

provided that $v(t)$ is continuous on $[a,b]$. Particularly,

$$\Lambda(a,b) = s(b) - s(a) = \int_a^b v(t)\mathrm{d}t.$$

Example 11.2.18 Consider the curve in $\mathbb{R}^2$ given by $\boldsymbol{r}(t) = (t, f(t))$ for $t \in [a,b]$ where f is a scalar-valued function differentiable on $[a,b]$. Then the arc-length function is given by

$$s(t) = \int_a^t v(u)\mathrm{d}u = \int_a^t \sqrt{1+(f'(u))^2}\mathrm{d}u.$$

Definition 11.2.19 Let $\boldsymbol{T}$ denote the unit tangent vector and s denote the arc-length function of a rectifiable curve described by $\boldsymbol{r}(t)$ for $t \in [a,b]$. We call $\dfrac{\mathrm{d}\boldsymbol{T}}{\mathrm{d}s}$ the **curvature vector** of the curve provided it's well-defined. Note

$$\frac{\mathrm{d}\boldsymbol{T}}{\mathrm{d}s} = \frac{\mathrm{d}\boldsymbol{T}}{\mathrm{d}t}\cdot\frac{\mathrm{d}t}{\mathrm{d}s} = \frac{\boldsymbol{T}'(t)}{v(t)} = \frac{\|\boldsymbol{T}'(t)\|\,\boldsymbol{N}(t)}{v(t)}.$$

This means that the curvature vector is parallel to the principal normal vector $\boldsymbol{N}$. We call

$$\kappa(t) := \left\| \frac{\mathrm{d}\boldsymbol{T}}{\mathrm{d}s} \right\| = \frac{\|\boldsymbol{T}'(t)\|}{v(t)}$$

the **curvature number** of the curve at point $\boldsymbol{r}(t)$.

Definition 11.2.20 When $\kappa(t) \neq 0$, we call $\rho(t) := \dfrac{1}{\kappa(t)}$ the **radius of curvature**. Then then circle in the osculating plane with radius $\rho(t)$ and centered at $\boldsymbol{r}(t) + \dfrac{1}{\kappa(t)}\boldsymbol{N}(t)$ is called the **osculating circle**. Note the osculating circle is tangent to the curve at point $\boldsymbol{r}(t)$ and has the same curvature as the curve at point $\boldsymbol{r}(t)$.

Theorem 11.2.21 For motion along a curve in $\mathbb{R}^2$ or $\mathbb{R}^3$ with velocity $\boldsymbol{v}(t)$, speed $v(t)$, acceleration $\boldsymbol{a}(t)$ and curvature κ. Then we have

$$\boldsymbol{a}(t) = v'(t)\boldsymbol{T}(t) + \kappa(t)v^2(t)\boldsymbol{N}(t).$$

and

$$\kappa(t) = \frac{\|\boldsymbol{a}(t) \times \boldsymbol{v}(t)\|}{v^3(t)}$$

3 Curves in Polar Coordinates

Sometimes it's more natural to use polar coordinates to describe a curve in $\mathbb{R}^2$:

$$\boldsymbol{r} = (r\cos\theta, r\sin\theta) = r(\cos\theta, \sin\theta).$$

Definition 11.3.1 We define the unit radial vector $\boldsymbol{e}_r$ and the tangential vector $\boldsymbol{e}_\theta$ as

$$\boldsymbol{e}_r = (\cos\theta, \sin\theta) \quad \text{and} \quad \boldsymbol{e}_\theta = \frac{\mathrm{d}\boldsymbol{e}_r}{\mathrm{d}\theta} = (-\sin\theta, \cos\theta).$$

Note $\|\boldsymbol{e}_r\| = \|\boldsymbol{e}_\theta\| = 1$ and $\boldsymbol{e}_r \cdot \boldsymbol{e}_\theta = 0$.

Example 11.3.2 Let $\boldsymbol{r}(t) = f(t), \theta(t) = g(t)$ for $t \in [a, b]$. Then

$$\boldsymbol{r}(t) = r\boldsymbol{e}_r = f(t)\boldsymbol{e}_r + 0 \cdot \boldsymbol{e}_\theta.$$

Thus the velocity $\boldsymbol{v}(t)$ is given by

$$\boldsymbol{v}(t) = \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \frac{\mathrm{d}r}{\mathrm{d}t}\boldsymbol{e}_r + r\frac{\mathrm{d}\boldsymbol{e}_r}{\mathrm{d}t} = \frac{\mathrm{d}r}{\mathrm{d}t}\boldsymbol{e}_r + r\frac{\mathrm{d}\boldsymbol{e}_r}{\mathrm{d}\theta}\frac{\mathrm{d}\theta}{\mathrm{d}t} = f'(t)\boldsymbol{e}_r + [f(t)g'(t)]\,\boldsymbol{e}_\theta.$$

We call $f'(t)$ the **radial component** of the velocity $\boldsymbol{v}(t)$ and $f(t)g'(t)$ the **transverse component** of the velocity $\boldsymbol{v}(t)$. Note the speed $v(t)$ is given by

$$v(t) = \sqrt{\boldsymbol{v}(t) \cdot \boldsymbol{v}(t)} = \sqrt{(f'(t))^2 + (f(t)g'(t))^2}$$

and the acceleration $\boldsymbol{a}(t)$ is given by

$$\boldsymbol{a}(t) = \left[\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} - r\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2\right]\boldsymbol{e}_r + \left[r\frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + 2\frac{\mathrm{d}r}{\mathrm{d}t}\frac{\mathrm{d}\theta}{\mathrm{d}t}\right]\boldsymbol{e}_\theta = \left[\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} - r\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2\right]\boldsymbol{e}_r + \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)\boldsymbol{e}_\theta.$$

We say the acceleration $\boldsymbol{a}(t)$ is **radical** if the transverse component is always zero, i.e., $r^2\dfrac{\mathrm{d}\theta}{\mathrm{d}t}$ is a constant.

Chapter 12 Differential Calculus of Scalar and Vector Fields

1 Basic Concepts

Definition 12.1.1 A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is called

- a **real-valued function** $f(t)$ if $n = m = 1$;
- a **vector-valued function** (of a real variable) $\boldsymbol{f}(t)$ if $n = 1$ and $m \geq 2$;
- a **scalar field** $f(\boldsymbol{x})$ if $n \geq 2$ and $m = 1$;
- a **vector field** $\boldsymbol{f}(\boldsymbol{x})$ if $n \geq 2$ and $m \geq 2$. We often write it as $\boldsymbol{f} = (f_1(\boldsymbol{x}), \cdots, f_m(\boldsymbol{x}))$ where $f_i(\boldsymbol{x})$ denotes the i-th component of $\boldsymbol{f}$.

Definition 12.1.2 Let $\boldsymbol{a} \in \mathbb{R}^n$ and $r > 0$. The set of all points $\boldsymbol{x} \in \mathbb{R}^n$ such that $\|\boldsymbol{x} - \boldsymbol{a}\| < r$ is called the **open** (n-)**ball** of radius r centered at $\boldsymbol{a}$ and is denoted by $B(\boldsymbol{a}; r)$ or simply $B(\boldsymbol{a})$.

Definition 12.1.3 Let $S \subseteq \mathbb{R}^n$ and $\boldsymbol{a} \in S$.

- We call $\boldsymbol{a}$ an **interior point** of S if there exists an $r > 0$ such that $B(\boldsymbol{a}; r) \subseteq S$.
- The set of all interior points of S is called the **interior** of S and is denoted by $\mathrm{int}(S)$.
- The set S is called **open** if every point of S is an interior point, i.e. $\mathrm{int}(S) = S$.
- A point $\boldsymbol{b} \in \mathbb{R}^n$ is said to be **exterior** to S if there is an open n-ball $B(\boldsymbol{b})$ such that $B(\boldsymbol{b}) \cap S = \varnothing$.
- The set of all points in $\mathbb{R}^n$ exterior to S is called the **exterior** of S and is denoted by $\mathrm{ext}(S)$.
- A point $\boldsymbol{c} \in \mathbb{R}^n$ is called a **boundary point** of S if c is $\boldsymbol{c} \notin \mathrm{int}(S)$ and $\boldsymbol{c} \notin \mathrm{ext}(S)$, i.e. for any open n-ball $B(\boldsymbol{c})$, we have $B(\boldsymbol{c}) \nsubseteq S$ and $B(\boldsymbol{c}) \cap S \neq \varnothing$.
- The set of all boundary points of S is called the **boundary** of S and is denoted by ∂S.
- If $\partial S \subseteq S$, then S is called a **closed** set.

Example 12.1.4 (0) $\mathbb{R}^n$ is open and closed. The empty set $\varnothing$ is also open and closed.

(1) An open n-ball is open. Note

$$\mathrm{ext}(B(\boldsymbol{a}; r)) = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x} - \boldsymbol{a}\| > r\}, \quad \partial B(\boldsymbol{a}; r) = \{\boldsymbol{x} \in \mathbb{R}^n : \|\boldsymbol{x} - \boldsymbol{a}\| = r\}.$$

(2) Let $A_1 = (a, b)$ and $A_2 = (c, d)$ where $a < b$, $c < d$ and $A_1, A_2 \subseteq \mathbb{R}$. Then $A_1 \times A_2 = \{(x, y) : x \in A_1, y \in A_2\}$ is an open set in $\mathbb{R}^2$.

Proposition 12.1.5 Let $S \subseteq \mathbb{R}^n$. Then $\operatorname{int}(S), \operatorname{ext}(S)$ and ∂S are mutually disjoint and $\mathbb{R}^n = \operatorname{int}(S) \cup \operatorname{ext}(S) \cup \partial S$.

Definition 12.1.6 Consider $S \subseteq \mathbb{R}^n$ and function $\boldsymbol{f} : S \to \mathbb{R}^m$.

- If $\boldsymbol{a} \in \mathbb{R}^n$ and $\boldsymbol{b} \in \mathbb{R}^m$, we write

$$\lim_{\boldsymbol{x}\to\boldsymbol{a}} \boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{b} \quad \text{or} \quad \boldsymbol{f}(\boldsymbol{x}) \to \boldsymbol{b} \text{ as } \boldsymbol{x} \to \boldsymbol{a}$$

  to mean that

$$\lim_{\|\boldsymbol{x}-\boldsymbol{a}\|\to 0} \|\boldsymbol{f}(\boldsymbol{x}) - \boldsymbol{b}\| = 0.$$

- $\boldsymbol{f}$ is said to be **continuous** at $\boldsymbol{a}$ if f is defined at $\boldsymbol{a}$ and $\lim_{\boldsymbol{x}\to\boldsymbol{a}} \boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{f}(\boldsymbol{a})$.
- $\boldsymbol{f}$ is said to be **continuous** on S if $\boldsymbol{f}$ is continuous at every point of S.

Theorem 12.1.7 If $\lim_{\boldsymbol{x}\to\boldsymbol{a}} \boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{b}$ and $\lim_{\boldsymbol{x}\to\boldsymbol{a}} \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{c}$, then

- For $\alpha, \beta \in \mathbb{R}$, $\lim_{\boldsymbol{x}\to\boldsymbol{a}} (\alpha\boldsymbol{f} + \beta\boldsymbol{g})(\boldsymbol{x}) = \alpha\boldsymbol{b} + \beta\boldsymbol{c}$;
- $\lim_{\boldsymbol{x}\to\boldsymbol{a}} (\boldsymbol{f} \cdot \boldsymbol{g})(\boldsymbol{x}) = \boldsymbol{b} \cdot \boldsymbol{c}$;
- $\lim_{\boldsymbol{x}\to\boldsymbol{a}} \|\boldsymbol{f}(\boldsymbol{x})\| = \|\boldsymbol{b}\|$.

Theorem 12.1.8 Suppose $\boldsymbol{f}$ and $\boldsymbol{g}$ are functions such that the composition $\boldsymbol{f} \circ \boldsymbol{g}$ is well-defined at $\boldsymbol{a}$. If $\boldsymbol{g}$ is continuous at $\boldsymbol{a}$ and $\boldsymbol{f}$ is continuous at $\boldsymbol{g}(\boldsymbol{a})$, then $\boldsymbol{f} \circ \boldsymbol{g}$ is continuous at $\boldsymbol{a}$.

Example 12.1.9

1. Suppose $\boldsymbol{f} = (f_1, \ldots, f_m) : \mathbb{R}^n \to \mathbb{R}^m$ is a function. Then $\boldsymbol{f}$ is continuous at $\boldsymbol{a}$ if and only if each component function f_i is continuous at $\boldsymbol{a}$ for $i = 1, \ldots, m$.
2. $\boldsymbol{f}(\boldsymbol{x}) = \boldsymbol{x}$ is called the **identity function**.
3. Polynomials in n variable

$$P(\boldsymbol{x}) = \sum_{i_1,\ldots,i_n} c_{i_1,\ldots,i_n} x_1^{i_1} \cdots x_n^{i_n}$$

   are continuous for all $\boldsymbol{x}$.
4. Scalar fields of the form $f(\boldsymbol{x}) = \dfrac{P(\boldsymbol{x})}{Q(\boldsymbol{x})}$ where P and Q are polynomials and $Q(\boldsymbol{x}) \neq 0$ are continuous at any $\boldsymbol{a}$ with $Q(\boldsymbol{a}) \neq 0$.
5. The function

$$f(x, y) = \begin{cases} 0, & (x, y) = (0, 0); \\ \dfrac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0) \end{cases}$$

   is not continuous at $(0, 0)$ since $\lim_{x\to 0} f(x, x) = \dfrac{1}{2}$ but $f(0, 0) = 0$. This indicates that continuity needs to be checked along all possible paths approaching the point, not just along some specific paths.

Theorem 12.1.10 If $\lim_{(x,y)\to(a,b)} f(x,y) = L$, and if the one-dimensional limits

$$\lim_{x\to a} f(x,y) \qquad \text{and} \qquad \lim_{y\to b} f(x,y)$$

both exist, then

$$\lim_{x\to a}\left[\lim_{y\to b} f(x,y)\right] = \lim_{y\to b}\left[\lim_{x\to a} f(x,y)\right] = L.$$

Remark 12.1.11 The two limits in the above equation are called **iterated limits**; this theorem shows that the existence of the two-dimensional limit and of the two one-dimensional limits implies the existence and equality of the two iterated limits. However, the converse is not true, which will be illustrated by the following example.

Example 12.1.12 Let

$$f(x,y) = \frac{x^2y^2}{x^2y^2 + (x-y)^2} \qquad \text{whenever} \qquad x^2y^2 + (x-y)^2 \neq 0.$$

Then

$$\lim_{x\to 0}\left[\lim_{y\to 0} f(x,y)\right] = \lim_{y\to 0}\left[\lim_{x\to 0} f(x,y)\right] = 0.$$

However, along the path $y = x$, we have

$$\lim_{x\to 0} f(x,x) = \lim_{x\to 0} \frac{x^4}{x^4+0} = 1 \neq 0.$$

2 The Derivative of a Scalar Field

Definition 12.2.1 Given a scalar field $f : S \to \mathbb{R}$ where $S \subseteq \mathbb{R}^n$. Let $\boldsymbol{a} \in \operatorname{int} S$ and $\boldsymbol{y} \in \mathbb{R}^n$.

- The **derivative of f at $\boldsymbol{a}$ with respect to $\boldsymbol{y}$** is denoted by $f'(\boldsymbol{a};\boldsymbol{y})$ and is defined as

$$f'(\boldsymbol{a};\boldsymbol{y}) := \lim_{h\to 0} \frac{f(\boldsymbol{a}+h\boldsymbol{y}) - f(\boldsymbol{a})}{h},$$

  provided the limit exists.

- In particular, if $\boldsymbol{y}$ is a unit vector, we also call $f'(\boldsymbol{a};\boldsymbol{y})$ a **directional derivative** of f at $\boldsymbol{a}$ in the direction $\boldsymbol{y}$.

- In particular, if $\boldsymbol{y} = \boldsymbol{e}_k$ is the k-th standard basis vector, we call $f'(\boldsymbol{a};\boldsymbol{e}_k)$ the **partial derivative** of f at $\boldsymbol{a}$ with respect to x_k and denote it by $\dfrac{\partial f}{\partial x_k}(\boldsymbol{a})$ or $D_k f(\boldsymbol{a})$.

Remark 12.2.2 The partial derivatives themselves are also scalar fields, so we can also calculate the partial derivatives of the partial derivatives, which are called the **second-order partial derivatives**.

Notation 12.2.3 In Leibniz's notation, we adopt the following convention for the second-order partial derivatives:

$$\frac{\partial^2 f}{\partial x^2} := \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right), \quad \frac{\partial^2 f}{\partial x \partial y} := \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right), \quad \frac{\partial^2 f}{\partial y \partial x} := \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right).$$

Example 12.2.4 The function

$$f(x,y) = \begin{cases} 0, & (x,y) = (0,0); \\ \dfrac{x^2y}{x^4+y^2}, & (x,y) \neq (0,0) \end{cases}$$

is not continuous at $(0,0)$ since approaching $(0,0)$ along the path $y = x^2$ gives

$$\lim_{x\to 0} f(x,x^2) = \lim_{x\to 0} \frac{x^4}{2x^4} = \frac{1}{2} \neq f(0,0) = 0.$$

However, the derivative $f'(\mathbf{0};\boldsymbol{y})$ exists for any $\boldsymbol{y} = (y_1,y_2)$ since

$$f'(\mathbf{0};\boldsymbol{y}) = \lim_{h\to 0} \frac{f(hy_1,hy_2) - f(0,0)}{h} = \lim_{h\to 0} \frac{h^3y_1^2y_2}{h^4y_1^4 + h^2y_2^2} = \lim_{h\to 0} \frac{y_1^2y_2}{h^2y_1^4 + y_2^2}.$$

If $y_2 \neq 0$, then

$$f'(\mathbf{0};\boldsymbol{y}) = \frac{y_1^2}{y_2}.$$

If $y_2 = 0$, then $f(hy_1, 0) = 0$, so

$$f'(\mathbf{0};\boldsymbol{y}) = 0.$$

This indicates that the existence of directional derivatives does not necessarily imply the continuity of the function. However, if we still want differentiability to imply continuity like in the case of $f : \mathbb{R} \to \mathbb{R}$, we need a stronger definition of differentiability.

Theorem 12.2.5 For a fixed $\boldsymbol{a}, \boldsymbol{y} \in \mathbb{R}^n$, let $g(t) := f(\boldsymbol{a} + t\boldsymbol{y})$ for $t \in \mathbb{R}$. Then $g'(t)$ exists if and only if $f'(\boldsymbol{a} + t\boldsymbol{y};\boldsymbol{y})$ exists, in which case $g'(t) = f'(\boldsymbol{a} + t\boldsymbol{y};\boldsymbol{y})$. Particularly, $f'(\boldsymbol{a};\boldsymbol{y}) = g'(0)$.

Theorem 12.2.6 Suppose $f'(\boldsymbol{a} + t\boldsymbol{y};\boldsymbol{y})$ exists for any $t \in [0,1]$. Then $f(\boldsymbol{a} + \boldsymbol{y}) - f(\boldsymbol{a}) = f'(\boldsymbol{a} + \theta\boldsymbol{y};\boldsymbol{y})$ for some $\theta \in (0,1)$.

Definition 12.2.7 Consider a scalar field $f : S \to \mathbb{R}$ where $S \subseteq \mathbb{R}^n$ and $\boldsymbol{a} \in \operatorname{int} S$. We say f is **differentiable** at $\boldsymbol{a}$ if there exists a linear map $T_{\boldsymbol{a}} : \mathbb{R}^n \to \mathbb{R}$ and a function $E : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ such that

$$f(\boldsymbol{a} + \boldsymbol{v}) = f(\boldsymbol{a}) + T_{\boldsymbol{a}}(\boldsymbol{v}) + \|v\|E(\boldsymbol{a},\boldsymbol{v})$$

for any $\boldsymbol{v}$ with $\boldsymbol{a} + \boldsymbol{v} \in B(\boldsymbol{a};r) \subseteq S$ with $r > 0$, where $E(\boldsymbol{a},\boldsymbol{v}) \to 0$ as $\boldsymbol{v} \to \mathbf{0}$.
The linear map $T_{\boldsymbol{a}}$ is called the **total derivative** of f at $\boldsymbol{a}$. The above equation is called the **first-order Taylor expansion** of $f(\boldsymbol{a} + \boldsymbol{v})$, whose error term is $\|v\|E(\boldsymbol{a},\boldsymbol{v}) = o(\|\boldsymbol{v}\|)$.

Remark 12.2.8 Note we cannot define derivative through quotient like in the case of $f : \mathbb{R} \to \mathbb{R}$ since we do not have a notion of division in $\mathbb{R}^n$. Instead, inspired by Taylor's expansion, we define the derivative through linear approximation, which is more general and can be applied to functions between general normed vector spaces.

Theorem 12.2.9 Suppose f is differentiable at $\boldsymbol{a}$ with total derivative $T_{\boldsymbol{a}}$. Then $f'(\boldsymbol{a};\boldsymbol{y})$ exists for any $\boldsymbol{y} \in \mathbb{R}^n$ and $T_{\boldsymbol{a}}(\boldsymbol{y}) = f'(\boldsymbol{a};\boldsymbol{y})$. Furthermore,

$$f'(\boldsymbol{a};\boldsymbol{y}) = \sum_{k=1}^{n} y_k \frac{\partial f}{\partial x_k}(\boldsymbol{a}).$$

Notation 12.2.10 The vector $\left(\frac{\partial f}{\partial x_1}(\boldsymbol{a}), \dots, \frac{\partial f}{\partial x_n}(\boldsymbol{a})\right)$ is called the **gradient** of f at $\boldsymbol{a}$ and is denoted by $\nabla f(\boldsymbol{a})$. Then we can write

$$f'(\boldsymbol{a};\boldsymbol{y}) = \nabla f(\boldsymbol{a}) \cdot \boldsymbol{y}.$$

With this notation, we can also write the first-order Taylor expansion as

$$f(\boldsymbol{a}+\boldsymbol{v}) = f(\boldsymbol{a}) + \nabla f(\boldsymbol{a}) \cdot \boldsymbol{v} + o(\|\boldsymbol{v}\|).$$

This is quite similar to the first-order Taylor expansion for $f : \mathbb{R} \to \mathbb{R}$ given by

$$f(a+h) = f(a) + f'(a)h + o(h).$$

Theorem 12.2.11 If a scalar field is differentiable at $\boldsymbol{a}$, then it is continuous at $\boldsymbol{a}$.

Theorem 12.2.12 If all the partial derivatives $D_1 f, \dots, D_n f$ exist in some open $n-$ball $B(\boldsymbol{a})$ and continuous at $\boldsymbol{a}$, then f is differentiable at $\boldsymbol{a}$, in which case we also say f is **continuously differentiable** at $\boldsymbol{a}$.

Theorem 12.2.13 Suppose f is a scalar field on an open set $S \subseteq \mathbb{R}^n$ and $\boldsymbol{r}$ is a vector-valued function mapping $J \subseteq \mathbb{R}$ to S. Define the composite function $g = f \circ \boldsymbol{r}$ on J as $g(t) = f(\boldsymbol{r}(t))$ for every $t \in J$. If f is differentiable at $\boldsymbol{r}(t)$ and $\boldsymbol{r}$ is differentiable at t, then g is differentiable at t and

$$g'(t) = \nabla f(\boldsymbol{r}(t)) \cdot \boldsymbol{r}'(t).$$

Proof. Suppose $\boldsymbol{r}'(t)$ exists for $t \in J$ and let $\boldsymbol{a} = \boldsymbol{r}(t)$. Since $a \in S$ and S is open, there is an open n-ball $B(\boldsymbol{a}) \subseteq S$. Note $\boldsymbol{r}$ is continuous at t, so we can consider $h \neq 0$ but with $\|h\|$ small enough so that $\boldsymbol{r}(t+h) \in B(\boldsymbol{a})$. Let $\boldsymbol{y} = \boldsymbol{r}(t+h) - \boldsymbol{r}(t)$, then

$$g(t+h) - g(t) = f(\boldsymbol{r}(t+h)) - f(\boldsymbol{r}(t)) = f(\boldsymbol{a}+\boldsymbol{y}) - f(\boldsymbol{a}) = \nabla f(\boldsymbol{a}) \cdot \boldsymbol{y} + o(\|\boldsymbol{y}\|).$$

Then

$$\frac{g(t+h) - g(t)}{h} = \nabla f(\boldsymbol{a}) \cdot \frac{\boldsymbol{y}}{h} + o\left(\frac{\|\boldsymbol{y}\|}{|h|}\right) = \nabla f(\boldsymbol{a}) \cdot \frac{\boldsymbol{r}(t+h) - \boldsymbol{r}(t)}{h} + o\left(\frac{\|\boldsymbol{r}(t+h) - \boldsymbol{r}(t)\|}{|h|}\right).$$

Since $\boldsymbol{r}$ is differentiable at t, we have $\lim_{h\to 0} \frac{\boldsymbol{r}(t+h) - \boldsymbol{r}(t)}{h} = \boldsymbol{r}'(t)$ and thus $\lim_{h\to 0} o\left(\frac{\|\boldsymbol{r}(t+h) - \boldsymbol{r}(t)\|}{|h|}\right) = 0$. Thus

$$g'(t) = \lim_{h\to 0} \frac{g(t+h) - g(t)}{h} = \nabla f(\boldsymbol{a}) \cdot \boldsymbol{r}'(t).$$

□

Example 12.2.14 If $\boldsymbol{r}$ describes a curve in $\mathbb{R}^2$ or $\mathbb{R}^3$ and $f : \mathbb{R}^2 \to \mathbb{R}$, $g = f \circ \boldsymbol{r}$, then

$$g'(t) = \nabla f(\boldsymbol{r}(t)) \cdot \boldsymbol{r}'(t) = \|\boldsymbol{r}'(t)\| \left(\nabla f(\boldsymbol{r}(t)) \cdot T(t)\right)$$

where T is the unit tangent vector of the curve. Here the term $\nabla f(\boldsymbol{r}(t)) \cdot T(t)$ is called the **directional derivative** of f along the curve provided it's well-defined.

3 The Derivatives of a Vector Field

Definition 12.3.1 Given a vector field $\boldsymbol{f} : S \to \mathbb{R}^m$ where $S \subseteq \mathbb{R}^n$. Let $\boldsymbol{a} \in \operatorname{int} S$ and $\boldsymbol{y} \in \mathbb{R}^n$.

- The **derivative of $\boldsymbol{f}$ at $\boldsymbol{a}$ with respect to $\boldsymbol{y}$** is denoted by $\boldsymbol{f}'(\boldsymbol{a};\boldsymbol{y})$ and is defined as

$$\boldsymbol{f}'(\boldsymbol{a};\boldsymbol{y}) := \lim_{h\to 0} \frac{\boldsymbol{f}(\boldsymbol{a}+h\boldsymbol{y}) - \boldsymbol{f}(\boldsymbol{a})}{h} \in \mathbb{R}^m,$$

  provided the limit exists. Let f_k denote the k-th component of $\boldsymbol{f}$, then $\boldsymbol{f}'(\boldsymbol{a};\boldsymbol{y})$ exists if and only if $f_k'(\boldsymbol{a};\boldsymbol{y})$ exists for $k = 1, \ldots, m$, in which case

$$\boldsymbol{f}'(\boldsymbol{a};\boldsymbol{y}) = (f_1'(\boldsymbol{a};\boldsymbol{y}), \ldots, f_m'(\boldsymbol{a};\boldsymbol{y})) = \sum_{k=1}^{m} f_k'(\boldsymbol{a};\boldsymbol{y})\boldsymbol{e}_k.$$

- We say that $\boldsymbol{f}$ is **differentiable** at $\boldsymbol{a}$ if there exists a linear map $\boldsymbol{T_a} : \mathbb{R}^n \to \mathbb{R}^m$ and a function $\boldsymbol{E} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\boldsymbol{f}(\boldsymbol{a}+\boldsymbol{v}) = \boldsymbol{f}(\boldsymbol{a}) + \boldsymbol{T_a}(\boldsymbol{v}) + \|v\|\boldsymbol{E}(\boldsymbol{a},\boldsymbol{v})$$

  for any $\boldsymbol{v}$ with $\boldsymbol{a}+\boldsymbol{v} \in B(\boldsymbol{a};r) \subseteq S$ with $r > 0$, where $\boldsymbol{E}(\boldsymbol{a},\boldsymbol{v}) \to \boldsymbol{0}$ as $\boldsymbol{v} \to \boldsymbol{0}$. The linear map $\boldsymbol{T_a}$ uniquely exists and is called the **total derivative** of $\boldsymbol{f}$ at $\boldsymbol{a}$.

Theorem 12.3.2 If $\boldsymbol{f}$ is differentiable at $\boldsymbol{a}$ with total derivative $\boldsymbol{T_a}$, then $\boldsymbol{f}'(\boldsymbol{a};\boldsymbol{y})$ exists for any $\boldsymbol{y} \in \mathbb{R}^n$ and $\boldsymbol{T_a}(\boldsymbol{y}) = \boldsymbol{f}'(\boldsymbol{a};\boldsymbol{y})$. Furthermore, if we write $\boldsymbol{f} = (f_1, \ldots, f_m)$ and $\boldsymbol{y} = (y_1, \ldots, y_n)$, then

$$\boldsymbol{T_a}(\boldsymbol{y}) = \boldsymbol{f}'(\boldsymbol{a};\boldsymbol{y}) = (\nabla f_1(\boldsymbol{a}) \cdot \boldsymbol{y}, \ldots, \nabla f_m(\boldsymbol{a}) \cdot \boldsymbol{y}) = \sum_{k=1}^{n} (\nabla f_k(\boldsymbol{a}) \cdot \boldsymbol{y})\, \boldsymbol{e}_k.$$

Remark 12.3.3 Note we can also write $\boldsymbol{T_a}(\boldsymbol{y})$ as a matrix-vector product as follows:

$$\boldsymbol{T_a}(\boldsymbol{y}) = \begin{pmatrix} \nabla f_1(\boldsymbol{a}) \\ \vdots \\ \nabla f_m(\boldsymbol{a}) \end{pmatrix} \boldsymbol{y} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(\boldsymbol{a}) & \frac{\partial f_1}{\partial x_2}(\boldsymbol{a}) & \cdots & \frac{\partial f_1}{\partial x_n}(\boldsymbol{a}) \\ \frac{\partial f_2}{\partial x_1}(\boldsymbol{a}) & \frac{\partial f_2}{\partial x_2}(\boldsymbol{a}) & \cdots & \frac{\partial f_2}{\partial x_n}(\boldsymbol{a}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(\boldsymbol{a}) & \frac{\partial f_m}{\partial x_2}(\boldsymbol{a}) & \cdots & \frac{\partial f_m}{\partial x_n}(\boldsymbol{a}) \end{pmatrix} \boldsymbol{y}.$$

This $m \times n$ matrix is called the **Jacobian matrix** of $\boldsymbol{f}$ at $\boldsymbol{a}$ and is denoted by $\mathbf{D}_{\boldsymbol{f}(\boldsymbol{a})}$. The kj-th entry is actually

$$\left[\mathbf{D}_{\boldsymbol{f}(\boldsymbol{a})}\right]_{kj} = \frac{\partial f_k}{\partial x_j}(\boldsymbol{a}) = D_j f_k(\boldsymbol{a}).$$

We often simply denote the total derivative $\boldsymbol{T_a}$ by $\boldsymbol{f}'(\boldsymbol{a})$, which is a linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$. Also we know $\boldsymbol{f}'(\boldsymbol{a};\boldsymbol{y}) = \mathbf{D}_{\boldsymbol{f}(\boldsymbol{a})}\boldsymbol{y}$ for any $\boldsymbol{y} \in \mathbb{R}^n$. Thus $\mathbf{D}_{\boldsymbol{f}(\boldsymbol{a})}$ is the matrix representation of the linear map $\boldsymbol{f}'(\boldsymbol{a})$ with respect to the standard bases of $\mathbb{R}^n$ and $\mathbb{R}^m$. We also have

$$\boldsymbol{f}(\boldsymbol{a}+\boldsymbol{v}) = \boldsymbol{f}(\boldsymbol{a}) + \mathbf{D}_{\boldsymbol{f}(\boldsymbol{a})}\boldsymbol{v} + \|\boldsymbol{v}\|\, \boldsymbol{E}(\boldsymbol{a},\boldsymbol{v}),$$

where $\boldsymbol{E}(\boldsymbol{a},\boldsymbol{v}) \to \mathbf{0}$ as $\boldsymbol{v} \to \mathbf{0}$.

Theorem 12.3.4 If a vector field $\boldsymbol{f}$ is differentiable at $\boldsymbol{a}$, then it is continuous at $\boldsymbol{a}$.

Lemma 12.3.5 Suppose $\boldsymbol{g}$ is differentiable at $\boldsymbol{a}$. Then $\|\boldsymbol{g}(\boldsymbol{a})(\boldsymbol{v})\| \le M(\boldsymbol{a})\|\boldsymbol{v}\|$ where $M(\boldsymbol{a}) = \sum_{k=1}^{m} \|\nabla g_k(\boldsymbol{a})\|$

Proof.

$$\|\boldsymbol{g}(\boldsymbol{a})(\boldsymbol{v})\| = \left\|\sum_{k=1}^{m} (\nabla g_k(\boldsymbol{a}) \cdot \boldsymbol{v})\, \boldsymbol{e}_k\right\| \le \sum_{k=1}^{m} |\nabla g_k(\boldsymbol{a}) \cdot \boldsymbol{v}| \le \sum_{k=1}^{m} \|\nabla g_k(\boldsymbol{a})\|\|\boldsymbol{v}\|.$$

□

Theorem 12.3.6 Suppose $\boldsymbol{f}$ and $\boldsymbol{g}$ are vector fields such that the composition $\boldsymbol{h} = \boldsymbol{f} \circ \boldsymbol{g}$ is defined in an opne n-ball $B(\boldsymbol{a})$. If $\boldsymbol{g}$ is differentiable at $\boldsymbol{a}$ with total derivative $\boldsymbol{g}'(\boldsymbol{a})$ and $\boldsymbol{f}$ is differentiable at $\boldsymbol{b} = \boldsymbol{g}(\boldsymbol{a})$ with total derivative $\boldsymbol{f}'(\boldsymbol{b})$, then $\boldsymbol{h} = \boldsymbol{f} \circ \boldsymbol{g}$ is differentiable at $\boldsymbol{a}$ and

$$\boldsymbol{h}'(\boldsymbol{a}) = (\boldsymbol{f} \circ \boldsymbol{g})'(\boldsymbol{a}) = \boldsymbol{f}'(\boldsymbol{b}) \circ \boldsymbol{g}'(\boldsymbol{a}).$$

Proof. We shall consider $\boldsymbol{h}(\boldsymbol{a}+\boldsymbol{v})$ for $\boldsymbol{y}$ with small $\|\boldsymbol{y}\|$. Note $\boldsymbol{b} = \boldsymbol{g}(\boldsymbol{a})$, let $\boldsymbol{v} = \boldsymbol{g}(\boldsymbol{a}+\boldsymbol{y}) - \boldsymbol{g}(\boldsymbol{a}) = \boldsymbol{g}(\boldsymbol{a}+\boldsymbol{y}) - \boldsymbol{b}$, then $\boldsymbol{h}(\boldsymbol{a}+\boldsymbol{y}) - \boldsymbol{h}(\boldsymbol{a}) = \boldsymbol{f}(\boldsymbol{b}+\boldsymbol{v}) - \boldsymbol{f}(\boldsymbol{b})$. Since f is differentiable at $\boldsymbol{b} = \boldsymbol{g}(\boldsymbol{a})$, we have

$$\begin{aligned}
\boldsymbol{f}(\boldsymbol{b}+\boldsymbol{v}) - \boldsymbol{f}(\boldsymbol{b}) &= \boldsymbol{f}'(\boldsymbol{b})(\boldsymbol{v}) + \|\boldsymbol{v}\|\boldsymbol{E}_f(\boldsymbol{b},\boldsymbol{v}) \\
&= \boldsymbol{f}'(\boldsymbol{b})(\boldsymbol{g}'(\boldsymbol{a})(\boldsymbol{y}) + \boldsymbol{E}_g(\boldsymbol{a},\boldsymbol{y})) + \|\boldsymbol{v}\|\boldsymbol{E}_f(\boldsymbol{b},\boldsymbol{v}) \\
&= \boldsymbol{f}'(\boldsymbol{b}) \circ \boldsymbol{g}'(\boldsymbol{a})(\boldsymbol{y}) + \|\boldsymbol{y}\|\boldsymbol{f}'(\boldsymbol{b})(\boldsymbol{E}_g(\boldsymbol{a},\boldsymbol{y})) + \|\boldsymbol{v}\|\boldsymbol{E}_f(\boldsymbol{b},\boldsymbol{v}) \\
&= \boldsymbol{f}'(\boldsymbol{b}) \circ \boldsymbol{g}'(\boldsymbol{a})(\boldsymbol{y}) + \|\boldsymbol{y}\|\boldsymbol{E}(\boldsymbol{a},\boldsymbol{y}),
\end{aligned}$$

where

$$\boldsymbol{E}(\boldsymbol{a},\boldsymbol{y}) = \begin{cases} \boldsymbol{f}'(\boldsymbol{b})(\boldsymbol{E}_g(\boldsymbol{a},\boldsymbol{y})) + \dfrac{\|\boldsymbol{v}\|}{\|\boldsymbol{y}\|}\boldsymbol{E}_f(\boldsymbol{b},\boldsymbol{v}), & \boldsymbol{y} \ne \mathbf{0}; \\ \mathbf{0}, & \boldsymbol{y} = \mathbf{0}. \end{cases}$$

We still need to show that if $\boldsymbol{y} \to \mathbf{0}$ (with $\boldsymbol{y} \ne \mathbf{0}$), then $\boldsymbol{E}(\boldsymbol{a},\boldsymbol{y}) \to \mathbf{0}$. Note as $\boldsymbol{y} \to \mathbf{0}$, the first term $\boldsymbol{f}'(\boldsymbol{b})(\boldsymbol{E}_g(\boldsymbol{a},\boldsymbol{y})) \to \mathbf{0}$ since $\boldsymbol{E}_g(\boldsymbol{a},\boldsymbol{y}) \to \mathbf{0}$ due to the differentiability at $\boldsymbol{a}$ and $\boldsymbol{f}'(\boldsymbol{b})$ is a linear map and thus continuous. For the second term $\frac{\|\boldsymbol{v}\|}{\|\boldsymbol{y}\|}\boldsymbol{E}_f(\boldsymbol{b},\boldsymbol{v})$, we will show that the factor $\frac{\|\boldsymbol{v}\|}{\|\boldsymbol{y}\|}$ is bounded as $\boldsymbol{y}\to\boldsymbol{0}$. Note

$$\|\boldsymbol{v}\| = \|\boldsymbol{g}(\boldsymbol{a}+\boldsymbol{y})-\boldsymbol{g}(\boldsymbol{a})\| = \|\boldsymbol{g}'(\boldsymbol{a})(\boldsymbol{y})+\|\boldsymbol{y}\|\boldsymbol{E}_g(\boldsymbol{a},\boldsymbol{y})\| \le \|\boldsymbol{g}'(\boldsymbol{a})(\boldsymbol{y})\|+\|\boldsymbol{y}\|\,\|\boldsymbol{E}_g(\boldsymbol{a},\boldsymbol{y})\|\,.$$

By the above lemma, we have $\|\boldsymbol{g}'(\boldsymbol{a})(\boldsymbol{y})\| \le M(\boldsymbol{a})\|\boldsymbol{y}\|$ for some constant $M(\boldsymbol{a})$. Therefore,

$$\frac{\|\boldsymbol{v}\|}{\|\boldsymbol{y}\|} \le M(\boldsymbol{a})+\|\boldsymbol{E}_g(\boldsymbol{a},\boldsymbol{y})\|\,.$$

Since $\boldsymbol{E}_g(\boldsymbol{a},\boldsymbol{y})\to\boldsymbol{0}$ as $\boldsymbol{y}\to\boldsymbol{0}$, we know the factor is bounded as long as $\boldsymbol{y}$ is not so far away from $\boldsymbol{0}$. Thus $\boldsymbol{E}(\boldsymbol{a},\boldsymbol{y})\to\boldsymbol{0}$ as $\boldsymbol{y}\to\boldsymbol{0}$ and we have $\boldsymbol{E}(\boldsymbol{a},\boldsymbol{y})=o(\|\boldsymbol{y}\|)$. Therefore,

$$\boldsymbol{h}(\boldsymbol{a}+\boldsymbol{y})-\boldsymbol{h}(\boldsymbol{a}) = \boldsymbol{f}'(\boldsymbol{b})\circ\boldsymbol{g}'(\boldsymbol{a})(\boldsymbol{y})+\|\boldsymbol{y}\|\boldsymbol{E}(\boldsymbol{a},\boldsymbol{y}) = \boldsymbol{f}'(\boldsymbol{b})\circ\boldsymbol{g}'(\boldsymbol{a})(\boldsymbol{y})+o(\|\boldsymbol{y}\|),$$

which means $\boldsymbol{h}$ is differentiable at $\boldsymbol{a}$ with total derivative $\boldsymbol{h}'(\boldsymbol{a})=\boldsymbol{f}'(\boldsymbol{b})\circ\boldsymbol{g}'(\boldsymbol{a})$. □

Remark 12.3.7 Note the corresponding matrix form of a linear map composition $\boldsymbol{h}=\boldsymbol{f}\circ\boldsymbol{g}$ is given by the matrix product of the Jacobian matrices of $\boldsymbol{f}$ and $\boldsymbol{g}$:

$$\mathbf{D}_{\boldsymbol{h}(\boldsymbol{a})} = \mathbf{D}_{\boldsymbol{f}(\boldsymbol{b})}\mathbf{D}_{\boldsymbol{g}(\boldsymbol{a})}.$$

Additionally, suppose $\boldsymbol{a}\in\mathbb{R}^p$, $\boldsymbol{b}\in\mathbb{R}^n$ and $\boldsymbol{f}(\boldsymbol{b})\in\mathbb{R}^m$, then $\boldsymbol{h}(\boldsymbol{a})\in\mathbb{R}^m$ and the Jacobian matrix of $\boldsymbol{h}$ at $\boldsymbol{a}$ is given by

$$\left[\mathbf{D}_{\boldsymbol{h}(\boldsymbol{a})}\right]_{kj} = \left[\mathbf{D}_{\boldsymbol{f}(\boldsymbol{b})}\mathbf{D}_{\boldsymbol{g}(\boldsymbol{a})}\right]_{kj} = \sum_{i=1}^{n}\left[\mathbf{D}_{\boldsymbol{f}(\boldsymbol{b})}\right]_{ki}\left[\mathbf{D}_{\boldsymbol{g}(\boldsymbol{a})}\right]_{ij} = \sum_{i=1}^{n}\frac{\partial f_k}{\partial x_i}(\boldsymbol{b})\frac{\partial g_i}{\partial x_j}(\boldsymbol{a}),$$

for $k=1,\dots,m$ and $j=1,\dots,p$. Equivalently, we can write the chain rule as

$$D_j h_k(\boldsymbol{a}) = \sum_{i=1}^{n} D_i f_k(\boldsymbol{b}) D_j g_i(\boldsymbol{a}).$$

Example 12.3.8 For the special case $\boldsymbol{a}\in\mathbb{R}^p$, $\boldsymbol{b}=\boldsymbol{g}(\boldsymbol{a})\in\mathbb{R}^n$ and $\boldsymbol{f}(\boldsymbol{b})\in\mathbb{R}$, the chain rule simplifies to

$$D_j h(\boldsymbol{a}) = \sum_{i=1}^{n} D_i f(\boldsymbol{b}) D_j g_i(\boldsymbol{a}), \quad j=1,\dots,p.$$

(1) In Section 5.2, we cover the case where $p=1$:

$$h'(t) = \sum_{i=1}^{n} D_i f(\boldsymbol{b}) g_i'(t) = \nabla f(\boldsymbol{b})\cdot\boldsymbol{g}'(t).$$

(2) Consider $p=2$ and $n=2$, $\boldsymbol{a}=(s,t)$, $\boldsymbol{b}=\boldsymbol{g}(\boldsymbol{a})=(x,y)$ with $x=g_1(s,t)$ and $y=g_2(s,t)$, then

$$D_1 h(s,t) = D_1 f(x,y) D_1 g_1(s,t) + D_2 f(x,y) D_1 g_2(s,t),$$

or equivalently,

$$\frac{\partial h}{\partial s} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s}+\frac{\partial f}{\partial y}\frac{\partial y}{\partial s}.$$

Theorem 12.3.9 Suppose F is a scalar field differentiable on an open set $T \subseteq \mathbb{R}^n$ and that the equation $F(x_1, \ldots, x_n) = 0$ defines x_n implicitly as a differentiable function of $x_1, \ldots, x_{n-1}$, say $x_n = f(x_1, \ldots, x_{n-1})$ for all $(x_1, \ldots, x_{n-1})$ in some open set $S \subseteq \mathbb{R}^{n-1}$. Then for each $k = 1, 2, \ldots, n-1$, the partial derivative $D_k f$ or $\dfrac{\partial f}{\partial x_k}$ exists and is given by

$$\frac{\partial f}{\partial x_k} = -\frac{D_k F(x_1, \ldots, x_{n-1}, f(x_1, \ldots, x_{n-1}))}{D_n F(x_1, \ldots, x_{n-1}, f(x_1, \ldots, x_{n-1}))} = -\frac{\partial F(x_1, \ldots, x_{n-1}, f(x_1, \ldots, x_{n-1}))/\partial x_k}{\partial F(x_1, \ldots, x_{n-1}, f(x_1, \ldots, x_{n-1}))/\partial x_n}.$$

Chapter 13 Applications of Differential Calculus

1 Applications to Geometry

Definition 13.1.1 Suppose f is a scalar field on $S \subseteq \mathbb{R}^n$ and $c \in \mathbb{R}$ is a constant. We call the set $L(c) := \{\boldsymbol{x} \in S : f(\boldsymbol{x}) = c\}$ a **level set** of f. In particular, if $n = 2$, we call $L(c)$ a **level curve** of f; if $n = 3$, we call $L(c)$ a **level surface** of f.

Example 13.1.2 Suppose $f : \mathbb{R}^2 \to \mathbb{R}$. For any curve $\boldsymbol{r}(t)$ in $\mathbb{R}^2$, if we let g be defined as $g(t) = f[\boldsymbol{r}(t)]$, then

$$g'(t) = \nabla f(\boldsymbol{r}(t)) \cdot \boldsymbol{r}'(t) = \|\boldsymbol{r}'(t)\| \left(\nabla f(\boldsymbol{r}(t)) \cdot T(t)\right).$$

Particularly, if this curve happens to be the level curve given by $f(x, y) = c$. Suppose level curve $L(c) := \{(x, y) : f(x, y) = c\}$ is also described by $\boldsymbol{r}_c(t)$. Then $g(t) = f[\boldsymbol{r}_c(t)] = c$ for any t, which means $g'(t) = 0$ for any t. Thus

$$\|\boldsymbol{r}_c'(t)\| \left(\nabla f(\boldsymbol{r}_c(t)) \cdot T(t)\right) = 0.$$

On this level curve, $\nabla f \cdot T = 0$, and we say: the gradient vector is always normal to the curve C; the directional derivative of f along any level curve of f is zero. By Cauchy-Schwarz inequality, we have

$$|\nabla f(x, y) \cdot \boldsymbol{y}| \leq \|\nabla f(x, y)\| \|\boldsymbol{y}\|,$$

maximized when $\boldsymbol{y}$ is parallel to $\nabla f(x, y)$, i.e., $\boldsymbol{y}$ is perpendicular to the tangent vector of a level curve.

Example 13.1.3 Suppose $f : \mathbb{R}^3 \to \mathbb{R}$ differentiable. Suppose $L(c)$ is a level surface and $\boldsymbol{a}$ is a point on the level surface. Let $\boldsymbol{r}(t)$ describes a curve that passes through $\boldsymbol{a}$ and lies completely with in $L(c)$. If we consider all curves passing through $\boldsymbol{a}$ and living within $L(c)$, then the tangent vectors of all therse curves at $\boldsymbol{a}$ determin/live in a plane that passes throught $\boldsymbol{a}$ and has a normal vector equal to $\nabla f(\boldsymbol{a})$ provided that $\nabla f(\boldsymbol{a}) \neq \boldsymbol{0}$. This plane is can be described by the equation

$$\nabla f(\boldsymbol{a}) \cdot (\boldsymbol{x} - \boldsymbol{a}) = 0.$$

2 Applications to Optimization Problems

Definition 13.2.1 A scalar field is said to

- have an **absolute (or global maximum)** at $\boldsymbol{a} \in S \subseteq \mathbb{R}^n$ if $f(\boldsymbol{a}) \geq f(\boldsymbol{x})$ for any $\boldsymbol{x} \in S$, in which case we also call $f(\boldsymbol{a})$ the **absolute maximum value** of f on S;
- have an **absolute (or global minimum)** at $\boldsymbol{a} \in S$ if $f(\boldsymbol{a}) \leq f(\boldsymbol{x})$ for any $\boldsymbol{x} \in S$, in which case we also

call $f(\boldsymbol{a})$ the **absolute minimum value** of f on S;

- have a **relative (or local maximum)** at $\boldsymbol{a} \in S$ if there exists $r > 0$ such that $f(\boldsymbol{a}) \geq f(\boldsymbol{x})$ for any $\boldsymbol{x} \in B(\boldsymbol{a}; r) \subseteq S$, in which case we also call $f(\boldsymbol{a})$ the **local maximum value** of f at $\boldsymbol{a}$;
- have a **relative (or local minimum)** at $\boldsymbol{a} \in S$ if there exists $r > 0$ such that $f(\boldsymbol{a}) \leq f(\boldsymbol{x})$ for any $\boldsymbol{x} \in B(\boldsymbol{a}; r) \subseteq S$, in which case we also call $f(\boldsymbol{a})$ the **local minimum value** of f at $\boldsymbol{a}$.

If f has either a relative maximum or a relative minimum at $\boldsymbol{a}$, then we say f has an **extremum** at $\boldsymbol{a}$.

Theorem 13.2.2 If f has an extremum at $\boldsymbol{a} \in \operatorname{int} S$ and f is differentiable at $\boldsymbol{a}$, then $\nabla f(\boldsymbol{a}) = \mathbf{0}$.

Remark 13.2.3 If f is differentiable at $\boldsymbol{a}$ and $\nabla f(\boldsymbol{a}) = \mathbf{0}$, then f does not necessarily have an extremum at $\boldsymbol{a}$. Assume $z = f(x, y) = x^2 - y^2$, then $\nabla f(0, 0) = \mathbf{0}$ but f does not have an extremum at $(0, 0)$ since f can be positive or negative in any neighborhood of $(0, 0)$.

Definition 13.2.4 Suppose scalar field f is differentiable at $\boldsymbol{a}$. If $\nabla f(\boldsymbol{a}) = \mathbf{0}$, then we say $\boldsymbol{a}$ is a **stationary point** of f. A stationary point is called a **saddle point** if every open n-ball $B(\boldsymbol{a})$ contains both points $\boldsymbol{x}$ with $f(\boldsymbol{x}) > f(\boldsymbol{a})$ and points $\boldsymbol{y}$ with $f(\boldsymbol{y}) < f(\boldsymbol{a})$.

Remark 13.2.5 Suppose f has a stationary point at $\boldsymbol{a}$, then the nature of this stationary point is determined by the sign of $f(\boldsymbol{x}) - f(\boldsymbol{a})$ for $\boldsymbol{x}$ close to $\boldsymbol{a}$. Suppose $\boldsymbol{x} = \boldsymbol{a} + \boldsymbol{y}$, then

$$f(\boldsymbol{a} + \boldsymbol{y}) - f(\boldsymbol{a}) = \nabla f(\boldsymbol{a}) \cdot \boldsymbol{y} + \|\boldsymbol{y}\| E(\boldsymbol{a}, \boldsymbol{y}),$$

where $E(\boldsymbol{a}, \boldsymbol{y}) \to 0$ as $\boldsymbol{y} \to \mathbf{0}$.

Theorem 13.2.6 Suppose f is a scalar field with continuous second-order partial derivatives $D_{ij} f$ in an open n-ball $B(\boldsymbol{a})$. Then for all $\boldsymbol{y} \in \mathbb{R}^n$ such that $(\boldsymbol{a} + \boldsymbol{y}) \in B(\boldsymbol{a})$, we have

$$f(\boldsymbol{a} + \boldsymbol{y}) - f(\boldsymbol{a}) = \nabla f(\boldsymbol{a}) \cdot \boldsymbol{y} + \frac{1}{2!} \boldsymbol{y} H(\boldsymbol{a} + c\boldsymbol{y}) \boldsymbol{y}^T,$$

where $c \in (0, 1)$ and $H(\boldsymbol{x})$ is called the **Hessian matrix** whose ij-th entry is given by $D_{ij} f(\boldsymbol{x}) = \dfrac{\partial^2 f}{\partial x_i \partial x_j}(\boldsymbol{x})$. Note that here $\boldsymbol{y}$ is treated as a row vector. We can also write the second-order Taylor expansion as

$$f(\boldsymbol{a} + \boldsymbol{y}) = f(\boldsymbol{a}) + \nabla f(\boldsymbol{a}) \cdot \boldsymbol{y} + \frac{1}{2!} \boldsymbol{y} H(\boldsymbol{a}) \boldsymbol{y}^T + \|\boldsymbol{y}\|^2 E_2(\boldsymbol{a}, \boldsymbol{y}),$$

where $E_2(\boldsymbol{a}, \boldsymbol{y}) \to \mathbf{0}$ as $\boldsymbol{y} \to \mathbf{0}$.

Example 13.2.7 Consider the function

$$f(x, y) = \begin{cases} xy \dfrac{x^2 - y^2}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}.$$

By definition,

$$D_1 f(0,y) = \lim_{h\to 0} \frac{f(h,y)-f(0,y)}{h} = \lim_{h\to 0} y\frac{h^2-y^2}{h^2+y^2} = -y,$$
$$D_{2,1} f(0,0) = D_2(D_1 f)(0,0) = \lim_{k\to 0} \frac{D_1 f(0,k) - D_1 f(0,0)}{k} = -1.$$

Analogously, we have

$$D_2 f(x,0) = x, \quad D_{1,2} f(0,0) = 1.$$

Though $D_{1,2}f(0,0)$ and $D_{2,1}f(0,0)$ both exist, they are not equal since we do not have the continuity of $D_{1,2}f$ and $D_{2,1}f$ at $(0,0)$: note

$$D_{2,1} f(x,y) = \frac{x^6+9x^4y^2-9x^2y^4-y^6}{(x^2+y^2)^3}, \qquad (x,y)\neq(0,0),$$

and we have

$$D_{2,1} f(0,0) = -1 \neq 1 = \frac{x^6}{x^6} = \lim_{x\to 0} D_{2,1} f(x,0) \quad \text{along } y=0.$$

This example shows that the continuity of second-order partial derivatives is necessary for the equality of mixed partial derivatives and motivates the following theorem.

Theorem 13.2.8 Suppose f is a scalar field on an open set $S\subseteq\mathbb{R}^n$, and suppose the first-order partial derivatives $D_i f$ and $D_j f$ exist on S. Assume further that the second-order partial derivatives $D_{ij}f$ and $D_{ji}f$ exist on S and are continuous at $\boldsymbol{a}\in S$. Then

$$D_{ij} f(\boldsymbol{a}) = D_{ji} f(\boldsymbol{a}).$$

Remark 13.2.9 One can actually prove a **stronger** version of the above theorem, which only requires the continuity of **one of the two** second order partial derivatives $D_{i,j}f$ and $D_{j,i}f$ at $\boldsymbol{a}$.

Lemma 13.2.10 By Theorem 13.2.8, we know the symmetric Hessian matrix has an orthonormal eigenbasis and $\boldsymbol{y}H(\boldsymbol{a}+c\boldsymbol{y})\boldsymbol{y}^T$ is a quadratic form associated with a unique symmetric matrix. If we know all the eigenvalues of $H(\boldsymbol{a})$ are positive, then

$$f(\boldsymbol{a}+\boldsymbol{y}) - f(\boldsymbol{a}) = \frac{1}{2}\boldsymbol{y}H(\boldsymbol{a})\boldsymbol{y}^T + \|\boldsymbol{y}\|^2 E_2(\boldsymbol{a},\boldsymbol{y}) > 0$$

for $\boldsymbol{y}$ close enough to $\mathbf{0}$.

Proof. Fix $\boldsymbol{y}$ and define $g(u) = f(\boldsymbol{a}+u\boldsymbol{y})$ for $u\in[-1,1]$. Then

$$f(\boldsymbol{a}+\boldsymbol{y}) = g(1) - g(0) = g'(0) + \frac{1}{2}g''(c)$$

where $c\in(0,1)$ by the second-order Taylor formula for g on $[0,1]$ with the Lagrange form of the remainder. Let $\boldsymbol{r} = \boldsymbol{a}+u\boldsymbol{y}$, then $\boldsymbol{r}'(u) = \boldsymbol{y}$ and

$$g'(u) = \nabla f(\boldsymbol{r})\cdot\boldsymbol{r}'(u) = \sum_{j=1} D_j f(\boldsymbol{r}) y_j.$$

Thus
$$g'(0) = \sum_{j=1}^{n=1} D_j f(\boldsymbol{a}) y_j = \nabla f(\boldsymbol{a}) \cdot \boldsymbol{y} = 0.$$
□

Lemma 13.2.11 Suppose $A \in \mathbf{M}_{n\times n}(\mathbb{R})$ is a symmetric matrix and $\boldsymbol{y}$ is a row vector in $\mathbb{R}^n$. Let $Q(\boldsymbol{y}) = \boldsymbol{y}A\boldsymbol{y}^T$. Then

1. $Q(\boldsymbol{y}) > 0$ for any $\boldsymbol{y} \neq \mathbf{0}$ if and only if all the eigenvalues of A are positive;
2. $Q(\boldsymbol{y}) < 0$ for any $\boldsymbol{y} \neq \mathbf{0}$ if and only if all the eigenvalues of A are negative;
3. $Q(\boldsymbol{y})$ takes both positive and negative values for $\boldsymbol{y} \neq \mathbf{0}$ if and only if A has both positive and negative eigenvalues.

Theorem 13.2.12 Suppose f is a scalar field with continuous second-order partial derivatives $D_{ij}f$ in an open n-ball $B(\boldsymbol{a})$ and Let $H(\boldsymbol{a})$ denote the Hessian matrix of f at $\boldsymbol{a}$, which is a stationary point of f. Then

1. If all the eigenvalues of $H(\boldsymbol{a})$ are positive, i.e., $H(\boldsymbol{a})$ is positive definite, then f has a local minimum at $\boldsymbol{a}$;
2. If all the eigenvalues of $H(\boldsymbol{a})$ are negative, i.e., $H(\boldsymbol{a})$ is negative definite, then f has a local maximum at $\boldsymbol{a}$;
3. If $H(\boldsymbol{a})$ has both positive and negative eigenvalues, then f has a saddle point at $\boldsymbol{a}$.

Corollary 13.2.13 Let $\boldsymbol{a}$ be a stationary point of a scalar field $f(x_1, x_2)$ with continuous second order partial derivatives in a 2-ball $B(\boldsymbol{a})$. Let
$$A = D_{1,1}f(\boldsymbol{a}), \qquad B = D_{1,2}f(\boldsymbol{a}), \qquad C = D_{2,2}f(\boldsymbol{a}),$$
and let
$$\Delta = \det H(\boldsymbol{a}) = \det \begin{pmatrix} A & B \\ B & C \end{pmatrix} = AC - B^2.$$
Then we have:

(a) If $\Delta > 0$ and $A > 0$, then f has a local minimum at $\boldsymbol{a}$;

(b) If $\Delta > 0$ and $A < 0$, then f has a local maximum at $\boldsymbol{a}$;

(c) If $\Delta < 0$, then f has a saddle point at $\boldsymbol{a}$;

(d) If $\Delta = 0$, then the test is inconclusive.

Remark 13.2.14 The determinant test is only applicable for $n = 2$. For $n \geq 3$, we need to check the signs of all the eigenvalues of the Hessian matrix to determine the nature of a stationary point.

Method 13.2.15 (The method of Lagrange's multipliers) If a scalar field $f(x_1, \ldots, x_n)$ has a relative extremum when it is subject to m constraints, say
$$g_1(x_1, \ldots, x_n) = 0, \qquad \ldots, \qquad g_m(x_1, \ldots, x_n) = 0$$

where $m < n$, then there exist m scalars $\lambda_1, \ldots, \lambda_m$ such that

$$\nabla f = \lambda_1 \nabla g_1 + \cdots + \lambda_m \nabla g_m$$

at each extremum point.

Remark 13.2.16 The idea of Lagrange multipliers is that, at a constrained extremum, f cannot change to first order along any allowable tangent direction $\mathbf{v}$. Thus

$$\nabla f \cdot \mathbf{v} = 0$$

for every tangent vector $\mathbf{v}$ to the constraint set. Hence ∇f must lie in the normal space of the constraint set. Since each ∇g_i is normal to the constraint $g_i = 0$, the normal space is generated by the constraint gradients, so

$$\nabla f = \lambda_1 \nabla g_1 + \cdots + \lambda_m \nabla g_m.$$

Geometrically, the level surface of f is tangent to the constraint set at the extremum point.

Theorem 13.2.17 Given $\boldsymbol{a} = (a_1, \ldots, a_n), \boldsymbol{b} = (b_1, \ldots, b_n) \in \mathbb{R}^n$ with $a_i < b_i$ for $i = 1, \ldots, n$. We define $[\boldsymbol{a}, \boldsymbol{b}] := [a_1, b_1] \times \cdots \times [a_n, b_n]$. If a scalar field f is continuous at every $\boldsymbol{x} \in [\boldsymbol{a}, \boldsymbol{b}]$, then

- f is bounded on $[\boldsymbol{a}, \boldsymbol{b}]$, i.e., there exists $M > 0$ such that $|f(\boldsymbol{x})| \leq M$ for any $\boldsymbol{x} \in [\boldsymbol{a}, \boldsymbol{b}]$;
- There exists $\boldsymbol{c}$ and $\boldsymbol{d}$ such that

$$f(\boldsymbol{c}) = \sup_{\boldsymbol{x} \in [\boldsymbol{a}, \boldsymbol{b}]} f(\boldsymbol{x}), \quad f(\boldsymbol{d}) = \inf_{\boldsymbol{x} \in [\boldsymbol{a}, \boldsymbol{b}]} f(\boldsymbol{x}).$$

Definition 13.2.18 Suppose $[\boldsymbol{a}, \boldsymbol{b}]$ and P_k is a partition of $[a_k, b_k]$ for $k = 1, \ldots, n$. We call $P := P_1 \times \cdots \times P_n$ a **partition** of $[\boldsymbol{a}, \boldsymbol{b}]$.

Definition 13.2.19 Let f be a continuous scalar field on $[\boldsymbol{a}, \boldsymbol{b}] \subseteq \mathbb{R}^n$ and let $M(f)$ and $m(f)$ denote the absolute maximum and minimum values of f on $[\boldsymbol{a}, \boldsymbol{b}]$, respectively. We call $M(f) - m(f)$ the **span** of f on $[\boldsymbol{a}, \boldsymbol{b}]$.

Theorem 13.2.20 Suppose f is a scalar field continuous on $[\boldsymbol{a}, \boldsymbol{b}]$. Then, for every $\varepsilon > 0$, there is a partition of $[\boldsymbol{a}, \boldsymbol{b}]$ such that the span of f on every subinterval of this partition is less than ε.

Chapter 14 Line Integrals

1 Basic Definitions and Properties

Definition 14.1.1 A function $\boldsymbol{r} : [a, b] \to \mathbb{R}^n$ that is continuous on $[a, b]$ is also called a **path** in $\mathbb{R}^n$. If $\boldsymbol{r}'$ exists and is continuous on (a, b), then we say $\boldsymbol{r}$ is a **smooth path**. If there is a partition of $[a, b]$ such that $\boldsymbol{r}$ is smooth on each subinterval of this partition, then we say $\boldsymbol{r}$ is a **piecewise smooth path**.

Definition 14.1.2 Let $\boldsymbol{r} : [a, b] \to \mathbb{R}^n$ be a piecewise smooth path and $\boldsymbol{f}$ be a vector field defined and bounded on the image of $\boldsymbol{r}$. The line integral of $\boldsymbol{f}$ along $\boldsymbol{r}$ is defined as

$$\int_{\boldsymbol{r}} \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{r} := \int_a^b \boldsymbol{f}(\boldsymbol{r}(t)) \cdot \boldsymbol{r}'(t)\mathrm{d}t$$

provided the limit on the right-hand side exists.

Remark 14.1.3 Suppose curve C is described by $\boldsymbol{r}(t)$ for $t \in [a, b]$ and $\boldsymbol{r}$ is a piecewise smooth path, then we can also write $\displaystyle\int \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{r}$ as

1. $\displaystyle\int_C \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{r}$;
2. $\displaystyle\int_{\boldsymbol{a}}^{\boldsymbol{b}} \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{r}$ where $\boldsymbol{a} = \boldsymbol{r}(a)$ and $\boldsymbol{b} = \boldsymbol{r}(b)$;
3. $\displaystyle\oint \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{r}$ or $\displaystyle\oint_C \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{r}$ if C is a closed curve, i.e., $\boldsymbol{r}(a) = \boldsymbol{r}(b)$;
4. $\displaystyle\int_C f_1(x, y)\mathrm{d}x + f_2(x, y)\mathrm{d}y$ if $\boldsymbol{f} = (f_1(x, y), f_2(x, y))$ with $x = r_1(t)$ and $y = r_2(t)$.

Remark 14.1.4 Since we define line integrals in terms of regular integrals of a scalar function on $[a, b]$, line integrals have the following basic properties of regular integral.

Theorem 14.1.5 Suppose $\boldsymbol{f}$ and $\boldsymbol{g}$ are vector fields defined and bounded on the image of a piecewise smooth path $\boldsymbol{r}$. Then

(1) Linearity:

$$\int (\alpha\boldsymbol{f} + \beta\boldsymbol{g}) \cdot \mathrm{d}\boldsymbol{r} = \alpha \int \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{r} + \beta \int \boldsymbol{g} \cdot \mathrm{d}\boldsymbol{r}$$

for any scalar α, β;

(2) Additivity: if curve C can be regarded as two pieces C_1 and C_2 joined together, then

$$\int_C \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{r} = \int_{C_1} \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{r} + \int_{C_2} \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{r};$$

Definition 14.1.6 Suppose $\boldsymbol{\beta} : [a,b] \to \mathbb{R}^n$ is a continuous path and $u : \mathbb{R} \to \mathbb{R}$ is differentiable on $[c,d]$ with $\{u(s) : s \in [c,d]\} = [a,b]$ and $u'(s) \neq 0$ for any $s \in [c,d]$. Then $\boldsymbol{\gamma} : [c,d] \to \mathbb{R}^n$ defined as $\boldsymbol{\gamma}(s) := \boldsymbol{\beta}[u(s)]$ for $s \in [c,d]$ is also a continuous path with the same image as $\boldsymbol{\beta}$. We call $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$ **equivalent paths** and u a **change of parameter function**. Namely, γ and $\boldsymbol{\beta}$ describe the same curve but with different parametrizations. If $u'(s) > 0$ for $s \in [c,d]$, then we say $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ trace out C in the same direction.

Definition 14.1.7 Suppose $\boldsymbol{\alpha}$ is a path with $\boldsymbol{\alpha}'$ continuous on $[a,b]$. Then $\boldsymbol{\alpha}$ describes a **rectifiable curve** C whose arc-length function s is given by

$$s(t) = \int_a^t \|\boldsymbol{\alpha}'(\tau)\|\mathrm{d}\tau, \quad t \in [a,b].$$

We also know $s'(t) = \|\boldsymbol{\alpha}'(t)\|$. Suppose f is a scalar function defined and bounded on C. The line integral of f with respect to arc-length along C is defined as

$$\int_C f\mathrm{d}s := \int_a^b f(\boldsymbol{\alpha}(t))s'(t)\mathrm{d}t.$$

Example 14.1.8 Let $\boldsymbol{g}$ be a vector field (e.g. velocity filed of the fluid) defined on a curve C described by $\boldsymbol{\alpha}(t)$ on $[a,b]$. Then $\varphi[\boldsymbol{\alpha}(t)] := \boldsymbol{g}[\boldsymbol{\alpha}(t)] \cdot \boldsymbol{T}(t)$ is a scalar field defined on C and

$$\int_C \varphi\mathrm{d}s = \int_a^b \varphi[\boldsymbol{\alpha}(t)]s'(t)\mathrm{d}t = \int_a^b \boldsymbol{g}[\boldsymbol{\alpha}(t)] \cdot \boldsymbol{T}(t)s'(t)\mathrm{d}t = \int_a^b \boldsymbol{g}[\boldsymbol{\alpha}(t)] \cdot \boldsymbol{\alpha}'(t)\mathrm{d}t = \int_C \boldsymbol{g} \cdot \mathrm{d}\boldsymbol{r}.$$

Then $\int_C \varphi\mathrm{d}s = \int_C \boldsymbol{g} \cdot \boldsymbol{T}\mathrm{d}s$ is called the **flow integral** of $\boldsymbol{g}$ along C. Particularly, when C is a closed curve, it is also called the **circulation** of $\boldsymbol{g}$ along C.

2 Fundamental Theorems of Calculus for Line Integrals

Definition 14.2.1 Let S be an open set in $\mathbb{R}^n$. We say S is **connected** if every pair of points in S can be joined by a piecewise smooth path whose image is totally contained in S.

Theorem 14.2.2 (Second Fundamental Theorem of Calculus for Line Integrals) Let f be a differentiable scalar field with continuous gradient ∇f on an open, connected set $S \subseteq \mathbb{R}^n$. Then for points $\boldsymbol{a}, \boldsymbol{b} \in S$ that are joined by a piecewise smooth path $\boldsymbol{\alpha}$ in S, we have

$$\int_{\boldsymbol{\alpha}}^{\boldsymbol{\beta}} \nabla f \cdot \mathrm{d}\boldsymbol{\alpha} = f(\boldsymbol{b}) - f(\boldsymbol{a}).$$

Proof. Let $\boldsymbol{a}, \boldsymbol{b} \in S$ and $\boldsymbol{\alpha}$ be a path with $\boldsymbol{\alpha}(a) = \boldsymbol{a}$ and $\boldsymbol{\alpha}(b) = \boldsymbol{b}$ and $\boldsymbol{\alpha}(t) \in S$ for $t \in [a, b]$.
Special Case: $\boldsymbol{\alpha}$ is smooth on $[a, b]$. We pick $g(t) := f[\boldsymbol{\alpha}(t)]$ for $t \in [a, b]$. Then g is differentiable on $[a, b]$ and

$$g'(t) = \nabla f[\boldsymbol{\alpha}(t)] \cdot \boldsymbol{\alpha}'(t)$$

for $t \in [a, b]$. By the **FTC2** for regular integrals, we have

$$\int \nabla f \cdot \mathrm{d}\boldsymbol{\alpha} = \int_a^b \nabla f[\boldsymbol{\alpha}(t)] \cdot \boldsymbol{\alpha}'(t)\mathrm{d}t = g(b) - g(a) = f(\boldsymbol{b}) - f(\boldsymbol{a}).$$

General Case: $\boldsymbol{\alpha}$ is piecewise smooth on $[a, b]$ with partition $P = \{t_0, t_1, \ldots, t_r\}$ where $a = t_0 < t_1 < \cdots < t_r = b$ such that $\boldsymbol{\alpha}$ is smooth on each subinterval $[t_{i-1}, t_i]$ for $i = 1, 2, \ldots, r$. Then by the additive property

$$\int_{\boldsymbol{\alpha}(a)}^{\boldsymbol{\alpha}(b)} \nabla f \cdot \mathrm{d}\boldsymbol{\alpha} = \sum_{i=1}^{r} \int_{\boldsymbol{\alpha}(t_{i-1})}^{\boldsymbol{\alpha}(t_i)} \nabla f \cdot \mathrm{d}\boldsymbol{\alpha} = \sum_{i=1}^{r} [f(\boldsymbol{\alpha}(t_i)) - f(\boldsymbol{\alpha}(t_{i-1}))] = f(\boldsymbol{b}) - f(\boldsymbol{a}).$$

□

Remark 14.2.3 The result of Theorem 14.2.2 can be interpreted as follows:

- the line integral's value of a gradient field only depends on the endpoints $\boldsymbol{a}$ and $\boldsymbol{b}$, i.e., **independent of path from $\boldsymbol{a}$ to $\boldsymbol{b}$**;
- $\oint \nabla f \cdot \mathrm{d}\boldsymbol{r} = 0$ for any closed curve in S.

Theorem 14.2.4 (First Fundamental Theorem of Calculus for Line Integrals) Let f be a vector field that is continuous on an open connected set $S \subseteq \mathbb{R}^n$ and assume that the line integral of f is independent of path in S. Let $\boldsymbol{a} \in S$ and define a scalar field φ on S by

$$\varphi(\boldsymbol{x}) := \int_{\boldsymbol{a}}^{\boldsymbol{x}} \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{\alpha}, \quad \boldsymbol{x} \in S$$

where $\boldsymbol{\alpha}$ is any piecewise smooth path in S from $\boldsymbol{a}$ to $\boldsymbol{x}$. Then the gradient of φ exists and $\nabla\varphi(\boldsymbol{x}) = \boldsymbol{f}(\boldsymbol{x})$ on S.

Proof. Suppose $\boldsymbol{f}(\boldsymbol{x}) = (f_1(\boldsymbol{x}), f_2(\boldsymbol{x}), \ldots, f_n(\boldsymbol{x}))$. We will prove $D_k\varphi(\boldsymbol{x}) = f_k(\boldsymbol{x})$ for $k = 1, 2, \ldots, n$ and any $\boldsymbol{x} \in S$. Suppose $B(\boldsymbol{x}; r) \subseteq S$ and $\boldsymbol{y}$ is a unit vector. Then $\boldsymbol{x} + h\boldsymbol{y} \in B(\boldsymbol{x}; r)$ for any $0 < |h| < r$.

$$\frac{\varphi(\boldsymbol{x} + h\boldsymbol{y}) - \varphi(\boldsymbol{x})}{h} = \frac{1}{h}\left[\int_{\boldsymbol{a}}^{\boldsymbol{x}+h\boldsymbol{y}} \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{\alpha} - \int_{\boldsymbol{a}}^{\boldsymbol{x}} \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{\alpha}\right] = \frac{1}{h}\int_{\boldsymbol{x}}^{\boldsymbol{x}+h\boldsymbol{y}} \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{\alpha}.$$

Particularly, consider the path $\boldsymbol{\alpha}(t) = \boldsymbol{x} + th\boldsymbol{y}$ for $t \in [0, 1]$. Then $\boldsymbol{\alpha}'(t) = h\boldsymbol{y}$ and

$$\frac{1}{h}\int_{\boldsymbol{x}}^{\boldsymbol{x}+h\boldsymbol{y}} \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{\alpha} = \int_0^1 \boldsymbol{f}(\boldsymbol{x} + th\boldsymbol{y}) \cdot \boldsymbol{y}\mathrm{d}t = \int_0^1 f_k(\boldsymbol{x} + th\boldsymbol{e}_k)\mathrm{d}t,$$

where $\boldsymbol{y} = \boldsymbol{e}_k$ is the k-th standard unit vector. We do a change of variable $u = th$ and get

$$\int_0^1 f_k(\boldsymbol{x} + th\boldsymbol{e}_k)\mathrm{d}t = \frac{1}{h}\int_0^h f_k(\boldsymbol{x} + u\boldsymbol{e}_k)\mathrm{d}u = \frac{\int_0^h f_k(\boldsymbol{x} + u\boldsymbol{e}_k)\mathrm{d}u - \int_0^0 f_k(\boldsymbol{x} + u\boldsymbol{e}_k)\mathrm{d}u}{h - 0}$$

which converges to $f_k(\boldsymbol{x})$ as $h \to 0$ by the **FTC1** for regular integrals. Hence,

$$\lim_{h\to 0} \frac{\varphi(\boldsymbol{x}+h\boldsymbol{y})-\varphi(\boldsymbol{x})}{h} = f_k(\boldsymbol{x}) = D_k\varphi(\boldsymbol{x}).$$

$\square$

Remark 14.2.5 Suppose $\boldsymbol{g}$ is a continuous vector field on an open connected set $S \subseteq \mathbb{R}^n$. Then by Theorem 14.2.4, if the line integral of $\boldsymbol{g}$ is independent of path in S, then there exists a scalar field φ such that $\nabla\varphi = \boldsymbol{g}$. Conbining with Theorem 14.2.2, we have the necessary and sufficient condition for a vector field to be a gradient field, which is stated in the following theorem.

Theorem 14.2.6 Suppose $\boldsymbol{g}$ is a continuous vector field on an open connected set $S \subseteq \mathbb{R}^n$. Then the following statements are equivalent:

(1) $\boldsymbol{g}$ is the gradient of some scalar field φ in S, i.e., $\boldsymbol{g} = \nabla\varphi$, where φ is called a **potential function** of $\boldsymbol{g}$;

(2) the line integral of $\boldsymbol{g}$ is independent of path in S;

(3) $\displaystyle\oint_C \boldsymbol{g}\cdot \mathrm{d}\boldsymbol{r} = 0$ for any piecewise smooth closed path C in S.

Proof. By Theorem 14.2.4, we have (2) $\Rightarrow$ (1). By Theorem 14.2.2, we have (1) $\Rightarrow$ (2) and (1) $\Rightarrow$ (3).
We will only show (3) $\Rightarrow$ (2). Suppose C_1 and C_2 are any two piecewise smooth paths in S sharing the same endpoints. Let C_1 be described by $\boldsymbol{\alpha}$ on $[a,b]$ and C_2 be described by $\boldsymbol{\beta}$ on $[c,d]$ with $\boldsymbol{\alpha}(a) = \boldsymbol{\beta}(c)$ and $\boldsymbol{\alpha}(b) = \boldsymbol{\beta}(d)$.
Define $\boldsymbol{\gamma}$ as

$$\boldsymbol{\gamma}(t) = \begin{cases} \boldsymbol{\alpha}(t), & t \in [a,b]; \\ \boldsymbol{\beta}(b+d-t), & t \in [b, b+d-c]. \end{cases}$$

Note $\boldsymbol{\gamma}(a) = \boldsymbol{\alpha}(a) = \boldsymbol{\beta}(c) = \boldsymbol{\gamma}(b+d-c)$. Then $\boldsymbol{\gamma}$ is a piecewise smooth path describing a closed curve in S. Then by (3), we know

$$0 = \oint_{\boldsymbol{\gamma}(a)}^{\boldsymbol{\gamma}(b+d-c)} \boldsymbol{g}\cdot\mathrm{d}\boldsymbol{r} = \int_{\boldsymbol{\alpha}(a)}^{\boldsymbol{\alpha}(b)} \boldsymbol{g}\cdot\mathrm{d}\boldsymbol{r} + \int_{\boldsymbol{\beta}(c)}^{\boldsymbol{\beta}(d)} \boldsymbol{g}\cdot(-\mathrm{d}\boldsymbol{r}) = \int_{\boldsymbol{\alpha}(a)}^{\boldsymbol{\alpha}(b)} \boldsymbol{g}\cdot\mathrm{d}\boldsymbol{r} - \int_{\boldsymbol{\beta}(c)}^{\boldsymbol{\beta}(d)} \boldsymbol{g}\cdot\mathrm{d}\boldsymbol{r}.$$

Hence, $\displaystyle\int_{\boldsymbol{\alpha}(a)}^{\boldsymbol{\alpha}(b)} \boldsymbol{g}\cdot\mathrm{d}\boldsymbol{r} = \int_{\boldsymbol{\beta}(c)}^{\boldsymbol{\beta}(d)} \boldsymbol{g}\cdot\mathrm{d}\boldsymbol{r}$, which means the line integral of $\boldsymbol{g}$ is independent of path in S. $\square$

Remark 14.2.7 Actually, Theorem 14.2.4 not only proves (2) $\Rightarrow$ (1), it also tells us we can fin a potential function φ (such that $\nabla\varphi = \boldsymbol{f}$) by picking $\varphi(\boldsymbol{x}) := \displaystyle\int_{\boldsymbol{a}}^{\boldsymbol{x}} \boldsymbol{f}\cdot\mathrm{d}\boldsymbol{\alpha}$ for any $\boldsymbol{a}\in S$ and any piecewise smooth path $\boldsymbol{\alpha}$ from $\boldsymbol{a}$ to $\boldsymbol{x}$.

Example 14.2.8 We call $S \subseteq \mathbb{R}^n$ a **convex set** if every pair of points in S can be joined by a line segment fully contained in S. Note if S is open convex, then S is open connected. Suppose S is open convex, we can pick

$$\varphi(\boldsymbol{x}) := \int_{\boldsymbol{a}}^{\boldsymbol{x}} \boldsymbol{f}\cdot\mathrm{d}\boldsymbol{\alpha} = \int_0^1 \boldsymbol{f}[\boldsymbol{a}+t(\boldsymbol{x}-\boldsymbol{a})]\cdot(\boldsymbol{x}-\boldsymbol{a})\mathrm{d}t.$$

Particularly, if $\mathbf{0} \in S$, then $\varphi(\boldsymbol{x}) = \displaystyle\int_0^1 \boldsymbol{f}(t\boldsymbol{x}) \cdot \boldsymbol{x} \mathrm{d}t$.

Example 14.2.9 Let $\boldsymbol{f} = (f_1, f_2, f_3)$, we know

$$\frac{\partial \varphi}{\partial x} = f_1, \quad \frac{\partial \varphi}{\partial y} = f_2, \quad \frac{\partial \varphi}{\partial z} = f_3.$$

Then we can pick

$$\varphi(x,y,z) = \int f_1(x,y,z)\mathrm{d}x + C_1(y,z) = \int f_2(x,y,z)\mathrm{d}y + C_2(x,z) = \int f_3(x,y,z)\mathrm{d}z + C_3(x,y).$$

Note that the functions C_1, C_2, C_3 are not arbitrary constants, but functions of the remaining variables. In practice, we collect all terms from the three integrals $\displaystyle\int f_1(x,y,z)\mathrm{d}x$, $\displaystyle\int f_2(x,y,z)\mathrm{d}y$ and $\displaystyle\int f_3(x,y,z)\mathrm{d}z$ and add the missing terms into $C_1(y,z)$, $C_2(x,z)$, or $C_3(x,y)$. This provides a systematic way to find the potential function φ of $\boldsymbol{f}$ if $\boldsymbol{f}$ is a gradient field.

Theorem 14.2.10 Let $\boldsymbol{f} = (f_1, \ldots, f_n)$ be a continuously differentiable vector field on an open set S in $\mathbb{R}^n$. If $\boldsymbol{f}$ is a gradient on S, then the partial derivatives of the components of $\boldsymbol{f}$ are related by the equations

$$D_i f_j(\boldsymbol{x}) = D_j f_i(\boldsymbol{x})$$

for $i, j = 1, 2, \ldots, n$ and every $\boldsymbol{x}$ in S.

Remark 14.2.11 This is a necessary condition for a vector field to be a gradient field. However, it is not a sufficient condition, i.e., there exist vector fields $\boldsymbol{f}$ such that $D_i f_j(\boldsymbol{x}) = D_j f_i(\boldsymbol{x})$ for $i, j = 1, 2, \ldots, n$ and every $\boldsymbol{x}$ in S, but $\boldsymbol{f}$ is not a gradient field on S.

Example 14.2.12 Let S be the set of all $(x,y) \neq (0,0)$ in $\mathbb{R}^2$, and let $\boldsymbol{f}$ be the vector field defined on S by the equation

$$\boldsymbol{f}(x,y) = \frac{-y}{x^2+y^2}\boldsymbol{i} + \frac{x}{x^2+y^2}\boldsymbol{j}$$

Note that

$$D_1 f_2(x,y) = \frac{\partial}{\partial x}\left(\frac{x}{x^2+y^2}\right) = \frac{y^2-x^2}{(x^2+y^2)^2} = D_2 f_1(x,y).$$

However, $\boldsymbol{f}$ is not a gradient on S. We pick $\boldsymbol{\alpha} = (\cos t, \sin t)$ for $t \in [0, 2\pi]$ describing the unit circle. Then

$$\oint_{\boldsymbol{\alpha}} \boldsymbol{f} \cdot \mathrm{d}\boldsymbol{r} = \int_0^{2\pi} \boldsymbol{f}(\cos t, \sin t) \cdot (-\sin t, \cos t)\mathrm{d}t = \int_0^{2\pi} 1\mathrm{d}t = 2\pi \neq 0.$$

Hence, the loop integral of $\boldsymbol{f}$ along the unit circle is not zero, which means $\boldsymbol{f}$ is not a gradient field on S. Actually, note that

$$\theta(x,y) = \arctan\frac{y}{x}, \qquad \nabla\theta(x,y) = \frac{-y}{x^2+y^2}\boldsymbol{i} + \frac{x}{x^2+y^2}\boldsymbol{j} = \boldsymbol{f}(x,y),$$

which seems to suggest $\boldsymbol{f}$ is a gradient field locally. However, θ cannot be defined as a continuous single-valued function on all $\mathbb{R}^2 \setminus \{(0,0)\}$. Hence, $\boldsymbol{f}$ is not a gradient field on S.

Chapter 15 Multiple Integrals

1 Double Integrals

In Chapter 02, we define the integral of a general function $f : [a, b] \to \mathbb{R}$ by approximating f by step functions. In this section, we follow the same idea to define the double integral of a function $f : Q \to \mathbb{R}$ where Q is a rectangle in $\mathbb{R}^2$.

Definition 15.1.1 Given two closed intervals $[a, b]$ and $[c, d]$ in $\mathbb{R}$, we call

$$Q := [a, b] \times [c, d] = \{(x, y) \in \mathbb{R}^2 : a \le x \le b,\ c \le y \le d\}$$

a **rectangle** in $\mathbb{R}^2$.

1. Suppose $P_1 = \{x_0, x_1, \ldots, x_n\}$ and $P_2 = \{y_0, y_1, \ldots, y_m\}$ are partitions of $[a, b]$ and $[c, d]$, respectively, with
$$a = x_0 < x_1 < \cdots < x_n = b, \qquad c = y_0 < y_1 < \cdots < y_m = d.$$
Then we call
$$P := P_1 \times P_2 = \{[x_{i-1}, x_i] \times [y_{j-1}, y_j] : 1 \le i \le n,\ 1 \le j \le m\}$$
a **partition** of Q, which decomposes Q into nm **subrectangles**.
2. The Cartesian product $(x_{i-1}, x_i) \times (y_{j-1}, y_j)$, where $1 \le i \le n$ and $1 \le j \le m$, is called an **open subrectangle** of P, or of Q.
3. Another partition $\widetilde{P}$ is said to be **finer than** P, or a **refinement** of P, if $P \subseteq \widetilde{P}$, i.e., all the points in P are also in $\widetilde{P}$.
4. A scalar field $f\colon Q \to \mathbb{R}$ is called a **step function** if there exists a partition P of Q such that f is constant on each open subrectangle of P.

Remark 15.1.2 For a rectangle Q in $\mathbb{R}^2$, the set of all step functions defined on Q is a vector space over $\mathbb{R}$: if f and g are step functions defined on Q with respect to partitions P_f and P_g, respectively, then $\alpha f + \beta g$ is a step function defined on Q with respect to the common refinement $P_f \cup P_g$ of P_f and P_g for any $\alpha, \beta \in \mathbb{R}$.

Definition 15.1.3 Suppose $Q = [a, b] \times [c, d]$ with a partition $P = P_1 \times P_2$ where $P_1 = \{x_0, x_1, \ldots, x_n\}$ and $P_2 = \{y_0, y_1, \ldots, y_m\}$. Let f be a step function defined on Q that takes constant value c_{ij} on the open subinterval $(x_{i-1}, x_i) \times (y_{j-1}, y_j)$ for $1 \le i \le n$ and $1 \le j \le m$. Then we define the **double integral** of f over Q by

$$\iint_Q f(x, y)\mathrm{d}x\mathrm{d}y := \sum_{i=1}^{n} \sum_{j=1}^{m} c_{ij} \cdot (x_i - x_{i-1})(y_j - y_{j-1}).$$

Note if we replace P with a finer partition $\widetilde{P}$, then the value of the double integral does not change.

Notation 15.1.4 We can also write the double integral of f over Q as $\iint_Q f$.

Remark 15.1.5 Since $f(x, y) = c_{ij}$ on $(x_{i-1}, x_i) \times (y_{j-1}, y_j)$, if we let $Q_{ij} := [x_{i-1}, x_i] \times [y_{j-1}, y_j]$, then we have

$$\begin{aligned}\iint_{Q_{ij}} f(x, y)\mathrm{d}x\mathrm{d}y &= c_{ij} \cdot (x_i - x_{i-1})(y_j - y_{j-1}) \\ &= f(x, y) \int_{x_{i-1}}^{x_i} \mathrm{d}x \int_{y_{j-1}}^{y_j} \mathrm{d}y \\ &= \left(\int_{x_{i-1}}^{x_i} f(x, y)\mathrm{d}x\right)\left(\int_{y_{j-1}}^{y_j} \mathrm{d}y\right) \\ &= \int_{y_{j-1}}^{y_j} \left[\int_{x_{i-1}}^{x_i} f(x, y)\mathrm{d}x\right] \mathrm{d}y.\end{aligned}$$

Analogously, we also have

$$\iint_{Q_{ij}} f(x, y)\mathrm{d}x\mathrm{d}y = \int_{x_{i-1}}^{x_i} \left[\int_{y_{j-1}}^{y_j} f(x, y)\mathrm{d}y\right] \mathrm{d}x.$$

Summing up all Q_{ij}, we get

$$\begin{aligned}\iint_Q f(x, y)\mathrm{d}x\mathrm{d}y &= \sum_{i=1}^{n}\sum_{j=1}^{m} \iint_{Q_{ij}} f(x, y)\mathrm{d}x\mathrm{d}y \\ &= \sum_{i=1}^{m} \int_{y_{j-1}}^{y_j} \left[\sum_{i=1}^{n} \int_{x_{i-1}}^{x_i} f(x, y)\mathrm{d}x\right] \mathrm{d}y \\ &= \int_c^d \left[\int_a^b f(x, y)\mathrm{d}x\right] \mathrm{d}y \\ &= \int_a^b \left[\int_c^d f(x, y)\mathrm{d}y\right] \mathrm{d}x.\end{aligned}$$

Theorem 15.1.6 Suppose s and t are step functionson rectangle $Q \subseteq \mathbb{R}^2$. Then we have the following properties:

(1) Linearity: for any $\alpha, \beta \in \mathbb{R}$,

$$\iint_Q (\alpha s + \beta t)(x, y)\mathrm{d}x\mathrm{d}y = \alpha \iint_Q s(x, y)\mathrm{d}x\mathrm{d}y + \beta \iint_Q t(x, y)\mathrm{d}x\mathrm{d}y,$$

(2) Additivity: if Q can be subdivided to rectangles Q_1 and Q_2, then

$$\iint_Q s(x, y)\mathrm{d}x\mathrm{d}y = \iint_{Q_1} s(x, y)\mathrm{d}x\mathrm{d}y + \iint_{Q_2} s(x, y)\mathrm{d}x\mathrm{d}y,$$

(3) Comparison: if $s(x, y) \leq t(x, y)$ for all $(x, y) \in Q$, then

$$\iint_Q s(x, y)\mathrm{d}x\mathrm{d}y \leq \iint_Q t(x, y)\mathrm{d}x\mathrm{d}y.$$

Definition 15.1.7 Let f be a scalar field defined and bounded on a rectangle Q. If there exists a unique number $I \in \mathbb{R}$ such that

$$\iint_Q s(x, y)\mathrm{d}x\mathrm{d}y \leq I \leq \iint_Q t(x, y)\mathrm{d}x\mathrm{d}y$$

for every pair of step functions s and t defined on Q with $s(x, y) \leq f(x, y) \leq t(x, y)$ for all $(x, y) \in Q$, then we call f **integrable** on Q and call I the **double integral** of f over Q, denoted by

$$I := \iint_Q f(x, y)\mathrm{d}x\mathrm{d}y.$$

Remark 15.1.8 If $s(x, y) \leq f(x, y)$ for all $(x, y) \in Q$, then we say s is **below** f and often write $s \leq f$. Similarly, if $f(x, y) \leq t(x, y)$ for all $(x, y) \in Q$, then we say t is **above** f and often write $f \leq t$.

Definition 15.1.9 Let f be a scalar field defined and bounded on a rectangle $Q \subseteq \mathbb{R}^2$. Let

$$S := \left\{ \iint_Q s(x, y)\mathrm{d}x\mathrm{d}y : s \text{ is a step function defined on } Q \text{ with } s \leq f \right\},$$
$$T := \left\{ \iint_Q t(x, y)\mathrm{d}x\mathrm{d}y : t \text{ is a step function defined on } Q \text{ with } f \leq t \right\}.$$

We call $\sup S$ the **lower integral** of f over Q, denoted by $\underline{I}(f)$, and call $\inf T$ the **upper integral** of f over Q, denoted by $\overline{I}(f)$.

Theorem 15.1.10 Let f be a scalar field defined and bounded on a rectangle $Q \subseteq \mathbb{R}^2$. Then

1. $\displaystyle\iint_Q s\,\mathrm{d}x\mathrm{d}y \leq \underline{I}(f) \leq \overline{I}(f) \leq \iint_Q t\,\mathrm{d}x\mathrm{d}y$ for any step functions $s \leq f \leq t$;
2. f is integrable on Q if and only if $\underline{I}(f) = \overline{I}(f)$, in which case $\displaystyle\iint_Q f\mathrm{d}x\mathrm{d}y = \underline{I}(f) = \overline{I}(f)$.

Theorem 15.1.11 Let f be a scalar field defined, bounded and integrable on a rectangle $Q = [a, b] \times [c, d]$. For every fixed $y \in [c, d]$, suppose $\displaystyle\int_a^b f(x, y)\mathrm{d}x$ exists and denote it by $A(y)$. If $\displaystyle\int_c^d A(y)\mathrm{d}y$ exists, then we have

$$\iint_Q f(x, y)\mathrm{d}x\mathrm{d}y = \int_c^d A(y)\mathrm{d}y = \int_c^d \left[\int_a^b f(x, y)\mathrm{d}x \right] \mathrm{d}y.$$

Proof. For any step functions $s \leq f \leq t$ on Q, we have

$$\int_a^b s(x, y)\mathrm{d}x \leq \int_a^b f(x, y)\mathrm{d}x = A(y) \leq \int_a^b t(x, y)\mathrm{d}x.$$

Then integrating both sides gives

$$\int_c^d \left[\int_a^b s(x,y)\mathrm{d}x\right] \mathrm{d}y \leq \int_c^d A(y)\mathrm{d}y \leq \int_c^d \left[\int_a^b t(x,y)\mathrm{d}x\right] \mathrm{d}y$$

for any arbitrary step functions $s \leq f \leq t$ on Q. By definition, we have

$$\iint_Q f(x,y)\mathrm{d}x\mathrm{d}y = \int_c^d A(y)\mathrm{d}y = \int_c^d \int_a^b f(x,y)\mathrm{d}x\mathrm{d}y.$$

$\square$

Remark 15.1.12 Alternatively, we could also evaluate $A(x) = \displaystyle\int_c^d f(x,y)\mathrm{d}y$ and then calculate $\displaystyle\int_a^b A(x)\mathrm{d}x$ to get the same result.

Definition 15.1.13 Suppose f is non-negative on a rectangle $Q \subseteq \mathbb{R}^2$, the set

$$\mathcal{S} := \{(x,y,z) \in \mathbb{R}^3 : (x,y) \in Q, 0 \leq z \leq f(x,y)\}$$

is called the **ordinate set** of f over Q.

Remark 15.1.14 If f is non-negative and integrable on Q, then $\displaystyle\iint_Q f(x,y)\mathrm{d}x\mathrm{d}y = \mathrm{vol}(\mathcal{S})$, where $\mathrm{vol}(\mathcal{S})$ is the volume of $\mathcal{S}$. Particularly, for any $y_0 \in [c,d]$, $A(y_0) = \displaystyle\int_a^b f(x,y_0)\mathrm{d}x$ describes the area of the cross section of $\mathcal{S}$ at $y = y_0$. Moreover, $\displaystyle\iint_Q f(x,y)\mathrm{d}x\mathrm{d}y = \int_c^d A(y)\mathrm{d}y$ can be interpreted that the volume of $\mathcal{S}_f$ is the integral of the cross-sectional area function $A(y)$ from $y = c$ to $y = d$.

Theorem 15.1.15 (Fubini's Theorem) If a scalar field f is continuous on a rectangle $Q = [a,b] \times [c,d]$, then f is integrable on Q. Moreover, we have

$$\iint_Q f(x,y)\mathrm{d}x\mathrm{d}y = \int_c^d \left[\int_a^b f(x,y)\mathrm{d}x\right] \mathrm{d}y = \int_a^b \left[\int_c^d f(x,y)\mathrm{d}y\right] \mathrm{d}x.$$

Proof of Integrability. Since f is continuous on Q, f is also bounded on Q and by Theorem 13.2.20, we know given $\varepsilon > 0$, there is a partition P of Q such that the span of f on each of the closed subrectangles in P is less than ε. For convenience, suppose P decomposes Q into $Q_1, \ldots, Q_N$ subrectangles and the span of f on Q_k is denoted by $M_k(f) - m_k(f)$ and satisfies $M_k(f) - m_k(f) < \varepsilon$ for $k = 1, \ldots, N$. Then we can define a pair of step functions s and t on Q as:

- for $x \in \operatorname{int} Q_k$, let $s(x) := m_k(f)$ and $t(x) := M_k(f)$;
- for $x \in \partial Q_k$, let $s(x) := m$ and $t(x) := M$, where m and M are the absolute minimum and maximum values of f on Q, respectively.

Then we have $s \le f \le t$ on Q and

$$\iint_Q s = \sum_{k=1}^{N} m_k(f) \cdot A(Q_k), \qquad \iint_Q t = \sum_{k=1}^{N} M_k(f) \cdot A(Q_k),$$

where $A(\cdot)$ denotes the area function. Thus, we have

$$0 \le \iint_Q t - \iint_Q s = \sum_{k=1}^{N} (M_k(f) - m_k(f)) \cdot A(Q_k) < \varepsilon \sum_{k=1}^{N} A(Q_k) = \varepsilon A(Q).$$

Note

$$0 \le \overline{I}(f) - \underline{I}(f) \le \iint_Q t - \iint_Q s < \varepsilon A(Q),$$

which implies $\overline{I}(f) = \underline{I}(f)$ since ε can be arbitrarily small. Therefore, f is integrable on Q. □

Proof. For any fixed $y \in [c, d]$, $f(x, y)$ is a continuous function of x on $[a, b]$, so $A(y) := \int_a^b f(x, y)\mathrm{d}x$ is well-defined. Next, we need to show $A(y)$ is integrable on $[c, d]$. We will do this by showing that $A(y)$ is continuous on $[c, d]$. For any $y, y_0 \in [c, d]$, we have

$$|A(y) - A(y_0)| = \left| \int_a^b f(x, y)\mathrm{d}x - \int_a^b f(x, y_0)\mathrm{d}x \right| \le \int_a^b |f(x, y) - f(x, y_0)|\mathrm{d}x \le (b-a) \cdot |f(x_0, y) - f(x_0, y_0)|$$

for some $x_0 \in [a, b]$. Thus

$$|A(y) - A(y_0)| \to 0 \quad \text{as } y \to y_0,$$

which implies $A(y)$ is continuous at every $y_0 \in [c, d]$. Then

$$\iint_Q f(x, y)\mathrm{d}x\mathrm{d}y = \int_c^d \left[\int_a^b f(x, y)\mathrm{d}x \right] \mathrm{d}y.$$

□

Remark 15.1.16 There is some functions $f : \mathbb{R}^2 \to \mathbb{R}$ not continuous on Q but the set of discontinuities of f is "not too large", then f still remains integrable on Q.

Definition 15.1.17 Let B be a bounded set in $\mathbb{R}^2$. B is said to **have content zero** if, for every $\varepsilon > 0$, there is a finite collection of rectangles whose union contains B and the sum of whose areas does not exceed ε. Formally, there exists rectangles $R_1, \ldots, R_k$ such that

$$B \subseteq \bigcup_{i=1}^{k} R_i, \qquad \sum_{i=1}^{k} A(R_i) < \varepsilon,$$

where $A(\cdot)$ denotes the area function.

Example 15.1.18
1. A set containing finitely many points in $\mathbb{R}^2$ has content zero.
2. A line segment in $\mathbb{R}^2$ has content zero.

3. If $A_1, A_2, \ldots, A_n$ all have content zero, then $\bigcup_{i=1}^{n} A_i$ has content zero.
4. If A has content zero and $B \subseteq A$, then B has content zero.

Theorem 15.1.19 If a scalar field f is defined and bounded on a rectangle Q and the set of discontinuities of f has content zero, then f is integrable on Q.

Proof. Let $M > 0$ be such that $|f(x,y)| \leq M$ for every $(x,y) \in Q$. Let

$$D = \{(x,y) \in Q : f \text{ is not continuous at } (x,y)\}$$

be the set of discontinuities of f. Since D has content zero, for every $\varepsilon > 0$, we can pick a partition P of Q such that:

- suppose $\{Q_1, \ldots, Q_d\}$ denotes the closed subrectangles of P such that

$$Q_k \cap D \neq \varnothing, \qquad D \subseteq \bigcup_{k=1}^{d} Q_k \qquad \text{with } \sum_{k=1}^{d} A(Q_k) < \varepsilon.$$

- suppose $\{Q_{d+1}, \ldots, Q_N\}$ denotes the rest of the closed subrectangles in P. Then f is continuous on each Q_k for $k = d+1, \ldots, N$. Furthermore, by Theorem 13.2.20, the span of f on each Q_k, denoted by $M_k(f) - m_k(f)$, is less than ε for $k = d+1, \ldots, N$.

Then we can define a pair of step functions s and t on Q as

$$s(x) := \begin{cases} -M & \text{if } x \in \operatorname{int} Q_k \text{ for } 1 \leq k \leq d, \\ m_k(f) & \text{if } x \in \operatorname{int} Q_k \text{ for } d+1 \leq k \leq N, \\ -M & \text{if } x \in \partial Q_k \text{ for } 1 \leq k \leq N, \end{cases} \quad t(x) := \begin{cases} M & \text{if } x \in \operatorname{int} Q_k \text{ for } 1 \leq k \leq d, \\ M_k(f) & \text{if } x \in \operatorname{int} Q_k \text{ for } d+1 \leq k \leq N, \\ M & \text{if } x \in \partial Q_k \text{ for } 1 \leq k \leq N. \end{cases}$$

Then $s \leq f \leq t$ on Q and

$$0 \leq \overline{I}(f) - \underline{I}(f) \leq \iint_Q t - \iint_Q s = \sum_{k=1}^{d} 2M \cdot A(Q_k) + \sum_{k=d+1}^{N} (M_k(f) - m_k(f)) \cdot A(Q_k) < 2M\varepsilon + \varepsilon A(Q)$$

for any $\varepsilon > 0$. Thus, $\overline{I}(f) = \underline{I}(f)$, which implies f is integrable on Q. $\square$

2 Double Integrals over more General Regions

Definition 15.2.1 Suppose $S \subseteq \mathbb{R}^2$ is a bounded region and there is a rectangle $Q = [a,b] \times [c,d]$ such that $S \subseteq Q$. Let f be a scalar field defined and bounded on S. We define the **etended scalar field** $\widetilde{f}$ of f on Q by

$$\widetilde{f}(x,y) := \begin{cases} f(x,y) & \text{if } (x,y) \in S, \\ 0 & \text{if } (x,y) \in Q \setminus S. \end{cases}$$

If $\widetilde{f}$ is integrable on Q, then we say f is **integrable** on S and define the **double integral** of f over S by

$$\iint_S f(x,y)\mathrm{d}x\mathrm{d}y := \iint_Q \widetilde{f}(x,y)\mathrm{d}x\mathrm{d}y.$$

Remark 15.2.2 We need to show that the definition does not depend on the choice of the rectangle Q. Otherwise, the definition is not well-defined. Suppose $Q_1 = [a_1, b_1] \times [c_1, d_1]$ and $Q_2 = [a_2, b_2] \times [c_2, d_2]$ are two arbitrary rectangles containing S. Now define $\widetilde{f}_1$ and $\widetilde{f}_2$ to be the extended scalar fields of f on Q_1 and Q_2, respectively. Note $S \subseteq Q_1 \cap Q_2$. Thus if $\widetilde{f}_1$ is integrable on Q_1, then $\widetilde{f}_1$ is also integrable on $Q_1 \cap Q_2$ and

$$\iint_{Q_1} \widetilde{f}_1 = \iint_{Q_1\cap Q_2} \widetilde{f}_1 = \iint_{Q_1\cap Q_2} \widetilde{f}_2 = \iint_{Q_2} \widetilde{f}_2.$$

Therefore, the definition of the double integral of f over S is invariant under the choice of the rectangle Q.

Definition 15.2.3 We define two types of regions in $\mathbb{R}^2$ as follows:

(1) a subset S_I of $\mathbb{R}^2$ is called a **type I region** if there are two continuous functions $\varphi_1, \varphi_2 : [a, b] \to \mathbb{R}$ satisfying $\varphi_1(x) \le \varphi_2(x)$ for all $x \in [a, b]$ such that

$$S_I = \{(x, y) \in \mathbb{R}^2 : a \le x \le b, \varphi_1(x) \le y \le \varphi_2(x)\};$$

(2) a subset S_{II} of $\mathbb{R}^2$ is called a **type II region** if there are two continuous functions $\psi_1, \psi_2 : [c, d] \to \mathbb{R}$ satisfying $\psi_1(y) \le \psi_2(y)$ for all $y \in [c, d]$ such that

$$S_{II} = \{(x, y) \in \mathbb{R}^2 : c \le y \le d, \psi_1(y) \le x \le \psi_2(y)\}.$$

Remark 15.2.4 We will consider integrals $\displaystyle\iint_S f$ for regions $S \subseteq \mathbb{R}^2$ that are of type I, type II, or that can be partitioned into finitely many subregions of type I or type II.

Theorem 15.2.5 If $\varphi : [a, b] \to \mathbb{R}$ is a continuous function, then the graph of φ, i.e., the set $\{(x, y) \in \mathbb{R}^2 : y = \varphi(x), a \le x \le b\}$, has content zero in $\mathbb{R}^2$.

Theorem 15.2.6 Let $S \subseteq \mathbb{R}^2$ be a region of type I between the graphs of two continuous functions $\varphi_1, \varphi_2 : [a, b] \to \mathbb{R}$. Suppose $f : S \to \mathbb{R}$ is a scalar field bounded on S and continuous on $\operatorname{int} S$. Then f is integrable on S and its value is given by

$$\iint_S f(x,y)\mathrm{d}x\mathrm{d}y = \int_a^b \left[\int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)\mathrm{d}y\right]\mathrm{d}x.$$

Corollary 15.2.7 Let $S \subseteq \mathbb{R}^2$ be a region of type II between the graphs of two continuous functions $\psi_1, \psi_2 : [c, d] \to \mathbb{R}$. Suppose $f : S \to \mathbb{R}$ is a scalar field bounded on S and continuous on $\operatorname{int} S$. Then f is integrable on

S and its value is given by

$$\iint_S f(x,y)\mathrm{d}x\mathrm{d}y = \int_c^d \left[\int_{\psi_1(y)}^{\psi_2(y)} f(x,y)\mathrm{d}x\right]\mathrm{d}y.$$

Example 15.2.8 The solid enclosed by the ellipsoid $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ is given by the set

$$E_{a,b,c} := \left\{(x,y,z)\in\mathbb{R}^3 : \frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}\le 1\right\}.$$

Let

$$S_{a,b} = \left\{(x,y)\in\mathbb{R}^2 : \frac{x^2}{a^2}+\frac{y^2}{b^2}\le 1\right\}$$

and define $f_{a,b,c} : S_{a,b}\to\mathbb{R}$ by

$$f(x,y) = c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}.$$

Then $f_{a,b,c}$ is continuous on $S_{a,b}$ and the ordinate set of f over $S_{a,b}$ is exactly $E_{a,b,c}$. Thus, we have

$$\begin{aligned}
\mathrm{vol}(E_{a,b,c}) &= \iint_{S_{a,b}} [f(x,y)-(-f(x,y))]\,\mathrm{d}x\mathrm{d}y \\
&= 2\int_{-a}^{a}\left[\int_{-b\sqrt{1-(x/a)^2}}^{b\sqrt{1-(x/a)^2}} c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}\mathrm{d}y\right]\mathrm{d}x \\
&= 2\int_{-a}^{a}\left[\int_{-bA}^{bA} c\sqrt{A^2-\frac{y^2}{b^2}}\mathrm{d}y\right]\mathrm{d}x \quad \text{where } A=\sqrt{1-\frac{x^2}{a^2}} \\
&= 2\int_{-a}^{a} A^2bc\left[\int_{-\pi/2}^{\pi/2}\cos^2\theta\mathrm{d}\theta\right]\mathrm{d}x \quad \text{where } y=bA\sin\theta, \theta\in[-\pi/2,\pi/2] \\
&= 2\int_{-a}^{a} A^2bc\frac{\pi}{2}\mathrm{d}x \\
&= \pi abc\int_{-a}^{a}\left(1-\frac{x^2}{a^2}\right)\mathrm{d}x \\
&= \frac{4}{3}\pi abc.
\end{aligned}$$

3 Green's Theorem

Definition 15.3.1 Suppose C is a curve described by a continuous vector-valued function $\boldsymbol{\alpha} : [a,b]\to\mathbb{R}^n$. Then

(1) we say C is a **closed curve** if $\boldsymbol{\alpha}(a)=\boldsymbol{\alpha}(b)$;

(2) we say C is a **simple closed curve** if C is a closed curve and $\boldsymbol{\alpha}(t_1)\ne\boldsymbol{\alpha}(t_2)$ for any $t_1,t_2\in[a,b)$ with $t_1\ne t_2$.

(3) we say C is a **Jordan curve** if it is a simple closed curve in a plane, i.e., $\alpha : [a,b]\to\mathbb{R}^2$.

(4) The Jordan Curve Theorem (the proof is beyond the scope of this text) states that if C is a Jordan curve, then $\mathbb{R}^2\setminus C$ is the disjoint union of two connected open subsets. One of these components is bounded and

is called the **inner region**. The other component is unbounded and is called the **outer region**. Moreover, C is the common boundary of the inner and outer regions.

Theorem 15.3.2 (Green's Theorem in the Plane) Let P and Q be two scalar fields defined and continuously differentiable on an open open set $S \subseteq \mathbb{R}^2$. Let C be a piecewise smooth Jordan curve and let R be the region enclosed by C (containing C). If $R \subseteq S$, then we have

$$\oint_C P\mathrm{d}x + Q\mathrm{d}y = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}x\mathrm{d}y.$$

Example 15.3.3 Let C be a piecewise smooth Jordan curve and let R be the region enclosed by C. Then we have

$$A(R) = \iint_R 1\mathrm{d}x\mathrm{d}y = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}x\mathrm{d}y = \oint_C P\mathrm{d}x + Q\mathrm{d}y = \frac{1}{2}\oint_C x\mathrm{d}y - y\mathrm{d}x,$$

where we take $Q(x, y) = \dfrac{x}{2}$ and $P(x, y) = -\dfrac{y}{2}$ to apply Green's Theorem. In particular, if C is described by the parameterization $x = X(t)$ and $y = Y(t)$ for $t \in [a, b]$, then

$$A(R) = \frac{1}{2}\int_a^b X(t)Y'(t) - Y(t)X'(t)\mathrm{d}t = \frac{1}{2}\int_a^b \det\begin{pmatrix} X(t) & Y(t) \\ X'(t) & Y'(t)\end{pmatrix} \mathrm{d}t.$$

4 Change of Variables in Double Integrals

Theorem 15.4.1 Let S and T be two bounded regions in $\mathbb{R}^2$ and let $f : S \to \mathbb{R}$ be a scalar field defined, bounded and integrable on S. Suppose there exists a function $\boldsymbol{X} : T \to S$ where $\boldsymbol{X} = (X_1, X_2)$ satisfies the following conditions:

(1) X_1 and X_2 are continuously differentiable on T;

(2) $\boldsymbol{X}$ is invertible;

(3) The Jacobian determinant

$$J(u, v) := \det\begin{pmatrix} \dfrac{\partial X_1}{\partial u} & \dfrac{\partial X_2}{\partial u} \\ \dfrac{\partial X_1}{\partial v} & \dfrac{\partial X_2}{\partial v}\end{pmatrix} \neq 0, \qquad \text{for every } (u, v) \in T.$$

Then we have

$$\iint_S f(x, y)\mathrm{d}x\mathrm{d}y = \iint_T f(X_1(u, v), X_2(u, v))|J(u, v)|\mathrm{d}u\mathrm{d}v.$$

Remark 15.4.2 Recall the 1d version

$$\int_{u(a)}^{u(b)} f(x)\mathrm{d}x = \int_a^b f(u(t))u'(t)\mathrm{d}t,$$

where $u : [a, b] \to \mathbb{R}$ is a continuously differentiable function. Note there's no absolute value sign in the 1d version.

Moreover, the conclusion of Theorem 15.4.1 still holds if we allow $\boldsymbol{X}$ to be non-invertible on a set of content zero, or $J(u,v)$ to vanish on such a subset.

Example 15.4.3 In the polar coordinate system, we have

$$\begin{cases} x = r\cos\theta, \\ y = r\sin\theta. \end{cases} \qquad r \geq 0, \theta \in [0, 2\pi).$$

Note $\boldsymbol{X}(r,\theta) = (r\cos\theta, r\sin\theta) : [0,\infty)\times[0,2\pi) \to \mathbb{R}^2$ is not bijective on $0 \times [0, 2\pi)$, which is a set of content zero. Thus we can still apply Theorem 15.4.1 to get

$$J(r,\theta) = \det\begin{pmatrix} \cos\theta & \sin\theta \\ -r\sin\theta & r\cos\theta \end{pmatrix} = r.$$

Therefore, for any scalar field f defined, bounded and integrable on a region $S \subseteq \mathbb{R}^2$, we have

$$\iint_S f(x,y)\mathrm{d}x\mathrm{d}y = \iint_T f(r\cos\theta, r\sin\theta) r\mathrm{d}r\mathrm{d}\theta,$$

where $T = \{(r,\theta) : r \geq 0, \theta \in [0,2\pi), (r\cos\theta, r\sin\theta) \in S\}$.

Example 15.4.4 We want to evaluate the Gaussian integral

$$\int_{-\infty}^{\infty} e^{-x^2}\mathrm{d}x = \sqrt{\pi}.$$

For any $R > 0$, let $I(R) := \displaystyle\int_{-R}^{R} e^{-x^2}\mathrm{d}x$. Then we have

$$I^2 = \int_{-R}^{R} e^{-x^2}\mathrm{d}x \cdot \int_{-R}^{R} e^{-y^2}\mathrm{d}y = \iint_{[-R,R]^2} e^{-(x^2+y^2)}\mathrm{d}x\mathrm{d}y.$$

Let

$$D_R = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq R^2\}, \quad D_{\sqrt{2}R} = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 2R^2\}.$$

Then we have $D_R \subseteq [-R,R]^2 \subseteq D_{\sqrt{2}R}$ and

$$\iint_{D_R} e^{-(x^2+y^2)}\mathrm{d}x\mathrm{d}y \leq I^2 \leq \iint_{D_{\sqrt{2}R}} e^{-(x^2+y^2)}\mathrm{d}x\mathrm{d}y.$$

Applying the change of variables in polar coordinates, for $\lambda > 0$, we have

$$\iint_{D_\lambda} e^{-(x^2+y^2)}\mathrm{d}x\mathrm{d}y = \int_0^{2\pi}\left[\int_0^{\lambda} e^{-r^2} r\mathrm{d}r\right]\mathrm{d}\theta = 2\pi\int_0^{\lambda} e^{-r^2} r\mathrm{d}r = \pi(1 - e^{-\lambda^2}) \to \pi \quad \text{as } \lambda \to \infty.$$

Thus, by taking $\lambda = R$ and $\lambda = \sqrt{2}R$ and letting $R \to \infty$, we have

$$\lim_{R\to\infty} (I(R))^2 = \pi.$$

However, we still need to show that

$$\lim_{R\to\infty}(I(R))^2 = \left(\lim_{R\to\infty} I(R)\right)^2$$

by showing that $\displaystyle\int_{-\infty}^{\infty} e^{-x^2}\mathrm{d}x$ exists. By symmetry, it suffices to show $\displaystyle\int_{0}^{\infty} e^{-x^2}\mathrm{d}x$ exists. Note $\displaystyle\int_{0}^{1} e^{-x^2}\mathrm{d}x$ is finite since e^{-x^2} is continuous on $[0,1]$. Thus we only need to show $\displaystyle\int_{1}^{\infty} e^{-x^2}\mathrm{d}x$ exists. By comparison test, we have

$$0 \le \int_1^R e^{-x^2}\mathrm{d}x \le \int_1^R xe^{-x^2}\mathrm{d}x = \frac{1}{2}\left(\frac{1}{e} - e^{-R^2}\right) \to \frac{1}{2e} \quad \text{as } R\to\infty.$$

Hence,

$$\lim_{R\to\infty} I(R) = \sqrt{\lim_{R\to\infty}(I(R))^2} = \sqrt{\pi}.$$

5 Multiple Integrals in Higher Dimensions

The idea of a double integral can be extended to higher dimensions. The following definitions and results are the natural generalizations of the corresponding ones in the 2d case. We will state the definitions and results without proofs since the proofs are analogous.

Definition 15.5.1 Let $f:\mathbb{R}^n\to\mathbb{R}$ be a scalar field defined and bounded on a bounded region $S\subseteq\mathbb{R}^n$ for any $n\ge 2$. In particular, if $n=3$, we call the integral a **triple integral** and denote it by

$$\iiint_S f(x,y,z)\mathrm{d}x\mathrm{d}y\mathrm{d}z \quad \text{or} \quad \iiint_S f(x,y,z).$$

In general, we call the integral a n**-fold integral** and denote it by

$$\int\cdots\int_S f(x_1,\ldots,x_n)\mathrm{d}x_1\cdots\mathrm{d}x_n \quad \text{or} \quad \int\cdots\int_S f(x_1,\ldots,x_n).$$

Definition 15.5.2 Given n closed intervals $[a_1,b_1],\ldots,[a_n,b_n]$, we define the **hyperrectangle** Q in $\mathbb{R}^n$ by

$$Q=[a_1,b_1]\times\cdots\times[a_n,b_n]=\{(x_1,\ldots,x_n)\in\mathbb{R}^n : a_i\le x_i\le b_i \text{ for } i=1,\ldots,n\}.$$

Definition 15.5.3 Let $Q:=[\boldsymbol{a},\boldsymbol{b}]=[a_1,b_1]\times\cdots\times[a_n,b_n]$ be a hyperrectangle in $\mathbb{R}^n$ and $P=P_1\times\cdots\times P_n$ be a partition of Q. Let f be a step function defined on Q that takes a constant value c_i on each open subinterval $Q_i:=(a_1^{(i)},b_1^{(i)})\times\cdots\times(a_n^{(i)},b_n^{(i)})$ of P. Then we define **the integral of a step function** f on Q by

$$\int\cdots\int_Q f(x_1,\ldots,x_n)\mathrm{d}x_1\cdots\mathrm{d}x_n := \sum_i\left(c_i\prod_{j=1}^n(b_j^{(i)}-a_j^{(i)})\right).$$

Definition 15.5.4 Suppose $f:Q\to\mathbb{R}$ is a scalar field defined and bounded on a hyperrectangle $Q\subseteq\mathbb{R}^n$. We say f is **integrable** on Q if there is a unique $I \in \mathbb{R}$ such that

$$\int \cdots \int_Q s \leq I \leq \int \cdots \int_Q t$$

for all step functions s and t defined on Q satisfying $s \leq f \leq t$ on Q. In this case, we define and denote the **integral** of f on Q by

$$\int \cdots \int_Q f(x_1, \ldots, x_n) \mathrm{d}x_1 \cdots \mathrm{d}x_n := I.$$

Remark 15.5.5 In particular, if f is continuous on Q, then f is integrable on Q. Moreover, if f is defined and bounded on Q and the set of discontinuities of f has content zero, then f is integrable on Q.

Definition 15.5.6 Suppose that $S \subseteq \mathbb{R}^n$ is a bounded region and there is a hyperrectangle $Q = [\boldsymbol{a}, \boldsymbol{b}]$ such that $S \subseteq Q$. Let f be a scalar field defined and bounded on S. We define the **extended scalar field** $\widetilde{f}$ of f on Q by

$$\widetilde{f}(\boldsymbol{x}) := \begin{cases} f(\boldsymbol{x}) & \text{if } \boldsymbol{x} \in S, \\ 0 & \text{if } \boldsymbol{x} \in Q \setminus S. \end{cases}$$

We say that f is **integrable** on S if $\widetilde{f}$ is integrable on Q. In this case, we define and denote the **integral** of f on S by

$$\int \cdots \int_S f(x_1, \ldots, x_n) \mathrm{d}x_1 \cdots \mathrm{d}x_n := \int \cdots \int_Q \widetilde{f}(x_1, \ldots, x_n) \mathrm{d}x_1 \cdots \mathrm{d}x_n.$$

Theorem 15.5.7 Let $S, T \subseteq \mathbb{R}^n$ be two bounded regions and let $f : S \to \mathbb{R}$ be a scalar field defined, bounded and integrable on S. Suppose there exists a function $\boldsymbol{X} : T \to S$ where $\boldsymbol{X} = (X_1, \ldots, X_n)$ satisfies the following conditions:

(1) $X_1, \ldots, X_n$ are continuously differentiable on T;

(2) $\boldsymbol{X}$ is invertible;

(3) The Jacobian determinant

$$J(u_1, \ldots, u_n) := \det \begin{pmatrix} \dfrac{\partial X_1}{\partial u_1} & \cdots & \dfrac{\partial X_n}{\partial u_1} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial X_1}{\partial u_n} & \cdots & \dfrac{\partial X_n}{\partial u_n} \end{pmatrix} \neq 0, \qquad \text{for every } (u_1, \ldots, u_n) \in T.$$

Then we have

$$\int \cdots \int_S f(x_1, \ldots, x_n) \mathrm{d}x_1 \cdots \mathrm{d}x_n = \int \cdots \int_T f(\boldsymbol{X}(u_1, \ldots, u_n)) |J(u_1, \ldots, u_n)| \mathrm{d}u_1 \cdots \mathrm{d}u_n.$$

Moreover, the conclusion still holds if we allow $\boldsymbol{X}$ to be non-invertible on a set of content zero, or $J(u_1, \ldots, u_n)$ to vanish on such a subset.

Remark 15.5.8 Note this matrix is the transpose of the Jacobian matrix of $\boldsymbol{X}$. However, the determinant is invariant under transposition, so we can still call it the Jacobian determinant of $\boldsymbol{X}$.

Corollary 15.5.9 In the cylindrical coordinate system, we have

$$\begin{cases} x = r\cos\theta, \\ y = r\sin\theta, \\ z = z. \end{cases} \qquad r \geq 0, \theta \in [0, 2\pi), z \in \mathbb{R}.$$

Then

$$J(r, \theta, z) = \det \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -r\sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = r.$$

Therefore, for any scalar field f defined, bounded and integrable on a region $S \subseteq \mathbb{R}^3$, we have

$$\iiint_S f(x, y, z)\mathrm{d}x\mathrm{d}y\mathrm{d}z = \iiint_T f(r\cos\theta, r\sin\theta, z) \cdot r \cdot \mathrm{d}r\mathrm{d}\theta\mathrm{d}z,$$

where $T = \{(r, \theta, z) : r \geq 0, \theta \in [0, 2\pi), z \in \mathbb{R}, (r\cos\theta, r\sin\theta, z) \in S\}$.

Corollary 15.5.10 In the spherical coordinate system, we have

$$\begin{cases} x = \rho\sin\varphi\cos\theta, \\ y = \rho\sin\varphi\sin\theta, \\ z = \rho\cos\varphi. \end{cases} \qquad \rho \geq 0, \theta \in [0, 2\pi), \varphi \in [0, \pi].$$

Then

$$J(\rho, \theta, \varphi) = \det \begin{pmatrix} \sin\varphi\cos\theta & \sin\varphi\sin\theta & \cos\varphi \\ -\rho\sin\varphi\sin\theta & \rho\sin\varphi\cos\theta & 0 \\ \rho\cos\varphi\cos\theta & \rho\cos\varphi\sin\theta & -\rho\sin\varphi \end{pmatrix} = -\rho^2\sin\varphi.$$

Therefore, for any scalar field f defined, bounded and integrable on a region $S \subseteq \mathbb{R}^3$, we have

$$\iiint_S f(x, y, z)\mathrm{d}x\mathrm{d}y\mathrm{d}z = \iiint_T f(\rho\sin\varphi\cos\theta, \rho\sin\varphi\sin\theta, \rho\cos\varphi) \cdot \rho^2\sin\varphi \cdot \mathrm{d}\rho\mathrm{d}\theta\mathrm{d}\varphi,$$

where $T = \{(\rho, \theta, \varphi) : \rho \geq 0, \theta \in [0, 2\pi), \varphi \in [0, \pi], (\rho\sin\varphi\cos\theta, \rho\sin\varphi\sin\theta, \rho\cos\varphi) \in S\}$.

Example 15.5.11 Let $E_{a,b,c}$ be the solid enclosed by the ellipsoid $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$. Applying the change of variables

$$\begin{cases} x = a\rho\sin\varphi\cos\theta, \\ y = b\rho\sin\varphi\sin\theta, \\ z = c\rho\cos\varphi, \end{cases} \qquad \rho \geq 0, \theta \in [0, 2\pi), \varphi \in [0, \pi],$$

we have

$$\begin{aligned}
\operatorname{vol}(E_{a,b,c}) &= \iiint_{E_{a,b,c}} 1\mathrm{d}x\mathrm{d}y\mathrm{d}z \\
&= \iiint_T abc\rho^2 \sin\varphi \mathrm{d}\rho\mathrm{d}\theta\mathrm{d}\varphi \\
&= abc \int_0^1 \rho^2 \mathrm{d}\rho \cdot \int_0^{2\pi} \mathrm{d}\theta \cdot \int_0^{\pi} \sin\varphi \mathrm{d}\varphi \\
&= abc \cdot \frac{1}{3} \cdot 2\pi \cdot 2 = \frac{4}{3}abc.
\end{aligned}$$

This results is consistent with the one we obtained in Example 15.2.8 but with much less computational effort.

Lemma 15.5.12 For any positive integer n and any $a > 0$, we have

$$V_n(a) = a^n V_n(1),$$

where $V_n(a)$ is the volume of the n-ball $B_n(a) = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_1^2 + \cdots + x_n^2 \le a^2\}$.

Proof. Let $\boldsymbol{X} : B_n(1) \to B_n(a)$ be the function defined by $\boldsymbol{X}(x_1, \ldots, x_n) = (ax_1, \ldots, ax_n)$. Then $\boldsymbol{X}$ is a bijection and

$$J(x_1, \ldots, x_n) = \det \operatorname{diag}(a, \ldots, a) = a^n.$$

Thus, by the change of variables in higher dimensions, we have

$$V_n(a) = \int \cdots \int_{B_n(a)} 1\mathrm{d}x_1 \cdots \mathrm{d}x_n = \int \cdots \int_{B_n(1)} a^n \mathrm{d}x_1 \cdots \mathrm{d}x_n = a^n V_n(1).$$

□

Lemma 15.5.13 For any positive integer n, we have

$$V_{n+2}(1) = \frac{2\pi}{n+2} V_n(1).$$

Proof. Note

$$\sum_{i=1}^{n+2} x_i^2 \le 1 \Leftrightarrow \begin{cases} x_{n+1}^2 + x_{n+2}^2 \le 1 \\ (x_1, \ldots, x_n) \in B_n(\sqrt{1 - x_{n+1}^2 - x_{n+2}^2}) \end{cases}$$

Thus, by the change of variables in higher dimensions, we have

$$\begin{aligned}
V_{n+2}(1) &= \int \cdots \int_{B_{n+2}(1)} 1 \mathrm{d}x_1 \cdots \mathrm{d}x_{n+2} \\
&= \iint_{x_{n+1}^2 + x_{n+2}^2 \leq 1} \left[\int \cdots \int_{B_n(\sqrt{1-x_{n+1}^2-x_{n+2}^2})} 1 \mathrm{d}x_1 \cdots \mathrm{d}x_n \right] \mathrm{d}x_{n+1} \mathrm{d}x_{n+2} \\
&= \iint_{x_{n+1}^2 + x_{n+2}^2 \leq 1} V_n(\sqrt{1-x_{n+1}^2-x_{n+2}^2}) \mathrm{d}x_{n+1} \mathrm{d}x_{n+2} \\
&= \iint_{x_{n+1}^2 + x_{n+2}^2 \leq 1} \left(\sqrt{1-x_{n+1}^2-x_{n+2}^2} \right)^n V_n(1) \mathrm{d}x_{n+1} \mathrm{d}x_{n+2} \\
&= V_n(1) \int_0^{2\pi} \left[\int_0^1 (1-r^2)^{n/2} r \mathrm{d}r \right] \mathrm{d}\theta \quad \text{where } x_{n+1} = r\cos\theta, x_{n+2} = r\sin\theta \\
&= \frac{2\pi}{n+2} V_n(1).
\end{aligned}$$

□

Proposition 15.5.14 Suppose $a > 0$ and $V_n(a)$ is the volume of the n-ball

$$B_n(a) = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_1^2 + \cdots + x_n^2 \leq a^2\}.$$

If n is even, i.e., $n = 2m$ for some positive integer m, then

$$V_n(a) = \frac{\pi^m}{m!} a^n.$$

Proof. By Lemma 15.5.12, it suffices to show $V_{2m}(1) = \frac{\pi^m}{m!}$ for any positive integer m. We prove this by induction on m.

Base case: when $m = 1$, we have $V_2(1) = \pi$ since $B_2(1)$ is the unit disk in $\mathbb{R}^2$. Thus, the conclusion holds for $m = 1$.

Induction step: assume the conclusion holds for $m = k$ for some positive integer k, i.e., $V_{2k}(1) = \frac{\pi^k}{k!}$. Then, by Lemma 15.5.13, we have

$$V_{2(k+1)}(1) = \frac{2\pi}{2k+2} V_{2k}(1) = \frac{\pi^{k+1}}{(k+1)!}.$$

Thus, the conclusion holds for $m = k + 1$. Hence, the conclusion holds for all positive integer m. □

Proposition 15.5.15 Suppose $a > 0$ and $V_n(a)$ is the volume of the n-ball

$$B_n(a) = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_1^2 + \cdots + x_n^2 \leq a^2\}.$$

If n is odd, i.e., $n = 2m - 1$ for some non-negative integer m, then

$$V_{2m-1}(a) = \frac{2((m-1)!)(4\pi)^{m-1}}{(2m-1)!} a^{2m-1}.$$

Proof. By Lemma 15.5.12, it suffices to show $V_{2m-1}(1) = \dfrac{2((m-1)!)(4\pi)^{m-1}}{(2m-1)!}$ for any non-negative integer m. We prove this by induction on m.

Base case: when $m = 0$, we have $V_1(1) = 2$ since $B_1(1)$ is the interval $[-1, 1]$ in $\mathbb{R}$. Thus, the conclusion holds for $m = 0$.

Induction step: assume the conclusion holds for $m = k$ for some non-negative integer k, i.e., $V_{2k-1}(1) = \dfrac{2((k-1)!)(4\pi)^{k-1}}{(2k-1)!}$. Then, by Lemma 15.5.13, we have

$$V_{2k+1}(1) = \frac{2\pi}{2k+1} V_{2k-1}(1) = \frac{2\pi}{2k+1} \cdot \frac{2((k-1)!)(4\pi)^{k-1}}{(2k-1)!} = \frac{2(k!)(4\pi)^k}{(2k+1)!}.$$

Thus, the conclusion holds for $m = k + 1$. Hence, the conclusion holds for all non-negative integer m. □

Chapter 16 Surface Integrals

1 Parametric Surfaces

Definition 16.1.1 Suppose $T \subseteq \mathbb{R}^2$ has nonempty and connected interior and $\widetilde{T} \supseteq T$ is an open set of $\mathbb{R}^2$. Let $\boldsymbol{r} : \widetilde{T} \to \mathbb{R}^3$ be a continuous vector function. Then we call the image $S = \boldsymbol{r}(T)$ of the restricted $\boldsymbol{r}\big|_T$ a **parametric surface** with respect to $(\boldsymbol{r}, T)$ and $(\boldsymbol{r}, T)$ a **parametrization** of S. Furthermore, if $\boldsymbol{r}\big|_T$ is injective, then we call S a **simple parametric surface**.

We typically regard S as lying in $xyz-$space, and $\widetilde{T}$ and T as lying in the $uv-$plane. Formally, there exists functions $X, Y, Z : \widetilde{T} \to \mathbb{R}$ such that

$$x = X(u, v), \quad y = Y(u, v), \quad z = Z(u, v).$$

Then the vector equation for S is

$$\boldsymbol{r}(u, v) = X(u, v)\boldsymbol{i} + Y(u, v)\boldsymbol{j} + Z(u, v)\boldsymbol{k}, \quad (u, v) \in \widetilde{T} \supseteq T.$$

Remark 16.1.2 Here, we need restriction of $\boldsymbol{r}$ from $\widetilde{T}$ to T to ensure that the partial derivatives of $\boldsymbol{r}$ is well-defined on the boundary of T. Note the parametrization of a surface is not unique.

Example 16.1.3 Particularly, if a surface S is given explicitly by an equation of the form $z = f(x, y)$, then we can parametrize S by

$$X(u, v) = u, \quad Y(u, v) = v, \quad Z(u, v) = f(u, v).$$

Conversely, suppose that we can solve the equations

$$x = X(u, v), \quad y = Y(u, v)$$

for u and v in terms of x and y, then we can express z as a function of x and y by substituting the expressions for u and v into the equation

$$z = Z(u, v).$$

Definition 16.1.4 Suppose S is a parametric surface with parametrization $(\boldsymbol{r} : \widetilde{T} \to \mathbb{R}^3, T)$ where

$$\boldsymbol{r}(u, v) = X(u, v)\boldsymbol{i} + Y(u, v)\boldsymbol{j} + Z(u, v)\boldsymbol{k}, \quad (u, v) \in \widetilde{T} \supseteq T,$$

where X, Y, Z are differentiable on $\widetilde{T}$. Then the **fundamental vector product** of $\boldsymbol{r}$ is defined by the cross product

$$\frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v} = \left(\frac{\partial X}{\partial u}, \frac{\partial Y}{\partial u}, \frac{\partial Z}{\partial u}\right) \times \left(\frac{\partial X}{\partial v}, \frac{\partial Y}{\partial v}, \frac{\partial Z}{\partial v}\right) = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \frac{\partial X}{\partial u} & \frac{\partial Y}{\partial u} & \frac{\partial Z}{\partial u} \\ \frac{\partial X}{\partial v} & \frac{\partial Y}{\partial v} & \frac{\partial Z}{\partial v} \end{vmatrix}.$$

If $(u_0, v_0) \in T$ is a point such that the partial derivatives $\dfrac{\partial \boldsymbol{r}}{\partial u}$ and $\dfrac{\partial \boldsymbol{r}}{\partial v}$ both exist, and are both continuous at (u_0, v_0) and

$$\left(\frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v}\right)\bigg|_{(u_0, v_0)} \neq \mathbf{0},$$

then we say $\boldsymbol{r}$ is **regular** at (u_0, v_0). If $\boldsymbol{r}$ is not regular at $(u_0, v_0) \in \widetilde{T}$, then we say that $\boldsymbol{r}$ **is singular** at (u_0, v_0). If $\boldsymbol{r}$ is regular at every $(u_0, v_0) \in \widetilde{T}$, then we say that $(\boldsymbol{r}, T)$ is a **smooth parametric representation**.
If $(x, y, z) \in S$ is a point such that, for every $(u_0, v_0) \in T$ with $\boldsymbol{r}(u_0, v_0) = (x, y, z)$, the parametric representation $\boldsymbol{r}$ is regular at (u_0, v_0), then we say that (x, y, z) **is a regular point** of S **(with respect to** $(\boldsymbol{r}, T)$**)**. If $(x, y, z) \in S$ is not a regular point of S (**with respect to** $(\boldsymbol{r}, T)$), then we say that (x, y, z) **is a singular point** of S (**with respect to** $(\boldsymbol{r}, T)$). The parametric surface S is called **smooth** (**with respect to** $(\boldsymbol{r}, T)$) if all of its points are regular points (**with respect to** $(\boldsymbol{r}, T)$).

Remark 16.1.5 Note a regular point for one parametrization of a surface may be a singular point for another parametrization of the same surface. Thus the regularity **does** depend on the choice of parametrization. Moreover, $\boldsymbol{r}$ may not be injective. Thus the definition of a regular point requires that all points in the preimage of (x, y, z) are regular points of $\boldsymbol{r}$.

Example 16.1.6 Follow the setup in Definition 16.1.1 and suppose $f : \widetilde{T} \to \mathbb{R}$ is a differentiable function and S is the surface given explicitly by the equation $z = f(x, y)$ for $(x, y) \in T \subseteq \widetilde{T}$. Then we can parametrize S by

$$\boldsymbol{r}(u, v) = u\boldsymbol{i} + v\boldsymbol{j} + f(u, v)\boldsymbol{k}.$$

Then the fundamental vector product of $\boldsymbol{r}$ is

$$\frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v} = -\frac{\partial f}{\partial x}\boldsymbol{i} - \frac{\partial f}{\partial y}\boldsymbol{j} + \boldsymbol{k}$$

for all $(u, v) \in T$. Thus all the singular points are those points (x, y, z) on S such that one of $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ does not exist.

Example 16.1.7 Suppose the hemisphere

$$S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = a^2, z \geq 0\}$$

is parametrized by

$$\boldsymbol{r}(\theta, \phi) = a(\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta), \quad (\theta, \phi) \in \mathbb{R}^2,$$

and $T = \{(\theta,\phi) : 0 \le \phi < 2\pi, 0 \le \theta \le \frac{\pi}{2}\}$. Then the norm of the fundamental vector product of $\boldsymbol{r}$ is

$$\left\|\frac{\partial \boldsymbol{r}}{\partial \theta} \times \frac{\partial \boldsymbol{r}}{\partial \phi}\right\| = a^2 \left\|-\sin\theta(\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta)\right\| = a^2\sin\theta.$$

Note that, with respect to this parametrization, the parameter values $(0,\phi)$ are singular. However, parametrizing S by

$$\boldsymbol{R}(u,v) = \begin{cases}(u,v,\sqrt{a^2-u^2-v^2}), & (u,v)\in T' \\ (u,v,0), & (u,v)\in\mathbb{R}^2\setminus T'\end{cases},$$

where $T' = \{(u,v) : u^2+v^2 \le a^2\}$, we have

$$\frac{\partial \boldsymbol{R}}{\partial u}(0,0) \times \frac{\partial \boldsymbol{R}}{\partial v}(0,0) = (1,0,0)\times(0,1,0) = (0,0,1) \neq \mathbf{0},$$

showing that the north pole is a regular point of S. Thus the regularity of a point depends on the choice of parametrization. In this case, the set of singular points is $\{(x,y,0) : x^2+y^2 = a^2\}$ since the partial derivatives of $\boldsymbol{R}$ fails to exist and be continuous at these points.(Note they are regular points with respect to the first parametrization.)

Notation 16.1.8 Expanding the fundamental vector product by the first row, we have

$$\frac{\partial \boldsymbol{r}}{\partial u}\times\frac{\partial \boldsymbol{r}}{\partial v} = \left(\frac{\partial Y}{\partial u}\frac{\partial Z}{\partial v} - \frac{\partial Y}{\partial v}\frac{\partial Z}{\partial u}\right)\boldsymbol{i} - \left(\frac{\partial X}{\partial u}\frac{\partial Z}{\partial v} - \frac{\partial X}{\partial v}\frac{\partial Z}{\partial u}\right)\boldsymbol{j} + \left(\frac{\partial X}{\partial u}\frac{\partial Y}{\partial v} - \frac{\partial X}{\partial v}\frac{\partial Y}{\partial u}\right)\boldsymbol{k}.$$

For convenience, we adopt the following notation

$$\frac{\partial(Y,Z)}{\partial(u,v)} := \frac{\partial Y}{\partial u}\frac{\partial Z}{\partial v} - \frac{\partial Y}{\partial v}\frac{\partial Z}{\partial u} = \det\begin{pmatrix}\frac{\partial Y}{\partial u} & \frac{\partial Y}{\partial v} \\ \frac{\partial Z}{\partial u} & \frac{\partial Z}{\partial v}\end{pmatrix}.$$

Then we can write

$$\frac{\partial \boldsymbol{r}}{\partial u}\times\frac{\partial \boldsymbol{r}}{\partial v} = \frac{\partial(Y,Z)}{\partial(u,v)}\boldsymbol{i} - \frac{\partial(X,Z)}{\partial(u,v)}\boldsymbol{j} + \frac{\partial(X,Y)}{\partial(u,v)}\boldsymbol{k}.$$

Proposition 16.1.9 Let S be a parametric surface with parametrization $(\boldsymbol{r} : \widetilde{T} \to \mathbb{R}^3, T)$ where

$$\begin{array}{rccc}\boldsymbol{r}: & \widetilde{T} & \longrightarrow & S\subseteq\mathbb{R}^3 \\ & (u,v) & \longmapsto & (X(u,v),Y(u,v),Z(u,v))\end{array}$$

and $C \subseteq T \subseteq \widetilde{T}$ be a smooth curve with parametrization

$$\begin{array}{rccc}\boldsymbol{\alpha}: & [a,b] & \longrightarrow & C\subseteq T\subseteq\widetilde{T} \\ & t & \longmapsto & (u(t),v(t))\end{array}$$

Then the composite $\boldsymbol{\rho} = \boldsymbol{r}\big|_C \circ \boldsymbol{\alpha} : [a,b] \to \boldsymbol{r}(C) \subseteq S$ is a smooth parametrization of a curve $\boldsymbol{r}(C)$ on S.

Moreover, for each $t \in [a, b]$, the vectors

$$\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}, \quad \left(\frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v}\right)\Bigg|_{(u,v)=\boldsymbol{\alpha}(t)}$$

are orthogonal to each other.

Proof. We adopt the notation in the statement of the proposition. Then the composite $\boldsymbol{\rho}$ is given by

$$\boldsymbol{\rho}(t) = X(u(t), v(t))\boldsymbol{i} + Y(u(t), v(t))\boldsymbol{j} + Z(u(t), v(t))\boldsymbol{k}.$$

Since $\boldsymbol{\alpha}$ and $\boldsymbol{r}$ are both continuously differentiable, $\boldsymbol{\rho}$ is also continuously differentiable. Thus $\boldsymbol{r}(C)$ is a smooth curve. Differentiating $\boldsymbol{\rho}$ with respect to t, we have

$$\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t} = u'(t)\left(\frac{\partial X}{\partial u}\boldsymbol{i} + \frac{\partial Y}{\partial u}\boldsymbol{j} + \frac{\partial Z}{\partial u}\boldsymbol{k}\right) + v'(t)\left(\frac{\partial X}{\partial v}\boldsymbol{i} + \frac{\partial Y}{\partial v}\boldsymbol{j} + \frac{\partial Z}{\partial v}\boldsymbol{k}\right) = u'(t)\frac{\partial \boldsymbol{r}}{\partial u} + v'(t)\frac{\partial \boldsymbol{r}}{\partial v}.$$

Note the fundamental vector product $\frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v}$ is normal to both $\frac{\partial \boldsymbol{r}}{\partial u}$ and $\frac{\partial \boldsymbol{r}}{\partial v}$. Thus $\frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v}$ is normal to $\frac{\mathrm{d}\boldsymbol{\rho}}{\mathrm{d}t}$. □

Remark 16.1.10 By Proposition 16.1.9, the fundamental vector product $\frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v}$ is said to be normal to the surface $S = \boldsymbol{r}(T)$. By definition, at each regular point $P \in \boldsymbol{r}(T)$,

$$\left(\frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v}\right)\Bigg|_P \neq \boldsymbol{0},$$

and the plane passing through P with normal vector $\left(\frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v}\right)\Big|_P$ is called the **tangent plane** to $\boldsymbol{r}(T)$ at P with respect to $(\boldsymbol{r}, T)$.

Definition 16.1.11 Suppose T is a region in the uv-plane and $S \subseteq \mathbb{R}^3$ be a parametric surface with parametrization $(\boldsymbol{r} : \widetilde{T} \to \mathbb{R}^3, T)$. The **area** of S is defined by the double integral

$$a(S) = \iint_T \left\|\frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v}\right\| \mathrm{d}u\mathrm{d}v.$$

Moreover, if the boundary of T is content zero, then we only require differentiability of $\boldsymbol{r}$ on $\operatorname{int} T$ and define $a(S)$ as the corresponding double integral over $\operatorname{int} T$.

Remark 16.1.12 It's possible that the area of a parametric surface $S = \boldsymbol{r}(T)$ may not exist. However, if T is a bounded region and $\boldsymbol{r}$ is continuously differentiable, then $a(S)$ exists. Later we'll see that under some natural conditions, the area of a parametric surface is independent of the choice of parametrization $\boldsymbol{r}$.

Example 16.1.13 Following the parametrization of the surface S in Example 16.1.6, we have

$$a(S) = \iint_T \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\,\mathrm{d}x\mathrm{d}y.$$

In particular, when S lies in a plane parallel to the xy-plane, i.e., $f(x,y) = c$ for some constant c, we have

$$a(S) = \iint_T 1\mathrm{d}x\mathrm{d}y,$$

which agrees with our definition of area for a region in the plane.

Example 16.1.14 Following the parametrization of the semisphere S in Example 16.1.7, we have

$$a(S) = \iint_T a^2 \sin\theta \mathrm{d}\theta\mathrm{d}\phi = a^2 \int_0^{\pi/2} \sin\theta \mathrm{d}\theta \int_0^{2\pi} \mathrm{d}\phi = 2\pi a^2.$$

Example 16.1.15 Let $a > b > 0$ and consider the torus S with vector equation

$$\boldsymbol{r}(u,v) = (a + b\cos v)\cos u\boldsymbol{i} + (a + b\cos v)\sin u\boldsymbol{j} + b\sin v\boldsymbol{k}, \quad (u,v) \in \mathbb{R}^2,$$

where $T = \{(u,v) : 0 \le u < 2\pi, 0 \le v < 2\pi\}$. Then the fundamental vector product of $\boldsymbol{r}$ is

$$\frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v} = b(a + b\cos v)\big(\cos v\cos u, \cos v\sin u, \sin v\big).$$

Thus the area of S is given by

$$a(S) = \iint_T b(a + b\cos v)\mathrm{d}u\mathrm{d}v = b\int_0^{2\pi} \mathrm{d}u \int_0^{2\pi} (a + b\cos v)\mathrm{d}v = 4\pi^2 ab.$$

2 Surface Integrals

Definition 16.2.1 Let S be a parametric surface with paramatrization $(\boldsymbol{r} : \widetilde{T} \to \mathbb{R}^3, T)$ where $\boldsymbol{r}$ is differentiable. Let $f : S \to \mathbb{R}$ be a bounded function. Then the **surface integral of** f **over** S is defined by the double integral

$$\iint_S f(x,y,z)\mathrm{d}S := \iint_T f(\boldsymbol{r}(u,v)) \left\| \frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v} \right\| \mathrm{d}u\mathrm{d}v.$$

Morevoer, if the boundary of T is content zero, then we only require differentiability of $\boldsymbol{r}$ on $\operatorname{int} T$ and define the surface integral as the corresponding double integral over $\operatorname{int} T$.

Notation 16.2.2 The surface integral of f over S is also denoted by

$$\iint_S f\mathrm{d}S.$$

Remark 16.2.3 To show that the definition of surface integral is well-defined, we need to verify that the integral does not depend on the choice of parametrization.

Definition 16.2.4 Let $(\boldsymbol{r}:\widetilde{T}\to\mathbb{R}^3, T)$ and $(\boldsymbol{s}:\widetilde{U}\to\mathbb{R}^3, U)$ be two parametrizations of the same surface $S=\boldsymbol{r}(T)=\boldsymbol{s}(U)$ such that $\boldsymbol{r}$ and $\boldsymbol{s}$ are both differentiable functions. If there exists a bijective and continuously differentiable function $\boldsymbol{\phi}:\widetilde{U}\to\widetilde{T}$ with nonvanishing Jacobian determinant such that $\boldsymbol{s}=\boldsymbol{r}\circ\boldsymbol{\phi}$ and $\boldsymbol{\phi}(U)=T$, then the parametrizations $(\boldsymbol{r},T)$ and $(\boldsymbol{s},U)$ are said to be **smoothly equivalent**.

Theorem 16.2.5 Let $(\boldsymbol{r}:\widetilde{T}\to\mathbb{R}^3, T)$ and $(\boldsymbol{s}:\widetilde{U}\to\mathbb{R}^3, U)$ be two parametrizations of the same surface $S=\boldsymbol{r}(T)=\boldsymbol{s}(U)$ such that $\boldsymbol{r}$ and $\boldsymbol{s}$ are both differentiable functions. Let $f:S\to\mathbb{R}$ be a bounded function. Suppose $(\boldsymbol{r},T)$ and $(\boldsymbol{s},U)$ are smoothly equivalent. If one of the double integrals

$$\iint_T f(\boldsymbol{r}(u,v))\left\|\frac{\partial\boldsymbol{r}}{\partial u}\times\frac{\partial\boldsymbol{r}}{\partial v}\right\|\mathrm{d}u\mathrm{d}v,\quad\text{or}\quad\iint_U f(\boldsymbol{s}(\xi,\eta))\left\|\frac{\partial\boldsymbol{s}}{\partial\xi}\times\frac{\partial\boldsymbol{s}}{\partial\eta}\right\|\mathrm{d}\xi\mathrm{d}\eta$$

exists, then both of them exist and they are equal.

Proof. By definition, there exists a bijective and continuously differentiable function $\boldsymbol{\phi}:\widetilde{U}\to\widetilde{T}$ with nonvanishing Jacobian determinant such that $\boldsymbol{s}=\boldsymbol{r}\circ\boldsymbol{\phi}$ and $\boldsymbol{\phi}(U)=T$. Writing $\boldsymbol{\phi}(\xi,\eta)=(u(\xi,\eta),v(\xi,\eta))$, we have

$$\begin{cases}\dfrac{\partial\boldsymbol{s}}{\partial\xi}&=\dfrac{\partial\boldsymbol{r}}{\partial u}\dfrac{\partial u}{\partial\xi}+\dfrac{\partial\boldsymbol{r}}{\partial v}\dfrac{\partial v}{\partial\xi},\\ \dfrac{\partial\boldsymbol{s}}{\partial\eta}&=\dfrac{\partial\boldsymbol{r}}{\partial u}\dfrac{\partial u}{\partial\eta}+\dfrac{\partial\boldsymbol{r}}{\partial v}\dfrac{\partial v}{\partial\eta}.\end{cases}$$

Thus the fundamental vector product of $\boldsymbol{s}$ is given by

$$\frac{\partial\boldsymbol{s}}{\partial\xi}\times\frac{\partial\boldsymbol{s}}{\partial\eta}=\left(\frac{\partial u}{\partial\xi}\frac{\partial v}{\partial\eta}-\frac{\partial u}{\partial\eta}\frac{\partial v}{\partial\xi}\right)\left(\frac{\partial\boldsymbol{r}}{\partial u}\times\frac{\partial\boldsymbol{r}}{\partial v}\right).$$

Since the Jacobian determinant of $\boldsymbol{\phi}$ is nonvanishing, we have

$$\left\|\frac{\partial\boldsymbol{s}}{\partial\xi}\times\frac{\partial\boldsymbol{s}}{\partial\eta}\right\|=\left|\frac{\partial u}{\partial\xi}\frac{\partial v}{\partial\eta}-\frac{\partial u}{\partial\eta}\frac{\partial v}{\partial\xi}\right|\left\|\frac{\partial\boldsymbol{r}}{\partial u}\times\frac{\partial\boldsymbol{r}}{\partial v}\right\|=\left|\frac{\partial(u,v)}{\partial(\xi,\eta)}\right|\left\|\frac{\partial\boldsymbol{r}}{\partial u}\times\frac{\partial\boldsymbol{r}}{\partial v}\right\|.$$

Hence, by the change of variables formula for double integrals, if one of the following integrals exists, then both of them exist and they are equal:

$$\begin{aligned}&\iint_T f(\boldsymbol{r}(u,v))\left\|\frac{\partial\boldsymbol{r}}{\partial u}\times\frac{\partial\boldsymbol{r}}{\partial v}\right\|\mathrm{d}u\mathrm{d}v\\ =&\iint_U f(\boldsymbol{r}(u(\xi,\eta),v(\xi,\eta)))\left\|\frac{\partial\boldsymbol{r}}{\partial u}\times\frac{\partial\boldsymbol{r}}{\partial v}\right\|\left|\frac{\partial(u,v)}{\partial(\xi,\eta)}\right|\mathrm{d}\xi\mathrm{d}\eta\\ =&\iint_U f(\boldsymbol{s}(\xi,\eta))\left\|\frac{\partial\boldsymbol{s}}{\partial\xi}\times\frac{\partial\boldsymbol{s}}{\partial\eta}\right\|\mathrm{d}\xi\mathrm{d}\eta.\end{aligned}$$

□

Remark 16.2.6 In above discussions, we define the surface integral of a scalar function over a parametric surface. We can also define the surface integral of a vector field over a parametric surface.

Definition 16.2.7 Let $(\boldsymbol{r} : \widetilde{T} \to \mathbb{R}^3, T)$ be a simple and smooth parametrization of a parametric surface S. Then restricting $\boldsymbol{r}$ to T gives a bijective function $\boldsymbol{r}\big|_T : T \to S$. Since $\boldsymbol{r}\big|_T$ is bijective, we define the **unit normal vector** of S at $\boldsymbol{r}(u,v)$ by

$$\boldsymbol{n}(x,y,z) = \left.\frac{\dfrac{\partial \boldsymbol{r}}{\partial u} \times \dfrac{\partial \boldsymbol{r}}{\partial v}}{\left\| \dfrac{\partial \boldsymbol{r}}{\partial u} \times \dfrac{\partial \boldsymbol{r}}{\partial v} \right\|}\right|_{(u,v)=\boldsymbol{r}^{-1}(x,y,z)}$$

having the same direction as the fundamental vector product.
Let $\boldsymbol{F} : \widetilde{S} \to \mathbb{R}^3$ be a continuous vector function where $\widetilde{S}$ is an open subset of $\mathbb{R}^3$ containing S. Then the **surface integral of $\boldsymbol{F}$ over** S, also called the **flux of $\boldsymbol{F}$ through** S, is defined by the scalar surface integral

$$\iint_S \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{S} := \iint_S (\boldsymbol{F} \cdot \boldsymbol{n})\, \mathrm{d}S.$$

Remark 16.2.8 We write the components of $\boldsymbol{F}$ and $\boldsymbol{r}$ as follows:

$$\begin{aligned} \boldsymbol{F}(x,y,z) &= P(x,y,z)\boldsymbol{i} + Q(x,y,z)\boldsymbol{j} + R(x,y,z)\boldsymbol{k}, \\ \boldsymbol{r}(u,v) &= X(u,v)\boldsymbol{i} + Y(u,v)\boldsymbol{j} + Z(u,v)\boldsymbol{k}. \end{aligned}$$

Then the surface integral of $\boldsymbol{F}$ over S can be written as

$$\iint_S \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{S} = \iint_T \left(P\frac{\partial(Y,Z)}{\partial(u,v)} + Q\frac{\partial(Z,X)}{\partial(u,v)} + R\frac{\partial(X,Y)}{\partial(u,v)} \right) \mathrm{d}u\mathrm{d}v.$$

It is common to write

$$\begin{aligned} \iint_T P(\boldsymbol{r}(u,v))\frac{\partial(Y,Z)}{\partial(u,v)}\mathrm{d}u\mathrm{d}v &=: \iint_S P\mathrm{d}y \wedge \mathrm{d}z, \\ \iint_T Q(\boldsymbol{r}(u,v))\frac{\partial(Z,X)}{\partial(u,v)}\mathrm{d}u\mathrm{d}v &=: \iint_S Q\mathrm{d}z \wedge \mathrm{d}x, \\ \iint_T R(\boldsymbol{r}(u,v))\frac{\partial(X,Y)}{\partial(u,v)}\mathrm{d}u\mathrm{d}v &=: \iint_S R\mathrm{d}x \wedge \mathrm{d}y. \end{aligned}$$

Thus, we have

$$\iint_S \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{S} = \iint_S P\mathrm{d}y \wedge \mathrm{d}z + Q\mathrm{d}z \wedge \mathrm{d}x + R\mathrm{d}x \wedge \mathrm{d}y.$$

Example 16.2.9 Following the parametrization of the surface S in Example 16.1.6, we have

$$\frac{\partial \boldsymbol{r}}{\partial u} \times \frac{\partial \boldsymbol{r}}{\partial v} = -\frac{\partial f}{\partial x}\boldsymbol{i} - \frac{\partial f}{\partial y}\boldsymbol{j} + \boldsymbol{k}.$$

Let $\widetilde{S}$ be an open set of $\mathbb{R}^3$ containing S, and let $\boldsymbol{F} : \widetilde{S} \to \mathbb{R}^3$ be a continuous vector field given by

$$\boldsymbol{F}(x,y,z) = P(x,y,z)\boldsymbol{i} + Q(x,y,z)\boldsymbol{j} + R(x,y,z)\boldsymbol{k}.$$

Then the surface integral of $\boldsymbol{F}$ over S is given by

$$\iint_S \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{S} = \iint_T \left(-P(x,y,f(x,y))\frac{\partial f}{\partial x} - Q(x,y,f(x,y))\frac{\partial f}{\partial y} + R(x,y,f(x,y)) \right) \mathrm{d}x\mathrm{d}y.$$

3 Curl and Divergence of Vector Fields

Definition 16.3.1 Let $\boldsymbol{F}$ be a differentiable vector field given by

$$\boldsymbol{F}(x, y, z) = P(x, y, z)\boldsymbol{i} + Q(x, y, z)\boldsymbol{j} + R(x, y, z)\boldsymbol{k}.$$

Then the **curl** of $\boldsymbol{F}$ is defined by

$$\operatorname{curl} \boldsymbol{F} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\boldsymbol{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\boldsymbol{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\boldsymbol{k}.$$

The **divergence** of $\boldsymbol{F}$ is defined by

$$\operatorname{div} \boldsymbol{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

Remark 16.3.2 It is often convenient to introduce the differential operator ∇ defined by

$$\nabla = \frac{\partial}{\partial x}\boldsymbol{i} + \frac{\partial}{\partial y}\boldsymbol{j} + \frac{\partial}{\partial z}\boldsymbol{k}.$$

Then we can write

$$\operatorname{curl} \boldsymbol{F} = \nabla \times \boldsymbol{F}, \quad \operatorname{div} \boldsymbol{F} = \nabla \cdot \boldsymbol{F}.$$

Here the dot and cross products are interpreted formally using the standard rules of vector algebra, treating ∇ as a vector of differential operators acting on the components of $\boldsymbol{F}$. In particular, we have

$$\operatorname{curl} \boldsymbol{F} = \det \begin{pmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{pmatrix}.$$

Proposition 16.3.3 Let $\boldsymbol{F}$ be a differentiable vector field given by

$$\boldsymbol{F}(x, y, z) = P(x, y, z)\boldsymbol{i} + Q(x, y, z)\boldsymbol{j} + R(x, y, z)\boldsymbol{k}.$$

Let

$$J_{\boldsymbol{F}} = \begin{pmatrix} \frac{\partial P}{\partial x} & \frac{\partial P}{\partial y} & \frac{\partial P}{\partial z} \\ \frac{\partial Q}{\partial x} & \frac{\partial Q}{\partial y} & \frac{\partial Q}{\partial z} \\ \frac{\partial R}{\partial x} & \frac{\partial R}{\partial y} & \frac{\partial R}{\partial z} \end{pmatrix}$$

be the Jacobian matrix of $\boldsymbol{F}$. Then we have

$$\operatorname{div} \boldsymbol{F} = \operatorname{tr}(J_{\boldsymbol{F}}).$$

Moreover, if $J_{\boldsymbol{F}}$ is symmetric, then $\operatorname{curl} \boldsymbol{F} = \boldsymbol{0}$.

Example 16.3.4 Let

$$S = \{(x,y,z) \in \mathbb{R}^3 : x^2 + y^2 \neq 0\}$$

and $\boldsymbol{F} : S \to \mathbb{R}^3$ be a vector field given by

$$\boldsymbol{F}(x,y,z) = -\frac{y}{x^2+y^2}\boldsymbol{i} + \frac{x}{x^2+y^2}\boldsymbol{j}.$$

Note S is an open subset of $\mathbb{R}^3$ and $\boldsymbol{F}$ is a differentiable vector field on S. Then the Jacobian matrix of $\boldsymbol{F}$ is given by

$$J_{\boldsymbol{F}} = \begin{pmatrix} \frac{2xy}{(x^2+y^2)^2} & \frac{y^2-x^2}{(x^2+y^2)^2} & 0 \\ \frac{y^2-x^2}{(x^2+y^2)^2} & -\frac{2xy}{(x^2+y^2)^2} & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus we have $\operatorname{div} \boldsymbol{F} = 0$ and $\operatorname{curl} \boldsymbol{F} = \boldsymbol{0}$.

Proposition 16.3.5 Let $\boldsymbol{F}$ and $\boldsymbol{G}$ be two twicely differentiable vector fields and f be a twice continuously differentiable scalar field. Then we have

(1) for any $\alpha, \beta \in \mathbb{R}$,

$$\operatorname{curl}(\alpha\boldsymbol{F} + \beta\boldsymbol{G}) = \alpha \operatorname{curl} \boldsymbol{F} + \beta \operatorname{curl} \boldsymbol{G}, \quad \operatorname{div}(\alpha\boldsymbol{F} + \beta\boldsymbol{G}) = \alpha \operatorname{div} \boldsymbol{F} + \beta \operatorname{div} \boldsymbol{G};$$

(2)

$$\operatorname{div}(f\boldsymbol{F}) = f \operatorname{div} \boldsymbol{F} + \boldsymbol{F} \cdot \nabla f;$$

(3)

$$\operatorname{curl}(\nabla f) = \boldsymbol{0}, \quad \operatorname{div}(\nabla \times \boldsymbol{F}) = 0.$$

(4)

$$\operatorname{div}(\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} =: \Delta f,$$

where Δ is called the **Laplacian** operator.

4 Stoke's Theorem

Theorem 16.4.1 (Stoke's Theorem) Let $T \subseteq \mathbb{R}^2$ where $\operatorname{int} T$ is the inner region of a piecewise smooth Jordan curve Γ. Suppose S is a simple smooth parametric surface with parametrization $\boldsymbol{r} : T \to \mathbb{R}^3$ such that $\boldsymbol{r}$ is a twice continuously differentiable function on an open set $\widetilde{T} \supseteq T \cup \Gamma$ and Γ is the boundary of T. Let $C = \boldsymbol{r}(\Gamma)$ be the boundary of S and $\boldsymbol{F} : S \to \mathbb{R}^3$ be a continuously differentiable function on an open set $\widetilde{S} \supseteq \mathbb{R}^3$ with $\widetilde{S} \supseteq \boldsymbol{r}(T \cup \Gamma)$. Suppose Γ is parametrized by $\boldsymbol{\gamma} : [a,b] \to \Gamma$, traversing the Jordan curve Γ in the counterclockwise direction, and C is parametrized by $\boldsymbol{\alpha} = \boldsymbol{r}\big|_{\Gamma} \circ \boldsymbol{\gamma} : [a,b] \to \mathbb{R}^3$. Then we have

$$\iint_S (\nabla \times \boldsymbol{F}) \cdot \mathrm{d}\boldsymbol{S} = \oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{\alpha}.$$

Proof. We write the components of $\boldsymbol{F}$ as

$$\boldsymbol{F}(x,y,z) = P(x,y,z)\boldsymbol{i} + Q(x,y,z)\boldsymbol{j} + R(x,y,z)\boldsymbol{k}.$$

Then the surface integral of $\nabla \times \boldsymbol{F}$ over S is given by

$$\iint_S \nabla \times \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{S} = \iint_S \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right) \mathrm{d}y \wedge \mathrm{d}z + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right) \mathrm{d}z \wedge \mathrm{d}x + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \mathrm{d}x \wedge \mathrm{d}y.$$

Moreover, the line integral of $\boldsymbol{F}$ over C is given by

$$\oint_C \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{\alpha} = \int_C P\mathrm{d}x + Q\mathrm{d}y + R\mathrm{d}z.$$

Therefore, it suffices to show that

$$\begin{cases} \displaystyle\iint_S -\frac{\partial P}{\partial y}\mathrm{d}x \wedge \mathrm{d}y + \frac{\partial P}{\partial z}\mathrm{d}z \wedge \mathrm{d}x = \int_C P\mathrm{d}x \\ \displaystyle\iint_S -\frac{\partial Q}{\partial z}\mathrm{d}y \wedge \mathrm{d}z + \frac{\partial Q}{\partial x}\mathrm{d}x \wedge \mathrm{d}y = \int_C Q\mathrm{d}y \\ \displaystyle\iint_S -\frac{\partial R}{\partial x}\mathrm{d}z \wedge \mathrm{d}x + \frac{\partial R}{\partial y}\mathrm{d}y \wedge \mathrm{d}z = \int_C R\mathrm{d}z \end{cases}$$

and add them together. Note all of these integrals exist because

- $T \cup \Gamma \subseteq \widetilde{T}$, where $\boldsymbol{r}$ is a twice continuously differentiable function on $\widetilde{T}$ and $T \cup \Gamma$ is compact;
- $\boldsymbol{r}(T \cup \Gamma) \subseteq \widetilde{S}$, where $\boldsymbol{F}$ is a continuously differentiable function on $\widetilde{S}$ and $\boldsymbol{r}(T \cup \Gamma)$ is compact.

We only show the first equality holds since the other two are analogous. Note we can write the surface integral over S as a double integral over T and apply Green's theorem to write the double integral as a line integral over Γ. First, we have

$$\iint_S -\frac{\partial P}{\partial y}\mathrm{d}x \wedge \mathrm{d}y + \frac{\partial P}{\partial z}\mathrm{d}z \wedge \mathrm{d}x = \iint_T -\frac{\partial P}{\partial y}\frac{\partial(X,Y)}{\partial(u,v)} + \frac{\partial P}{\partial z}\frac{\partial(Z,X)}{\partial(u,v)}\mathrm{d}u\mathrm{d}v.$$

Let

$$U = \boldsymbol{r}^{-1}(\widetilde{S}) := \{(u,v) \in \widetilde{T} : \boldsymbol{r}(u,v) \in \widetilde{S}\}.$$

Since $\boldsymbol{r} : \widetilde{T} \to \mathbb{R}^3$ is continuous and $\widetilde{S} \supseteq \boldsymbol{r}(T \cup \Gamma)$, we know $U \subseteq \mathbb{R}^2$ is open and $U \supseteq T \cup \Gamma$. We restrict $\boldsymbol{r}$ to U:

$$\begin{aligned} \boldsymbol{r}\big|_U : \quad U &\longrightarrow \widetilde{S} \subseteq \mathbb{R}^3 \\ (u,v) &\longmapsto (X(u,v), Y(u,v), Z(u,v)) \end{aligned}$$

and define

$$\begin{aligned} p := P \circ \boldsymbol{r}\big|_U : \quad U &\longrightarrow \mathbb{R} \\ (u,v) &\longmapsto P(X(u,v), Y(u,v), Z(u,v)) \end{aligned}$$

which is a continuously differentiable function on U. We claim that

$$-\frac{\partial P}{\partial y}\frac{\partial(X,Y)}{\partial(u,v)} + \frac{\partial P}{\partial z}\frac{\partial(Z,X)}{\partial(u,v)} = \frac{\partial}{\partial u}\left(p\frac{\partial X}{\partial v}\right) - \frac{\partial}{\partial v}\left(p\frac{\partial X}{\partial u}\right).$$

To see this, we first expand the right-hand side:

$$\frac{\partial}{\partial u}\left(p\frac{\partial X}{\partial v}\right)-\frac{\partial}{\partial v}\left(p\frac{\partial X}{\partial u}\right)=\frac{\partial p}{\partial u}\frac{\partial X}{\partial v}-\frac{\partial p}{\partial v}\frac{\partial X}{\partial u}.$$

Then we apply the chain rule to write $\dfrac{\partial p}{\partial u}$ and $\dfrac{\partial p}{\partial v}$ as follows:

$$\begin{cases}\dfrac{\partial p}{\partial u}=\dfrac{\partial P}{\partial x}\dfrac{\partial X}{\partial u}+\dfrac{\partial P}{\partial y}\dfrac{\partial Y}{\partial u}+\dfrac{\partial P}{\partial z}\dfrac{\partial Z}{\partial u},\\ \dfrac{\partial p}{\partial v}=\dfrac{\partial P}{\partial x}\dfrac{\partial X}{\partial v}+\dfrac{\partial P}{\partial y}\dfrac{\partial Y}{\partial v}+\dfrac{\partial P}{\partial z}\dfrac{\partial Z}{\partial v}.\end{cases}$$

Substituting these expressions into the previous expansion, we have

$$\begin{aligned}\frac{\partial}{\partial u}\left(p\frac{\partial X}{\partial v}\right)-\frac{\partial}{\partial v}\left(p\frac{\partial X}{\partial u}\right)&=\left(\frac{\partial P}{\partial x}\frac{\partial X}{\partial u}+\frac{\partial P}{\partial y}\frac{\partial Y}{\partial u}+\frac{\partial P}{\partial z}\frac{\partial Z}{\partial u}\right)\frac{\partial X}{\partial v}-\left(\frac{\partial P}{\partial x}\frac{\partial X}{\partial v}+\frac{\partial P}{\partial y}\frac{\partial Y}{\partial v}+\frac{\partial P}{\partial z}\frac{\partial Z}{\partial v}\right)\frac{\partial X}{\partial u}\\&=\frac{\partial P}{\partial y}\left(\frac{\partial Y}{\partial u}\frac{\partial X}{\partial v}-\frac{\partial Y}{\partial v}\frac{\partial X}{\partial u}\right)+\frac{\partial P}{\partial z}\left(\frac{\partial Z}{\partial u}\frac{\partial X}{\partial v}-\frac{\partial Z}{\partial v}\frac{\partial X}{\partial u}\right)\\&=-\frac{\partial P}{\partial y}\frac{\partial(X,Y)}{\partial(u,v)}+\frac{\partial P}{\partial z}\frac{\partial(Z,X)}{\partial(u,v)}.\end{aligned}$$

Hence, we have

$$\iint_S -\frac{\partial P}{\partial y}\mathrm{d}x\wedge\mathrm{d}y+\frac{\partial P}{\partial z}\mathrm{d}z\wedge\mathrm{d}x=\iint_T\left(\frac{\partial}{\partial u}\left(p\frac{\partial X}{\partial v}\right)-\frac{\partial}{\partial v}\left(p\frac{\partial X}{\partial u}\right)\right)\mathrm{d}u\mathrm{d}v.$$

Note p is continuously differentiable on U and X is twice continuously differentiable on U. Therefore, $p\dfrac{\partial X}{\partial v}$ and $p\dfrac{\partial X}{\partial u}$ are continuously differentiable on $U\supseteq T\cup\Gamma$. Thus, by Green's theorem, we have

$$\iint_T\left(\frac{\partial}{\partial u}\left(p\frac{\partial X}{\partial v}\right)-\frac{\partial}{\partial v}\left(p\frac{\partial X}{\partial u}\right)\right)\mathrm{d}u\mathrm{d}v=\int_\Gamma p\frac{\partial X}{\partial v}\mathrm{d}v+p\frac{\partial X}{\partial u}\mathrm{d}u.$$

Finally, we write $\boldsymbol{\gamma}=u\boldsymbol{i}+v\boldsymbol{j}$ and by definition of the line integral and chain rule, we have

$$\int_\Gamma p\frac{\partial X}{\partial v}\mathrm{d}v+p\frac{\partial X}{\partial u}\mathrm{d}u=\int_a^b p(\boldsymbol{\gamma}(t))\left(\left.\frac{\partial X}{\partial u}\right|_{\boldsymbol{\gamma}(t)}u'(t)+\left.\frac{\partial X}{\partial v}\right|_{\boldsymbol{\gamma}(t)}v'(t)\right)\mathrm{d}t=\int_a^b p(\boldsymbol{\gamma}(t))\frac{\mathrm{d}}{\mathrm{d}t}(X\circ\boldsymbol{\gamma})(t)\mathrm{d}t.$$

Moreover, since $\boldsymbol{\alpha}=\boldsymbol{r}\big|_\Gamma\circ\boldsymbol{\gamma}$ on $[a,b]$ and $p=P\circ\boldsymbol{r}$ on Γ, we have

$$p\circ\boldsymbol{\gamma}=P\circ\boldsymbol{r}\big|_\Gamma\circ\boldsymbol{\gamma}=P\circ\boldsymbol{\alpha},$$

and thus

$$\int_a^b p(\boldsymbol{\gamma}(t))\frac{\mathrm{d}}{\mathrm{d}t}(X\circ\boldsymbol{\gamma})(t)\mathrm{d}t=\int_a^b P(\boldsymbol{\alpha}(t))\frac{\mathrm{d}}{\mathrm{d}t}(X\circ\boldsymbol{\gamma})(t)\mathrm{d}t.$$

Furthermore, $\boldsymbol{\alpha}:[a,b]\to C$ parametrizes C and $X\circ\gamma$ is precisely the x-component of α. Hence, we have

$$\int_a^b P(\boldsymbol{\alpha}(t))\frac{\mathrm{d}}{\mathrm{d}t}(X\circ\boldsymbol{\gamma})(t)\mathrm{d}t=\int_C P\mathrm{d}x.$$

Therefore, we have shown that

$$\iint_S -\frac{\partial P}{\partial y}\mathrm{d}x \wedge \mathrm{d}y + \frac{\partial P}{\partial z}\mathrm{d}z \wedge \mathrm{d}x = \int_C P\mathrm{d}x.$$

Adding the other two equalities together, we finish the proof of Stoke's theorem. □

Example 16.4.2 Let C be the curve of intersection of the sphere $x^2+y^2+z^2=a^2$ and the plane $x+y+z=0$. We want to evaluate the line integral

$$\int_C y\mathrm{d}x + z\mathrm{d}y + x\mathrm{d}z.$$

Let

$$\begin{array}{rccc} \boldsymbol{F}: & \mathbb{R}^3 & \longrightarrow & \mathbb{R}^3 \\ & (x,y,z) & \longmapsto & y\boldsymbol{i}+z\boldsymbol{j}+x\boldsymbol{k} \end{array}$$

We will find a simple smooth parametrization $(\boldsymbol{r}:\widetilde{T}\to\mathbb{R}^3, T)$ of a surface S such that $\partial S = C$ and apply Stoke's theorem to evaluate the line integral.
Note S should be a disk in the $x+y+z=0$ plane. Let $\widetilde{T}=\mathbb{R}^2$ and $T=\{(u,v)\in\mathbb{R}^2 : u^2+v^2\le a^2\}\subseteq\widetilde{T}$. Note T is closed with nonempty connected interior, and its boundary $\Gamma=\{(u,v)\in\mathbb{R}^2 : u^2+v^2=a^2\}$ is a smooth Jordan curve.
We define

$$\begin{array}{rccc} \boldsymbol{r}: & \widetilde{T} & \longrightarrow & \mathbb{R}^3 \\ & (u,v) & \longmapsto & u\boldsymbol{b}_1+v\boldsymbol{b}_2 \end{array}$$

where $\{\boldsymbol{b}_1,\boldsymbol{b}_2\}$ is an orthonormal basis for the plane $x+y+z=0$. In particular, we can take

$$\boldsymbol{b}_1=\frac{1}{\sqrt{6}}(1,1,-2),\quad \boldsymbol{b}_2=\frac{1}{\sqrt{2}}(1,-1,0).$$

Then we have $\boldsymbol{r}(T)=S$ since

$$\|\boldsymbol{r}(u,v)\|^2=u^2+v^2\le a^2.$$

Letting $(x,y,z)=\boldsymbol{r}(u,v)$ be the corresponding point on the plane $x+y+z=0$ for any $(u,v)\in\widetilde{T}$, we have

$$(x,y,z)\in S \Leftrightarrow x^2+y^2+z^2\le a^2 \Leftrightarrow u^2+v^2\le a^2 \Leftrightarrow (u,v)\in T.$$

Moreover, we have

$$\frac{\partial\boldsymbol{r}}{\partial u}\times\frac{\partial\boldsymbol{r}}{\partial v}=-\frac{1}{\sqrt{3}}(1,1,1).$$

Thus, $(\boldsymbol{r}:\widetilde{T}\to\mathbb{R}^3,T)$ is a simple smooth parametrization of S. We parametrize the boundary Γ of T by

$$\begin{array}{rccc} \boldsymbol{\gamma}: & [0,2\pi] & \longrightarrow & \Gamma \\ & t & \longmapsto & a\cos t\boldsymbol{i}+a\sin t\boldsymbol{j}. \end{array}$$

Then we parametrize C by $\boldsymbol{\alpha}=\boldsymbol{r}\big|_\Gamma\circ\boldsymbol{\gamma}:[0,2\pi]\to C$ and we have $r(\partial T)=C$. The curl of $\boldsymbol{F}$ is given by

$$\nabla\times\boldsymbol{F}=\left(\frac{\partial}{\partial y}x-\frac{\partial}{\partial z}z\right)\boldsymbol{i}+\left(\frac{\partial}{\partial z}y-\frac{\partial}{\partial x}x\right)\boldsymbol{j}+\left(\frac{\partial}{\partial x}z-\frac{\partial}{\partial y}y\right)\boldsymbol{k}=-\boldsymbol{i}-\boldsymbol{j}-\boldsymbol{k}.$$

Thus, by Stoke's theorem, we have

$$\int_C \boldsymbol{F}\cdot \mathrm{d}\boldsymbol{\alpha} = \iint_S (\nabla\times\boldsymbol{F})\cdot \mathrm{d}\boldsymbol{S} = \int_T (\nabla\times\boldsymbol{F})\big|_{(x,y,z)=\boldsymbol{r}(u,v)}\cdot\left(\frac{\partial \boldsymbol{r}}{\partial u}\times\frac{\partial \boldsymbol{r}}{\partial v}\right)\mathrm{d}u\mathrm{d}v = \int_T \sqrt{3}\mathrm{d}u\mathrm{d}v = \sqrt{3}\pi a^2.$$

5 Orientable and Closed Surfaces

Definition 16.5.1 A subset $S\subseteq\mathbb{R}^3$ is said to be a **piecewise smooth surface** if it can be expressed as a finite union

$$S=\bigcup_{i=1}^{n} S_i$$

where each S_i admits a simple smooth parametrization $(\boldsymbol{r}_i, T_i)$ and the intersection of any two distinct S_i and S_j is either empty, a finite set of points, or a finite number of smooth curves; we call these intersections **common boundaries**.

Remark 16.5.2 There are two parametrizations of the northern hemisphere S of radius $a>0$ in Example 16.1.7. However, neither of them is a simple smooth parametrization of S. We can combine them to make a simple smooth parametrization of S by the following example.

Example 16.5.3 Suppose S is the northern hemisphere of radius $a>0$. Let

$$\cos\theta_0=\frac{b}{a},\quad 0<b<a,\quad 0<\theta_0<\frac{\pi}{2}.$$

We parametrize the top cap S_1 of S by

$$\begin{array}{rccl} \boldsymbol{r}_1: & \widetilde{T}_1 & \longrightarrow & \mathbb{R}^3 \\ & (x,y) & \longmapsto & (x,y,\sqrt{a^2-x^2-y^2}) \end{array},$$

where $\widetilde{T}_1=\{(x,y)\in\mathbb{R}^2: x^2+y^2<a^2\}$ and $T_1=\{(x,y)\in\mathbb{R}^2: x^2+y^2\le b^2\}\subseteq\widetilde{T}_1$. We parametrize the bottom part S_2 by

$$\begin{array}{rccl} \boldsymbol{r}_2: & \widetilde{T}_2 & \longrightarrow & \mathbb{R}^3 \\ & (\theta,\phi) & \longmapsto & (a\sin\theta\cos\phi, a\sin\theta\sin\phi, a\cos\theta) \end{array},$$

where $\widetilde{T}_2=\mathbb{R}^2$ and $T_2=\{(\theta,\phi)\in\mathbb{R}^2: 0\le\theta\le\theta_0, 0\le\phi<2\pi\}\subseteq\widetilde{T}_2$. Then $S=S_1\cup S_2$ is piecewise smooth with respect to $(\boldsymbol{r}_1,T_1)$ and $(\boldsymbol{r}_2,T_2)$.

Definition 16.5.4 Let $S=\bigcup_{i=1}^{n} S_i$ be a piecewise smooth surface. We say S is **orientable** if it is possible to choose simple smooth parametrizations $(\boldsymbol{r}_i, T_i)$ of S_i such that, for every distince pair of adjacent patches S_i and S_j, sharing a common boundary curve $C_{ij}=S_i\cap S_j$, the induced direction on C from S_i is opposite to that induced from S_j.

Example 16.5.5 Note the northern hemisphere is orientable in Example 16.5.3. However, the Möbius strip is not orientable.

Definition 16.5.6 An orientable piecewise smooth surface S is said to be **closed** if its entire outer boundary is empty, i.e., $\partial S = \varnothing$.

6 Gauss's Theorem

Theorem 16.6.1 Let V be a closed and bounded solid in $\mathbb{R}^3$ whose boundary $S = \partial V$ is a piecewise smooth closed surface. Let $\boldsymbol{n}$ be the outer unit normal vector of S pointing away from the inner region V. Suppose $\boldsymbol{F}$ is a continuously differentiable vector field on an open set containing V. Then we have

$$\iint_S \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{S} = \iiint_V \operatorname{div} \boldsymbol{F} \mathrm{d}V.$$

Proof. We only prove the case when V is convex. We write the components of $\boldsymbol{F}$ and $\boldsymbol{n}$ as

$$\begin{cases} \boldsymbol{F}(x,y,z) & = P(x,y,z)\boldsymbol{i} + Q(x,y,z)\boldsymbol{j} + R(x,y,z)\boldsymbol{k}, \\ \boldsymbol{n}(x,y,z) & = \cos(\alpha(x,y,z))\boldsymbol{i} + \cos(\beta(x,y,z))\boldsymbol{j} + \cos(\gamma(x,y,z))\boldsymbol{k}. \end{cases}$$

Then the desired equality can be written as

$$\iiint_V \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) \mathrm{d}V = \iint_S (P\cos\alpha + Q\cos\beta + R\cos\gamma)\, \mathrm{d}S.$$

Therefore, it suffices to show that

$$\begin{cases} \displaystyle\iiint_V \frac{\partial P}{\partial x} \mathrm{d}V = \iint_S P\cos\alpha \mathrm{d}S \\ \displaystyle\iiint_V \frac{\partial Q}{\partial y} \mathrm{d}V = \iint_S Q\cos\beta \mathrm{d}S \\ \displaystyle\iiint_V \frac{\partial R}{\partial z} \mathrm{d}V = \iint_S R\cos\gamma \mathrm{d}S \end{cases}$$

We only prove the third equality since the other two are analogous. Since V is convex, we can write

$$V = \{(x,y,z) \in \mathbb{R}^3 : (x,y) \in D, g_1(x,y) \le z \le g_2(x,y)\}$$

where $D \subseteq \mathbb{R}^2$ is a closed, connected and bounded region and $g_1, g_2 : D \to \mathbb{R}$ are continuous functions. Then we have

$$\iiint_V \frac{\partial R}{\partial z} \mathrm{d}V = \iint_D \left(\int_{g_1(x,y)}^{g_2(x,y)} \frac{\partial R}{\partial z} \mathrm{d}z \right) \mathrm{d}x\mathrm{d}y = \iint_D R(x,y,g_2(x,y)) - R(x,y,g_1(x,y))\mathrm{d}x\mathrm{d}y.$$

Then, we evaluate the surface integral on the right-hand side by deviding S into three parts:

1. the bottom surface S_1, where $z = g_1(x,y)$;
2. the top surface S_2, where $z = g_2(x,y)$;
3. the side surface S_3.

On the bottom surface S_1, we have

$$\boldsymbol{r}_1(x,y) = x\boldsymbol{i} + y\boldsymbol{j} + g_1(x,y)\boldsymbol{k}, \quad (x,y) \in D.$$

Then the fundamental vector product points in the opposite direction of the outer unit normal vector $\boldsymbol{n}$ and we have

$$\iint_{S_1} R\cos\gamma \mathrm{d}S = -\iint_{S_1} R\mathrm{d}x \wedge \mathrm{d}y = -\iint_D R(x,y,g_1(x,y))\mathrm{d}x\mathrm{d}y.$$

On the top surface S_2, we have

$$\boldsymbol{r}_2(x,y) = x\boldsymbol{i} + y\boldsymbol{j} + g_2(x,y)\boldsymbol{k}, \quad (x,y) \in D.$$

Then the fundamental vector product points in the same direction of the outer unit normal vector $\boldsymbol{n}$ and we have

$$\iint_{S_2} R\cos\gamma \mathrm{d}S = \iint_{S_2} R\mathrm{d}x \wedge \mathrm{d}y = \iint_D R(x,y,g_2(x,y))\mathrm{d}x\mathrm{d}y.$$

On S_3, the fundamental vector product is perpendicular to $\boldsymbol{k}$ so $\gamma = \pi/2$ and we have

$$\iint_{S_3} R\cos\gamma \mathrm{d}S = 0.$$

Summing up the three parts, we have

$$\iint_S R\cos\gamma \mathrm{d}S = \iint_D R(x,y,g_2(x,y)) - R(x,y,g_1(x,y))\mathrm{d}x\mathrm{d}y.$$

This completes the proof of Gauss's theorem for convex solids. □

Example 16.6.2 Let $V = [0,1]^3$ be the unit cube in $\mathbb{R}^3$ and $S = \partial V$ be the boundary of V. Let

$$\begin{array}{rccc} \boldsymbol{F}: & \mathbb{R}^3 & \longrightarrow & \mathbb{R}^3 \\ & (x,y,z) & \longmapsto & x^2\boldsymbol{i} + y^2\boldsymbol{j} + z^2\boldsymbol{k} \end{array}$$

Note the surface integral of $\boldsymbol{F}$ over S, $\iint_S \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{S}$, is the sum of the surface integrals over the six faces of the cube. It is quite annoying if we compute the surface integral directly over six faces. However, we can apply Gauss's theorem to compute the surface integral by evaluating the triple integral of $\operatorname{div}\boldsymbol{F}$ over V. We have

$$\operatorname{div}\boldsymbol{F} = \frac{\partial}{\partial x}x^2 + \frac{\partial}{\partial y}y^2 + \frac{\partial}{\partial z}z^2 = 2x + 2y + 2z.$$

Thus, by Gauss's theorem, we have

$$\iint_S \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{S} = \iiint_V \operatorname{div}\boldsymbol{F}\mathrm{d}V = \iiint_{[0,1]^3} 2x + 2y + 2z\mathrm{d}x\mathrm{d}y\mathrm{d}z = 3.$$

Corollary 16.6.3 Let $V(t)$ be the solid sphere of radius $t > 0$ centered at a point $\boldsymbol{a} \in \mathbb{R}^3$ and $S(t) = \partial V(t)$ be the boundary of $V(t)$. Suppose $\boldsymbol{F}$ is a continuously differentiable vector field on some open set containing $V(t)$

for all sufficiently small $t > 0$. Then

$$\operatorname{div} \boldsymbol{F}(\boldsymbol{a}) = \lim_{t \to 0^+} \frac{1}{|V(t)|} \iint_{S(t)} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{S},$$

where $|V(t)|$ is the volume of $V(t)$.

Proof. Write $\varphi = \operatorname{div} \boldsymbol{F}$. Let $\varepsilon > 0$. Then it suffices to show that there exists $\delta > 0$ such that for any $t \in (0, \delta)$, we have

$$\left| \frac{1}{|V(t)|} \iint_{S(t)} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{S} - \varphi(\boldsymbol{a}) \right| < \varepsilon.$$

Since φ is continuous at $\boldsymbol{a}$, there exists $\delta > 0$ such that

$$|\varphi(\boldsymbol{x}) - \varphi(\boldsymbol{a})| < \frac{\varepsilon}{2}, \quad \text{whenever } \|\boldsymbol{x} - \boldsymbol{a}\| < \delta.$$

Then, for any $t \in (0, \delta)$, since $\varphi(\boldsymbol{a})$ is constant, we have

$$\varphi(\boldsymbol{a})|V(t)| = \varphi(\boldsymbol{a}) \iiint_{V(t)} 1 \mathrm{d}V = \iiint_{V(t)} \varphi(\boldsymbol{a}) \mathrm{d}V.$$

By Gauss's theorem we have

$$\iiint_{V(t)} \varphi(\boldsymbol{a}) \mathrm{d}V = \iiint_{V(t)} \varphi(\boldsymbol{x}) + \iiint_{V(t)} \varphi(\boldsymbol{a}) - \varphi(\boldsymbol{x}) \mathrm{d}V = \iint_{S(t)} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{S} + \iiint_{V(t)} \varphi(\boldsymbol{a}) - \varphi(\boldsymbol{x}) \mathrm{d}V$$

and thus

$$\left| \varphi(\boldsymbol{a})|V(t)| - \iint_{S(t)} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{S} \right| = \left| \iiint_{V(t)} \varphi(\boldsymbol{a}) - \varphi(\boldsymbol{x}) \mathrm{d}V \right| \leq \iiint_{V(t)} |\varphi(\boldsymbol{a}) - \varphi(\boldsymbol{x})| \mathrm{d}V \leq |V(t)| \cdot \frac{\varepsilon}{2} < \varepsilon |V(t)|.$$

Dividing both sides by $|V(t)|$, we have

$$\left| \frac{1}{|V(t)|} \iint_{S(t)} \boldsymbol{F} \cdot \mathrm{d}\boldsymbol{S} - \varphi(\boldsymbol{a}) \right| < \varepsilon.$$

This completes the proof. □

Remark 16.6.4 The corollary above gives us a coordinate free formula for the divergence of a vector field at a point. It also suggests the geometric interpretation of divergence: the divergence of a vector field at a point is the infinitesimal net outward flux per unit volume at that point.